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## Long-term hormone therapy for perimenopausal and postmenopausal women (Review)

Bofill Rodriguez M, Yong LN, Mirkov S, Bekos C, Lethaby A, Farquhar C

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## [Intervention Review]

# Long-term hormone therapy for perimenopausal and postmenopausal women

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## ABSTRACT

### Background

Hormone therapy is widely provided to control menopausal symptoms and has been used for the management and prevention of cardiovascular disease, osteoporosis and dementia in older women. This is an updated version of a Cochrane review first published in 2005.

### Objectives

To assess the long-term effects of prolonged use (at least one year) of hormone therapy on mortality, cardiovascular outcomes, cancer, gallbladder disease, fractures and cognition in perimenopausal and postmenopausal women.

### Search methods

We used the Cochrane Gynaecology and Fertility Group Specialised Register, CENTRAL, MEDLINE, three other databases and two trial registers, together with reference checking, citation searching and contact with study authors to identify the studies included in the review. The latest search date was 26 September 2024.

### Selection criteria

We included randomised, double-blind trials in which peri- or postmenopausal women took hormone therapy or placebo for at least one year. We included various oestrogen formulations, with or without progestogens. We focused on studies assessing hormone therapy's effects on long-term clinical outcomes, including death, coronary events and cancer. Hormone therapy's efficacy in managing menopausal symptoms was beyond the scope of this review, and is assessed in other Cochrane reviews.

### Data collection and analysis

Two review authors independently selected studies, assessed risk of bias and extracted data. We calculated risk ratios (RRs) for dichotomous data and mean differences (MDs) for continuous data, along with 95% confidence intervals (CIs). We assessed the certainty of the evidence using GRADE.

### Main results

We included 24 studies – with two newly added in this update – involving 45,660 participants. We derived nearly 70% of the data from two well-conducted studies: the Heart and Estrogen/progestin Replacement Study (HERS 1998) and the large, multi-component Women's Health Initiative research programme, which included two hormone therapy arms (WHI 1998). Across all the studies, most participants

were postmenopausal American women with one or more comorbidities. The mean participant age in most studies was over 60 years. Only one included study focused on perimenopausal women.

We present full results for all included studies with available data in the main review. The results presented below are drawn from WHI 1998, in which the combined hormone therapy arm and the oestrogen-only arm were run concurrently, with women assigned to the appropriate trial based on their uterus status.

One study with 16,608 postmenopausal women with an intact uterus compared combined continuous hormone therapy (conjugated equine oestrogen and medroxyprogesterone acetate) to placebo, and measured outcomes at an average of 5.6 years of follow-up. Based on this study, combined continuous hormone therapy probably makes little to no difference to the risk of a coronary event (RR 1.17, 95% CI 0.95 to 1.44; moderate-certainty evidence). It may increase the risk of stroke (RR 1.39, 95% CI 1.09 to 2.09; low-certainty evidence) and venous thromboembolism (RR 2.03, 95% CI 1.55 to 6.64; low-certainty evidence). Compared to placebo, combined continuous hormone therapy probably increases the risk of breast cancer (RR 1.27, 95% CI 1.03 to 1.56; moderate-certainty evidence) and probably makes little to no difference to the risk of lung cancer (RR 1.06, 95% CI 0.77 to 1.46; moderate-certainty evidence). It may increase gallbladder disease requiring surgery (RR 1.64, 95% CI 1.30 to 2.06; 14,203 participants; low-certainty evidence), and probably reduces the risk of all clinical fractures (RR 0.78, 95% CI 0.71 to 0.86; moderate-certainty evidence).

One study including 10,739 postmenopausal women who had undergone a hysterectomy compared oestrogen-only (conjugated equine oestrogen) hormone therapy to placebo, and measured outcomes at an average of seven years' follow-up. Based on this study, oestrogen-only hormone therapy probably makes little to no difference to the risk of coronary events (RR 0.94, 95% CI 0.78 to 1.13), venous thromboembolism (RR 1.32, 95% CI 1.00 to 1.74) and breast cancer (RR 0.79, 95% CI 0.61 to 1.01), all with moderate-certainty evidence. It may make little to no difference to the risk of lung cancer (RR 1.04, 95% CI 0.73 to 1.48; low-certainty evidence). Oestrogen-only hormone therapy probably increases the risk of stroke (RR 1.33, 95% CI 1.06 to 1.67) and gallbladder disease requiring surgery (RR 1.78, 95% CI 1.42 to 2.24), and probably reduces the risk of all clinical fractures (RR 0.73, 95% CI 0.65 to 0.80), all with moderate-certainty evidence.

We judged most included studies to have a low risk of bias for most domains. The overall certainty of evidence for the main comparisons was moderate. The main limitation was that only about 30% of women were 50 to 59 years old at baseline, the age group most likely to consider hormone therapy for vasomotor symptoms.

### Authors' conclusions

Long-term follow-up of women using hormone therapy suggests that the risk profiles vary between combined hormone therapy and oestrogen-only therapy.

Oestrogen-only hormone therapy probably makes little to no difference to coronary events, and probably increases the risk of stroke and gallbladder disease. It probably makes little to no difference in the risk of breast cancer, and probably reduces the risk of all fractures. Combined hormone therapy may increase the risk of thromboembolism and probably increases the risk of breast cancer.

These results should be interpreted with caution as they are based on one study using oral hormone therapy, which may not represent the risks of the hormone therapy currently used in clinical practice.

## PLAIN LANGUAGE SUMMARY

### In perimenopausal and postmenopausal women, what are the long-term effects of using hormone therapy for at least one year?

#### Key messages

- The effects of hormone therapy on women's long-term health differ depending on whether they take combined hormone therapy (both oestrogen and progestogen) or oestrogen alone.
- We suggest using caution when considering the results of this review, given the evolution in the doses and methods for taking hormone therapy over time. Findings summarised here come from studies that gave hormone therapy as tablets taken by mouth.
- Future studies could explore the long-term effects of hormone therapy in women who are under 50 years, or whose ovaries have stopped working because of medical treatment.

#### What are menopause and perimenopause?

Menopause occurs naturally (usually between 44 and 55 years) when the ovaries stop releasing eggs and reduce (almost stop) hormone production. Menopause is diagnosed when these changes lead to 12 consecutive months without a menstrual period. Menopause also occurs after surgery to remove both ovaries, and does not require this 12-month timeframe.

Perimenopause is the transitional phase before menopause, when hormone levels decline and periods become irregular. It may last several years and causes symptoms like hot flushes, night sweats and mood changes.

Lower hormone levels may affect women's heart, blood vessels and bones, increasing the risk of high cholesterol, weight and fat changes, high blood pressure, diabetes, heart disease, stroke, osteoporosis and reduced strength or mobility.

## What is hormone therapy?

Hormone therapy uses oestrogen, with or without another hormone called progestogen, to relieve menopause symptoms. Women without a womb (uterus) can take oestrogen alone, while those with a womb need both (oestrogen and progestogen) to protect against cancer of the womb lining (endometrial cancer).

Oestrogen may be given orally or through the skin (transdermally), and progestogen may be given orally, through the vagina (transvaginally) or by using an intrauterine device (T-shaped device inside the womb). Combined hormone therapy can be taken daily or cyclically (oestrogen is given daily and progestogen is given for part of the month, producing withdrawal vaginal bleeds).

## What did we want to find out?

We wanted to find out if long-term use of hormone therapy affects women's health in the long term.

## What did we do?

We searched for studies that compared menopausal women taking hormone therapy for at least one year to those taking an inactive 'dummy' medicine (placebo). We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

## What did we find?

We found 24 studies involving 45,660 participants. Most data came from two large, high-quality studies conducted during the 1990s and reported in the early 2000s. The results presented below are drawn from one of those studies. We present full results in the main review.

### Combined continuous hormone therapy compared to placebo

Based on evidence from one study in 16,608 women that measured outcomes at roughly 5.5 years of follow-up, combined continuous hormone therapy **probably**:

- made little or no difference to the risk of having a heart attack or developing lung cancer;
- increased the chance of developing breast cancer (from about 19 to 24 women in every 1000);
- reduced the risk of bone fractures (from about 111 to 87 women in every 1000).

Combined continuous hormone therapy **may** have increased the chances of having:

- a stroke (from about 13 to 18 women in every 1000);
- a blood clot in a vein (from about 10 to 20 women in every 1000);
- gallbladder disease requiring surgery (from about 16 to 27 women in every 1000).

### Oestrogen-only therapy compared to placebo

Based on evidence from one study in 10,739 women who had undergone a hysterectomy (surgery to remove the womb) that measured outcomes at 7 years of follow-up, oestrogen-only hormone therapy **probably**:

- made little or no difference to the risk of having a heart attack, a blood clot in a vein, or developing breast cancer;
- reduced the chance of having a bone fracture (from about 141 to 103 women in every 1000);
- increased the chance of having a stroke (from about 24 to 32 women in every 1000) and gallbladder disease requiring surgery (from about 27 to 47 women in every 1000).

Oestrogen-only hormone therapy **may** have made little or no difference to the risk of developing lung cancer.

We did not have enough data to assess the risk of long-term hormone therapy use in women younger than 50 years.

## What are the limitations of the evidence?

Only about 30% of the women were 50 to 59 years old at baseline, the age group most likely to consider hormone therapy for hot flushes and night sweats. In addition, hormone therapy has changed over time, with new ways to take it, different hormone types, and updated doses. The study providing most of the data gave women hormone therapy in pill form, which may have different risks to the types currently used in clinical practice.

## How current is this evidence?

The evidence is current to September 2024.

## SUMMARY OF FINDINGS

### Summary of findings 1. Combined continuous hormone therapy compared with placebo for peri- and postmenopausal women

#### Combined continuous hormone therapy compared with placebo for perimenopausal and postmenopausal women

**Population:** relatively healthy postmenopausal women

**Setting:** community

**Intervention:** combined continuous hormone therapy (moderate-dose oestrogen) – CEE 0.625 mg + MPA 2.5 mg

**Comparison:** placebo

| Outcomes                                                                  | Illustrative comparative risks* (95% CI) |                                                             | Relative effect (95% CI)         | No. of participants (studies) | Certainty of the evidence (GRADE)   | Analyses |
|---------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|----------------------------------|-------------------------------|-------------------------------------|----------|
|                                                                           | Assumed risk with placebo*               | Corresponding risk with combined continuous hormone therapy |                                  |                               |                                     |          |
| <b>Coronary events (MI or cardiac death)</b><br>Follow-up: mean 5.6 years | <b>20 per 1000</b>                       | <b>23 per 1000</b><br>(19 to 29)                            | <b>RR 1.17</b><br>(0.95 to 1.44) | 16,608<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate<sup>a</sup></b> | 1.19.4   |
| <b>Stroke</b><br>Follow-up: mean 5.6 years                                | <b>13 per 1000</b>                       | <b>18 per 1000</b><br>(14 to 23)                            | <b>RR 1.39</b><br>(1.09 to 1.77) | 16,608<br>(1 study)           | ⊕⊕⊖⊖<br><b>Low<sup>ab</sup></b>     | 1.23.4   |
| <b>Venous thromboembolism (DVT or PE)</b><br>Follow-up: mean 5.6years     | <b>10 per 1000</b>                       | <b>20 per 1000</b><br>(16 to 26)                            | <b>RR 2.03</b><br>(1.55 to 2.64) | 16,608<br>(1 study)           | ⊕⊕⊖⊖<br><b>Low<sup>ab</sup></b>     | 1.30.4   |
| <b>Breast cancer</b><br>Follow-up: median 5.6 years                       | <b>19 per 1000</b>                       | <b>24 per 1000</b><br>(20 to 30)                            | <b>RR 1.27</b> (1.03 to 1.56)    | 16,608<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate<sup>a</sup></b> | 6.3.5    |
| <b>Any lung cancer</b><br>Follow-up: median 5.6 years                     | <b>9 per 1000</b>                        | <b>10 per 1000</b><br>(7 to 13)                             | <b>RR 1.06</b><br>(0.77 to 1.46) | 16,608<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate<sup>a</sup></b> | 6.12.1   |
| <b>Gallbladder disease</b><br>Follow-up: mean 5.6 years                   | <b>16 per 1000</b>                       | <b>27 per 1000</b><br>(21 to 34)                            | <b>RR 1.64</b><br>(1.30 to 2.06) | 14,203<br>(1 study)           | ⊕⊕⊖⊖<br><b>Low<sup>ab</sup></b>     | 6.21.3   |
| <b>All clinical fractures</b><br>Follow-up: mean 5.6 years                | <b>111 per 1000</b>                      | <b>87 per 1000</b><br>(79 to 96)                            | <b>RR 0.78</b><br>(0.71 to 0.86) | 16,608<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate<sup>a</sup></b> | 6.32.3   |

\*The basis for the **assumed risk** is the mean risk in the control group. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**CEE:** conjugated equine oestrogens; **CI:** confidence interval; **DVT:** deep vein thrombosis; **MI:** myocardial infarction; **MPA:** medroxyprogesterone acetate; **PE:** pulmonary embolism; **RR:** risk ratio

#### GRADE Working Group grades of evidence

**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.

**Moderate certainty:** we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

**Low certainty:** our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

**Very low certainty:** we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

<sup>a</sup>Downgraded one level for questionable applicability: only 33% (5520 participants) of the study sample were 50 to 59 years of age at baseline (i.e. the age women are most likely to consider hormone therapy for vasomotor symptoms); mean participant age was 63 years.

<sup>b</sup>Downgraded one level for imprecision (either few events or wide confidence intervals)

## Summary of findings 2. Oestrogen-only hormone therapy compared with placebo for peri- and postmenopausal women

### Oestrogen-only hormone therapy compared with placebo for perimenopausal and postmenopausal women

**Population:** relatively healthy postmenopausal women (without a uterus)

**Setting:** community

**Intervention:** oestrogen-only hormone therapy CEE 0.625 mg (moderate dose)

**Comparison:** placebo

| Outcomes                                                                               | Illustrative comparative risks* (95% CI) |                                                        | Relative effect (95% CI)         | No. of participants (studies) | Certainty of the evidence (GRADE)    | Analyses |
|----------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------|----------------------------------|-------------------------------|--------------------------------------|----------|
|                                                                                        | Assumed risk with placebo                | Corresponding risk with oestrogen-only hormone therapy |                                  |                               |                                      |          |
| <b>Coronary events (MI or cardiac death)</b><br>Follow-up: mean 7.1 years <sup>a</sup> | <b>41 per 1000</b>                       | <b>38 per 1000</b><br>(32 to 46)                       | <b>RR 0.94</b><br>(0.78 to 1.13) | 10,739<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 1.18.3   |
| <b>Stroke</b><br>Follow-up: mean 7.1 years <sup>a</sup>                                | <b>24 per 1000</b>                       | <b>32 per 1000</b><br>(25 to 40)                       | <b>RR 1.33</b><br>(1.06 to 1.67) | 10,739<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 1.22.2   |
| <b>Venous thromboembolism (DVT or PE)</b><br>Follow-up: mean 7.1 years <sup>a</sup>    | <b>16 per 1000</b>                       | <b>21 per 1000</b><br>(16 to 28)                       | <b>RR 1.32</b><br>(1.00 to 1.74) | 10,739<br>(1 study)           | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 1.29.3   |

|                                                                         |                     |                                    |                                  |                     |                                      |        |
|-------------------------------------------------------------------------|---------------------|------------------------------------|----------------------------------|---------------------|--------------------------------------|--------|
| <b>Breast cancer</b><br>Follow-up: mean 7.1 years <sup>a</sup>          | <b>25 per 1000</b>  | <b>20 per 1000</b><br>(15 to 25)   | <b>RR 0.79</b><br>(0.61 to 1.01) | 10,739<br>(1 study) | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 6.1.6  |
| <b>Lung cancer</b><br>Follow-up: mean 7.9 years                         | <b>6 per 1000</b>   | <b>6 per 1000</b><br>(4 to 10)     | <b>RR 1.04</b><br>(0.73 to 1.48) | 10,739<br>(1 study) | ⊕⊕⊖⊖<br><b>Low</b> <sup>bc</sup>     | 6.11.1 |
| <b>Gallbladder disease</b><br>Follow-up: mean 7.1 years <sup>a</sup>    | <b>27 per 1000</b>  | <b>47 per 1000</b><br>(38 to 60)   | <b>RR 1.78</b><br>(1.42 to 2.24) | 8376<br>(1 study)   | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 6.20.2 |
| <b>All clinical fractures</b><br>Follow-up: mean 7.1 years <sup>a</sup> | <b>141 per 1000</b> | <b>103 per 1000</b><br>(92 to 113) | <b>RR 0.73</b><br>(0.65 to 0.80) | 10,739<br>(1 study) | ⊕⊕⊕⊖<br><b>Moderate</b> <sup>b</sup> | 6.29.3 |

\*The basis for the **assumed risk** is the mean risk in the control group. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**CEE:** conjugated equine oestrogens; **CI:** confidence interval; **DVT:** deep vein thrombosis; **MI:** myocardial infarction; **PE:** pulmonary embolism; **RR:** risk ratio

#### GRADE Working Group grades of evidence

**High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.

**Moderate certainty:** we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

**Low certainty:** our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

**Very low certainty:** we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

<sup>a</sup> Median use of CEE 5.9 years ([LaCroix 2011](#)).

<sup>b</sup> Downgraded one level for questionable applicability: only 31% of the study participants were 50 to 59 years of age at baseline (i.e. the age women are most likely to consider hormone therapy for vasomotor symptoms); mean participant age was 63 years.

<sup>c</sup> Downgraded for imprecision (either few events or wide confidence intervals)

## BACKGROUND

### Description of the condition

Menopause – a natural event for middle-aged women – is defined as the cessation of menstrual cycles. It is caused by reduced secretion of the ovarian hormones oestrogen and progesterone (Nelson 2008). Most women experience menopause (the last menstrual period) after a phase of changing ovarian function (the perimenopause or menopause transition) that may last several years and is characterised by irregular menstrual cycles (Greendale 1999; Nelson 2008). Women are said to be postmenopausal when menstruation has ceased for 12 months.

The median age at onset of menopause varies across geographical regions. In Europe, it ranges from about 50 to 53 years, in North America from 50 to 51 years, in Latin America from 44 to 53 years, and in Asia from 42 to 51 years (Palacios 2010). The experience of menopause differs between different races and ethnicities, as well as by menopausal stage (Avis 2001; El Khoudary 2019; Palacios 2010).

Symptoms tend to fluctuate, and their severity varies greatly between individuals, with some reporting intense discomfort and a substantial reduction in quality of life. Many (although not all) perimenopausal and postmenopausal women report a variety of symptoms, related to mental health, physiological systems and functions. Symptoms include hot flushes and night sweats, sleep disturbances, depression and anxiety, cognitive performance difficulties, vaginal dryness, urinary incontinence and sexual functional decline. Changes affecting physiological systems include alterations in the cardiovascular and cardiometabolic systems (e.g. lipids, body fat distribution and metabolic syndrome risk), loss of bone mineral density and declines in physical function (El Khoudary 2019).

In this review, we use the term 'women' throughout to refer to the population studied. We acknowledge that not all people with female reproductive systems identify as women. However, for clarity, consistency and because the included studies recruited participants described as women, we use 'women' as the standard term in this review.

### Description of the intervention

Hormone therapy consists of oestrogen alone (oestrogen-only hormone therapy) or oestrogen combined with a progestogen (combined hormone therapy). Oestrogen is available in various formulations and doses, administered orally, vaginally, intranasally or via implant, skin patch, cream or gel (Deady 2004).

The addition of a progestogen reduces the risk of endometrial hyperplasia associated with the use of oestrogen alone in people with a uterus (Furness 2012). However, progestogens have adverse effects on blood lipids and may have the potential to cause symptoms such as headache, bloating and breast tenderness (McKinney 1998). Progestogens used for hormone therapy include synthetic derivatives of progesterone, synthetic derivatives of testosterone and natural progesterones derived from plants. These differ in their metabolic action and potential for adverse effects, and it is currently unclear which type of progestogen has the best risk-benefit profile for use in hormone therapy. In combined hormone therapy, progestogen can be taken continuously (every day, orally or via an intrauterine progestogen-releasing device) or

cyclically (for part of each month or less frequently). Clinical effects vary according to the type of hormone therapy and its duration of use.

Hormone therapy has been used for over 70 years to treat menopausal symptoms, with well-established efficacy demonstrated in a Cochrane review of 24 randomised controlled trials conducted between 1971 and 2000 (MacLennan 2004). Hormone therapy has also been used for the management or prevention of chronic disease. Oestrogens and progestogens affect most body systems and have been proposed as causing or preventing a wide range of conditions, as outlined in the next section (How the intervention might work). Recommendations for hormone therapy have changed over time, but during the 1990s, expert consensus generally held that most postmenopausal women could benefit from hormone therapy (Hemminki 2000a). This view was based on strong and consistent observational evidence that hormone therapy reduced the risk of coronary heart disease (CHD) by at least 30%. A meta-analysis of 25 cohort, case-control and angiographic studies published up to 1997 revealed a risk ratio of 0.70 (95% confidence interval (CI) 0.65 to 0.75) for CHD amongst oestrogen users compared with never-users (Barrett-Connor 1998). Other benefits reported in observational studies of hormone therapy included strong evidence of a reduction in osteoporotic fractures, a possible preventive or delaying effect on cognitive decline or dementia, and even a reduction in overall mortality for current users (Barrett-Connor 1998).

In the early 2000s, a study in women aged 50 to 97 years old reported an increased risk of breast cancer, stroke, pulmonary embolism and myocardial infarction in women with an intact uterus using hormone therapy, although a lower incidence of fracture due to osteoporosis was also observed (Anderson 2004). The study led to a marked decrease in hormone therapy use, driven by concerns amongst both patients and healthcare providers (Kohn 2019). Subsequent studies have linked the benefits and risks of hormone therapy to its type, dose, duration, route of administration and timing of initiation (Canonico 2007; NAMS 2017).

### How the intervention might work

Oestrogen has a favourable effect on some biomarkers, including indicators of cardiovascular disease and bone metabolism disorders. It has been shown to improve endothelial vasodilator function, promote angiogenesis and modulate autonomic function. Thus, the cardioprotective benefits of oestrogen have some biological plausibility (Chakrabarti 2014; Miller 2008). However, biomarkers interact via multiple complex pathways, and the overall effect of oestrogen on clinical outcomes cannot be predicted with any certainty. Therefore, trials with clinical endpoints such as myocardial infarction are necessary (Banks 2009a).

### Overview of the evolving evidence and ongoing debates

Observational studies have suggested a range of adverse effects of hormone therapy, including doubling or tripling of the risk of thromboembolic events (Barlow 1997), a large increase in endometrial cancer risk amongst people taking oestrogen without progestogen, an increased incidence of gallbladder disease (Uhler 2000), and a possible link between hormone therapy and breast cancer (Bhupathiraju 2017; Douketis 2005). The suggestion that hormone therapy might increase the risk of breast cancer was

supported by evidence of an increase in breast density in a high proportion of women taking oestrogen, but findings were initially inconsistent and controversial (Barrett-Connor 1998). The results of a very large observational study conducted in the UK raised concerns that current users of both combined and oestrogen-only hormone therapy were at increased risk of both incident and fatal breast cancer after relatively short periods of use (Beral 2003). The increase in risk was greatest amongst users of combined hormone therapy, with no large variations reported between the effects of specific oestrogens or specific progestogens. Risks were greater if hormone therapy use started at around the time of menopause than if it started later. Breast cancer rates were highest amongst current users of combined hormone therapy who began use within five years of menopause (Beral 2011).

Coronary heart disease (CHD) is the leading cause of morbidity and mortality among older postmenopausal women. Early observational studies suggested that hormone therapy might substantially reduce CHD risk, potentially outweighing its adverse effects (Kaplan 1985; Knopp 1988). However, these uncontrolled studies were prone to significant selection and compliance biases: women who used oestrogen were typically healthier, better educated, and more adherent to medical advice, with a lower baseline risk of cardiovascular disease. Recognising these biases, researchers called for randomised controlled trials (RCTs) to clarify whether hormone therapy genuinely reduced CHD risk (Barrett-Connor 2001; Hemminki 2000a). It has been suggested that wide prescribing of hormone therapy in the 1990s – despite the lack of randomised evidence of its efficacy and safety – might reflect a conflict between commercial and professional interest groups and good public policy (Hemminki 2000b). Moreover, as the next paragraph explains, RCTs have so far not demonstrated the marked CHD benefits of hormone therapy seen in the observational studies, and have raised questions about hormone therapy's overall risk-benefit profile.

In 2002, the Heart and Estrogen/progestin Replacement Study (HERS and HERS II) found that combined hormone therapy did not reduce the risk of cardiovascular events in women with CHD (Grady 2002; Hulley 1998; Hulley 2002). Moreover, the Women's Health Initiative (WHI) study concluded that combined hormone therapy promotes CHD, stroke, venous thromboembolism and pulmonary embolism (VTE/PE), breast cancer and dementia in postmenopausal women (Shumaker 2003; Shumaker 2004; WHI 1998; WHI 2002). For this reason, the American Food and Drug Administration (FDA) recommended labelling changes for hormone therapy, including 'black box' warnings that oestrogens increase the risk of heart disease, heart attack, stroke, breast cancer, blood clots and dementia (FDA 2021; Gottlieb 2003; Stefanick 2005). The recommendations included lower doses for both oestrogens and progestogens, the shortest possible treatment duration and alternative non-oestrogen treatments for the prevention of postmenopausal osteoporosis. As a result, hormone therapy use declined sharply – from 40% of postmenopausal women in their 50s and 30% in their 60s in 2000 to only 7% in both age groups by 2010 (Sprague 2012).

However, relatively recent evidence derived from subgroup analyses of data from the WHI trials suggest that some subsets of women may benefit from hormone therapy, while others may be at increased risk of harm (WHI 1998). These findings support the 'timing hypothesis', demonstrating that oestrogens may have a

beneficial effect on the heart if started in early menopause, but may be harmful if started in late menopause, when coronary arteries are more likely to show signs of atherosclerosis (Bassuk 2016; Manson 2015; Manson 2020). In addition, analysis of long-term follow-up data from the WHI trials indicate that oestrogen use in women with a prior hysterectomy is associated with a lower risk of breast cancer and breast cancer-associated mortality (Chlebowski 2020). In line with these developments, in November 2025, the FDA announced that it was lifting the black box warnings on hormone therapy products (Health and Human Services 2025).

In a 2019 systematic review assessing the route, timing, and duration of hormone therapy, transdermal oestrogen (< 50 µg/day) combined with micronised progesterone emerged as safer than oral oestrogens for thrombotic and stroke risk. The review further indicated a possible protective role for vaginal hormone therapy in reducing stroke and myocardial infarction risk, though the authors called for further investigation of this result (Oliver-Williams 2019).

### Why it is important to do this review

This is an updated version of a Cochrane review first published in 2005 (Farquhar 2005), and subsequently updated in 2009 (Farquhar 2009), 2012 (Marjoribanks 2012a), and 2017 (Marjoribanks 2017). Its importance lies in the fact that it synthesises the currently available evidence on a wide range of key long-term clinical outcomes, with the aim of assisting women and clinicians to make informed judgements about hormone therapy treatment.

It complements and extends the extensive and growing literature on various aspects of hormone therapy. This literature includes at least 17 Cochrane reviews, addressing hormone therapy and: cardiovascular disease (Boardman 2015); dementia and cognitive function (Hogervorst 2009; Lethaby 2008; Marjoribanks 2017); endometrial hyperplasia (Furness 2012; Gaudard 2016; Kim 2025; Lethaby 2016; Mittermeier 2020); hot flushes (Gaudard 2016; MacLennan 2004; Scheffers 2015); pelvic organ prolapse (Ismail 2010); sexual function (Lara 2023; Nastri 2013); urinary incontinence (Cody 2009; Cody 2012); vaginal atrophy (Lethaby 2016; Suckling 2006); and weight and body fat distribution (Kongnyuy 1999). This review is not intended to replace these other Cochrane reviews, which remain an important source of evidence on individual aspects of hormone therapy and will continue to be updated regularly.

The role of hormone therapy in managing menopausal symptoms has been extensively studied, and its efficacy in relieving vasomotor symptoms is well established (Gaudard 2016; MacLennan 2004). However, the management of perimenopausal and menopausal symptoms is not the focus of this review. Accordingly, we made an a priori decision to exclude studies with treatment durations of less than one year and to omit outcomes related to symptom control, early-onset adverse effects of hormone therapy and surrogate endpoints, such as endometrial hyperplasia and bone mineral density. Hormone therapy's longer-term clinical effects on women's health – whether protective or associated with increased risks – have become clearer with extended follow-up. This review examines these long-term outcomes.

### OBJECTIVES

To assess the long-term effects of prolonged use (at least one year) of hormone therapy on mortality, cardiovascular outcomes, cancer,

gallbladder disease, fractures and cognition in perimenopausal and postmenopausal women.

## METHODS

### Criteria for considering studies for this review

#### Types of studies

We included randomised, double-blind studies, defined as those in which participants, investigators and/or outcome assessors were blinded to treatment allocation. We excluded quasi-randomised studies. For cross-over studies, we intended to use only results from the end of the first phase (before the treatment cross-over) because of the potential carry-over effect of hormone therapy from the first treatment phase. However, we identified no cross-over studies eligible for inclusion.

#### Types of participants

Eligible participants were perimenopausal or postmenopausal women recruited from any healthcare setting or a population-based sample.

We defined perimenopausal women as those who had not yet had their final menstrual period but were in the transitional period between more-or-less regular cycles of ovulation and menstruation and complete cessation of these cycles. Postmenopausal women included those with surgical menopause (removal of both ovaries) and women with spontaneous menopause and amenorrhoea for longer than 12 months.

We included studies in women both with and without a prior history of disease (e.g. cardiovascular disease, fracture, osteoporosis, dementia).

#### Types of interventions

We included all oestrogens, with and without progestogens, administered by oral, transdermal, subcutaneous or intranasal routes, and given as perimenopausal or postmenopausal therapy for any reason for 12 months or longer, compared with placebo. We excluded trials in which participants took hormone therapy for less than 12 months, as we deemed such durations to be unlikely to capture intervention-related clinical events.

We excluded studies with co-interventions that might potentially affect the outcomes being measured, as well as studies of topical vaginal oestrogens such as creams, topical tablets and rings. These interventions are covered in another Cochrane review ([Lethaby 2016](#)). Given the vast number of studies investigating hormone therapy for menopause, we restricted inclusion to studies reporting at least one of our outcomes of interest because hormone therapy may be studied in the same population for different purposes, and we wished to ensure that we included only relevant studies.

#### Types of outcome measures

As outlined in [Why it is important to do this review](#), we focused on long-term clinical outcomes and did not include menopausal symptom control and early-onset side effects of hormone therapy as outcomes.

### Primary outcomes

We included studies reporting at least one of the following outcomes, at any time point. The asterisk (\*) indicates that the outcome is presented in the review's summary of findings tables (see [Summary of findings 1](#); [Summary of findings 2](#)).

- Death from any cause (total mortality)
- Cause-specific mortality
- Coronary events (myocardial infarction or coronary death)\*
- Stroke (ischaemic or haemorrhagic)\* or transient ischaemic attack (TIA)
- Venous thromboembolism\* (pulmonary embolism or deep vein thrombosis)
- Breast cancer\*
- Colorectal cancer
- Lung cancer\*
- Endometrial cancer
- Ovarian cancer
- Gallbladder disease requiring surgery\*
- Fractures (hip fracture, clinically diagnosed vertebral fracture, total clinically diagnosed fracture\*)
- Cognitive function (using global measures) or dementia (including Alzheimer's disease), as measured in the included studies

Although we measured endometrial cancer, we did not include it in the summary of finding tables as it is the focus of another Cochrane review ([Kim 2025](#)).

### Secondary outcomes

No secondary outcomes.

### Search methods for identification of studies

In consultation with the Gynaecology and Fertility Group Information Specialist, we searched for all published and unpublished double-blind RCTs of hormone therapy versus placebo, without language or date restrictions, on 26 September 2024.

#### Electronic searches

On 26 September 2024, we performed electronic searches of:

- the Cochrane Gynaecology and Fertility Group Specialised Register, ProCite platform, from the inception of the database to 23 November 2023. The Specialised Register has not been maintained since 2023 ([Appendix 1](#));
- the Cochrane Central Register of Controlled Trials (CENTRAL), EBM Reviews Ovid platform, to 26 September 2024 ([Appendix 2](#));
- MEDLINE, Ovid platform, searched from 1946 to 26 September 2024 ([Appendix 3](#));
- Embase, Ovid platform, searched from 1980 to 26 September 2024 ([Appendix 4](#));
- PsycINFO, Ovid platform, searched from 1806 to 26 September 2024 ([Appendix 5](#));
- the Cumulative Index to Nursing and Allied Health Literature (CINAHL), EBSCO platform, searched from 1982 to September 2016 ([Appendix 6](#)). Later CINAHL trials are now included in the CENTRAL 2024 search output.

We also searched the following trial registers for ongoing and registered trials:

- [ClinicalTrials.gov](https://clinicaltrials.gov) (a service of the US National Institutes of Health);
- the [International Clinical Trials Registry Platform \(ICTRP\)](https://www.ictwp.org/) (search portal of the World Health Organization International Trials Registry Platform).

We combined the MEDLINE search with the Cochrane highly sensitive search strategy for identifying randomised trials, which appears in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2024).

We combined the Embase and CINAHL searches with trial filters developed by the Scottish Intercollegiate Guidelines Network (SIGN).

### Searching other resources

For the 2025 update, we searched all studies on [RetractionWatch](https://retractionwatch.com/), and conducted a retraction assessment. We double-checked to ensure all included studies passed the assessment.

## Data collection and analysis

### Selection of studies

Working independently, two review authors (of AL, CB, MB, SM and LNY) screened the titles, abstracts or both of all publications obtained by the search strategy to identify potentially eligible studies. If the abstract suggested that the study might be eligible for inclusion, we obtained the full article. Two review authors (MB and SM or CF) checked potentially eligible studies against the review's inclusion criteria. We performed this assessment while unblinded, and resolved any uncertainty through discussion with a third author. If necessary, we sought additional information from the corresponding author of the study.

### Data extraction and management

When studies had multiple publications, we collated these reports and assigned each study a single study ID, so that each study, rather than each report, is the unit of interest in the review. Owing to the very large number of publications for some studies (e.g. [WHI 1998](#)), we referred in the text to specific secondary publications that are listed under the study ID, to make clear where specific outcomes and time frames are reported (e.g. see [WHI 1998](#), Chlebowski 2010).

When no events occurred in either comparison group for a particular outcome, we did not enter data in the [Data and analyses](#) tables for that outcome, in keeping with [Cochrane recommendations](#).

Extracted data included study characteristics and outcome data (see data extraction form for details; [Appendix 7](#)). We corresponded with study investigators to request additional data on methods, results or both, as required.

### Assessment of risk of bias in included studies

Two review authors independently assessed the included studies for risk of bias, using the original Cochrane risk of bias assessment tool (Higgins 2017), to examine selection bias (random sequence generation and allocation concealment), performance bias (blinding of participants and personnel), detection bias

(blinding of outcome assessors), attrition bias (incomplete outcome data), reporting bias (selective reporting) and other bias. We resolved disagreements through discussion, or sought assistance from a third review author. We described all judgements fully and presented our conclusions in the studies' risk of bias tables ([Characteristics of included studies](#)). We incorporated these judgements into our interpretation of review findings by performing sensitivity analyses (see [Sensitivity analysis](#)).

### Measures of treatment effect

We performed statistical analysis according to guidelines provided in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2024). We analysed treatment effects by comparing outcomes for each group measured at the end of therapy, during follow-up or at both time points.

For dichotomous data, we generated 2 × 2 tables for each study and expressed data as risk ratios (RRs) with 95% confidence intervals (CIs). If studies reported the number of events occurring in each comparison group at a mean follow-up time (i.e. not all women were followed up for that duration, and others were followed for longer), we made the simplifying assumption that risk was constant across the follow-up period and reported data as dichotomous data at a fixed time point. If risk varied significantly across the follow-up period, we noted this variation in the [Results](#) section.

We expressed continuous data as mean differences (MDs) with 95% CIs.

For outcomes for which studies reported no events in both the hormone therapy and placebo groups, we did not enter results into data tables.

### Unit of analysis issues

The unit of analysis was the individual participant. We did not attempt to pool data at different time points to avoid double counting.

### Dealing with missing data

We analysed data on an intention-to-treat basis as far as possible and attempted to obtain missing data from the original trialists. Where these could not be obtained, we analysed only available data.

### Assessment of heterogeneity

We considered whether clinical and methodological characteristics of the included studies were sufficiently similar for meta-analysis to provide a clinically meaningful summary, and if this was not the case, we planned to refrain from pooling the data. We assessed statistical heterogeneity by measuring the  $I^2$  statistic. We regarded an  $I^2$  statistic of more than 50% as indicating substantial heterogeneity.

### Assessment of reporting biases

In view of the difficulty of detecting and correcting for publication bias and other reporting biases, we aimed to minimise their potential impact by ensuring a comprehensive search for eligible studies and by staying alert to the duplication of data. If we had included 10 or more studies in an analysis, we planned to use a funnel plot to explore the possibility of small-study effects (the tendency of small studies to show larger intervention effects).

## Data synthesis

We combined data for meta-analysis using Review Manager (RevMan) software and the Peto-Mantel-Haenszel method (RevMan 2025). Because we could not reach a consensus on whether to use a fixed-effect or random-effects model, we performed both types of analysis. We treated this as a sensitivity analysis to assess how the choice of model affected the results, and we interpreted findings not robust across both models with caution. Our primary analysis forest plots display results obtained with the fixed-effect model.

We planned to combine continuous data for meta-analysis if such data were available for pooling. Meta-analytical methods for continuous outcomes assume that measurements are approximately normally distributed. We used the ratio of the mean to its standard deviation as a crude check for skew; for any study group with a ratio below 1.65, we considered the data potentially skewed. Unless original data were available for log transformation, we did not include skewed results in the analysis tables, but instead reported them in a table (Table 1). We also reported data in this table if they were clearly skewed and if investigators presented results as median values and ranges with non-parametric tests of significance.

We conducted separate analyses according to the type of hormone therapy used (oestrogen-only or combined hormone therapy). For some outcomes (death, cardiovascular disease, cognition scores), we anticipated that the intervention effect might vary with participants' clinical status. Therefore, we conducted separate analyses for studies of women without major health problems and for studies of women with specific health conditions. For other outcomes (cancer, cholecystic disease, fractures), we combined all available study results.

We synthesised dichotomous outcome data using risk ratios (RRs) with 95% confidence intervals (CIs). Where possible, we translated pooled RRs into absolute effects per 1000 participants using baseline risks from the summary of findings tables.

## Subgroup analysis and investigation of heterogeneity

If we detected substantial heterogeneity ( $> 50\%$ ), we planned to explore possible explanations for it by performing subgroup analyses (e.g. different follow-up lengths) and sensitivity analyses (e.g. by risk of bias). We planned to take any statistical heterogeneity into account when interpreting the results, especially if we noted variation in the direction of effect. We did not conduct any subgroup analysis.

## Sensitivity analysis

As noted above, we checked whether using a random-effects model for each analysis materially influenced our findings.

We planned to conduct sensitivity analyses to examine the effects of methodological differences between studies if we identified sufficient studies (i.e. more than five). These analyses might help to explain any substantial statistical heterogeneity detected. We planned to explore the following specific differences.

- Restricting analysis to studies with adequate methods: defined for this purpose as adequate allocation concealment, analysis by intention-to-treat approach and losses to follow-up of less than 10%.
- Differences amongst studies with respect to participants, interventions or clinical criteria for defining outcomes, although as noted above, we planned to refrain from combining studies that were obviously dissimilar in these respects.

We planned to conduct additional subgroup or sensitivity analyses if other possible sources of heterogeneity became evident during preparation of the review. We acknowledge that we would have had to interpret the results of any such post hoc analyses with great caution.

## Summary of findings and assessment of the certainty of the evidence

We prepared two summary of findings tables, following guidance in the *Cochrane Handbook* (Schünemann 2024). These tables present the two main comparisons: combined hormone therapy versus placebo, Summary of findings 1; and oestrogen-only hormone therapy versus placebo, Summary of findings 2. They present the most clinically relevant outcomes: coronary events, stroke, venous thromboembolism, breast cancer, lung cancer, gallbladder disease requiring surgery and clinical fractures. We assessed the certainty of the evidence for each outcome using the GRADE approach, considering five domains: risk of bias, inconsistency, indirectness, imprecision and publication bias. We rated the certainty of evidence as high, moderate, low or very low, and justified our judgements in footnotes to aid readers' interpretation. For randomised trials, evidence starts at high certainty and is downgraded based on concerns in any of the five domains.

Results in both summary of findings tables come from a single large study (WHI 1998).

# RESULTS

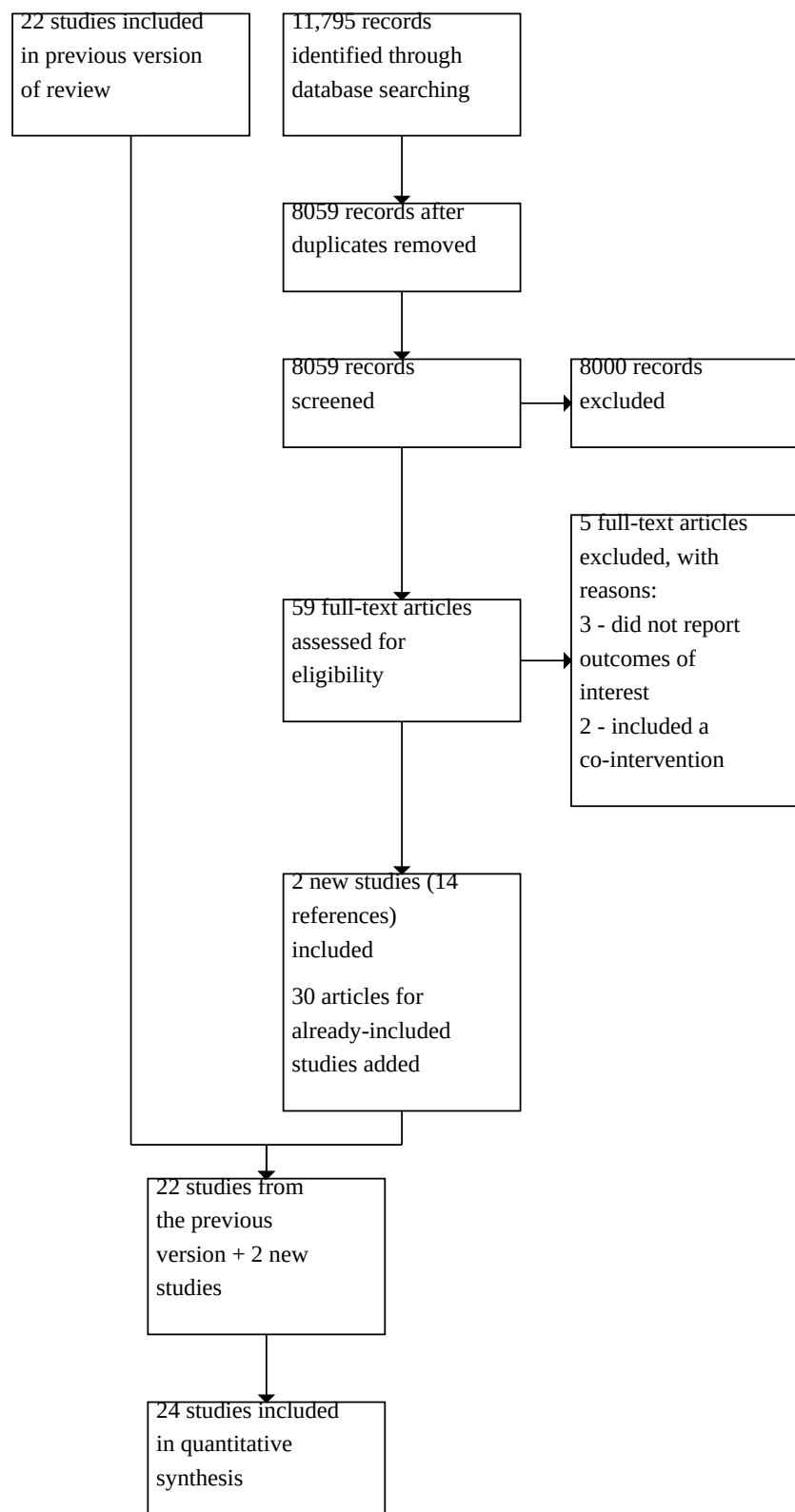
## Description of studies

### Results of the search

The update searches conducted in September 2024 yielded 11,795 records. After removing 3736 duplicates, we screened 8059 records and discarded 8000 as clearly irrelevant. We retrieved the full texts of the remaining 59 records for detailed screening. From these, we identified two new studies eligible for inclusion in the review: REPLENISH 2018 (12 references) and Ran 2017 (two references). We also identified many new publications for already-included studies, and added 30 publications contributing data to the review: one for ELITE 2014, two for HERS 1998, three for KEEPS 2012 and 24 for WHI 1998. Thus, we identified a total of 54 new publications (14 articles for two newly-included studies and 30 articles for previously-included ones).

We documented the updated search process in a PRISMA flow diagram (Moher 2009; see Figure 1). Search results and the study screening process for previous versions of the review are detailed in Farquhar 2005, Farquhar 2009, Marjoribanks 2012a and Marjoribanks 2017.

**Figure 1. PRISMA study flow diagram**



**Figure 1. (Continued)**

synthesis

### Included studies

We included 24 studies with 45,660 randomised participants: 24,455 randomised to some form of hormone therapy and 21,205 to placebo (treatment allocation was unclear for 16 women in one study ([Ferency 2002](#))). Two completed but unreported studies have been placed in [Characteristics of studies awaiting classification](#) ([ChiCTR-IPR-16008754](#); [Hodis 2003](#)), and there are two ongoing studies ([NCT04090957](#); [NCT05586724](#)).

[WISDOM 2007](#) included 1307 additional people who were randomised to a comparison of two active hormone therapies, but they are not included in this review. Investigators analysed results for more than 99% of the participants following the intention-to-treat (ITT) approach.

Although some studies used biological measures as their primary outcome (e.g. lumen of carotid artery; [WAVE 2002](#)), we included them because they also reported clinical endpoints relevant to this review as prespecified secondary outcomes.

The very large [WHI 1998](#) study incorporated randomised comparisons of two different hormone therapy regimens versus placebo and published these results separately. We refer to the combined oestrogen and progesterone versus placebo comparison as [WHI 1998](#) (combined hormone therapy arm), and the oestrogen-only hormone therapy versus placebo comparison as [WHI 1998](#) (oestrogen-only hormone therapy arm). [WHI 1998](#) also included a subgroup study known as the Women's Health Initiative Memory Study (WHIMS), which measured cognitive outcomes in older women (aged 65 to 79 years at study entry) from both arms of [WHI 1998](#). We refer to this study as [WHI 1998](#) (WHIMS) ([Shumaker 1998](#)). An additional ancillary study – [WHI 1998](#) (WHISCA) – enrolled women from [WHI 1998](#) (WHIMS) who were free of dementia to investigate the effects of hormone therapy on domain-specific cognitive function in older women ([Resnick 2004](#)).

The included studies varied dramatically in size. The largest was [WHI 1998](#), which randomised 27,347 participants, and the other studies varied in sample size from 40 ([Tierney 2009](#)) to 5692 participants ([WISDOM 2007](#)). Investigators included 8000 women in each group in [WHI 1998](#) (combined hormone therapy arm) and more than 5000 in each group in [WHI 1998](#) (oestrogen-only hormone therapy arm), along with more than 1400 in each group in the oestrogen-only hormone therapy arm of [WHI 1998](#) (WHIMS) and more than 2200 in each group in the combined arm of [WHI 1998](#) (WHIMS). [HERS 1998](#) included about 1380 women in each comparison group; [ESPRIT 2002](#) included more than 500 in each group; [EPHT 2006](#) included around 400 women in each group; and [KEEPS 2012](#) included 220 to 275 per group. Otherwise, none of the studies included more than 210 women in each group. Five of the smaller studies were single-centred ([ELITE 2014](#); [EPAT 2001](#); [Nachtigall 1979](#); [Obel 1993](#); [Tierney 2009](#)), and it is unclear whether one study enlisted more than one trial centre ([EVTET 2000](#)). The other 10 studies involved between seven and 40 trial centres.

Fifteen studies were conducted in the USA, and one in each of the following countries: UK, Estonia, Norway, Canada, Denmark and China. Three studies were international in scope (one in the USA and Canada; one in Canada and the Netherlands; and one in the UK, Australia and New Zealand). Two studies were originally planned as part of a larger international project, but planning was beset with delays ([EPHT 2006](#); [WISDOM 2007](#)). In the meantime, [WHI 1998](#) began in the USA, with the consequence that other countries were no longer prepared to commit funds to a second study with similar objectives. Both [EPHT 2006](#) and [WISDOM 2007](#) were prematurely closed as a result of publication of early [WHI 1998](#) findings.

We attempted to contact investigators for the following studies to request more information about their methods or outcomes: [Barakat 2006](#); [ELITE 2014](#); [EPAT 2001](#); [EPHT 2006](#); [ERA 2000](#); [EVTET 2000](#); [Ferency 2002](#); [HERS 1998](#); [KEEPS 2012](#); [Mulnard 2000](#); [Notelovitz 2002](#); [Obel 1993](#); [PEPI 1995](#); [WAVE 2002](#); [WEST 2001](#); [WHI 1998](#); [WISDOM 2007](#). Investigators from the following studies kindly supplied clarification or additional unpublished data, or both: [Barakat 2006](#); [ELITE 2014](#); [EPHT 2006](#); [ERA 2000](#); [HERS 1998](#); [Obel 1993](#); [PEPI 1995](#); [WAVE 2002](#); [WISDOM 2007](#).

### Participants

The participants included in these studies were predominantly postmenopausal, spontaneously or surgically. The age of participants ranged from 26 to 91 years, with participants' mean or median age in each study ranging from 48 to 76 years (no age was stated in [Obel 1993](#)). In more than half of the studies, the mean participant age was over 60 years. Inclusion criteria varied according to the primary objectives of individual studies. Some investigated the use of hormone therapy for treating menopausal symptoms or disease prevention, and thus enrolled women in reasonably good health. Others assessed whether hormone therapy was beneficial for women with a history of cancer or established disease, including heart disease, thromboembolic disease, stroke, Alzheimer's disease or long-term medical conditions requiring hospitalisation. These studies restricted entry to women who had received a diagnosis of the condition of interest.

### Studies in women without established medical conditions

Fifteen studies enrolled relatively healthy women ([ELITE 2014](#); [EPAT 2001](#); [EPHT 2006](#); [Ferency 2002](#); [Greenspan 2005](#); [KEEPS 2012](#); [Notelovitz 2002](#); [Obel 1993](#); [PEPI 1995](#); [Ran 2017](#); [REPLENISH 2018](#); [Tierney 2009](#); [WHI 1998](#); [WISDOM 2007](#); [Yaffe 2006](#)). Women in some of these studies had risk factors (such as raised cholesterol), and a small minority in individual studies had a history of cardiovascular disease, but most participants were fit women without overt disease. Most of these studies were interested in hormone therapy for disease prevention.

Three large studies investigated the use of hormone therapy to prevent cardiovascular disease, while also reporting a wide range of other endpoints ([PEPI 1995](#); [WHI 1998](#); [WISDOM 2007](#)). Researchers

provided highly detailed lists of inclusion and exclusion criteria. In [WHI 1998](#), enrolment was targeted to establish set fractions for baseline age categories and to achieve representation of racial and ethnic groups in the proportions recorded by the American census for individuals 50 to 79 years of age.

The [WHI 1998](#) (combined hormone therapy arm) investigators noted that prevalence of prior cardiovascular disease in participants was low: 4.4% had a history of myocardial infarction, coronary revascularisation, stroke or transient ischaemic attack. They also commented that levels of cardiovascular risk factors were consistent with a generally healthy population of postmenopausal women: 2.9% reported a history of angina, 36% were hypertensive (or were being treated for hypertension), 13% were being treated for high cholesterol, 4.4% were being treated for diabetes and 10.5% were current smokers ([Manson 2003](#)). Similarly, in [WHI 1998](#) (oestrogen-only hormone therapy arm), participants in general were considered healthy, although 4.1% had a history of myocardial infarction or coronary revascularisation, 5.8% had a history of angina, 1.4% had a history of stroke, 1.6% had a history of venous thrombosis, 48% were hypertensive (or were being treated for hypertension), 15% were receiving treatment for high cholesterol, 7.7% were being treated for diabetes and 10.5% were current smokers ([Stefanick 2003](#)).

[PEPI 1995](#) compared the characteristics of their cohort with values returned in large American surveys, and concluded that although the [PEPI 1995](#) cohort was generally in better health than the wider US population, these individuals were not so markedly different as to limit the generalisability of study results. Both [KEEPS 2012](#) and [ELITE 2014](#) were designed to test whether menopausal hormone therapy initiated soon after menopause could delay progression of atherosclerosis. Two other 'prevention' studies aimed to test the possible beneficial effects of hormone therapy on arterial wall density and bone density ([EPAT 2001](#) and [Notelovitz 2002](#), respectively). Four much smaller studies also enrolled women without stated health problems who were in early menopause ([Obel 1993](#)), or were postmenopausal and aimed to assess the effects of hormone therapy on endometrial safety ([Ferenczy 2002](#); [Obel 1993](#)) and other clinical outcomes ([Greenspan 2005](#); [Tierney 2009](#)).

[WISDOM 2007](#) recruited women with no known major health problems from general practice registers in countries with free or low-fee healthcare systems. Investigators designed recruitment to target older women first; as a result, the median participant age was 63 years and few women in younger age groups were included when the study closed prematurely.

#### Studies in women with established medical conditions or a history of cancer

Six studies included women with established cardiovascular disease ([ERA 2000](#); [ESPRIT 2002](#); [EVTET 2000](#); [HERS 1998](#); [WAVE 2002](#); [WEST 2001](#)). [ERA 2000](#) and [WAVE 2002](#) included women who had coronary artery stenosis evident on angiogram. [HERS 1998](#) and [ESPRIT 2002](#) randomised women who had had a myocardial infarction or, in the case of [HERS 1998](#), coronary artery surgery. [EVTET 2000](#) and [WEST 2001](#) included women who had had a thromboembolic (pulmonary embolism (PE) or deep vein thrombosis (DVT)) or cerebrovascular event (stroke or transient ischaemic attack (TIA)). The largest of these six studies, [HERS 1998](#), compared its cohort of women with a similar group of women

presumed to have coronary heart disease, who were participants in a survey designed to produce nationally representative data: the [HERS 1998](#) cohort included fewer smokers, women with hypertension and women with diabetes than the comparison group, but individuals were comparable with respect to blood pressure, body mass index, physical activity and cholesterol levels.

[Mulnard 2000](#) included women with Alzheimer's disease, and an older study included women with a range of medical conditions, including diabetes, need for custodial care, arteriosclerosis and chronic neurological disorders ([Nachtigall 1979](#)). All participants in this study were hospitalised for the duration of the 10-year study. One study enrolled women after surgery (including bilateral salpingo-oophorectomy) for early-stage endometrial cancer ([Barakat 2006](#)).

#### Interventions

The included studies used a wide variety of oestrogen-alone or oestrogen and progestogen combinations as interventions. Some included more than one intervention arm, each with a different dose, formulation or route of hormone therapy. Most comparisons used a moderate dose of oestrogen (e.g. oestradiol 1 mg, conjugated equine oestrogen (CEE) 0.625 mg daily or transdermal oestradiol 0.05 mg twice-weekly). [Nachtigall 1979](#) used a much higher dose than the other included studies, reflecting the fact that it was conducted many years earlier. The range of interventions used were as follows.

#### Oestrogen-only hormone therapies

- Oestradiol (17-B oestradiol), an oestrogen derived from Mexican wild yam, 1 mg orally ([ELITE 2014](#); [EPAT 2001](#); [WEST 2001](#)).
- Oestradiol valerate, which is a pro-drug for oestradiol (meaning that it is converted in the body into the active form); the dose used was 2 mg ([ESPRIT 2002](#); [Ran 2017](#)).
- Intranasal 17-B oestradiol, delivered by a puff via each nostril once a day, at a dose of 0.15 mg or 0.3 mg daily ([Nielsen 2006](#)).
- Conjugated equine oestrogen (CEE), a blend of equine oestrogens; 0.625 mg ([Barakat 2006](#); [ERA 2000](#); [Greenspan 2005](#); [Mulnard 2000](#); [PEPI 1995](#); [WAVE 2002](#); [WHI 1998](#) (oestrogen-only hormone therapy arm)) and 1.25 mg daily ([Mulnard 2000](#)). [Barakat 2006](#) allowed doubling of the dose for women who were symptomatic. [WISDOM 2007](#) also included an oestrogen-only arm, but the comparison group was taking combined therapy, and this comparison is not relevant to this review.

Most studies using oestrogen-only hormone therapy did not randomise women to this arm unless they had had a hysterectomy ([Greenspan 2005](#); [Mulnard 2000](#); [Nachtigall 1979](#); [Notelovitz 2002](#); [WAVE 2002](#); [WEST 2001](#); [WHI 1998](#) (oestrogen-only hormone therapy arm)).

#### Combined hormone therapy regimens

Combined regimens included one of the above types of oestrogen in combination with one of the following progestogens.

- Medroxyprogesterone acetate (MPA), a synthetic progestogen structurally related to progesterone.
- Dydrogesterone, a synthetic progestogen structurally related to progesterone.
- Norethisterone (norethindrone), a synthetic progestogen structurally related to testosterone.

- Micronised progesterone, a natural progestogen synthesised from plant sources and finely ground to improve its absorption.
- Drospirenone, a synthetic progestogen structurally related to spironolactone.
- Progesterone oral 50 or 100 mg daily.

#### Continuous combined regimens

These included the following.

- CEE 0.625 mg with MPA 2.5 mg daily (EPHT 2006; ERA 2000; Greenspan 2005; HERS 1998; PEPI 1995; WAVE 2002; WHI 1998 (combined arm); WISDOM 2007).
- CEE 2.5 mg with MPA 10 mg daily (Nachtigall 1979).
- Oestradiol 2 mg with 1 mg norethisterone daily (EVTET 2000).
- Oestretol (E4) and progesterone (Pg) in different combinations of daily oestradiol (mg) – progesterone (mg) (E4 1/100 Pg, E4 0.5/100 Pg, E4 0.5/50 Pg, or E4 0.25/50 Pg all combined) (REPLENISH 2018).

#### Combined sequential regimens

These included the following.

- Oestradiol 1 mg daily with MPA 5 mg for 12 days once a year (WEST 2001).
- Oestradiol valerate 1 mg daily with MPA 5 mg 12 days per month for 53 weeks (Ran 2017).
- Oestradiol 1 mg daily for 4 days, oestradiol 1 mg plus 0.35 mg norethindrone daily for 3 days each week (Tierney 2009).
- Oestradiol 2 mg days 1 to 22, 1 mg days 22 to 28, with norethisterone 1 mg days 13 to 22 (Obel 1993).
- Oestradiol 1 mg daily with dydrogesterone 5 mg or 10 mg days 14 to 28 (Ferenczy 2002).
- Oestradiol 2 mg daily with 10 to 20 mg dydrogesterone days 14 to 28 (Ferenczy 2002).
- Oestradiol 0.05 mg patch with cyclic micronised progesterone 200 mg daily for 12 days a month (KEEPS 2012).
- CEE 0.425 mg daily with cyclic micronised progesterone 200 mg daily for 12 days a month (KEEPS 2012).
- CEE 0.625 with MPA 10 mg days 1 to 12 (PEPI 1995).
- CEE 0.625 mg with micronised progesterone 200 mg days 1 to 12 (PEPI 1995).
- Oral oestradiol 1 mg daily, plus 40 mg cyclic micronised progesterone as 4% vaginal gel for 10 days per 30-day cycle for women with an intact uterus only (ELITE 2014).

The control arm of each study received placebo tablets, patches or nasal spray, as appropriate.

The duration of hormone therapy use varied, with the longest study lasting 10 years (Nachtigall 1979). Four studies reported outcomes after hormone therapy use for around one year (EVTET 2000; Mulnard 2000; WISDOM 2007; REPLENISH 2018); seven measured outcomes after two years (EPAT 2001; ESPRIT 2002; Ferenczy 2002; Notelovitz 2002; Obel 1993; Tierney 2009; Yaffe 2006), seven at around three years (Barakat 2006; EPHT 2006; ERA 2000; Greenspan 2005; PEPI 1995; WAVE 2002; WEST 2001), and one at four years (KEEPS 2012). HERS 1998 measured outcomes after 4.1 years and continued the study unblinded for 2.7 additional years. ELITE 2014 measured outcomes after 2.5 years and subsequently at five years

of hormone therapy use. Ran 2017 measured outcomes at five years.

Investigators planned that interventions in the WHI study would continue for 8.5 years, but both arms of the study were terminated early. WHI 1998 (combined hormone therapy arm) was stopped early owing to net harm. Researchers reported outcomes at 5.6 years and over four subsequent months of follow-up for primary and selected outcomes, incorporating events up to the date that participants were instructed to stop their study pills. WHI 1998 (oestrogen-only hormone therapy arm) was also stopped early when it was decided that the prospect of obtaining more precise evidence about effects of the intervention was unlikely to outweigh potential harms, although no predefined safety boundaries had been crossed. Investigators reported results in the oestrogen-only arm for a mean follow-up of 7.1 years for primary outcomes: the median time receiving treatment was 5.9 years in the intervention group and 5.8 years in the placebo group. Additional post-study follow-up occurred in WHI 1998, as noted below.

Two other studies also closed prematurely in response to WHI 1998 findings (EPHT 2006; WISDOM 2007).

#### Outcomes

The outcomes measured by individual studies varied according to study objectives. Major clinical events were not primary outcomes for several of these studies but were measured as adverse effects; for example, cardiovascular events or the incidence of cancer and fracture in the study population, or both. Eight studies used biological measures as their primary outcome (ELITE 2014; EPAT 2001; ERA 2000; KEEPS 2012; Notelovitz 2002; PEPI 1995; WAVE 2002; Yaffe 2006).

The largest study in the review, WHI 1998, was concerned mainly with the cardioprotective role of hormone therapy in relatively healthy women, and study authors reported cardiovascular clinical endpoints as the primary outcome. They designated invasive breast cancer as a primary adverse outcome and included the incidence of other cancers, fractures, gallbladder disease and death as secondary outcomes. Two other studies measured similar outcomes (EPHT 2006; WISDOM 2007).

WHI 1998 also conducted a number of analyses not specified in the study protocol. Lung cancer was not a prespecified outcome but was investigated in both arms of the study in post hoc analyses, which included additional follow-up periods after the planned completion date of the study.

After the intervention phase of WHI 1998 had been completed, investigators followed up major clinical outcomes in surviving participants (i.e. those who consented), comprising 78% of participants in the oestrogen-only arm and 83% in the combined hormone therapy arm. Median cumulative follow-up (intervention phase plus extended follow-up) was 13.2 years in the oestrogen-only arm (including median post-intervention follow-up of 6.6 years) and 13 years in the combined group (including median post-intervention follow-up of 6.6 years) (Manson 2013).

WHI 1998 (WHIMS) comprised a large subset of older women from WHI 1998 who were evaluated for probable dementia (the planned primary outcome) and for mild cognitive impairment (as a planned secondary outcome). Researchers also reported global cognitive function, although this was not a formally prespecified endpoint.

WHI 1998 (WHIMS) reported separate results for the two study arms and also pooled study results, but we did not include the pooled results in this review (see [Methods](#)).

Two smaller studies reported endometrial cancer as a primary outcome ([Barakat 2006](#); [Ferenczy 2002](#)). Two studies reported as primary outcomes clinical events that were not of interest for this review ([Obel 1993](#); [Tierney 2009](#)), but researchers measured review outcomes of interest as adverse events.

Five other studies were concerned with the effect of hormone therapy on established clinical disease. Four reported cardiovascular outcomes: primary outcomes were myocardial infarction or death ([ESPRIT 2002](#); [HERS 1998](#)), thromboembolism ([EVTET 2000](#)), and stroke ([WEST 2001](#)). The larger studies also measured a range of other major clinical events, such as the incidence of cancer, fracture and gallbladder disease ([ESPRIT 2002](#); [HERS 1998](#)). One study reported the effect of hormone therapy on global cognitive function ([Greenspan 2005](#)), and one on progression of symptoms in women with Alzheimer's disease ([Mulnard 2000](#)). Another study measured a wide range of clinical outcomes over a treatment period of 10 years with hormone therapy in women who were receiving long-term hospital care for a range of medical conditions ([Nachtigall 1979](#)).

The specific cause-mortality reported in the studies included death from: coronary heart disease, stroke, colorectal cancer, breast cancer, lung cancer and any cancer.

See [Characteristics of included studies](#) for details.

### Excluded studies

There are a total of 47 excluded studies documented in this review update, with five excluded in this update as well as 42 excluded in previous versions. The reasons for exclusion are:

- no outcomes of interest for this review (27 studies);
- ineligible design: not double-blind studies (five studies);
- did not meet study duration criterion (less than one year's duration) or reported only short-term (3-month) outcomes (six studies);
- ineligible comparison: no placebo group (seven studies);
- ineligible intervention: co-intervention used in the hormone therapy group (one study);
- ineligible population: hormone therapy used as a co-intervention for the pharmacological treatment of dementia (one study).

See [Excluded studies](#) for details as we excluded some studies for more than one reason.

### Risk of bias in included studies

See [Figure 2](#) and [Figure 3](#).

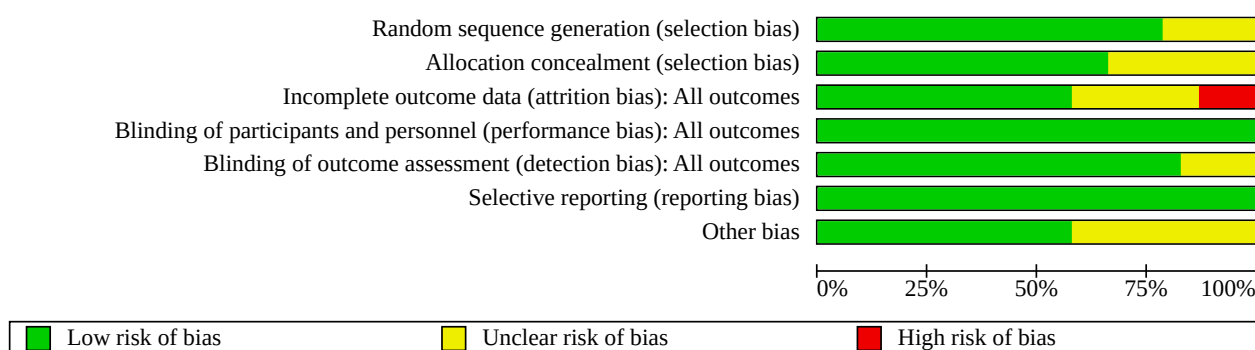
**Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.**

|                 | Random sequence generation (selection bias) | Allocation concealment (selection bias) | Incomplete outcome data (attrition bias): All outcomes | Blinding of participants and personnel (performance bias): All outcomes | Blinding of outcome assessment (detection bias): All outcomes | Selective reporting (reporting bias) | Other bias |
|-----------------|---------------------------------------------|-----------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------|------------|
| Barakat 2006    | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| ELITE 2014      | +                                           | +                                       | ?                                                      | +                                                                       | +                                                             | +                                    | +          |
| EPAT 2001       | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| EPHT 2006       | +                                           | +                                       | ?                                                      | +                                                                       | +                                                             | +                                    | ?          |
| ERA 2000        | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | ?          |
| ESPRIT 2002     | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| EVTET 2000      | +                                           | ?                                       | ?                                                      | +                                                                       | +                                                             | +                                    | +          |
| Ferenczy 2002   | ?                                           | ?                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| Greenspan 2005  | ?                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | ?          |
| HERS 1998       | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| KEEPS 2012      | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| Mulnard 2000    | +                                           | ?                                       | +                                                      | +                                                                       | ?                                                             | +                                    | ?          |
| Nachtigall 1979 | ?                                           | ?                                       | +                                                      | +                                                                       | +                                                             | +                                    | ?          |
| Notelovitz 2002 | ?                                           | ?                                       | +                                                      | +                                                                       | +                                                             | +                                    | +          |
| Obel 1993       | ?                                           | ?                                       | ?                                                      | +                                                                       | ?                                                             | +                                    | ?          |
| PEPI 1995       | +                                           | +                                       | +                                                      | +                                                                       | +                                                             | +                                    | ?          |
| Ran 2017        | +                                           | ?                                       | ?                                                      | +                                                                       | ?                                                             | +                                    | +          |

Figure 2. (Continued)

|                |   |   |   |   |   |   |   |
|----------------|---|---|---|---|---|---|---|
| Ran 2017       | + | ? | ? | + | ? | + | + |
| REPLENISH 2018 | + | ? | ? | + | + | + | + |
| Tierney 2009   | + | + | ? | + | ? | + | ? |
| WAVE 2002      | + | + | + | + | + | + | ? |
| WEST 2001      | + | + | + | + | + | + | + |
| WHI 1998       | + | + | + | + | + | + | ? |
| WISDOM 2007    | + | + | + | + | + | + | + |
| Yaffe 2006     | + | + | + | + | + | + | + |

Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.



### Allocation

Nineteen of the 24 included studies described a satisfactory method of randomisation, which in all cases was computer-generated. Eighteen described a satisfactory method of allocation concealment: namely, researchers entered information about an eligible participant, or they accomplished this via remote contact between the recruiting centre and the study co-ordinating centre or pharmacy. One of these studies randomised women who expressed an interest in participating, but did not open the randomisation envelope until their eligibility had been checked, and they had consented (EPHT 2006). Three studies described using computer-generated randomisation but did not provide details of the procedure for allocation to treatment (EVTET 2000; Mulnard 2000; REPLENISH 2018). Four studies did not provide sufficient information about the methods used for randomisation or allocation concealment (Ferenczy 2002; Nachtigall 1979; Notelovitz 2002; Obel 1993). One study reported using a random number table for sequence generation, but provided no details on the allocation (Ran 2017).

We rated 19 studies as having a low risk of bias for random sequence generation and 16 for allocation concealment. We rated the remaining studies as having an unclear risk of bias in these domains.

### Blinding

All included studies were described as (at least) double-blind. The authors of 18 studies either explicitly stated that all participants, clinical staff and outcome assessors or research staff were blinded to treatment allocation, or they reported 'hard' outcomes unlikely to be influenced by blinding. In the WHI study, 331 women randomised to receive active treatment were unblinded and changed arms from WHI 1998 (oestrogen-only hormone therapy arm) to WHI 1998 (combined hormone therapy arm) according to a change in protocol. Three studies apparently blinded participants and clinical staff but did not explicitly state if outcome assessors were also blinded (Mulnard 2000; Obel 1993; Tierney 2009).

The larger studies described an unblinded mechanism that was used when required for management of adverse effects. PEPI 1995 unblinded 39 women (4%) during the course of the study, 32 of whom were taking oestrogen-only hormone therapy. WHI 1998 (combined hormone therapy arm) reported that, during 5.6 years of follow-up, 3444 women in the combined hormone therapy group (40%) and 548 women in the placebo group (6%) were unblinded. In WHI 1998 (oestrogen-only hormone therapy arm), only 100 women in the active group (< 2%) and 83 in the placebo group (< 2%) were unblinded. Nachtigall 1979 reported that 13 women in the hormone therapy group and 17 in the control group were unblinded. Two women were unblinded in WISDOM 2007. The other studies did not report this information.

One randomised blinded study completed 4.1 years of follow-up and was then extended for a further 2.7 years, unblinded (HERS 1998).

We rated all studies as having a low risk of performance bias and 20 as having a low risk of detection bias. We rated four studies as having an unclear risk of detection bias.

### Incomplete outcome data

For the purposes of this review, we defined losses to follow-up as participants for whom outcomes of interest were unknown (and who may or may not have had outcomes imputed in statistical analysis). We defined dropouts as participants who stopped their allocated treatment (and in some cases changed to a different off-trial treatment) but had known clinical outcomes and were included in the analysis. Adherence to treatment refers to the number of tablets actually taken, which we often assessed by pill counts (Table 2). We defined intention-to-treat analysis as analysis of all randomised participants in the groups to which they were randomised.

Dropout rates were generally high, particularly in the active treatment groups, and they increased over time. In WHI 1998 (combined hormone therapy arm), 42% of the active treatment group and 38% of the placebo group were no longer taking their allocated treatment at five years, and a further 10.7% of the placebo group had crossed to active therapy. In WHI 1998 (oestrogen-only hormone therapy arm), 53% of participants overall were no longer taking their allocated treatment at 6.8 years, and a further 5.7% had initiated hormone use outside the study. See the Characteristics of included studies table and Table 2 for details on dropouts and non-adherence in other studies.

Losses to follow-up were low in most studies, with no women lost to follow-up in seven studies (EPAT 2001; ERA 2000; ESPRIT 2002; EVTET 2000; Mulnard 2000; Nachtigall 1979; WEST 2001), and 1% to 5.2% lost in five other studies, all of which were large and of long duration (3 to 6.8 years) (Greenspan 2005; HERS 1998; PEPI 1995; WAVE 2002; WHI 1998). Only five women (0.01%) were lost to follow-up in WISDOM 2007. The Estonian study monitored outcomes by means of linkages to a national health insurance database and national cancer registry, and study authors stated that the probability of missing data in these databases was small (EPHT 2006). However, different publications for this study reported slightly different numbers of randomised participants (EPHT 2006). In six smaller studies of one to five years' duration, a higher proportion of women (8.5% to 21%) were lost to follow-up (ELITE 2014; KEEPS 2012; Notelovitz 2002; Obel 1993; Tierney 2009; Yaffe 2006), and in Ferenczy 2002, results were unavailable for 34% of participants for the outcome of interest for this review. It was unclear whether any women were lost to follow-up in Barakat 2006 (see Description of studies).

Fourteen of the included studies supplied sufficient data to permit an intention-to-treat (ITT) analysis, at least for all reported outcomes of interest for this review (EPAT 2001; ERA 2000; ESPRIT 2002; EVTET 2000; Greenspan 2005; HERS 1998; KEEPS 2012; Mulnard 2000; Nachtigall 1979; Notelovitz 2002; WEST 2001; WHI 1998; WISDOM 2007; Yaffe 2006), or such data were extractable, and a further two studies analysed more than 97% of participants by intention to treat (PEPI 1995; WAVE 2002). Five studies did not include all participants in an ITT analysis for outcomes of

interest (ELITE 2014; EVTET 2000; Ferenczy 2002; Obel 1993; Tierney 2009). It was unclear whether one study used ITT analysis because investigators provided no description of participants other than those that were "eligible and assessable" (Barakat 2006), and one study had slightly differing participation rates across trial publications (EPHT 2006).

WHI 1998 (combined hormone therapy arm) and WHI 1998 (WHISCA) continued follow-up beyond the planned study completion date (March 2005) for women who consented to continue follow-up. All women had already been instructed to stop taking their assigned study medication in July 2002. Seventeen per cent of surviving women in WHI 1998 (combined hormone therapy arm) declined to provide re-consent, and their data were censored for the additional follow-up period. Baseline characteristics were similar in the two groups, and imputation analyses suggested that this loss to follow-up did not appear to influence study findings. Fifteen per cent of women in WHI 1998 (WHISCA) declined to continue follow-up. The study extension phase ran from April 2005 to September 2010. WHI 1998 (oestrogen-only hormone therapy arm) also conducted extended follow-up (in 78% of surviving participants) from April 2005 to September 2010; among women who provided additional consent, baseline characteristics were similar to those of the original randomised group.

We rated 14 studies as having a low risk of attrition bias, seven as having an unclear risk and three as having a high risk.

### Selective reporting

All studies reported all expected outcomes, and we rated them as having a low risk of selective reporting.

### Other potential sources of bias

Ten of the included studies had other potential sources of bias (EPHT 2006; ERA 2000; Greenspan 2005; Mulnard 2000; Nachtigall 1979; Obel 1993; PEPI 1995; Tierney 2009; WAVE 2002; WHI 1998); we rated them as having an unclear risk of this bias. In most cases, potential bias was related to baseline imbalance between participants in individual prognostic characteristics and did not appear likely to have a marked effect on outcomes. We rated the other studies as having a low risk of bias in this domain.

### Effects of interventions

See: [Summary of findings 1 Combined continuous hormone therapy compared with placebo for peri- and postmenopausal women](#); [Summary of findings 2 Oestrogen-only hormone therapy compared with placebo for peri- and postmenopausal women](#)

We structured the evidence synthesis below as follows: for **three** outcome domains – **death, cardiovascular events** (including myocardial infarction, stroke, transient ischaemic attack (TIA), deep vein thrombosis (DVT) or pulmonary embolism (PE)) and **cognitive measures** – we grouped the studies by population into five groups:

1. participants with no major health problems (seven studies, including WHI 1998 and WISDOM 2007);
2. participants with cardiovascular disease (five studies, including HERS 1998);
3. participants with Alzheimer's disease (one small study: Mulnard 2000);

4. participants who had undergone surgery for endometrial cancer (one small study: [Barakat 2006](#));
5. participants hospitalised with chronic illness or requiring custodial care (one small study: [Nachtigall 1979](#)).

Within each outcome domain, where data were available, we stratified the results by hormone therapy type, in this order:

1. oestrogen-only hormone therapy;
2. combined continuous hormone therapy regimens;
3. combined sequential hormone therapy regimens.

For **three** outcome domains – **cancer, gallbladder disease requiring surgery** and **fractures** – we grouped all the included studies together. This is not to say that we pooled heterogeneous populations for meta-analysis, but rather that we considered each of these outcome domains in turn, stratifying the results by the three hormone therapy types, as above. We pooled comparisons only if they used the same combination of oestrogen and progestogen for the same (or a similar) length of time. The number of studies assessing each outcome varied, from 14 studies for breast cancer to two studies for lung cancer. The evidence synthesis for these outcomes begins under subheading 6 below ('All peri- and postmenopausal women').

Finally, within these categories, we grouped interventions according to the oestrogen dose used, with equivalence between doses based on the Australian Menopause Society guide to equivalent hormone therapy doses ([AMS 2016](#)). It classifies hormone therapy as low dose (e.g. oral oestradiol 1 mg), medium dose (e.g. oral oestradiol 2 mg, transdermal oestradiol 50 µg, conjugated equine oestrogen 0.065 mg) or higher dose (e.g. transdermal oestradiol 75 µg).

The results presented in [Summary of findings 1](#) and [Summary of findings 2](#) are drawn only from the largest study included in the review, [WHI 1998](#), and its related publications.

### Meta-analysis

Although we have grouped comparisons with similar oestrogen doses together, we pooled comparisons (i.e. meta-analysed them) only if they used the same combination of oestrogen and progestogen for the same (or a similar) length of time. [WHI 1998](#) and [PEPI 1995](#) used the same hormone therapy regimen and reported several of the same clinical outcomes at three years, but in most cases, [PEPI 1995](#) reported no events in either arm. We combined three studies for some 3-year (2.8 to 3.2) outcomes ([ERA 2000](#); [HERS 1998](#); [WAVE 2002](#)). However, meta-analysis was inappropriate for most outcomes because the studies used different types or doses of oestrogen or progestogen, or both, and these do not necessarily have the same metabolic effects; or they used different durations of hormone therapy, which might have led to different effects as the result of trends over time.

Very few results were suitable for pooling; therefore, statistical heterogeneity was not a major issue in this review. One meta-analysis displayed high heterogeneity ( $I^2 = 66.2\%$ ), but it involved only two small studies with few events, and we attributed the heterogeneity to chance ([Analysis 2.20](#)).

### Time points for reporting results

In some cases, we rounded time points up or down for reporting results, as follows.

- [WHI 1998](#) (oestrogen-only hormone therapy arm) reported results after a mean follow-up of 7.1 or 7.9 years. Among women who consented (78% of those surviving), follow-up was extended (for a median of 6.6 years) after the predefined study termination date to achieve a cumulative median follow-up of 13.2 years. The median duration of active treatment in this arm of the study was 5.8 to 5.9 years ([LaCroix 2011](#)). We reported results at mean or median follow-up points as reported by the study publications.
- [WHI 1998](#) (combined hormone therapy arm) reported results after a mean of 5.6 years of active treatment (intervention phase) or at a mean of 7.9 years. The 7.9-year follow-up included 2.4 years of post-intervention follow-up and continued up to the predefined study termination date (31 March 2005). Among women who consented (83% of those surviving), follow-up was extended after the predefined study termination date for a median of 6.6 years to achieve a median cumulative follow-up of 13 years. This arm of the study also reported selected clinical outcomes for each year of follow-up: all women had been enrolled for at least 3.5 years at the time of the study publication, so we used these data to calculate outcomes on an ITT basis after one, two and three years of hormone therapy use, with all randomised participants inserted as the denominator ([Chlebowski 2009](#)). We reported results at mean or median follow-up points as reported by the study publications.
- [EPHT 2006](#) reported results for most outcomes at a mean follow-up of 3.43 years, with a range of two to five years. Results for quality of life were reported at a mean of 3.6 years. We reported results in our tables as if all participants underwent three years of follow-up.
- [WISDOM 2007](#) reported results after a median follow-up of 11.9 months (range 7.1 to 19.6). We reported results in our tables as if all women had one year of follow-up.
- [Barakat 2006](#) reported results after a median follow-up of 35.7 months. We reported results in our tables as if all women had three years of follow-up.
- [HERS 1998](#) reported results from the blinded portion of the study after a mean follow-up of 4.1 years, as mentioned above (see [Methods](#)). These results were presented as dichotomous data, and investigators reported selected clinical outcomes for each year of follow-up. All women had been enrolled for at least three years at the time of the report, so for this review, we used these data to calculate outcomes on an ITT basis after one, two and three years of hormone therapy use, with all randomised participants inserted as the denominator.

### Results

Most findings come from the two biggest studies, [HERS 1998](#) and [WHI 1998](#), both of which reported adequate methods of allocation concealment, analysed all participants using an intention-to-treat approach, and reported small losses to follow-up (1% to 5.2%).

## 1. Peri- or postmenopausal women without major health problems

### 1.1, 1.2, 1.3 Death from any cause (total mortality)

Seven studies, with a total of eight different interventions, reported this outcome in healthy women (ELITE 2014; EPHT 2006; EPAT 2001; KEEPS 2012; PEPI 1995; WHI 1998; WISDOM 2007). The comparisons comprised oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for varying durations from one year to nearly eight years, with extended follow-up to 10.7 years in WHI 1998 (oestrogen-only arm).

There was no clear evidence of a difference between hormone therapy and placebo for this outcome in any population group.

### 1.1 Death from any cause: oestrogen-only hormone therapy (Analysis 1.1)

1.1.1 Oestradiol 1 mg versus placebo at 2-year follow-up (RR 0.33, 95% CI 0.01 to 8.10; 1 study, 222 participants) (EPAT 2001)

1.1.2 CEE 0.625 mg versus placebo at 7.1 years of follow-up (RR 1.03, 95% CI 0.88 to 1.20; 1 study, 10,739 participants) (WHI 1998)

1.1.3 CEE 0.625 mg versus placebo at 10.7 years of follow-up (RR 1.02, 95% CI 0.91 to 1.13; 1 study, 10,739 participants) (WHI 1998)

1.1.4 CEE 0.625 mg versus placebo at 18-year follow-up (RR 0.94, 95% CI 0.89 to 1.0; 1 study, 10,739 participants) (WHI 1998)

### 1.2 Death from any cause: combined hormone therapy (Analysis 1.2)

1.2.1 CEE 0.625 mg + MPA 2.5 mg at 1-year follow-up (RR 1.31, 95% CI 0.76 to 2.27;  $I^2 = 0\%$ ; 2 studies, 20,993 participants) (WHI 1998; WISDOM 2007)

1.2.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 1.05, 95% CI 0.71 to 1.56; 1 study, 16,608 participants) (WHI 1998)

1.2.3 CEE 0.625 mg + progesterone (P; as per footnotes in analyses) at 3-year follow-up (RR 1.09, 95% CI 0.81 to 1.46;  $I^2 = 0\%$ ; 4 studies, 18,075 participants) (EPHT 2006; PEPI 1995 (2 comparisons); WHI 1998)

1.2.4 CEE 0.045 mg + 200 mg sequential progesterone at 4-year follow-up (RR 3.58, 95% CI 0.15 to 87.57; 1 study, 505 participants) (KEEPS 2012)

1.2.5 CEE 0.625 mg + MPA 2.5 mg at 5.6 years of follow-up (RR 1.00, 95% CI 0.84 to 1.19; 1 study, 16,608 participants) (WHI 1998)

1.2.6 CEE 0.625 mg + MPA 2.5 mg at 7.9 years of follow-up (RR 1.06, 95% CI 0.93 to 1.20; 1 study, 16,608 participants) (WHI 1998)

1.2.7 CEE 0.625 mg + MPA 2.5 mg at 13.2 years of follow-up (RR 1.00, 95% CI 0.92 to 1.08; 1 study, 16,608 participants) (WHI 1998)

1.2.8 CEE 0.625 mg + MPA 2.5 mg at 18 years of follow-up (RR 1.01, 95% CI 0.96 to 1.07; 1 study, 16,608 participants) (WHI 1998)

### 1.3 Death from any cause: oestrogen with or without sequential progesterone vaginal gel (Analysis 1.3)

1.3.1 Oestradiol 1 mg daily, with or without 4% vaginal progesterone gel after a median of 5 years (RR 0.99, 95% CI 0.06 to 15.77; 1 study, 643 participants) (ELITE 2014)

### Cause-specific mortality

### 1.4, 1.5, 1.6 Death from coronary heart disease

Four studies (EPAT 2001; Tierney 2009; WHI 1998; WISDOM 2007), with a total of five different interventions (comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo), reported this outcome in relatively healthy women. The studies' durations varied, from one year to nearly eight years, with extended follow-up to 10.7 years in WHI 1998 (oestrogen-only arm).

Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome in any population group.

### 1.4 Death from coronary heart disease: oestrogen-only hormone therapy (Analysis 1.4)

1.4.1 Oestradiol 1 mg daily at 2-year follow-up (RR 0.33, 95% CI 0.01 to 8.10; 1 study, 222 participants) (EPAT 2001)

1.4.2 CEE 0.625 mg at 7.1 years of follow-up (RR 0.98, 95% CI 0.69 to 1.38; 1 study, 10,739 participants) (WHI 1998)

1.4.3 CEE 0.625 mg at 10.7 years of follow-up (RR 0.92, 95% CI 0.71 to 1.19; 1 study, 10,739 participants) (WHI 1998)

1.4.4 CEE 0.625 mg at 18-year follow-up (RR 0.89, 95% CI 0.75 to 1.05; 1 study, 10,739 participants) (WHI 1998)

### 1.5 Death from coronary heart disease: combined continuous hormone therapy (Analysis 1.5)

1.5.1 CEE 0.625 mg + MPA 2.5 mg at a mean of 5.6 years of follow-up (RR 1.06, 95% CI 0.68 to 1.66; 1 study, 16,608 participants) (WHI 1998)

1.5.2 CEE 0.625 mg + MPA 2.5 mg at a mean of 7.9 years of follow-up (RR 0.99, 95% CI 0.72 to 1.38; 1 study, 16,608 participants) (WHI 1998)

1.5.3 CEE 0.625 mg + MPA 2.5 mg at a mean of 18 years of follow-up (RR 1.04, 95% CI 0.88 to 1.21; 1 study, 16,608 participants) (WHI 1998)

### 1.6 Death from coronary heart disease: combined sequential hormone therapy (Analysis 1.6)

1.6.1 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 0.34, 95% CI 0.01 to 8.27; 1 study, 142 participants) (Tierney 2009)

### 1.7, 1.8, 1.9 Death from stroke

Four comparisons of relatively healthy women taking combined continuous hormone therapy (WISDOM 2007; WHI 1998 (combined hormone therapy arm)), oestrogen-only hormone therapy (WHI 1998 (oestrogen-only hormone therapy arm)), or combined sequential hormone therapy (Tierney 2009), reported this outcome.

Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome.

### 1.7 Death from stroke: oestrogen-only hormone therapy (Analysis 1.7)

1.7.1 CEE 0.625 mg at 7.1 years of follow-up (RR 1.16, 95% CI 0.58 to 2.32; 1 study, 10,739 participants) (WHI 1998)

1.7.2 CEE 0.625 mg at 18-year follow-up (RR 0.98, 95% CI 0.77 to 1.24; 1 study, 10,739 participants) (WHI 1998)

### 1.8 Death from stroke: combined continuous hormone therapy (Analysis 1.8)

1.8.1 CEE 0.625 mg + MPA 2.5 mg at median 1-year follow-up (RR 2.99, 95% CI 0.12 to 73.37; 1 study, 4385 participants) (WISDOM 2007)

1.8.2 CEE 0.625 mg + MPA 2.5 mg at mean 5.6 years of follow-up (RR 1.04, 95% CI 0.46 to 2.35; 1 study, 16,608 participants) (WHI 1998)

1.8.3 CEE 0.625 mg + MPA 2.5 mg at mean 18-year follow-up (RR 1.11, 95% CI 0.90 to 1.37; 1 study, 16,608 participants) (WHI 1998)

### 1.9 Death from stroke: combined sequential hormone therapy (Analysis 1.9)

1.9.1 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone for 2-year follow-up (RR 3.08, 95% CI 0.13 to 74.46; 1 study, 142 participants) (Tierney 2009)

### 1.10, 1.11 Death from colorectal cancer

Investigators reported this outcome in relatively healthy women in the oestrogen-only group of WHI 1998 after a mean follow-up of 7.1 years, as well as in the WHI 1998 (combined hormone therapy arm) at a mean follow-up of 5.6 and 7.1 years. Researchers also reported on this outcome after 11.6 years' follow-up, including a mean of 5.6 years of intervention plus post-intervention follow-up after the study ended, in 83% of participants (WHI 1998; Simon 2012).

There was no clear evidence of a difference between hormone therapy and placebo for this outcome.

### 1.10 Death from colorectal cancer: oestrogen-only hormone therapy (Analysis 1.10)

1.10.1 CEE 0.625 mg at 7.1 years of follow-up (RR 1.28, 95% CI 0.66 to 2.46; 1 study, 10,739 participants) (WHI 1998)

1.10.2 CEE 0.625 mg at 18-year follow-up (RR 1.20, 95% CI 0.79 to 1.83; 1 study, 10,739 participants) (WHI 1998)

### 1.11 Death from colorectal cancer: combined continuous hormone therapy (Analysis 1.11)

1.11.1 CEE 0.625 mg + MPA 2.5 mg at mean 5.6 years of follow-up (RR 0.95, 95% CI 0.40 to 2.29; 1 study, 16,608 participants) (WHI 1998)

1.11.2 CEE 0.625 mg + MPA 2.5 mg at 7.1 years of follow-up (RR 1.01, 95% CI 0.52 to 1.96; 1 study, 16,608 participants) (WHI 1998)

1.11.3 CEE 0.625 mg + MPA 2.5 mg at 11.6 years of follow-up (RR 1.31, 95% CI 0.80 to 2.14; 1 study, 16,608 participants) (WHI 1998)

1.11.4 CEE 0.625 mg + MPA 2.5 mg at 18-year follow-up (RR 1.01, 95% CI 0.69 to 1.48; 1 study, 16,608 participants) (WHI 1998)

No data were available for combined sequential hormone therapy for death from colorectal cancer.

### 1.12, 1.13 Death from breast cancer

One study in comparatively healthy women taking oestrogen-only hormone therapy for a median of 7.2 years (WHI 1998 (oestrogen-only hormone therapy arm)), with post-intervention follow-up of a median of 4.7 years, reported this outcome. In one study of relatively healthy women taking combined continuous hormone therapy for 5.6 years (WHI 1998), follow-up for breast cancer outcomes was continued for a mean total of 11 years in women in WHI 1998 (combined hormone therapy arm) who agreed to continue follow-up after the planned study completion date (Chlebowski 2010c).

There was no clear evidence of a difference between hormone therapy and placebo for this outcome at one or 5.6 years.

Among peri- and postmenopausal people taking oestrogen-only hormone therapy, after a median of 11.8 years (7.2 years' intervention plus post-intervention follow-up), the death rate from breast cancer was lower in the hormone therapy arm (RR 0.38, 95% CI 0.15 to 0.98) (WHI 1998 (oestrogen-only hormone therapy)).

### 1.12 Death from breast cancer: oestrogen-only hormone therapy (Analysis 1.12).

1.12.1 CEE 0.625 mg at median 10.8 years of follow-up (RR 0.587, 95% CI 0.32 to 1.02; 1 study, 10,739 participants) (WHI 1998)

1.12.2 CEE 0.625 mg at 18 years of cumulative follow-up (RR 0.55, 95% CI 0.33 to 0.92; 1 study, 10,739 participants) (WHI 1998)

1.12.3 CEE 0.625 mg at 21 years of cumulative follow-up (RR 0.67, 95% CI 0.42 to 1.05; 1 study, 10,739 participants) (WHI 1998)

At 10.8 and 21 years' follow-up the WHI 1998 estrogen only therapy group reported little to no difference for breast cancer mortality compared to placebo.

At 18 years of follow-up participants in the estrogen only group reported a small decrease in breast cancer mortality (all ages combined). Individual results by age group did not report clear evidence of difference: for 50 to 59 years HR 0.46 (95%CI 0.16 to 1.32), 60 to 69 years HR 0.52 (95%CI 0.25 to 1.12) and 70 to 79 years HR 0.72 (0.28-1.90).

### 1.13 Death from breast cancer: combined continuous hormone therapy (Analysis 1.13)

1.13.1 CEE 0.625 mg + MPA 2.5 mg at mean 5.6 years of follow-up (RR 0.95, 95% CI 0.24 to 3.81; 1 study, 16,608 participants) (WHI 1998)

1.13.2 CEE 0.625 mg + MPA 2.5 mg at mean 12.5 years of follow-up (RR 1.48, 95% CI 0.98 to 2.25; 1 study, 16,608 participants) (WHI 1998)

1.13.3 CEE 0.625 mg + MPA 2.5 mg at 18 years of cumulative follow-up (RR 1.45, 95% CI 0.98 to 2.16; 1 study, 16,608 participants) (WHI 1998)

1.13.4 CEE 0.625 mg + MPA 2.5 mg at 20 years of cumulative follow-up (RR 1.28, 95% CI 0.90 to 1.82; 1 study, 16,608 participants) ([WHI 1998](#))

Mortality for breast cancer in participants receiving combined hormonal treatment up to 20 years of follow-up reported little to no difference compared to placebo.

#### 1.14, 1.15, 1.16 Death from lung cancer

[WHI 1998](#) reported this outcome in relatively healthy women in the oestrogen-only hormone therapy group in a post hoc analysis after a mean follow-up of 7.9 years (including 8 months' follow-up post-intervention) ([Chlebowski 2010b](#)), and in the combined hormone therapy arm of [WHI 1998](#) in a post hoc analysis after a mean follow-up of eight years (including 2.4 years' follow-up post-intervention) ([Chlebowski 2009](#)). Study authors reported lung cancer overall, non-small cell lung cancer and small cell lung cancer separately. One much smaller study reported this outcome in women taking combined sequential hormone therapy or placebo ([Tierney 2009](#)).

There was no clear evidence of a difference between hormone therapy and placebo for any of these outcomes among women in the oestrogen-only hormone therapy arm of [WHI 1998](#).

#### 1.14 Death from lung cancer at 7.9 years of follow-up: oestrogen-only hormone therapy (moderate dose) ([Analysis 1.14](#))

1.14.1 Death from lung cancer (non-small cell or small cell) (RR 1.05, 95% CI 0.65 to 1.70; 1 study, 10,739 participants) ([WHI 1998](#))

1.14.2 Death from non-small cell lung cancer (RR 0.88, 95% CI 0.52 to 1.50; 1 study, 10,739 participants) ([WHI 1998](#))

1.14.3 Death from small cell lung cancer (RR 2.04, 95% CI 0.62 to 6.79; 1 study, 10,739 participants) ([WHI 1998](#))

#### 1.15 Death from lung cancer: combined continuous hormone therapy (moderate dose) ([Analysis 1.15](#))

1.15.1 Death from lung cancer (non-small cell or small cell) at mean 7.9 years (RR 1.74, 95% CI 1.18 to 2.55; 1 study, 16,608 participants) ([WHI 1998](#))

1.15.2 Death from non-small cell lung cancer at mean 7.9 years (RR 1.91, 95% CI 1.24 to 2.93; 1 study, 16,608 participants) ([WHI 1998](#))

1.15.3 Death from small cell lung cancer at mean 7.9 years (RR 1.16, 95% CI 0.48 to 2.81; 1 study, 16,608 participants) ([WHI 1998](#))

1.15.4 Death from lung cancer (any type) at a median of 14 years (RR 1.10, 95% CI 0.88 to 1.39; 1 study, 16,608 participants) ([WHI 1998](#))

Review authors noted no clear evidence of difference between the combined sequential hormone therapy and placebo. However, in the combined hormone therapy arm of [WHI 1998](#), women in the intervention group were more likely to die of lung cancer overall (RR 1.74, 95% CI 1.18 to 2.55), or of non-small cell lung cancer (RR 1.91, 95% CI 1.24 to 2.93) than participants in the placebo arm. The absolute risk of lung cancer increased from five per 1000 in the control group to nine per 1000 (95% CI 6 to 13) in the hormone therapy group. This finding was independent of smoking status. There was no clear evidence of a difference in the mortality rate for small cell lung cancer between groups.

#### 1.16 Death from lung cancer: combined sequential hormone therapy ([Analysis 1.16](#))

1.16.1 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone for 2-year follow-up (RR 3.08, 95% CI 0.13 to 74.46; 1 study, 142 participants) ([Tierney 2009](#))

#### 1.17 Death from any cancer

Two studies in relatively healthy women taking continuous hormone therapy for three years ([EPHT 2006](#)) and for 5.2 years ([WHI 1998](#) (combined hormone therapy arm)) reported this outcome. There was no clear evidence of a difference between hormone therapy and placebo for this outcome.

#### 1.17 Death from any cancer: combined continuous hormone therapy ([Analysis 1.17](#))

1.17.1 CEE 0.625 mg daily + MPA 2.5 mg at 3 years of follow-up (RR 2.77, 95% CI 0.11 to 67.80; 1 study, 777 participants) ([EPHT 2006](#))

1.17.2 CEE 0.625 mg daily + MPA 2.5 mg for 5.2 years of follow-up (RR 1.16, 95% CI 0.87 to 1.53; 1 study, 16,608 participants) ([WHI 1998](#))

No data were available for oestrogen-only hormone therapy and combined sequential hormone therapy for death from any cancer.

#### 1.18, 1.19, 1.20, 1.21 Coronary events (myocardial infarction (MI) or cardiac death)

#### 1.18 Coronary events (MI or cardiac death): oestrogen-only hormone therapy

([Analysis 1.18](#))

Eight studies ([ELITE 2014](#); [EPAT 2001](#); [EPHT 2006](#); [KEEPS 2012](#); [PEPI 1995](#); [Tierney 2009](#); [WHI 1998](#), [WISDOM 2007](#)), with a total of nine different interventions (comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo), reported this outcome in relatively healthy women. The studies' durations ranged from one year to over seven years, with extended follow-up to 10.7 years in [WHI 1998](#) (oestrogen-only arm) ([LaCroix 2011](#)) and to 13.2 years in the combined hormone therapy arm ([Manson 2013](#)).

[WHI 1998](#) (oestrogen-only hormone therapy arm) showed little or no difference between the two groups for this outcome at 7.1 years (RR 0.94, 95% CI 0.78 to 1.13; 1 study, 10,739 participants; moderate certainty evidence) and 10.7 years of follow-up (RR 0.95, 95% CI 0.82 to 1.10; 1 study, 10,739 participants).

1.18.1 Oestradiol 1 mg at 2 years of follow-up (RR 0.50, 95% CI 0.05 to 5.43; 1 study, 222 participants) ([EPAT 2001](#))

1.18.2 CEE 0.625 mg at 3 years of follow-up (RR 2.98, 95% CI 0.12 to 72.72; 1 study, 349 participants) ([PEPI 1995](#))

1.18.3 CEE 0.625 mg at mean 7.1 years of follow-up (RR 0.94, 95% CI 0.78 to 1.13; 1 study, 10,739 participants) ([WHI 1998](#))

1.18.4 CEE 0.625 mg at 10.7 years of follow-up (RR 0.95, 95% CI 0.82 to 1.10; 1 study, 10,739 participants) ([WHI 1998](#))

#### 1.19: Coronary events (MI or cardiac death): combined continuous hormone therapy

([Analysis 1.19](#))

However, [WHI 1998](#) (combined hormone therapy arm) reported that relatively healthy women taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) were more likely to be at higher risk of a coronary event after taking hormone therapy for one, two and three years (at one year: RR 1.74 (95% CI 1.05 to 2.89); at two years: RR 1.49 (95% CI 1.05 to 2.12); at three years: RR 1.43 (95% CI 1.05 to 1.95)). At a mean follow-up of 5.6 years, there was little or no difference between groups (RR 1.17, 95% CI 0.95 to 1.44; moderate-certainty evidence), and investigators observed no differences between groups after extended follow-up to 13.2 years.

[WISDOM 2007](#) and [EPHT 2006](#) reported data for this outcome at one year and three years, respectively. Pooling these data with data from [WHI 1998](#) (combined hormone therapy arm) resulted in a risk ratio at one year of 1.89 (95% CI 1.15 to 3.10) and at three years of 1.45 (95% CI 1.07 to 1.98). The absolute risk of a coronary event increased after one year from two per 1000 in the control group to four per 1000 (95% CI 3 to 7) in the hormone therapy group; after two years from six per 1000 in the control group to nine per 1000 (95% CI 7 to 13) in the hormone therapy group; and after four years from eight per 1000 in the control group to 11 per 1000 (95% CI 8 to 13) in the hormone therapy group.

1.19.1 CEE 0.625 mg + MPA 2.5 mg at mean/median 1-year follow-up (RR 1.89, 95% CI 1.15 to 3.10;  $I^2 = 16\%$ ; 2 studies, 20,993 participants) ([WHI 1998](#); [WISDOM 2007](#))

1.19.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 1.49, 95% CI 1.05 to 2.12; 1 study, 16,608 participants) ([WHI 1998](#))

1.19.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (RR 1.45, 95% CI 1.07 to 1.98;  $I^2 = 0\%$ ; 2 studies, 17,385 participants) ([EPHT 2006](#); [WHI 1998](#))

1.19.4 CEE 0.625 mg + MPA 2.5 mg at mean 5.6 years of follow-up (RR 1.17, 95% CI 0.95 to 1.44; 1 study, 16,608 participants) ([WHI 1998](#))

1.19.5 CEE 0.625 mg + MPA 2.5 mg at 13.2 years of follow-up (RR 1.08, 95% CI 0.95 to 1.22; 1 study, 16,608 participants) ([WHI 1998](#))

1.19.6 Oestradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily at 1-year follow-up (RR 0.45, 95% CI 0.02 to 9.35; 1 study, 1835 participants) ([REPLENISH 2018](#))

## 1.20 Coronary events (MI or cardiac death): combined sequential hormone therapy

No other studies found clear evidence of a difference between combined sequential hormone therapy and placebo for this outcome ([Analysis 1.20](#) and [Analysis 1.21](#)).

1.20.1 CEE 0.625 mg daily + micronised progesterone 200 mg days 1–12 at 3-year follow-up (RR 4.89, 95% CI 0.24 to 101.09; 1 study, 352 participants) ([PEPI 1995](#))

1.20.2 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 0.34, 95% CI 0.01 to 8.27; 1 study, 142 participants) ([Tierney 2009](#))

1.20.3 Oestradiol patch 0.05 mg + 200 mg sequential progesterone at 4-year follow-up (RR 3.71, 95% CI 0.15 to 90.70; 1 study, 497 participants) ([KEEPS 2012](#))

## 1.21 Coronary events (MI or cardiac death): oestrogen with or without sequential progesterone vaginal gel

1.21.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of 5 years (RR 0.33, 95% CI 0.03 to 3.16; 1 study, 643 participants) ([ELITE 2014](#))

## 1.22, 1.23, 1.24 Stroke

Six studies ([EPAT 2001](#); [EPHT 2006](#); [KEEPS 2012](#); [PEPI 1995](#); [Tierney 2009](#); [WHI 1998](#)), with a total of seven different interventions (comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo), reported this outcome in relatively healthy women. The studies' durations ranged from one year to nearly eight years. [WHI 1998](#) (oestrogen-only arm) extended follow-up to 10.7 years, and [Manson 2013](#) extended follow-up to 13.2 years in the combined hormone therapy arm. One of the studies reported no events ([KEEPS 2012](#)).

Participants receiving hormone therapy in the [WHI 1998](#) (oestrogen-only hormone therapy arm) were more likely to have a stroke than the placebo group at 7.1 years' follow-up (RR 1.33, 95% CI 1.06 to 1.67; moderate-certainty evidence; [Analysis 1.22](#)). The absolute risk of a stroke increased from 23 per 1000 in the control group to 32 per 1000 (95% CI 25 to 40) in the hormone therapy group (oestrogen-only). Study authors noted that the excess in the intervention arm was due to an increased risk of ischaemic rather than haemorrhagic stroke, and that the excess risk became apparent after four years of follow-up ([Hendrix 2006](#)). However, the increased risk was not maintained during extended follow-up (overall 10.7 years; RR 1.17, 95% CI 0.97 to 1.40) ([LaCroix 2011](#)).

## 1.22 Stroke: unopposed oestrogen

([Analysis 1.22](#))

1.22.1 Oestradiol 1 mg at 2 years of follow-up (RR 3.00, 95% CI 0.12 to 72.86; 1 study, 222 participants) ([EPAT 2001](#))

1.22.2 CEE 0.625 mg at mean 7.1 years of follow-up (RR 1.33, 95% CI 1.06 to 1.67; 1 study, 10,739 participants) ([WHI 1998](#))

1.22.3 CEE 0.625 mg at 10.7 years of follow-up (RR 1.17, 95% CI 0.97 to 1.40; 1 study, 10,739 participants) ([WHI 1998](#))

The combined hormone therapy arm of [WHI 1998](#) reported no clear evidence of difference between groups in the incidence of stroke during the first two years of the study. However, women taking combined continuous hormone therapy were more likely to have a stroke after taking hormone therapy for three or more years (at three years: RR 1.47, 95% CI 1.02 to 2.11; at a mean of 5.6 years: RR 1.39, 95% CI 1.09 to 1.77; at a mean of 7.9 years: RR 1.29, 95% CI 1.06 to 1.56). There was no clear evidence of a difference between groups at 13.2 years (RR 1.15, 99% CI 0.99 to 1.33). [EPHT 2006](#) also reported data for this outcome at three years. Pooling these data with data from [WHI 1998](#) (combined hormone therapy arm) resulted in a risk ratio at three years of 1.46 (95% CI 1.02 to 2.09). The absolute risk of a stroke increased at three years from six per 1000 in the control group to eight per 1000 (95% CI 6 to 12) in the hormone therapy group; at 5.6 years from 14 per 1000 in the control group to 19 per 1000 (95% CI 15 to 24) in the hormone therapy group; and at 7.9 years from 21 per 1000 in the control group to 28 per 1000 (95% CI 23 to 34) in the hormone therapy group.

### 1.23 Stroke: combined continuous hormone therapy

(Analysis 1.23)

1.23.1 CEE 0.625 mg + MPA 2.5 mg at 1-year follow-up (RR 0.95, 95% CI 0.49 to 1.86; 1 study, 16,608 participants) (WHI 1998)

1.23.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 1.31, 95% CI 0.83 to 2.06; 1 study, 16,608 participants) (WHI 1998)

1.23.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (RR 1.46, 95% CI 1.02 to 2.09;  $I^2 = 0\%$ ; 2 studies, 17,385 participants) (EPHT 2006; WHI 1998)

1.23.4 CEE 0.625 mg + MPA 2.5 mg at a mean of 5.6 years of follow-up may increase the risk of stroke compared to placebo (RR 1.39, 95% CI 1.09 to 1.77; 1 study, 16,608 participants; low-certainty evidence) (WHI 1998).

1.23.5 CEE 0.625 mg + MPA 2.5 mg at mean 7.9 years of follow-up (RR 1.29, 95% CI 1.06 to 1.56; 1 study, 16,608 participants) (WHI 1998)

1.23.6 CEE 0.625 mg + MPA 2.5 mg at mean 13.2 years of follow-up (RR 1.15, 95% CI 0.99 to 1.33; 1 study, 16,608 participants) (WHI 1998)

### 1.24 Stroke: combined sequential hormone therapy

There was no clear evidence of a difference between combined sequential hormone therapy and placebo for stroke (Analysis 1.24).

1.24.1 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 3.08, 95% CI 0.13 to 74.46; 1 study, 142 participants) (Tierney 2009)

1.24.2 CEE 0.625 mg daily + MPA 10 mg days 1–12 at 3-year follow-up (RR 3.00, 95% CI 0.12 to 73.14; 1 study, 348 participants) (PEPI 1995)

1.24.3 CEE 0.625 mg daily + micronised progesterone 200 mg days 1–12 at 3-year follow-up (RR 2.93, 95% CI 0.12 to 71.51; 1 study, 352 participants) (PEPI 1995)

1.24.4 CEE 0.45 mg daily + micronised progesterone 200 mg daily for 12 days/month at 4-year follow-up (not estimable) (KEEPS 2012)

### 1.25, 1.26, 1.27 Transient ischaemic attack

Four studies with a total of five different interventions (ELITE 2014; EPAT 2001; PEPI 1995; Tierney 2009), comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, reported this outcome in relatively healthy women. There was no clear evidence of a difference between hormone therapy and placebo for this outcome.

### 1.25 Transient ischaemic attack: oestrogen-only hormone therapy

(Analysis 1.25)

1.25.1 Oestradiol 1 mg at 2-year follow-up (RR 3.00, 95% CI 0.12 to 72.86; 1 study, 222 participants) (EPAT 2001)

### 1.26 Transient ischaemic attack: combined sequential hormone therapy

(Analysis 1.26)

1.26.1 17-B-oestradiol 1 mg daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 1.03, 95% CI 0.07 to 16.13; 1 study, 142 participants) (Tierney 2009)

1.26.2 CEE 0.625 mg daily + MPA 10 mg days 1–12 at 3-year follow-up (RR 3.00, 95% CI 0.12 to 73.14; 1 study, 348 participants) (PEPI 1995)

### 1.27 Transient ischaemic attack: oestrogen with or without sequential progesterone vaginal gel

(Analysis 1.27)

1.27.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of five years (RR 0.50, 95% CI 0.05 to 5.44; 1 study, 643 participants) (ELITE 2014)

### 1.28 Stroke or transient ischaemic attack: combined continuous hormone therapy

(Analysis 1.28)

One study in relatively healthy women taking combined continuous hormone therapy or placebo for a median of one year reported stroke or TIA as a combined outcome (WISDOM 2007).

1.28.1 Results of the analysis show no clear evidence of a difference in this outcome between participants taking hormone therapy (CEE 0.625 mg + MPA 2.5 mg) and those taking placebo for a mean/median of one year (RR 0.73, 95% CI 0.37 to 1.46; 1 study, 4385 participants) (WISDOM 2007).

No data were available for oestrogen-only or combined sequential hormone therapy for stroke or transient ischaemic attack.

### 1.29, 1.30, 1.31, 1.32 Venous thromboembolism (DVT or PE)

Six studies with a total of five different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, reported this outcome in relatively healthy women (ELITE 2014; KEEPS 2012; PEPI 1995; Tierney 2009; WHI 1998; WISDOM 2007). The studies' durations ranged from one year to nearly eight years, with extended follow-up to 10.7 years in WHI 1998 (oestrogen-only arm).

WHI 1998 (oestrogen-only hormone therapy arm) reported that relatively healthy women taking oestrogen-only hormone therapy (CEE 0.625 mg) were more likely to present a thromboembolic event than those taking placebo. The risk was highest within the first two years (RR 2.22, 95% CI 1.12 to 4.39). The absolute risk of an event increased from two per 1000 in the control group to five per 1000 (95% CI 2 to 10) in the hormone therapy group. At a mean follow-up of seven years, the risk difference decreased and there was no clear evidence of a difference between groups (RR 1.32, 95% CI 1.00 to 1.74). The increased risk disappeared during extended follow-up (overall 10.7 years' follow-up: RR 1.04, 95% CI 0.84 to 1.29). When deep vein thrombosis was considered as a single outcome (without pulmonary embolism), the rate was lower in the hormone therapy group during extended follow-up (RR 0.63, 95% CI 0.41 to 0.98), although the rate over the entire 10.7 years of intervention and extended follow-up did not indicate clear evidence of a difference between the two groups.

### 1.29 Venous thromboembolism (DVT or PE): oestrogen-only hormone therapy

(Analysis 1.29)

1.29.1 CEE 0.625 mg at up to two years of follow-up (RR 2.22, 95% CI 1.12 to 4.39; 1 study, 10,739 participants) ([WHI 1998](#))

1.29.2 CEE 0.625 mg at 3-year follow-up (RR 6.96, 95% CI 0.36 to 133.75; 1 study, 349 participants) ([PEPI 1995](#))

1.29.3 CEE 0.625 mg at 7.1 years of follow-up may make little or no difference between the groups (RR 1.32, 95% CI 1.00 to 1.74; 1 study, 10,739 participants; moderate-certainty evidence) ([WHI 1998](#)).

1.29.4 CEE 0.625 mg at 10.7 years of follow-up (RR 1.04, 95% CI 0.84 to 1.29; 1 study, 10,739 participants) ([WHI 1998](#))

The combined hormone therapy arm of [WHI 1998](#) reported that relatively healthy women taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) were more likely to present a thromboembolic event than those taking placebo; this applied at one to nearly eight years of follow-up (at one year: RR 3.59, 95% CI 1.95 to 6.61; at two years: RR 2.98, 95% CI 1.88 to 4.71; at three years: RR 2.54, 95% CI 1.73 to 3.72; at a mean of 5.6 years: RR 2.03, 95% CI 1.55 to 2.64; at a mean of 7.9 years: RR 1.65, 95% CI 1.32 to 2.05). Analysis of this comparison revealed that the risk of venous thromboembolism diminishes over time. [WISDOM 2007](#) also reported data for this outcome at one year. Pooling these data with data from [WHI 1998](#) (oestrogen-only hormone therapy arm) resulted in a risk ratio at one year of 4.28 (95% CI 2.49 to 7.34). The absolute risk of an event increased at one year from two per 1000 in the control group to seven per 1000 (95% CI 4 to 11) in the hormone therapy group; at two years from three per 1000 in the control group to nine per 1000 (95% CI 6 to 14) in the hormone therapy group; and at 5.6 years from 10 per 1000 in the control group to 20 per 1000 (95% CI 15 to 26) in the hormone therapy group.

### 1.30 Venous thromboembolism (DVT or PE): combined continuous hormone therapy

([Analysis 1.30](#))

1.30.1 CEE 0.625 mg + MPA 2.5 mg at a mean/median of 1-year follow-up (RR 4.28, 95% CI 2.49 to 7.34;  $I^2 = 7\%$ ; 2 studies, 20,993 participants) ([WHI 1998](#); [WISDOM 2007](#))

1.30.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 2.98, 95% CI 1.88 to 4.71; 1 study, 16,608 participants) ([WHI 1998](#))

1.30.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (RR 2.54, 95% CI 1.73 to 3.72; 1 study, 16,608 participants) ([WHI 1998](#))

1.30.4 CEE 0.625 mg + MPA 2.5 mg at a mean of 5.6 years of follow-up may increase the risk of venous thromboembolism compared to placebo (RR 2.03, 95% CI 1.55 to 2.64; 1 study, 16,608 participants; low-certainty evidence) ([WHI 1998](#))

1.30.5 CEE 0.625 mg + MPA 2.5 mg at a mean of 7.9 years of follow-up (RR 1.65, 95% CI 1.32 to 2.05; 1 study, 16,707 participants) ([WHI 1998](#))

None of the other studies found any clear evidence of a difference between hormone therapy and placebo for this outcome.

### 1.31 Venous thromboembolism (DVT or PE): combined sequential hormone therapy

([Analysis 1.31](#))

1.31.1 1 mg 17-B-oestradiol daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 3.08, 95% CI 0.13 to 74.46; 1 study, 142 participants) ([Tierney 2009](#))

1.31.2 CEE 0.625 mg daily + MPA 10 mg days 1–12 at 3-year follow-up (RR 3.00, 95% CI 0.12 to 73.14; 1 study, 348 participants) ([PEPI 1995](#))

1.31.3 CEE 0.045 mg + 200 mg sequential progesterone at 4-year follow-up (RR 0.40, 95% CI 0.02 to 9.73; 1 study, 505 participants) ([KEEPS 2012](#))

1.31.4 Oestradiol patch 0.05 mg + 200 mg sequential progesterone at 4-year follow-up (RR 1.24, 95% CI 0.08 to 19.69; 1 study, 497 participants) ([KEEPS 2012](#))

1.31.5 Oestradiol 0.25 to 1 mg + 50 or 100 mg progesterone daily at 1-year follow-up (RR 0.27, 95% CI 0.01 to 6.61; 1 study, 1835 participants) ([REPLENISH 2018](#))

### 1.32 Venous thromboembolism (DVT or PE): oestrogen with or without sequential progesterone vaginal gel

([Analysis 1.32](#))

1.32.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of five years of follow-up (RR 1.49, 95% CI 0.25 to 8.83; 1 study, 643 participants) ([ELITE 2014](#))

### 1.33 Global cognitive function

Five studies reported this outcome. They compared the following active treatments to placebo: low-dose oestrogen patches for two years ([Yaffe 2006](#)); combined continuous CEE 0.625 mg with or without MPA 2.5 mg for three years ([Greenspan 2005](#)); oestradiol 1 mg daily with or without intermittent vaginal progesterone gel with follow-up at 2.5 and five years ([ELITE 2014](#)); 0.45 mg oral or 0.05 mg transdermal oestrogen with intermittent progesterone 200 mg for four years ([KEEPS 2012](#)); oestrogen-only hormone therapy for a mean of 5.6 years ([WHI 1998](#) (WHIMS)); and continuous CEE 0.625 mg + MPA 2.5 mg for a mean of 4.2 years ([WHI 1998](#) (WHIMS)). Researchers measured global cognitive function using a cognitive screening test known as the Modified Mini-Mental State Examination (3MSE), on which a higher score reflects better cognitive functioning. [KEEPS 2012](#) included women 42 to 58 years of age at randomisation; [WHI 1998](#) (WHIMS) included only women over 65 years old; and [Yaffe 2006](#) included only participants over 60 years of age.

Over the follow-up periods described above for [Yaffe 2006](#), [Greenspan 2005](#), [KEEPS 2012](#) and [ELITE 2014](#), investigators noted no clear evidence of a difference in cognitive function between intervention and placebo groups. Investigators in [ELITE 2014](#) subgrouped comparisons according to when oestradiol was initiated (within six years of menopause versus 10 or more years after menopause). Study results showed no evidence of differences between the two subgroups in the effect of hormone therapy on cognition at 2.5 years.

In both treatment groups and in both placebo groups of [WHI 1998](#) (WHIMS), mean 3MSE scores increased from baseline and continued to increase for three to five years before they started to decline. Results showed a pattern of higher increases from baseline in 3MSE scores in the placebo groups, which emerged after one to two years and were maintained throughout the study. The mean difference between groups in 3MSE score changes indicated no

clear evidence of a difference between both arms of the study (oestrogen-only hormone therapy arm: weighted mean difference (WMD) -0.26, 95% CI -0.52 to 0.00; combined hormone therapy arm: WMD -0.18, 95% CI -0.36 to 0.00).

In the [WHI 1998](#) (WHIMS) combined hormone therapy arm, a decline of 10 points or more in 3MSE scores (which represents > 2 standard deviations from baseline mean scores) was more likely to occur amongst women in the active treatment group (RR 1.57, 95% CI 1.10 to 2.24). The oestrogen-only hormone therapy group reported no clear evidence of a difference.

### 1.33 Global cognitive function

([Analysis 1.33](#))

#### Oestrogen-only hormone therapy

1.33.1 Transdermal oestradiol 0.014 mg at 2-year follow-up: MMSE scores (baseline MMSE  $\leq$  90) (MD -1.21, 95% CI -5.05 to 2.63; 1 study) ([Yaffe 2006](#))

1.33.2 Transdermal oestradiol 0.014 mg at 2-year follow-up: MMSE scores (baseline MMSE > 90) (MD -0.30, 95% CI -0.73 to 0.13; 1 study) ([Yaffe 2006](#))

1.33.3 CEE 0.625 mg (mod dose) for mean 5.2 years of follow-up: MMSE scores (MD -0.26, 95% CI -0.52 to 0.00; 1 study) ([WHI 1998](#))

#### Oestrogen-only or combined continuous hormone therapy

1.33.4 CEE 0.625 mg with or without 2.5 mg MPA at 3-year follow-up: MMSE scores (MD -0.10, 95% CI -0.35 to 0.15; 1 study) ([Greenspan 2005](#))

#### Combined continuous hormone therapy

1.33.5 Combined continuous CEE 0.625 mg + MPA 2.5 mg for mean 4.2 years of follow-up: MMSE scores (MD -0.18, 95% CI -0.36 to 0.00; 1 study) ([WHI 1998](#))

#### Oestrogen-only or combined sequential hormone therapy

1.33.6 Oestrogen with or without sequential progesterone vaginal gel (MD -0.03, 95% CI -0.21 to 0.15; 1 study) ([ELITE 2014](#))

### 1.34, 1.35 Probable dementia

This outcome was reported by [WHI 1998](#) (WHIMS), which included only postmenopausal women over 65 years of age and compared oestrogen-only hormone therapy (CEE 0.625 mg) to placebo for a mean of 5.6 years, and combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) to placebo for a mean of 4.2 years.

In the oestrogen-only hormone therapy arm, researchers noted no clear evidence of a difference between groups. Participants in the combined hormone therapy arm were more likely to present with dementia than the placebo group (RR 1.97, 95% CI 1.16 to 3.33). At 4.2 years, the absolute risk of probable dementia increased from nine per 1000 in the control group to 18 per 1000 (95% CI 11 to 30) in the hormone therapy group.

### 1.34 Probable dementia: oestrogen-only hormone therapy ([Analysis 1.34](#))

CEE 0.625 mg for 5.2 years of follow-up (RR 1.52, 95% CI 0.89 to 2.59; 1 study, 2947 participants) ([WHI 1998](#))

### 1.35 Probable dementia: combined continuous hormone therapy ([Analysis 1.35](#))

Combined continuous CEE 0.625 mg + MPA 2.5 mg for a mean of 4.05 years (RR 1.99, 95% CI 1.17 to 3.38; 1 study, 4532 participants) ([WHI 1998](#))

No data were available for combined sequential hormone therapy for probable dementia.

## 2. Women with cardiovascular disease

### 2.1, 2.2, 2.3 Death from any cause

Five studies in women with cardiovascular disease measured death from any cause ([ERA 2000](#); [ESPRIT 2002](#); [HERS 1998](#); [WAVE 2002](#); [WEST 2001](#)). The studies assessed four different interventions, comprising comparisons of oestrogen-only hormone therapy or combined continuous hormone therapy versus placebo for durations ranging from two to four years, with unblinded follow-up to 6.8 years ([HERS 1998](#)). Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome in any population group.

### 2.1 Death from any cause: oestrogen-only hormone therapy ([Analysis 2.1](#))

2.1.1 CEE 0.625 mg daily at 3-year follow-up (2.8 to 3.2) (RR 1.31, 95% CI 0.53 to 3.22;  $I^2 = 0\%$ ; 2 studies, 327 participants) ([ERA 2000](#); [WAVE 2002](#))

2.1.2 Oestradiol valerate 2 mg at 2-year follow-up (RR 0.81, 95% CI 0.51 to 1.27; 1 study, 1017 participants) ([ESPRIT 2002](#))

2.1.3 Oestradiol valerate 2 mg at 14 years of follow-up (RR 1.03, 95% CI 0.89 to 1.19; 1 study, 1017 participants) ([ESPRIT 2002](#))

### 2.2 Death from any cause: oestrogen-only or combined sequential hormone therapy ([Analysis 2.2](#))

2.2.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) at 2.8 years of follow-up (RR 1.14, 95% CI 0.77 to 1.67; 1 study, 664 participants) ([WEST 2001](#))

### 2.3 Death from any cause: combined continuous hormone therapy ([Analysis 2.3](#))

2.3.1 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (2.8-3.2) (RR 0.86, 95% CI 0.28 to 2.62;  $I^2 = 49\%$ ; 2 studies, 297 participants) ([ERA 2000](#); [WAVE 2002](#))

2.3.2 CEE 0.625 mg + MPA 2.5 mg at 4 years of follow-up (RR 1.06, 95% CI 0.84 to 1.34; 1 study, 2763 participants) ([HERS 1998](#))

2.3.3 CEE 0.625 mg + MPA 2.5 mg at 4 to 7 years of follow-up (unblinded) (RR 1.14, 95% CI 0.90 to 1.44; 1 study, 2321 participants) ([HERS 1998](#))

No data were available for combined sequential hormone therapy for death from any cause for participants with cardiovascular disease.

## 2.4, 2.5, 2.6 Death from coronary heart disease

Five studies in women with cardiovascular disease measured death from coronary heart disease (ERA 2000; ESPRIT 2002; HERS 1998; WAVE 2002; WEST 2001). The studies assessed four different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from two to four years, with unblinded follow-up to 6.8 years (HERS 1998). Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome in any population group.

### 2.4 Death from coronary heart disease: oestrogen-only hormone therapy (Analysis 2.4)

2.4.1 CEE 0.625 mg daily at 3-year follow-up (2.8-3.2) (RR 1.31, 95% CI 0.36 to 4.77;  $I^2 = 0\%$ ; 2 studies, 327 participants) (ERA 2000; WAVE 2002)

2.4.2 Oestradiol valerate 2 mg at 2-year follow-up (RR 0.69, 95% CI 0.40 to 1.18; 1 study, 1017 participants) (ESPRIT 2002)

### 2.5 Death from coronary heart disease: oestrogen-only or combined sequential hormone therapy (Analysis 2.5)

2.5.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 0.82, 95% CI 0.37 to 1.81; 1 study, 664 participants) (WEST 2001)

### 2.6 Death from coronary heart disease: combined continuous hormone therapy (Analysis 2.6)

2.6.1 CEE 0.625 mg daily + MPA 2.5 mg at 1-year follow-up (RR 1.55, 95% CI 0.73 to 3.29; 1 study, 2763 participants) (HERS 1998)

2.6.2 CEE 0.625 mg daily + MPA 2.5 mg at 2-year follow-up (RR 1.50, 95% CI 0.90 to 2.51; 1 study, 2763 participants) (HERS 1998)

2.6.3 CEE 0.625 mg daily + MPA 2.5 mg at 3-year follow-up (2.8-3.2) (RR 1.30, 95% CI 0.88 to 1.90;  $I^2 = 0\%$ ; 3 studies, 3060 participants) (ERA 2000; HERS 1998; WAVE 2002)

2.6.4 CEE 0.625 mg daily + MPA 2.5 mg at 4+ years of follow-up (median 4.1) (RR 1.19, 95% CI 0.85 to 1.67; 1 study, 2763 participants) (HERS 1998)

2.6.5 CEE 0.625 mg daily + MPA 2.5 mg at 4 to 6.8 years of follow-up (RR 0.99, 95% CI 0.71 to 1.39; 1 study, 2321 participants) (HERS 1998)

### 2.7 Death from stroke

One study in women with a history of stroke, who were taking oestrogen-only hormone therapy (with annual progesterone for those with an intact uterus) for 2.8 years, reported this outcome (WEST 2001). Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome.

### 2.7 Death from stroke: oestrogen-only or combined sequential hormone therapy (Analysis 2.7)

2.7.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) at 2.8 years of follow-up (RR 2.91, 95% CI 0.95 to 8.93; 1 study, 664 participants) (WEST 2001)

No data were available for combined continuous hormone therapy for death from stroke for participants with cardiovascular disease.

### 2.8 Death from any cancer

One study in women with cardiovascular disease taking combined continuous hormone therapy for 4.1 years, with unblinded follow-up to 6.8 years, reported this outcome (HERS 1998). Results of analysis showed no clear evidence of a difference between hormone therapy and placebo for this outcome.

### 2.8 Death from any cancer: combined continuous hormone therapy (Analysis 2.8)

2.8.1 CEE 0.625 mg daily + MPA 2.5 mg for 4+ years (median 4.1) (RR 0.88, 95% CI 0.49 to 1.57; 1 study, 2763 participants) (HERS 1998)

2.8.2 CEE 0.625 mg daily + MPA 2.5 mg for 4 to 6.8 years (unblinded) (RR 1.51, 95% CI 0.86 to 2.65; 1 study, 2321 participants) (HERS 1998)

No data were available for oestrogen-only or combined sequential hormone therapy for death from any cancer for participants with cardiovascular disease.

### 2.9, 2.10, 2.11 Coronary event (myocardial infarction (MI) or cardiac death)

#### 2.9 Coronary event (MI or cardiac death): oestrogen-only hormone therapy (Analysis 2.9)

2.9.1 CEE 0.625 mg daily at 2.8 to 3.2 years of follow-up (RR 1.13, 95% CI 0.54 to 2.40;  $I^2 = 0\%$ ; 2 studies, 327 participants) (ERA 2000; WAVE 2002)

2.9.2 Oestradiol valerate 2 mg at 2-year follow-up (RR 1.00, 95% CI 0.72 to 1.39; 1 study, 1017 participants) (ESPRIT 2002)

2.9.2 Oestradiol valerate 2 mg at 14 years of follow-up (RR 1.21, 95% CI 0.91 to 1.62; 1 study, 1017 participants) (ESPRIT 2002)

#### 2.10 Coronary event (MI or cardiac death): oestrogen-only or combined sequential hormone therapy (Analysis 2.10)

2.10.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 0.97, 95% CI 0.57 to 1.65; 1 study, 664 participants) (WEST 2001)

#### 2.11 Coronary event (MI or cardiac death): combined continuous hormone therapy (Analysis 2.11)

2.11.1 CEE 0.625 mg + MPA 2.5 mg at 1-year follow-up (RR 1.50, 95% CI 1.00 to 2.25; 1 study, 2763 participants) (HERS 1998)

2.11.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 1.20, 95% CI 0.91 to 1.58; 1 study, 2763 participants) (HERS 1998)

2.11.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (2.8-3.2) (RR 1.07, 95% CI 0.86 to 1.33;  $I^2 = 0\%$ ; 3 studies, 3060 participants) (ERA 2000; HERS 1998; WAVE 2002)

2.11.4 CEE 0.625 mg + MPA 2.5 mg at a median of 4.1 years of follow-up (RR 0.99, 95% CI 0.81 to 1.19; 1 study, 2763 participants) (HERS 1998)

2.11.5 CEE 0.625 mg + MPA 2.5 mg at 4 to 6.8 years of follow-up (unblinded) (RR 1.01, 95% CI 0.78 to 1.29; 1 study, 2321 participants) ([HERS 1998](#))

## 2.12, 2.13, 2.14 Stroke (first or recurrent)

Five studies measured this outcome in women with cardiovascular disease ([ESPRIT 2002](#); [EVTET 2000](#); [HERS 1998](#); [WAVE 2002](#); [WEST 2001](#)). The studies assessed five different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from one to four years, with unblinded follow-up to 6.8 years ([HERS 1998](#)). None of the studies found clear evidence of a difference between hormone therapy and placebo for this outcome.

## 2.12 Stroke (first or recurrent): oestrogen-only hormone therapy ([Analysis 2.12](#))

2.12.1 CEE 0.625 mg daily for 2.8 years of follow-up (RR 0.69, 95% CI 0.12 to 3.98; 1 study, 122 participants) ([WAVE 2002](#))

2.12.2 Oestradiol valerate 2 mg at 2-year follow-up (RR 1.64, 95% CI 0.60 to 4.47; 1 study, 1017 participants) ([ESPRIT 2002](#))

2.12.2 Oestradiol valerate 2 mg at 14 years of follow-up (RR 1.13, 95% CI 0.54 to 2.36; 1 study, 1017 participants) ([ESPRIT 2002](#))

## 2.13 Stroke (first or recurrent): oestrogen-only or combined sequential hormone therapy ([Analysis 2.13](#))

2.13.1 Oestradiol 1 mg (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 1.09, 95% CI 0.79 to 1.51; 1 study, 664 participants) ([WEST 2001](#))

## 2.14 Stroke (first or recurrent): combined continuous hormone therapy ([Analysis 2.14](#))

2.14.1 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 years of follow-up (RR 0.32, 95% CI 0.01 to 7.82; 1 study, 140 participants) ([EVTET 2000](#))

2.14.2 CEE 0.625 mg + MPA 2.5 mg for 2.8 years of follow-up (RR 5.23, 95% CI 0.26 to 105.85; 1 study, 88 participants) ([WAVE 2002](#))

2.14.3 CEE 0.625 mg + MPA 2.5 mg for a median 4.1 years of follow-up (RR 1.23, 95% CI 0.90 to 1.68; 1 study, 2763 participants) ([HERS 1998](#))

2.14.4 CEE 0.625 mg + MPA 2.5 mg for 4 to 6.8 years of follow-up (unblinded) (RR 1.05, 95% CI 0.71 to 1.57; 1 study, 2321 participants) ([HERS 1998](#))

## 2.15, 2.16, 2.17 Transient ischaemic attack

Three studies in women with cardiovascular disease measured this outcome ([ESPRIT 2002](#); [HERS 1998](#); [WEST 2001](#)). The studies assessed three different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from two to four years, with unblinded follow-up to 6.8 years ([HERS 1998](#)). Results of analysis showed no clear evidence of a difference between hormone therapy and placebo for this outcome.

## 2.15 Transient ischaemic attack: oestrogen-only hormone therapy ([Analysis 2.15](#))

2.15.1 Oestradiol valerate 2 mg at 2-year follow-up (RR 1.13, 95% CI 0.54 to 2.36; 1 study, 1017 participants) ([ESPRIT 2002](#))

## 2.16 Transient ischaemic attack: oestrogen-only or combined sequential hormone therapy ([Analysis 2.16](#))

2.16.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 1.16, 95% CI 0.70 to 1.94; 1 study, 664 participants) ([WEST 2001](#))

## 2.17 Transient ischaemic attack: combined continuous hormone therapy ([Analysis 2.17](#))

2.17.1 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 0.80, 95% CI 0.51 to 1.23; 1 study, 2763 participants) ([HERS 1998](#))

2.17.2 CEE 0.625 mg + MPA 2.5 mg for 4 to 6.8 years of follow-up (unblinded) (RR 0.95, 95% CI 0.49 to 1.84; 1 study, 2321 participants) ([HERS 1998](#))

## 2.18, 2.19 Stroke or transient ischaemic attack

One study in women with known coronary disease taking oestrogen-only hormone therapy, combined continuous therapy or placebo also reported this composite outcome at 3.2 years' mean follow-up ([ERA 2000](#)). Results of analysis show no clear evidence of a difference for this outcome between the groups.

## 2.18 Stroke or transient ischaemic attack: oestrogen-only hormone therapy ([Analysis 2.18](#))

2.18.1 CEE 0.625 mg daily for 3.2 years of follow-up (RR 0.88, 95% CI 0.28 to 2.78; 1 study, 205 participants) ([ERA 2000](#))

## 2.19 Stroke or transient ischaemic attack: combined continuous hormone therapy ([Analysis 2.19](#))

2.19.1 CEE 0.625 mg + MPA 2.5 mg for 3.2 years of follow-up (RR 1.01, 95% CI 0.34 to 3.03; 1 study, 209 participants) ([ERA 2000](#))

No data were available for combined sequential hormone therapy for stroke or transient ischaemic attack for women with cardiovascular disease.

## 2.20, 2.21 Venous thromboembolism (first or recurrent PE or DVT)

Five studies in women with cardiovascular disease measured this outcome ([ERA 2000](#); [ESPRIT 2002](#); [EVTET 2000](#); [HERS 1998](#); [WAVE 2002](#)). The studies assessed five different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from one to four years, with unblinded follow-up to 6.8 years ([HERS 1998](#)).

In [HERS 1998](#), participants with cardiovascular disease who were taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) for one to four years were more likely to experience venous thromboembolism (VTE) than those on placebo at one year (RR 3.26, 95% CI 1.06 to 9.96), two years (RR 3.51, 95% CI 1.42 to 8.66), three years (RR 3.01, 95% CI 1.50 to 6.04) and at a mean of 4.1 years (RR 2.62, 95% CI 1.39 to 4.94) ([Analysis 2.21](#)). The absolute risk of an event increased at one year from three per 1000 in the control group to nine per 1000 (95% CI 3 to 29) in the hormone therapy

group; at two years from four per 1000 in the control group to 15 per 1000 (95% CI 6 to 38) in the hormone therapy group; and at 4.1 years from nine per 1000 in the control group to 13 per 1000 (95% CI 6 to 28) in the hormone therapy group.

None of the other studies found any clear evidence of a difference between hormone therapy and placebo for this outcome.

## 2.20 VTE (first or recurrent PE or DVT): oestrogen-only hormone therapy (Analysis 2.20)

2.20.1 Oestradiol valerate 2 mg at 2-year follow-up (RR 1.23, 95% CI 0.33 to 4.55; 1 study, 1017 participants) ([ESPRIT 2002](#))

2.20.2 CEE 0.625 mg daily for 2.8 to 3.2 years of follow-up (RR 1.64, 95% CI 0.44 to 6.17;  $I^2 = 66\%$ ; 2 studies, 327 participants) ([ERA 2000](#); [WAVE 2002](#))

## 2.21 VTE (first or recurrent PE or DVT): combined continuous hormone therapy (Analysis 2.21)

2.21.1 CEE 0.625 mg + MPA 2.5 mg for 1 year follow-up (RR 3.26, 95% CI 1.06 to 9.96; 1 study, 2763 participants) ([HERS 1998](#))

2.21.2 Continuous oestradiol 2 mg + norethisterone acetate 1 mg for 1.3 years of follow-up (RR 6.80, 95% CI 0.86 to 53.85; 1 study, 140 participants) ([EVTET 2000](#))

2.21.3 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 3.51, 95% CI 1.42 to 8.66; 1 study, 2763 participants) ([HERS 1998](#))

2.21.4 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (2.8 to 3.2) (RR 3.01, 95% CI 1.50 to 6.04;  $I^2 = 0\%$ ; 3 studies, 3060 participants) ([ERA 2000](#); [HERS 1998](#); [WAVE 2002](#))

2.21.5 CEE 0.625 mg + MPA 2.5 mg for median 4.1 years of follow-up (RR 2.62, 95% CI 1.39 to 4.94; 1 study, 2763 participants) ([HERS 1998](#))

2.21.6 CEE 0.625 mg + MPA 2.5 mg for 4 to 7 years of follow-up (unblinded) (RR 1.37, 95% CI 0.63 to 2.98; 1 study, 2321 participants) ([HERS 1998](#))

No data were available for combined sequential hormone therapy for VTE for women with cardiovascular disease.

## 3. Women with dementia

One small study included women with mild to moderate Alzheimer's disease ([Mulnard 2000](#)). Researchers compared unopposed oestrogen for one year to placebo. The study's primary outcome was change in overall status on the Alzheimer's Disease Co-operative Study version of the Clinical Global Impression of Change Scale. Results of analysis show no clear evidence of a difference between the groups: unopposed CEE 0.625 mg or 1.25 mg daily at 1-year follow-up (RR 1.10, 95% CI 0.88 to 1.38; 1 study, 120 participants; [Analysis 3.1](#)). [Mulnard 2000](#) did not measure any other review outcomes of interest.

## 4. Women post-surgery for early-stage endometrial cancer

[Barakat 2006](#) was the only included study that enrolled participants who had undergone surgery for stage I or II endometrial cancer. It compared oestrogen-only hormone therapy (CEE 0.625 mg daily) to placebo in 1236 participants for a median of three years of follow-

up. It assessed three mortality outcomes, but no cardiovascular events or cognitive measures.

### 4.1 Death from any cause

Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome (RR 1.37, 95% CI 0.77 to 2.45; 1 study, 1236 participants; [Analysis 4.1](#)).

### 4.2 Death from endometrial cancer

Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome (RR 1.25, 95% CI 0.34 to 4.63; 1 study, 1236 participants; [Analysis 4.2](#)).

### 4.3 Death from CHD

Results of analysis show no clear evidence of a difference between hormone therapy and placebo for this outcome (RR 1.25, 95% CI 0.34 to 4.63; 1 study, 1236 participants; [Analysis 4.3](#)).

## 5. Women hospitalised with chronic illness

[Nachtigall 1979](#) was the only included study that enrolled participants hospitalised for chronic disease or because they required custodial care. It compared combined sequential hormone therapy (CEE 2.5 mg daily plus MPA 10 mg for seven days each cycle) to placebo in 168 participants for 10 years. [Nachtigall 1979](#) assessed death from any cause, coronary events and venous thromboembolism, but no cognitive measures or quality of life outcomes.

### 5.1 Death from any cause

[Nachtigall 1979](#) reported no clear evidence of a difference between hormone therapy and placebo for this outcome (RR 0.43, 95% CI 0.11 to 1.60; 1 study, 168 participants; [Analysis 5.1](#)).

### 5.2 Coronary events (myocardial infarction)

[Nachtigall 1979](#) found no clear evidence of a difference between hormone therapy and placebo (RR 0.33, 95% CI 0.04 to 3.14; 1 study, 168 participants; [Analysis 5.2](#)).

### 5.3 Venous thromboembolism (DVT or PE)

[Nachtigall 1979](#) found no clear evidence of a difference between hormone therapy and placebo (RR 0.33, 95% CI 0.01 to 8.07; 1 study, 168 participants; [Analysis 5.3](#)).

## 6. All peri- and postmenopausal women

### 6.1, 6.2, 6.3, 6.4, 6.5 Breast cancer

Nine studies reported this outcome in relatively healthy women ([ELITE 2014](#); [EPAT 2001](#); [EPHT 2006](#); [Greenspan 2005](#); [KEEPS 2012](#); [Notelovitz 2002](#); [PEPI 1995](#); [WHI 1998](#); [WISDOM 2007](#)). The studies assessed 11 different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from one year to nearly eight years. [WHI 1998](#) (combined hormone therapy arm) extended follow-up beyond the planned completion date to achieve a mean 11-year follow-up for this outcome in the 85% of participants who consented to stay in the study; [WHI 1998](#) (oestrogen-only arm) extended follow-up to a total of 10.7 years in the 78% of participants who agreed to continue.

Four studies in women with cardiovascular disease measured this outcome (ERA 2000; ESPRIT 2002; HERS 1998; WAVE 2002). The studies assessed four different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from two to four years, with unblinded follow-up to 7.1 years (HERS 1998).

Finally, Nachtigall 1979 also measured this outcome, comparing combined sequential hormone therapy to placebo for 10 years in women hospitalised for chronic disease or because they required custodial care.

WHI 1998 (oestrogen-only hormone therapy arm) reported no clear evidence of a difference in the risk of breast cancer at 7.1 years of follow-up among relatively healthy participants taking oestrogen-only hormone therapy (CEE 0.625 mg) compared with those taking placebo (RR 0.79, 95% CI 0.61 to 1.01). Follow-up continued for a median of 5.8 years after the intervention phase. The overall cumulative breast cancer incidence over the 10.7 years of mean follow-up (median 11.8 years) showed that participants in the hormone therapy group (RR 0.78, 95% CI 0.63 to 0.96) were less likely to have breast cancer. The absolute risk of breast cancer decreased over 10.7 years of follow-up from 37 per 1000 in the control group to 29 per 1000 (95% CI 23 to 35) in the hormone therapy group. The overall cumulative rate remained lower after a median of 13 years of follow-up (RR 0.80, 95% CI 0.65 to 0.97). Study authors noted that when event rates in the early and late post-intervention periods were compared, the hazard ratios (HRs) for breast cancer indicated clear evidence of a difference ( $P = 0.04$ ). The difference between groups in HR for breast cancer diminished over time and disappeared at approximately 4.5 years post-intervention (Chlebowski 2015a).

WHI 1998 (combined hormone therapy arm) reported this outcome at yearly intervals. Results showed no clear evidence of a difference between groups in the incidence of breast cancer during the first four years of follow-up, but the hormone therapy group was at a higher risk of breast cancer after taking hormone therapy for five or more years (at a mean of 5.6 years of follow-up: RR 1.27, 95% CI 1.03 to 1.56; at a mean of 7.9 years of follow-up: RR 1.27, 95% CI 1.07 to 1.52). The absolute risk of breast cancer increased at 5.6 years of follow-up from 19 per 1000 in the control group to 23 per 1000 (95% CI 19 to 29) in the hormone therapy group; and at 7.9 years of follow-up from 26 per 1000 in the control group to 33 per 1000 (95% CI 28 to 40) in the hormone therapy group. Analysis in this arm of WHI 1998 revealed a trend for increasing breast cancer risk over time in the group taking hormone therapy. WISDOM 2007 also reported data for this outcome at a median follow-up of one year. Pooling these data with data from WHI 1998 (combined hormone therapy arm) resulted in a reduced risk of breast cancer at one year in the hormone therapy arm (RR 0.52, 95% CI 0.28 to 0.96). However, at a mean of 11 years of follow-up in WHI 1998 (combined hormone therapy arm), the rate of invasive breast cancer was higher in the hormone therapy arm (RR 1.25, 95% CI 1.08 to 1.45). Rates remained higher in the intervention arm at a median of 13.2 years of follow-up (RR 1.28, 95% CI 1.11 to 1.47) (Analysis 6.3). Breast cancers diagnosed in the hormone therapy group were of similar histology and stage to those diagnosed amongst controls but were more likely to be node positive ( $P = 0.03$ ).

Results of analysis show no clear evidence of a difference between any other type of hormone therapy and placebo for this outcome, although (as noted above) relevant studies were small.

### 6.1 Breast cancer: oestrogen-only hormone therapy

#### (Analysis 6.1)

6.1.1 Oestrogen-only patch 0.025 mg daily at 2-year follow-up (RR 2.93, 95% CI 0.12 to 71.04; 1 study, 176 participants) (Notelovitz 2002)

6.1.2 Oestradiol 1 mg at 2-year follow-up (RR 0.33, 95% CI 0.01 to 8.10; 1 study, 222 participants) (EPAT 2001)

6.1.3 Oestradiol valerate 2 mg at 2-year follow-up (RR 0.98, 95% CI 0.25 to 3.91; 1 study, 1017 participants) (ESPRIT 2002)

6.1.4 Oestradiol patch 0.075 mg at 2-year follow-up (RR 2.93, 95% CI 0.12 to 71.04; 1 study, 176 participants) (Notelovitz 2002)

6.1.5 CEE 0.625 mg for 2.8 to 3.2 years of follow-up (RR 2.05, 95% CI 0.38 to 11.04;  $I^2 = 0\%$ ; 3 studies, 676 participants) (ERA 2000; PEPI 1995; WAVE 2002)

6.1.6 CEE 0.625 mg at 7.1 years of follow-up probably makes little or no difference between the groups (RR 0.79, 95% CI 0.61 to 1.01; 1 study, 10,739 participants; moderate-certainty evidence) (WHI 1998)

6.1.7 CEE 0.625 mg after 10.7 years of follow-up (RR 0.78, 95% CI 0.63 to 0.96; 1 study, 10,739 participants) (WHI 1998)

6.1.8 CEE 0.625 mg after 13 years of follow-up (RR 0.80, 95% CI 0.65 to 0.97; 1 study, 10,739 participants) (WHI 1998)

6.1.9 CEE 0.625 mg up to 21 years of follow-up (RR 0.82, 95% CI 0.70 to 0.97; 1 study, 10,739 participants) (WHI 1998)

### 6.2 Breast cancer: oestrogen-only or combined hormone therapy

#### (Analysis 6.2)

6.2.1 CEE 0.625 mg with or without 2.5 mg MPA at 3-year follow-up (OR 0.99, 95% CI 0.14 to 7.14; 1 study, 373 participants) (Greenspan 2005)

### 6.3 Breast cancer: combined continuous hormone therapy

#### (Analysis 6.3)

6.3.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1-year follow-up (RR 0.52, 95% CI 0.28 to 0.96;  $I^2 = 0\%$ ; 2 studies, 23,182 participants) (WHI 1998; WISDOM 2007)

6.3.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 0.71, 95% CI 0.47 to 1.08; 1 study, 16,608 participants) (WHI 1998)

6.3.3 CEE 0.625 mg + MPA 2.5 mg for 2.8 to 3.4 years of follow-up (RR 0.85, 95% CI 0.62 to 1.18;  $I^2 = 0\%$ ; 3 studies, 17733 participants) (EPHT 2006; PEPI 1995; WHI 1998)

6.3.4 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 1.36, 95% CI 0.82 to 2.27; 1 study, 2763 participants) (HERS 1998)

6.3.5 CEE 0.625 mg + MPA 2.5 mg for 5.6 years of follow-up probably increases the risk of breast cancer compared to placebo (RR 1.27,

95% CI 1.03 to 1.56; 1 study, 16,608 participants; moderate-certainty evidence) ([WHI 1998](#))

6.3.6 CEE 0.625 mg + MPA 2.5 mg for 4 to 7 years of follow-up (RR 1.08, 95% CI 0.52 to 2.23; 1 study, 2321 participants) ([HERS 1998](#))

6.3.7 CEE 0.625 mg + MPA 2.5 mg for 7.9 years of follow-up (RR 1.27, 95% CI 1.07 to 1.52; 1 study, 16607 participants) ([WHI 1998](#))

6.3.8 CEE 0.625 mg + MPA 2.5 mg after 11 years of follow-up (RR 1.25, 95% CI 1.08 to 1.45; 1 study, 16,608 participants) ([WHI 1998](#))

6.3.9 CEE 0.625 mg + MPA 2.5 mg after 13.2 years of follow-up (RR 1.28, 95% CI 1.11 to 1.47; 1 study, 16,608 participants) ([WHI 1998](#))

6.3.10 CEE 0.625 mg + MPA 2.5 mg after 20 years of cumulative follow-up (RR 1.24, 95% CI 1.10 to 1.40; 1 study, 16,608 participants) ([WHI 1998](#))

6.3.11 Oestradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily for one year (RR 1.17, 95% CI 0.07 to 20.72; 1 study, 1835 participants) ([REPLENISH 2018](#))

#### 6.4 Breast cancer: combined sequential hormone therapy

([Analysis 6.4](#))

6.4.1 CEE 0.625 mg daily + MPA 10 mg days 1–12 at 3-year follow-up (RR 2.00, 95% CI 0.18 to 21.85; 1 study, 348 participants) ([PEPI 1995](#))

6.4.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1–12 at 3-year follow-up (RR 3.91, 95% CI 0.44 to 34.64; 1 study, 352 participants) ([PEPI 1995](#))

6.4.3 CEE 0.045 mg daily + sequential progesterone 200 mg for 4 years of follow-up (RR 1.79, 95% CI 0.30 to 10.64; 1 study, 505 participants) ([KEEPS 2012](#))

6.4.4 Oestradiol patch 0.05 mg + sequential progesterone 200 mg for 4 years of follow-up (RR 1.86, 95% CI 0.31 to 11.02; 1 study, 497 participants) ([KEEPS 2012](#))

6.4.5 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle for 10 years of follow-up (RR 0.11, 95% CI 0.01 to 2.03; 1 study, 168 participants) ([Nachtigall 1979](#))

6.4.6 Oestradiol valerate 1 mg + MPA 5 mg 12 days/month at 5 years of follow-up (RR 0.48, 95% CI 0.04 to 5.19; 1 study, 153 participants) ([Ran 2017](#))

#### 6.5 Breast cancer: oestrogen with or without sequential progesterone vaginal gel

([Analysis 6.5](#))

6.5.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of 5 years of follow-up (RR 1.24, 95% CI 0.50 to 3.10; 1 study, 643 participants) ([ELITE 2014](#))

#### 6.6, 6.7, 6.8, 6.9, 6.10 Colorectal cancer

Seven studies reported this outcome ([ELITE 2014](#); [EPAT 2001](#); [HERS 1998](#); [Greenspan 2005](#); [PEPI 1995](#); [Tierney 2009](#); [WHI 1998](#), [WISDOM 2007](#)). The studies investigated seven different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging

from one year to nearly eight years, with extended follow-up to 10.7 years in [WHI 1998](#) (oestrogen-only arm). Investigators also reported this outcome after extended follow-up in [WHI 1998](#) (combined hormone therapy arm) at 11.6 years ([Simon 2012](#)) and at 13.2 years ([Manson 2013](#)).

[Nachtigall 1979](#) also measured this outcome, comparing combined sequential hormone therapy to placebo for 10 years in women hospitalised for chronic disease or because they required custodial care.

[WHI 1998](#) (combined hormone therapy arm) reported no clear evidence of a difference in the incidence of colorectal cancer among relatively healthy participants taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) compared to those taking placebo, at one to four years' follow-up. However, participants taking combined continuous hormone therapy had a lower incidence of colon cancer at a mean follow-up of 5.6 years (RR 0.64, 95% CI 0.44 to 0.91). The absolute risk of colorectal cancer decreased from nine per 1000 in the control group to six per 1000 (95% CI 4 to 8) in the hormone therapy group. Rates tended to favour the hormone therapy group over extended follow-up: there was no clear evidence of a difference at 7.9 years (RR 0.76, 95% CI 0.57 to 1.01) nor at 13.2 years (RR 0.80, 95% CI 0.63 to 1.01), but there was evidence of a difference at 11.6 years (RR 0.78, 95% CI 0.61 to 0.99).

Results of analysis show no clear evidence of a difference between any other type of hormone therapy and placebo for this outcome.

#### 6.6 Colorectal cancer: oestrogen-only hormone therapy

([Analysis 6.6](#))

6.6.1 CEE 0.625 mg at 3-year follow-up (RR 0.33, 95% CI 0.01 to 8.08; 1 study, 349 participants) ([PEPI 1995](#))

6.6.2 CEE 0.625 mg for 7.1 years of follow-up (RR 1.15, 95% CI 0.81 to 1.63; 1 study, 10,739 participants) ([WHI 1998](#))

6.6.3 CEE 0.625 mg for 10.7 years of follow-up (RR 1.11, 95% CI 0.82 to 1.49; 1 study, 10,739 participants) ([WHI 1998](#))

#### 6.7 Colorectal cancer: oestrogen-only or combined hormone therapy

([Analysis 6.7](#))

6.7.1 CEE 0.625 mg with or without 2.5 mg MPA at 3-year follow-up (OR 3.02, 95% CI 0.31 to 29.26; 1 study, 373 participants) ([Greenspan 2005](#))

#### 6.8 Colorectal cancer: combined continuous hormone therapy

([Analysis 6.8](#))

6.8.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1-year follow-up (RR 0.68, 95% CI 0.32 to 1.42;  $I^2 = 0\%$ ; 2 studies, 20,993 participants) ([WHI 1998](#); [WISDOM 2007](#))

6.8.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 0.83, 95% CI 0.46 to 1.50; 1 study, 16,608 participants) ([WHI 1998](#))

6.8.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (RR 0.81, 95% CI 0.49 to 1.34;  $I^2 = 0\%$ ; 2 studies, 16,956 participants) ([PEPI 1995](#); [WHI 1998](#))

6.8.4 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 0.69, 95% CI 0.32 to 1.48; 1 study, 2763 participants) ([HERS 1998](#))

6.8.5 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 years of follow-up (RR 0.64, 95% CI 0.44 to 0.91; 1 study, 16,608 participants) ([WHI 1998](#))

6.8.6 CEE 0.625 mg + MPA 2.5 mg for 4 to 6.8 years of follow-up (unblinded) (RR 0.81, 95% CI 0.46 to 1.44; 1 study, 2321 participants) ([HERS 1998](#))

6.8.7 CEE 0.625 mg + MPA 2.5 mg after 7.9 years of follow-up (RR 0.76, 95% CI 0.57 to 1.01; 1 study, 16,608 participants) ([WHI 1998](#))

6.8.8 CEE 0.625 mg + MPA 2.5 mg after 11.6 years of follow-up (RR 0.78, 95% CI 0.61 to 0.99; 1 study, 16,608 participants) ([WHI 1998](#))

6.8.9 CEE 0.625 mg + MPA 2.5 mg after 13.2 years of follow-up (RR 0.80, 95% CI 0.63 to 1.01; 1 study, 16,608 participants) ([WHI 1998](#))

### 6.9 Colorectal cancer: combined sequential hormone therapy

([Analysis 6.9](#))

6.9.1 CEE 0.625 mg daily + MPA 10 mg days 1–12 at 3-year follow-up (RR 0.33, 95% CI 0.01 to 8.13; 1 study, 348 participants) ([PEPI 1995](#))

6.9.10 CEE 0.625 mg daily + micronised progesterone 200 mg days 1–12 at 3-year follow-up (RR 0.33, 95% CI 0.01 to 7.95; 1 study, 352 participants) ([PEPI 1995](#))

6.9.11 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle for 10 years of follow-up (RR 1.00, 95% CI 0.06 to 15.73; 1 study, 168 participants) ([Nachtigall 1979](#))

### 6.10 Colorectal cancer: oestrogen with or without sequential progesterone vaginal gel

([Analysis 6.10](#))

6.10.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of 5 years of follow-up (RR 1.49, 95% CI 0.25 to 8.83; 1 study, 643 participants) ([ELITE 2014](#))

### 6.11, 6.12, 6.13 Lung cancer

[WHI 1998](#) (oestrogen-only arm) reported this outcome in relatively healthy women after a mean follow-up of 7.1 years; [WHI 1998](#) (combined hormone therapy arm) reported this outcome at 5.6 years and after extended follow-up at 7.9 years and 14 years. Results of analysis show no clear evidence of a difference between hormone therapy and placebo groups for this outcome.

### 6.11 Lung cancer: oestrogen-only hormone therapy

([Analysis 6.11](#))

6.11.1 CEE 0.625 mg may make little or no difference to any type of lung cancer (non-small cell or small cell) at 7.1 years of follow-up compared to placebo (RR 1.04, 95% CI 0.73 to 1.48; 1 study, 10,739 participants; low-certainty evidence) ([WHI 1998](#))

### 6.12 Lung cancer: combined continuous hormone therapy

([Analysis 6.12](#))

6.12.1 CEE 0.625 mg + MPA 2.5 mg probably makes little or no difference to any lung cancer (non-small cell or small cell) at 5.6 years of follow-up compared to placebo (RR 1.06, 95% CI 0.77 to

1.46; 1 study, 16,608 participants; moderate-certainty evidence) ([WHI 1998](#))

6.12.2 CEE 0.625 mg + MPA 2.5 mg may make little or no difference to any lung cancer at 7.9 years of follow-up (RR 1.22, 95% CI 0.92 to 1.62; 1 study, 16,608 participants) ([WHI 1998](#))

6.12.3 CEE 0.625 mg + MPA 2.5 mg probably makes little or no difference to any lung cancer after median 14 years of follow-up (RR 1.13, 95% CI 0.93 to 1.38; 1 study, 16,608 participants) ([WHI 1998](#))

### 6.13 Lung cancer: combined sequential hormone therapy

([Analysis 6.13](#))

6.13.1 17-B-oestradiol 1 mg daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (OR 3.13, 95% CI 0.13 to 78.13; 1 study, 142 participants) ([Tierney 2009](#))

### 6.14, 6.15, 6.16, 6.17 Endometrial cancer

Ten studies reported this outcome ([EPAT 2001](#); [ESPRIT 2002](#); [Ferency 2002](#); [HERS 1998](#); [KEEPS 2012](#); [Nachtigall 1979](#); [Obel 1993](#); [PEPI 1995](#); [WHI 1998](#) (combined hormone therapy arm); [Ran 2017](#)). The studies assessed 14 different interventions, comprising comparisons of oestrogen-only hormone therapy, combined continuous hormone therapy and combined sequential hormone therapy versus placebo, for durations ranging from one year to nearly 10 years. [WHI 1998](#) (combined hormone therapy arm) also reported this outcome after extended follow-up, at 13.2 years. The [Barakat 2006](#) study of oestrogen-only hormone therapy versus placebo in women who had undergone surgery for stage I or II endometrial cancer measured recurrent endometrial cancer. In comparisons of oestrogen-only hormone therapy versus placebo ([EPAT 2001](#); [ESPRIT 2002](#); [PEPI 1995](#)), all women with a uterus were monitored closely for endometrial hyperplasia, and two studies specified that study medications were withdrawn if atypical hyperplasia was detected ([ESPRIT 2002](#); [PEPI 1995](#)).

At 13 years' median follow-up, rates of endometrial cancer were lower in the combined hormone therapy group (RR 0.66, 95% CI 0.48 to 0.90) in [WHI 1998](#).

Results of analysis showed no clear evidence of a difference between hormone therapy and placebo for this outcome, and no clear evidence of a difference between groups in rates of recurrent endometrial cancer. One study reported no events ([Obel 1993](#)).

### 6.14 Endometrial cancer: oestrogen-only hormone therapy

[Analysis 6.14](#)

6.14.1 CEE 0.625 mg for 3 to 3.2 years of follow-up (RR 0.33, 95% CI 0.01 to 8.10; 1 study, 238 participants) ([PEPI 1995](#))

### 6.15 Endometrial cancer: combined continuous hormone therapy

[Analysis 6.15](#)

6.15.1 CEE 0.625 mg + MPA 2.5 mg at 1-year follow-up (RR 0.95, 95% CI 0.13 to 6.76; 1 study, 16,608 participants) ([WHI 1998](#))

6.15.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 0.95, 95% CI 0.31 to 2.95; 1 study, 16,608 participants) ([WHI 1998](#))

6.15.3 CEE 0.625 mg + MPA 2.5 mg for 3 to 3.2 years of follow-up (RR 0.80, 95% CI 0.35 to 1.82;  $I^2 = 0\%$ ; 2 studies, 16847 participants) (PEPI 1995; WHI 1998)

6.15.4 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 0.40, 95% CI 0.08 to 2.06; 1 study, 2763 participants) (HERS 1998)

6.15.5 CEE 0.625 mg + MPA 2.5 mg for 5.6 years of follow-up (RR 0.86, 95% CI 0.51 to 1.44; 1 study, 16,608 participants) (WHI 1998)

6.15.6 CEE 0.625 mg + MPA 2.5 mg for 4 to 6.8 years of follow-up (unblinded) (RR 0.14, 95% CI 0.01 to 2.78; 1 study, 2321 participants) (HERS 1998)

6.15.7 CEE 0.625 mg + MPA 2.5 mg for 7.9 years of follow-up (RR 0.81, 95% CI 0.54 to 1.20; 1 study, 16,608 participants) (WHI 1998)

6.15.8 CEE 0.625 mg + MPA 2.5 mg after a median of 13 years of follow-up (RR 0.66, 95% CI 0.48 to 0.90; 1 study, 16,608 participants) (WHI 1998)

6.15.9 Oestradiol 0.25 to 1 mg + 50 to 100 mg progesterone for one year at 1-year follow-up (not estimable) (REPLENISH 2018)

#### 6.16 Endometrial cancer: combined sequential hormone therapy

##### Analysis 6.16

6.16.1 17-B-oestradiol 1 mg + dydrogesterone 5 mg days 14 to 28 at 2-year follow-up (RR 1.90, 95% CI 0.08 to 45.95; 1 study, 163 participants) (Ferenczy 2002)

6.16.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1 to 12 at 3-year follow-up (RR 0.33, 95% CI 0.01 to 8.03; 1 study, 239 participants) (PEPI 1995)

6.16.3 Oestradiol 2 mg + dihydrogesterone 20 mg at 2-year follow-up (RR 3.30, 95% CI 0.16 to 67.59; 1 study, 159 participants) (Ferenczy 2002)

6.16.4 CEE 0.045 mg + 200 mg sequential progesterone for 4 years of follow-up (RR 5.97, 95% CI 0.29 to 123.81; 1 study, 505 participants) (KEEPS 2012)

6.16.5 Oestradiol patch 0.05 mg + 200 mg sequential progesterone for 4 years of follow-up (RR 3.71, 95% CI 0.15 to 90.70; 1 study, 497 participants) (KEEPS 2012)

6.16.6 CEE 2.5 mg daily + MPA 10 mg for 7 days each cycle for 10 years of follow-up (RR 0.33, 95% CI 0.01 to 8.07; 1 study, 168 participants) (Nachtigall 1979)

#### 6.17 Recurrent endometrial cancer: oestrogen-only hormone therapy

##### Analysis 6.17

6.17.1 CEE 0.625 mg for a median of 3 years of follow-up (RR 1.17, 95% CI 0.54 to 2.50; 1 study, 1236 participants) (Barakat 2006)

#### 6.18, 6.19 Ovarian cancer

WHI 1998 (combined hormone therapy arm), which used combined continuous CEE 0.625 mg + MPA 2.5 mg, reported ovarian cancer incidence at 5.6 years of mean follow-up and again after extended follow-up at 13.2 years (Manson 2013). ELITE 2014, which utilised oestrogen with or without sequential progesterone vaginal gel,

reported a single event. Results of analysis showed no clear evidence of a difference between groups for this outcome.

#### 6.18 Ovarian cancer: combined continuous hormone therapy

##### Analysis 6.18

6.18.1 CEE 0.625 mg + MPA 2.5 mg for 5.6 years of follow-up (RR 1.43, 95% CI 0.76 to 2.69; 1 study, 16,608 participants) (WHI 1998)

6.18.2 CEE 0.625 mg + MPA 2.5 mg after 13.2 years of follow-up (RR 1.23, 95% CI 0.82 to 1.85; 1 study, 16,608 participants) (WHI 1998)

#### 6.19 Ovarian cancer: oestrogen with or without sequential progesterone vaginal gel

##### Analysis 6.19

6.19.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after a median of 5 years of follow-up (RR 0.33, 95% CI 0.01 to 8.08; 1 study, 643 participants) (ELITE 2014)

#### 6.20, 6.21, 6.22 Gallbladder disease requiring surgery

Four studies reported gallbladder disease requiring surgery (ERA 2000; HERS 1998; PEPI 1995; WHI 1998). The studies compared oestrogen-only hormone therapy, combined continuous hormone therapy and sequential combined hormone therapy to placebo for three years to more than seven years. For this outcome, the two largest studies stated that they excluded from analysis women who had their gallbladder removed (HERS 1998), or those who reported a history of gallbladder disease, or both (WHI 1998).

Meta-analysis of the three studies comparing oestrogen-only hormone therapy to placebo indicated that participants in the hormone therapy group were more likely to present gallbladder disease requiring surgery (RR 1.75, 95% CI 1.40 to 2.19); these studies had a mean follow-up ranging from three to 7.1 years (ERA 2000; PEPI 1995; WHI 1998). The absolute risk of an event increased from 26 per 1000 in the control group to 45 per 1000 (95% CI 36 to 57) in the hormone therapy group. Meta-analysis of the four studies comparing combined continuous hormone therapy to placebo also indicated that participants receiving hormone therapy were more likely to present gallbladder disease requiring surgery (RR 1.55, 95% CI 1.29 to 1.86); these studies had a mean follow-up ranging from three to 5.6 years (ERA 2000; HERS 1998; PEPI 1995; WHI 1998). The absolute risk of an event increased from 27 per 1000 in the control group to 47 per 1000 (95% CI 38 to 60) in the hormone therapy group. Although these studies had differing lengths of follow-up, review authors noted no statistical heterogeneity in either meta-analysis. WHI 1998 investigators reported that hazard estimates for risk in active and placebo groups started to diverge during the first year of follow-up, with the oestrogen group separating earlier than the combined continuous hormone therapy group.

#### 6.20 Gallbladder disease requiring surgery: oestrogen-only hormone therapy

##### Analysis 6.20

6.20.1 CEE 0.625 mg for 3 to 3.2 years of follow-up (RR 0.77, 95% CI 0.17 to 3.39; 2 studies, 554 participants) (ERA 2000; PEPI 1995)

6.20.2 CEE 0.625 mg for 7.1 years of follow-up probably increased the risk of gallbladder disease requiring surgery compared to

placebo (RR 1.78, 95% CI 1.42 to 2.24; 1 study, 8376 participants; moderate-certainty evidence) (WHI 1998)

### 6.21 Gallbladder disease requiring surgery: combined continuous hormone therapy

#### Analysis 6.21

6.21.1 CEE 0.625 mg + 2.5 mg MPA at 3-year follow-up (RR 2.01, 95% CI 0.61 to 6.59;  $I^2 = 0\%$ ; 2 studies, 557 participants) (ERA 2000; PEPI 1995)

6.21.2 CEE 0.625 mg + 2.5 mg MPA for 4 years of follow-up (RR 1.35, 95% CI 0.98 to 1.85; 1 study, 2253 participants) (HERS 1998)

6.21.3 CEE 0.625 mg + 2.5 mg MPA for a mean of 5.6 years of follow-up may increase the risk of gallbladder disease requiring surgery compared to placebo (RR 1.64, 95% CI 1.30 to 2.06; 1 study, 14203 participants; low-certainty evidence) (WHI 1998)

### 6.22 Gallbladder disease requiring surgery: combined sequential hormone therapy

#### Analysis 6.22

6.22.1 CEE 0.625 mg daily + MPA 10 mg days 1 to 12 at 3-year follow-up (RR 2.00, 95% CI 0.37 to 10.78; 1 study, 348 participants) (PEPI 1995)

6.22.2 CEE 0.625 mg daily + micronised progesterone 200 mg days 1 to 12 at 3-year follow-up (RR 1.47, 95% CI 0.25 to 8.67; 1 study, 352 participants) (PEPI 1995)

### 6.23, 6.24, 6.25, 6.26 Hip fractures

Five studies reported the incidence of hip fracture. These studies compared combined continuous hormone therapy (HERS 1998; WHI 1998; WISDOM 2007), combined sequential hormone therapy (Tierney 2009; WEST 2001), and oestrogen-only hormone therapy (WEST 2001; WHI 1998) to placebo for between one and 7.9 years, with extended follow-up to 10.7 years in WHI 1998 (oestrogen-only arm), and to 13.2 years in both arms of WHI 1998.

Both arms of WHI 1998 reported clear evidence of a difference in the risk of hip fracture, with participants receiving hormone therapy being less likely to present with it. WHI 1998 (oestrogen-only hormone therapy arm) reported clear evidence of a difference in the risk of hip fracture, with participants taking hormone therapy being less likely to have a fracture than the group taking placebo (CEE 0.625 mg) at 7.1 years' mean follow-up (RR 0.66, 95% CI 0.46 to 0.95; Analysis 6.23). The absolute risk of a hip fracture decreased from 14 per 1000 in the control group to nine per 1000 (95% CI 6 to 13) in the hormone therapy group. Benefit derived from hormone therapy was not maintained during extended follow-up (to 13.2 years). WHI 1998 (combined hormone therapy arm) reported this outcome at yearly intervals, and found no clear evidence of a difference in the incidence of hip fracture during the first four years of follow-up, but at 5.6 years' mean follow-up, participants taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) had a reduction in the risk of hip fracture (RR 0.67, 95% CI 0.47 to 0.96; Analysis 6.25). The absolute risk of hip fracture decreased from nine per 1000 in the control group to six per 1000 (95% CI 4 to 9) in the hormone therapy group. This decreased risk remained in the hormone therapy group at a mean follow-up of 7.9 years (RR 0.77, 95% CI 0.60 to 0.99) and 13.2 years (RR 0.82, 95% CI 0.69 to 0.97).

However, HERS 1998 found no clear evidence of a difference between combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) and placebo for this outcome. The unblinded extension of this study reported clear evidence of difference, with the participants receiving hormone therapy being more likely to have a fracture at 4.1 to 6.8 years (post randomisation) (RR 2.10, 95% CI 1.06 to 4.16; Analysis 6.25).

Other studies found no clear evidence of differences between groups.

### 6.23 Hip fractures: oestrogen-only hormone therapy

#### (Analysis 6.23)

6.23.1 CEE 0.625 mg for 7.1 years of follow-up (RR 0.66, 95% CI 0.46 to 0.95; 1 study, 10,739 participants) (WHI 1998)

6.23.2 CEE 0.625 mg for 10.7 years of follow-up (RR 0.92, 95% CI 0.71 to 1.18; 1 study, 10,739 participants) (WHI 1998)

6.23.3 CEE 0.625 mg after 13.2 years of follow-up (RR 0.93, 95% CI 0.74 to 1.17; 1 study, 10,739 participants) (WHI 1998)

### 6.24 Hip fractures: oestrogen-only or combined sequential hormone therapy

#### (Analysis 6.24)

6.24.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 0.62, 95% CI 0.27 to 1.42; 1 study, 664 participants) (WEST 2001)

### 6.25 Hip fractures: combined continuous hormone therapy

#### (Analysis 6.25)

6.25.1 CEE 0.625 mg + MPA 2.5 mg for mean/median 1 year follow-up (RR 0.64, 95% CI 0.26 to 1.57;  $I^2 = 0\%$ ; 2 studies, 20,993 participants) (WHI 1998; WISDOM 2007)

6.25.2 CEE 0.625 mg + MPA 2.5 mg at 2-year follow-up (RR 0.61, 95% CI 0.31 to 1.18; 1 study, 16,608 participants) (WHI 1998)

6.25.3 CEE 0.625 mg + MPA 2.5 mg at 3-year follow-up (RR 0.70, 95% CI 0.42 to 1.17; 1 study, 16,608 participants) (WHI 1998)

6.25.4 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 1.16, 95% CI 0.55 to 2.42; 1 study, 2763 participants) (HERS 1998)

6.25.5 CEE 0.625 mg + MPA 2.5 mg for 5.6 years of follow-up (RR 0.67, 95% CI 0.47 to 0.96; 1 study, 16,608 participants) (WHI 1998)

6.25.6 CEE 0.625 mg + MPA 2.5 mg for 4–6.8 years of follow-up (unblinded) (RR 2.10, 95% CI 1.06 to 4.16; 1 study, 2321 participants) (HERS 1998)

6.25.7 CEE 0.625 mg + MPA 2.5 mg for 7.9 years of follow-up (RR 0.77, 95% CI 0.60 to 0.99; 1 study, 16,608 participants) (WHI 1998)

6.25.8 CEE 0.625 mg + MPA 2.5 mg after 13.2 years of follow-up (RR 0.82, 95% CI 0.69 to 0.97; 1 study, 16,608 participants) (WHI 1998)

### 6.26 Hip fractures: combined sequential hormone therapy

#### (Analysis 6.26)

6.26.1 17-B-oestradiol 1 mg daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 0.34, 95% CI 0.01 to 8.27; 1 study, 142 participants) ([Tierney 2009](#))

## 6.27, 6.28 Vertebral fractures

[WHI 1998](#) (oestrogen-only hormone therapy arm) reported the incidence of vertebral fracture at a follow-up of 7.1 years. Two studies of combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) versus placebo also reported the incidence of vertebral fracture at a follow-up of four to nearly eight years ([HERS 1998](#); [WHI 1998](#) (combined hormone therapy arm)).

At a mean of 7.1 years' follow-up, [WHI 1998](#) (oestrogen-only hormone therapy arm) reported that participants in the hormone therapy group were less likely to have a vertebral fracture than the ones in the placebo group (RR 0.64, 95% CI 0.44 to 0.94; [Analysis 6.27](#)). The absolute risk of a clinical vertebral fracture decreased from 13 per 1000 in the control group to eight per 1000 (95% CI 6 to 12) in the hormone therapy group. Similarly, at a mean of 5.6 years' follow-up, [WHI 1998](#) (combined hormone therapy arm) reported that participants receiving hormone therapy were less likely to have a fracture than those in the placebo group (RR 0.68, 95% CI 0.49 to 0.96; [Analysis 6.28](#)). The absolute risk of a clinical vertebral fracture decreased from 10 per 1000 in the control group to seven per 1000 (95% CI 5 to 10) in the hormone therapy group. At a mean of 7.9 years' follow-up in [WHI 1998](#) (combined hormone therapy arm), there was no clear evidence of differences between groups. [HERS 1998](#) found no clear evidence of a difference between groups during follow-up.

## 6.27 Vertebral fractures: oestrogen-only hormone therapy

([Analysis 6.27](#))

6.27.1 CEE 0.625 mg for 7.1 years of follow-up (RR 0.64, 95% CI 0.44 to 0.94; 1 study, 10,739 participants) ([WHI 1998](#))

## 6.28 Vertebral fractures: combined continuous hormone therapy

([Analysis 6.28](#))

6.28.1 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 0.74, 95% CI 0.37 to 1.47; 1 study, 2763 participants) ([HERS 1998](#))

6.28.2 CEE 0.625 mg + MPA 2.5 mg for mean 5.6 years of follow-up (RR 0.68, 95% CI 0.49 to 0.96; 1 study, 16,608 participants) ([WHI 1998](#))

6.28.3 CEE 0.625 mg + MPA 2.5 mg for 4–6.8 years of follow-up (unblinded) (RR 1.10, 95% CI 0.49 to 2.48; 1 study, 2321 participants) ([HERS 1998](#))

6.28.4 CEE 0.625 mg + MPA 2.5 mg for mean 7.9 years of follow-up (RR 0.78, 95% CI 0.60 to 1.01; 1 study, 16,608 participants) ([WHI 1998](#))

No data were available for combined sequential hormone therapy for vertebral fractures.

## 6.29, 6.30, 6.31, 6.32, 6.33 All clinical fractures

Nine studies reported the incidence of any fracture ([EPHT 2006](#); [ERA 2000](#); [ESPRIT 2002](#); [Greenspan 2005](#); [HERS 1998](#); [Tierney 2009](#); [WEST 2001](#); [WHI 1998](#); [WISDOM 2007](#)). The studies compared oestrogen-only hormone therapy, combined continuous hormone

therapy and combined sequential hormone therapy to placebo for one year to nearly eight years of follow-up.

Both arms of [WHI 1998](#) reported clear evidence of a difference favouring hormone therapy in terms of reducing the risk of fracture. Investigators reported this at a mean of 5.6 and 7.9 years of follow-up in participants taking combined continuous hormone therapy (CEE 0.625 mg + MPA 2.5 mg) (at 5.6 years: RR 0.78, 95% CI 0.71 to 0.86; at 7.9 years: RR 0.82, 95% CI 0.76 to 0.89) and at 7.1 years' mean follow-up in participants taking oestrogen-only hormone therapy (CEE 0.625 mg) (RR 0.73, 95% CI 0.65 to 0.80) ([Analysis 6.29](#); [Analysis 6.32](#)). At 5.6 years, in [WHI 1998](#) (combined hormone therapy arm), the absolute risk of any fracture decreased from 111 per 1000 in the control group to 86 per 1000 (95% CI 79 to 94) in the hormone therapy group. In [WHI 1998](#) (oestrogen-only hormone therapy arm) at 7.1 years, the absolute risk of any fracture decreased from 140 per 1000 in the control group to 102 per 1000 (95% CI 91 to 112) in the hormone therapy group. None of the other studies found any clear evidence of a difference between hormone therapy and placebo for this outcome.

## 6.29 All clinical fractures: oestrogen-only hormone therapy

([Analysis 6.29](#))

6.29.1 Oestradiol valerate 2 mg at 2-year follow-up (RR 0.60, 95% CI 0.29 to 1.26; 1 study, 1017 participants) ([ESPRIT 2002](#))

6.29.2 CEE 0.625 mg daily for 3.2 years of follow-up (RR 0.42, 95% CI 0.17 to 1.04; 1 study, 205 participants) ([ERA 2000](#))

6.29.3 CEE 0.625 mg for 7.1 years of follow-up probably reduces the risk of all clinical fractures compared to placebo (RR 0.73, 95% CI 0.65 to 0.80; 1 study, 10,739 participants; moderate-certainty evidence) ([WHI 1998](#))

## 6.30 All clinical fractures: oestrogen-only or combined hormone therapy

([Analysis 6.30](#))

6.30.1 CEE 0.625 mg with or without 2.5 mg MPA at 3-year follow-up (RR 0.56, 95% CI 0.25 to 1.23; 1 study, 373 participants) ([Greenspan 2005](#))

## 6.31 All clinical fractures: oestrogen-only or combined sequential hormone therapy

([Analysis 6.31](#))

6.31.1 Oestradiol 1 mg daily (if no uterus) + MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years of follow-up (RR 1.23, 95% CI 0.68 to 2.19; 1 study, 664 participants) ([WEST 2001](#))

## 6.32 All clinical fractures: combined continuous hormone therapy

([Analysis 6.32](#))

6.32.1 CEE 0.625 mg + MPA 2.5 mg for median 1-year follow-up (RR 0.69, 95% CI 0.46 to 1.02; 1 study, 4385 participants) ([WISDOM 2007](#))

6.32.2 CEE 0.625 mg + MPA 2.5 mg for 3.2 to 3.4 years of follow-up (RR 0.52, 95% CI 0.32 to 0.87;  $I^2 = 0\%$ ; 2 studies, 986 participants) ([EPHT 2006](#); [ERA 2000](#))

6.32.3 CEE 0.625 mg + MPA 2.5 mg for a mean of 5.6 years of follow-up probably reduces the risk of all clinical fractures compared to

placebo (RR 0.78, 95% CI 0.71 to 0.86; 1 study, 16,608 participants; moderate-certainty evidence) ([WHI 1998](#))

6.32.4 CEE 0.625 mg + MPA 2.5 mg for 4 years of follow-up (RR 0.95, 95% CI 0.76 to 1.18; 1 study, 2763 participants) ([HERS 1998](#))

6.32.5 CEE 0.625 mg + MPA 2.5 mg for 4 to 6.8 years of follow-up (unblinded) (RR 1.23, 95% CI 0.91 to 1.65; 1 study, 2321 participants) ([HERS 1998](#))

6.32.6 CEE 0.625 mg + MPA 2.5 mg for mean 7.9 years of follow-up (RR 0.82, 95% CI 0.76 to 0.89; 1 study, 16,608 participants) ([WHI 1998](#))

### 6.33 All clinical fractures: combined sequential hormone therapy

([Analysis 6.33](#))

6.33.1 17-B-oestradiol 1 mg daily + (3 days weekly) 0.35 mg norethindrone at 2-year follow-up (RR 0.44, 95% CI 0.12 to 1.64; 1 study, 142 participants) ([Tierney 2009](#))

## DISCUSSION

### Summary of main results

In general, results from this review need to be interpreted with caution, as participants differed in the time elapsed since menopause when hormone therapy was initiated, as well as in the dosage, formulation and route of administration. One very large study ([WHI 1998](#)), which has considerably weighted many results in this review, was carried out under the hypothesis that hormone therapy would help prevent cardiovascular disease and dementia, and enrolled participants up to 79 years of age at baseline. These characteristics of the review's included participants make it difficult to draw conclusions for women in perimenopause or early menopause.

We included 24 studies, with 24,455 participants randomised to some form of hormone therapy and 21,205 to placebo. The studies varied dramatically in size, from 40 participants to over 27,000. Fifteen studies were conducted in the USA, and one in each of the following countries: UK, Estonia, Norway, Canada, Denmark and China; three studies were international in scope (one in the USA and Canada, one in Canada and the Netherlands and one in the UK, Australia and New Zealand). The participants included in these studies were predominantly postmenopausal (either spontaneously or surgically). Participants were from 26 to 91 years old, with the mean or median participant age in each study ranging from 48 to 76 years. Fifteen studies enrolled relatively healthy women. Six studies included women with cardiovascular disease: two involved women with single-vessel coronary artery stenosis; one included women who survived a myocardial infarction and were discharged within 31 days; one focused on women with a history of venous thromboembolism; one enrolled women with coronary disease such as prior myocardial infarction or coronary artery bypass surgery; and one included women with a previous stroke or transient ischaemic attack. One study included participants with endometrial cancer. One included participants with probable Alzheimer's disease. One included participants with chronic conditions.

The oestrogen-only hormone therapies included four different regimes, and the combined hormone therapies included four

continuous and 11 sequential regimes. Most studies were at a low risk of bias for selection, attrition, performance, detection and reporting bias. Three studies were at a high risk of attrition bias.

In relatively healthy women (generally fit, postmenopausal participants without overt disease), the use of combined continuous hormone therapy at 5.6 years of follow-up probably makes little to no difference to the risk of a coronary event, and it may increase the risk of stroke and venous thromboembolism. It probably increases the risk of breast cancer and may increase gallbladder disease requiring surgery. It may make little to no difference in lung cancer. Combined continuous hormone therapy probably reduces the risk of all clinical fractures.

In relatively healthy women (generally fit, postmenopausal participants without overt disease), the use of oestrogen-only hormone therapy at seven years' follow-up probably makes little to no difference in coronary events, venous thromboembolism and breast cancer, and may make little to no difference to the risk of lung cancer. It probably increases the risk of stroke and gallbladder disease requiring surgery. Oestrogen-only hormone therapy probably reduces the risk of all clinical fractures.

### Coronary events

No evidence indicates that hormone therapy has a role in the treatment or prevention of coronary disease.

There is no clear evidence that hormone therapy modifies death from cardiovascular disease for peri- or postmenopausal women without major health problems (oestrogen-only: [EPAT 2001](#); [WHI 1998](#); combined continuous: [WHI 1998](#); or combined sequential: [Tierney 2009](#)) or women with cardiovascular disease ([ERA 2000](#); [ESPRIT 2002](#); [WAVE 2002](#); [HERS 1998](#); [WEST 2001](#)).

For women who are generally healthy on oestrogen-only hormone therapy, there is no clear evidence of a difference in coronary events at up to 10 years of follow-up ([WHI 1998](#); [EPAT 2001](#); [PEPI 1995](#)). The same is true for women with cardiovascular disease at up to 14 years of follow-up ([ERA 2000](#); [ESPRIT 2002](#); [WAVE 2002](#)).

Continuous hormone therapy probably increases the risk of coronary events after up to three years of treatment in women without major health problems ([WHI 1998](#); [EPHT 2006](#); [WISDOM 2007](#)), with a reduction over time (no difference from women not receiving treatment). [WHI 1998](#) investigators suggest that the apparent decline in cardiovascular risk in later years may be due to an acceleration of events during earlier years amongst susceptible women in the hormone therapy group. For women with known cardiovascular disease, there is no clear evidence of a difference in coronary events after up to eight years of follow-up ([ERA 2000](#); [HERS 1998](#)).

For women taking combined sequential hormone therapy, there is no clear evidence of a difference in the risk of coronary events at up to four years ([KEEPS 2012](#); [PEPI 1995](#); [Tierney 2009](#)). There were no data available for women with cardiovascular disease. There may be little to no difference in myocardial infarction for women hospitalised with chronic illness for up to 10 years ([Nachtigall 1979](#)).

[WHI 1998](#) (combined hormone therapy arm) conducted prespecified subgroup analyses to evaluate whether any clinical characteristics of the study population might plausibly modulate the coronary effects of hormone therapy: variables included

age, time since menopause, presence or absence of vasomotor symptoms, prior hormone use, coronary heart disease (CHD) risk factor status and presence or absence of pre-existing cardiovascular disease. However, none of these variables significantly affected the results.

It has been suggested that the vascular effects of hormone therapy may differ according to a person's age or time since the onset of menopause (WHI 1998). Thus, oestrogen may counteract the early stages of atherosclerosis in recently menopausal women by inhibiting lipid deposits within the endothelium. However, hormone therapy may have adverse effects on more advanced disease, by facilitating an increase in enzymes that tend to disrupt atherosclerotic lesions, and by encouraging clot formation (Manson 2013; Reslan 2012). Research findings on the effect of hormone therapy and menopause timing offer no clear evidence of a difference for coronary events (ELITE 2014; KEEPS 2012). Still, KEEPS 2012 reported that coronary atherosclerosis was significantly greater among women in the late-postmenopause stratum than among those in the early-postmenopause stratum, and ELITE 2014 reported improvement in some cardiovascular disease risk markers. These outcomes were not part of this review.

## Stroke

There is no clear evidence that hormone therapy modifies death from stroke for generally healthy women at up to 18 years of follow-up (Tierney 2009; WHI 1998; WISDOM 2007).

For women who are generally healthy taking oestrogen-only hormone therapy, the risk of stroke varied over time, with no clear evidence of a difference at up to two years (EPAT 2001), and with stroke more likely for the hormone therapy group at seven years of follow-up (WHI 1998).

For women who are generally healthy and taking combined continuous hormone therapy, the risk of stroke increased over time. There was no clear evidence of a difference at up to two years (EPHT 2006; WHI 1998), but stroke was more likely in the hormone therapy group from three to 7.9 years of follow-up (WHI 1998). For sequential continuous regimens, there is no evidence of a difference at up to two years of follow-up (Tierney 2009).

For women with cardiovascular disease using oestrogen-only hormone therapy, there is no clear evidence of a difference in death from stroke at up to 2.8 years of follow-up (WEST 2001), or in the incidence of stroke at up to 14 years of follow-up (ESPRIT 2002; WAVE 2002).

There is no clear evidence of a difference for women with cardiovascular disease using combined continuous hormone therapy in stroke events at up to 6.8 years of follow-up.

## Venous thromboembolism

Women taking oestrogen-only hormone therapy (moderate dose) were probably more likely to have a thromboembolic event for up to two years (WHI 1998), but there is no clear evidence of difference at longer-term follow-up (up to 10 years) (PEPI 1995; WHI 1998).

Women receiving combined continuous hormone therapy were more likely to have thromboembolism at all follow-up time points (to 7.9 years) (WHI 1998), but there is no clear evidence of a

difference for combined sequential hormone therapy regimens (KEEPS 2012; PEPI 1995; REPLENISH 2018; Tierney 2009).

Amongst women taking combined hormone therapy in WHI 1998, those with the factor V Leiden mutation (a blood coagulation disorder) were at higher risk of venous thromboembolism (Cushman 2004). Statistical power was insufficient to allow investigators to determine whether significant excess risk was associated with a history of venous thromboembolism (amongst women taking combined hormone therapy). The incidence of thromboembolism was higher amongst older and obese participants, although this was related to their higher baseline risk of an event, and their risk ratio did not differ from that of other participants taking combined hormone therapy.

## Breast cancer

The incidence of breast cancer in women taking combined continuous hormone therapy varied over time. Initially (in the first year), it was lower than in the placebo group (WHI 1998; WISDOM 2007). By the second year, there was no clear evidence of a difference. By the fifth year of use, there was evidence of an increased risk. At a mean of 11 years' follow-up in WHI 1998, women in the combined hormone therapy group were more likely to have invasive breast cancer than controls, and longer follow-ups (to a median of 20 years) showed no evidence of attenuation of risk (Chlebowski 2015a; Chlebowski 2020).

WHI 1998 investigators commented that breast cancers in the combined hormone therapy group were diagnosed at a similar grade but at a more advanced stage, and suggested that combined hormone therapy may stimulate breast cancer growth while delaying diagnosis. Evidence shows that combined hormone therapy increases the frequency of abnormal mammograms and indications for breast biopsy, but compromises the diagnostic performance of both of these tests (Chlebowski 2008). These factors would account for the lower incidence of breast cancer amongst women taking combined therapy during the first two years in WHI 1998. A published subgroup analysis of prior hormone use in WHI 1998 revealed that the cumulative incidence of breast cancer over time in women taking combined hormone therapy increased at a greater rate than in those taking placebo after about three years for prior hormone users and after about five years for those with no prior use. Interference with mammography precluded the possibility of defining with any reliability a time frame for the safe use of combined hormone therapy (Anderson 2006).

For women using oestrogen-only hormone therapy, there is no clear evidence of a difference at up to seven years of follow-up in the incidence of breast cancer (ERA 2000; PEPI 1995; WAVE 2002; WHI 1998), but WHI 1998 reported a decrease in the incidence of breast cancer in follow-up over 10 years (Chlebowski 2015a; Chlebowski 2020). Comparison of hazard ratios during early and late post-intervention periods showed that the lower risk of breast cancer in the oestrogen arm persisted for about 4.5 years after the intervention was provided, at which point there was no clear evidence of a difference (Chlebowski 2015a; Chlebowski 2015b). A published subgroup analysis showed fewer early cancers and fewer ductal carcinomas in the intervention group, although the incidence of lobular tumours did not suggest clear evidence of a difference. Results showed that the reduction in breast cancer risk was concentrated in women without benign breast disease ( $P = 0.01$ ) or a first-degree family history of breast cancer

( $P = 0.02$ ) (Anderson 2012). Oestrogen-only hormone therapy appears to increase the number of women needing to repeat mammography or breast biopsy but, in contrast to combined hormone therapy, does not appear to compromise breast cancer detection (Chlebowski 2010a).

WHI 1998 researchers stated that differences between participants across the two arms of the study did not explain differences in breast cancer incidence and suggested that increased risk in the combined group might be due to progestogen. A nested case-control study compared pair-matched controls of women who developed breast cancer in either arm of WHI 1998 (Zhao 2014). This study suggested that post-treatment changes in serum oestrogens and concentrations of sex hormone-binding globulin, or changes in the association of such concentrations with disease risk, might explain both the increased breast cancer risk noted with combined hormone therapy and the reduction in risk seen with unopposed oestrogen (Zhao 2014). It has been observed that exposure to oestrogen after a sustained period of oestrogen deprivation reduces the risk of breast cancer (Jordan 2015; Obiorah 2013).

### Lung cancer

Post hoc analysis of WHI 1998 data revealed that combined hormone therapy did not significantly increase the incidence of lung cancer over eight years of follow-up but did increase mortality from lung cancer, independent of smoking status. Study authors suggested that this might be so because combined hormone therapy stimulates the growth of pre-existing small cell lung cancers (Chlebowski 2009).

### Gallbladder disease

Researchers have noted a statistically significant association between hormone therapy and gallbladder disease, with excess risk related to both oestrogen-only and combined continuous hormone therapy. Although most of the statistical power for this outcome was derived from WHI 1998, findings with respect to combined continuous hormone therapy were strongly supported by data from both blinded and unblinded follow-up in HERS 1998. WHI 1998 investigators noted that the risk started to increase in the active group during the first year and appeared to increase over time. They calculated that for one excess case of gallbladder disease, 323 women would need to take oestrogen-only hormone therapy, or 500 women would need to take combined continuous hormone therapy for a year.

### Fractures

Evidence on hormone therapy and fractures is not consistent. WHI 1998 found a reduced risk of fractures in women taking combined continuous hormone therapy or oestrogen-only hormone therapy over nearly eight years of follow-up, but HERS 1998 reported no benefit for women taking continuous combined hormone therapy. Moreover, unblinded continuation of HERS 1998 revealed a significantly *increased* risk of hip fracture amongst women. Study authors attributed this finding to chance, noting that the effect was considerably smaller in the as-treated analysis, and that such a finding lacks biological plausibility (Hulley 2004). WHI 1998 (combined hormone therapy arm) investigators tested the hypothesis that the beneficial effect of hormone therapy on fracture incidence differed according to fracture risk factors. They found that the reduction in risk provided by hormone therapy was

no greater in women at high risk of fracture (Cauley 2003). However, WHI 1998 excluded women with severe osteoporosis and did not routinely collect bone mineral density data. Thus, we are uncertain of the benefits or harms of hormone therapy in women with severe osteoporosis.

WHI 1998 reported a reduced risk of hip fracture associated with oestrogen-only hormone therapy at up to seven years, but no evidence of a difference at longer follow-up (up to 13.2 years). Our analyses of hip fracture may have had insufficient power to reach conclusive findings. The risk of hip fracture rises steeply from the age of about 60 years, but is still under 0.5% amongst women in the UK at 65 to 69 years of age (Banks 2009b).

It is important to highlight that women who need treatment for low bone mineral density require lifelong therapy, but the highest risk of cardiovascular events with combined hormone therapy occurs during the first year of use.

### Cognitive outcomes

There was little or no difference in global cognitive function with hormone therapy at different dosages and time frames compared to placebo (Greenspan 2005; WHI 1998; Yaffe 2006).

For women with probable dementia, there was little to no difference with oestradiol-only hormone therapy compared to placebo (WHI 1998), but for women taking combined continuous hormone therapy, researchers reported an increased risk at up to five years of follow-up (WHI 1998).

These findings were unexpected and contrast sharply with findings reported in earlier research. WHI 1998 (WHIMS) investigators suggested that this might be due to the healthy user bias seen in observational studies (whereby hormone therapy users had a better prognosis at baseline than the control groups), the differential effects of hormone therapy on specific domains of cognition not measured individually by the Modified Mini-Mental State Examination (3MSE), or both. Alternatively, they suggested that hormone therapy might need to be initiated during a critical period, such as menopause, to protect cognitive function at a later age. The mean age of the WHIMS population was 71 years, and the study could not address this theory, although previous users of hormone therapy in WHIMS did not have higher scores (Espeland 2004). Moreover, results of extended follow-up in WHI 1998 (WHISCA) show no evidence of any benefit in domain-specific cognitive function from oestrogen-alone or combined hormone therapy.

A post hoc comparison of global cognitive function amongst younger women in WHI 1998 included participants randomised to either the active arm of WHI 1998 (CEE or combined hormone therapy) versus the placebo arm (Espeland 2013; Vaughan 2013). Scores were similar in the two groups ( $P = 0.66$ ). The study included 1376 women 50 to 55 years of age when randomised for WHI 1998. Cognitive testing was conducted an average of 7.2 years following trial completion, when participants had a mean age of 67.2 years, and was repeated one year later. Study investigators concluded that "CEE-based therapies produced no overall sustained benefit or risk to cognitive function when administered to women aged 50–55 years" (Espeland 2013). However, amongst 2880 women who had enrolled in WHI 1998 at the age of 65 to 79 years, long-term decrements in global cognitive function were noted in the hormone

therapy groups (CEE or combined hormone therapy) relative to the placebo group, which consisted of older women ( $P < 0.05$ ). Effects were small, and decrements were fairly stable. Findings did not vary according to type of hormone therapy, prior use or time since last menstrual period (Espeland 2016).

### Colorectal cancer

The reduced incidence of colorectal cancer in women taking combined continuous hormone therapy in WHI 1998 at 5.6 years of follow-up was offset by the finding that colorectal cancers diagnosed in these women tended to be more advanced, with greater likelihood of lymphatic or metastatic involvement. Moreover, the reduced incidence in the hormone therapy group did not lead to a reduced death rate from colorectal cancer over extended follow-up (7.1 years), although investigators noted that an even longer period of follow-up might be required to observe a mortality benefit from a reduction in the incidence of small, localised cancers. Women taking oestrogen-only hormone therapy in WHI 1998 did not have a reduced incidence of or death rate from colorectal cancer over seven years' follow-up, nor during extended follow-up to 10.7 years. Overall, no strong evidence suggests a clinically meaningful reduction in colorectal cancer rates with oestrogen alone or oestrogen plus progestogen. Findings in the WHI observational study supported this conclusion (Prentice 2009b).

### Endometrial and ovarian cancers

None of the included studies showed an increase in the incidence of endometrial cancer in the group taking hormone therapy. Three studies randomised women with a uterus to oestrogen-only hormone therapy (EPAT 2001; ESPRIT 2002; PEPI 1995). As endometrial cancer is well documented as an adverse effect of unopposed oestrogen (Kurman 1985), these participants were closely monitored for atypical endometrial hyperplasia and received treatment (and discontinuation of study medications) if it was detected. PEPI 1995 reported that participants in the oestrogen-only hormone therapy group were significantly more likely to develop atypical endometrial hyperplasia than those in the placebo group, whereas participants in the combined hormone therapy groups in the same study showed no increased risk of hyperplasia. After more than 13 years' extended follow-up, rates of endometrial cancer were lower in the combined hormone therapy group in WHI 1998.

The Barakat 2006 study of oestrogen-only therapy in women who had undergone surgery for stage I or II endometrial cancer was underpowered owing to early discontinuation, and could not conclusively refute or support the safety of this therapy with regard to risk of recurrence. Study authors noted that recurrence rates were low, at 1.9% in the placebo group and 2.3% in the intervention group (Barakat 2006).

WHI 1998 authors reported no clear evidence of a difference in the incidence of ovarian cancer with combined continuous hormone therapy at up to 13 years of follow-up.

A randomised study with 4-year follow-up of 130 women with a history of ovarian cancer found that oestrogen-only hormone therapy did not negatively affect disease-free or overall survival time compared with no hormone therapy (Guidozzi 1999). A similar randomised study showed that women with severe menopausal symptoms after ovarian cancer who took hormone therapy (of varying types, according to consultant preference) had improved

overall and relapse-free survival compared with controls not taking hormone therapy (AHT 2015). The present systematic review did not include these studies because they lacked a placebo control group.

## Overall completeness and applicability of evidence

### Type of hormone therapy

Nearly all significant findings described in this review were derived from the two biggest studies: HERS 1998 and WHI 1998. Both studies evaluated oral conjugated equine oestrogen (CEE) 0.625 mg, with or without continuous medroxyprogesterone (MPA 2.5 mg). Smaller studies using other types of hormone therapy reported very few or no major clinical events. We were generally unable to combine results from individual studies because they used different types of hormone therapy, which may not be equivalent in effect, or they differed with respect to the study population, or both. Controversy surrounds the degree to which the findings of WHI 1998 apply to any type of hormone therapy other than continuous combined oral CEE 0.625 mg with or without MPA 2.5 mg. Effects may vary with different oestrogens and progestogens, different time frames for hormone therapy use, and different doses and routes of administration. Currently, there are several regimens and routes of administration for hormone therapy, including but not limited to the following (Mukherjee 2025):

- oestrogen: oestradiol administered orally or transdermally (as a gel, pump, sachet or spray);
- progestogens: delivered orally (dydrogesterone, norethisterone acetate, medroxyprogesterone acetate, micronised progesterone), or via levonorgestrel-releasing intrauterine devices;
- progesterone-selective receptors of oestrogen, such as tibolone and bazedoxifene.

All this makes it very difficult to draw conclusions about the different types of hormone therapy.

### Population characteristics

It is important to consider any increased risk to health in absolute rather than relative terms. This review is inevitably dominated by the findings of WHI 1998, which was designed to evaluate the efficacy of hormone therapy in preventing major causes of morbidity and mortality amongst older postmenopausal women (Matthews 1997). It was not designed to evaluate the risks and benefits of hormone therapy for treatment of menopausal symptoms, and it specifically excluded women who reported menopausal symptoms severe enough to preclude assignment to placebo treatment (Anderson 2003). Moreover, WHI 1998 did not include women younger than 50 years of age, and study findings may not apply to young surgically menopausal women; for example, someone who has had both ovaries removed while in her forties (Kaunitz 2002).

Evidence is lacking on the long-term effects of hormone therapy on healthy younger women, who are most likely to use it for menopausal symptoms. They are likely to be in their early fifties, when the absolute risk of a life-threatening event is low. It has been estimated that the absolute risk for many diseases approximately doubles with each decade of age (Hulley 2004). A published subgroup analysis of women aged 50 to 59 years in WHI 1998 (combined hormone therapy arm) revealed that, for relatively

healthy women taking combined continuous hormone therapy, venous thrombosis was the only outcome with an increased risk. Risk in the hormone therapy group increased from eight venous thromboses per 10,000 participants per year to 19 per 10,000 participants per year. This increase in risk was highest during the first year of therapy but continued over five years of treatment, and it was particularly high in obese women (i.e. body mass index greater than 30), who had a 5-year risk of 1.4% compared with 0.5% amongst women of normal weight). In the oestrogen-only arm of WHI 1998, over the full 10.7 years of intervention and extended follow-up, younger women (aged 50 to 59 years) randomised to hormone therapy had significantly more favourable outcomes than those randomised to placebo. The hormone therapy group had lower hazard ratios for coronary heart disease, myocardial infarction and death when compared with the placebo group. Findings were similar for both coronary outcomes when data were stratified by time since menopause rather than by age. This contrasted with findings in older women, amongst whom those in the hormone therapy group showed a trend for higher rates of coronary heart disease, myocardial infarction and death, and significantly higher rates of colorectal cancer and chronic disease. WHI 1998 authors noted that study participants took unopposed oestrogen for a median duration of less than six years, and that study results cannot be extrapolated to longer or shorter treatment durations.

It has been suggested that the effects of hormone therapy may differ according to whether it is initiated soon after menopause or after a lengthy gap (Barret-Connor 2007). Analysis of randomised and observational data revealed that for most clinical outcomes, effects of hormone therapy did not vary by hormone therapy timing (randomised data: Prentice 2009a; observation data: WHI 1998); this applied to both oestrogen-only and combined hormone therapy. One exception was breast cancer, for which the risk was higher among women who initiated hormone therapy soon after menopause than in those who had a longer time gap. Their overall risk of cancer was also higher.

Hormone therapy appears to increase the risk of recurrence for women with a history of breast cancer. Two unblinded studies conducted in Sweden randomised breast cancer survivors with menopausal symptoms to hormone therapy or non-hormonal treatment. Both studies were terminated early owing to an increase in the incidence of recurrent breast cancer in the hormonal group in one of the studies (risk ratio (RR) 3.5, 95% confidence interval (CI) 1.5 to 8.1) (Chlebowski 2004; Holmberg 2004). After a median of four years' follow-up in this study, a clinically and statistically significantly increased risk of a new breast cancer event continued in the hormone therapy arm (RR 2.4, 95% CI 1.3 to 4.2) (Holmberg 2008). A similar study initiated in the UK terminated recruitment prematurely in January 2004 (ICR 2001).

For cardiovascular outcomes, results of HERS 1998 largely support the results of WHI 1998 (combined hormone therapy arm), suggesting that these findings can be generalised to older postmenopausal women taking combined continuous hormone therapy, whether or not they have known cardiovascular risk factors (although their findings differ with respect to fracture risk).

### Health benefits and risks after stopping hormone therapy

WHI 1998 (combined hormone therapy arm) reported health outcomes at a mean of 2.4 years' extended follow-up after the

planned intervention period (Heiss 2008). Follow-up data for this period were available for 95% of women, few of whom were using hormone therapy during extended follow-up (4.3% in the intervention arm and 1.2% in the placebo arm at one year after the study was stopped). Over the course of follow-up, the risk of coronary events, stroke and venous thromboembolism decreased in the group that had been randomised to combined hormone therapy, and reached a level comparable with that of the placebo group. Similarly, results showed no clear evidence of differences between groups in the risk of fractures or of colorectal cancer by the end of post-intervention follow-up. However, in the group that had been randomised to combined hormone therapy, the hazard ratio (HR) for the outcome "all cancer" increased from 1.03 (95% CI 0.92 to 1.15) during the intervention phase to 1.24 (95% CI 1.04 to 1.48) in the post-intervention period. This increase in risk was attributable in part to the disappearance of previous apparent protection from colorectal cancer, with some continued excess risk of breast cancer, along with an added risk of lung cancer in the hormone therapy group. Study authors noted that clinical vigilance appears to be warranted with regard to the sustained higher risk of malignancy following termination of combined hormone therapy.

WHI 1998 (oestrogen-only arm) reported health outcomes at a mean of 3.9 years' extended follow-up (LaCroix 2011). Follow-up data for this period were available for 78% of the participants, of whom only a small minority were using hormone therapy (up to 4.7% in the intervention arm and 3% in the placebo arm). Over the course of extended follow-up, results continued to show no significant difference between groups in rates of coronary events. Increases in risk of stroke and venous thromboembolism in the hormone therapy arm rapidly disappeared, as did the reduced risk of hip fracture in this group. As noted above, the lower incidence of breast cancer persisted for up to 10.7 years of follow-up (i.e. including both planned intervention and extended follow-up periods).

### Other considerations: oestrogen-only hormone therapy and risk of endometrial cancer

Most studies using oestrogen-only hormone therapy did not randomise participants to this intervention unless they had had a hysterectomy (Greenspan 2005; Mulnard 2000; Nachtigall 1979; Notelovitz 2002; WAVE 2002; WEST 2001; WHI 1998 (oestrogen-only hormone therapy arm)). The remaining studies (where we are uncertain or did not use the presence of a uterus as an indication for the type of hormone therapy) did not report on endometrial cancer. The risk of endometrial cancer in women with a uterus and oestrogen-only hormone therapy was beyond the scope of this review. The increased risk of endometrial cancer has been described for decades in the literature (Grady 1995). The UK's National Institute for Health and Care Excellence (NICE) guideline on menopause management reports that women with an intact uterus receiving oestrogen-only hormone therapy are at an increased risk of endometrial cancer, based on the UK statistics (NICE 2015; NICE 2024).

A recently updated Cochrane review, Kim 2025, assessing hormone therapy in postmenopausal women and the risk of endometrial hyperplasia, reported that, at one year, unopposed oestrogen increases the risk of endometrial hyperplasia (odds ratio (OR) 5.86, 95% CI 4.09 to 8.40; 12 studies, 2910 participants; high-certainty evidence), and at over one year, probably increases the risk (OR 8.97, 95% CI 6.78 to 11.87; 15 studies, 2539 participants; moderate-

certainty evidence). There were no data available for endometrial cancer at one year, and the review's authors are uncertain of the true risk at over one year due to very low-certainty evidence.

In industrialised countries, endometrial cancer is the most common malignancy of the female reproductive tract. Most endometrial cancers are endometrioid adenocarcinoma, and their precursor is endometrial hyperplasia (Nees 2022).

### Certainty of the evidence

Most of the included studies were at low risk of bias in most domains (Figure 2). We rated the overall certainty of the evidence as moderate. The main limitation concerned the applicability of the evidence, as most of the data were provided by WHI 1998, in which only about 33% of the study sample was 50 to 59 years of age at baseline (i.e. the age at which women are most likely to consider hormone therapy for vasomotor symptoms). The 50- to 59-year-old age group had 8833 participants, 5520 in the combined arm and 3313 in the oestrogen-only arm. The mean participant age in WHI 1998 was 63 years.

A high proportion of the participants in these studies did not receive the treatment to which they were randomised. In general, the number of women who discontinued their medication or took less than 80% of their medication was disproportionately high in the hormone therapy groups, presumably because of a higher incidence of adverse effects such as vaginal bleeding. Investigators for WHI 1998 noted that if discontinuation of treatment and initiation of non-study treatment occurred independently of risk factors for clinical outcomes, their intention-to-treat analysis underestimates both the harms and benefits of hormone therapy amongst women who adhere to treatment (WHI 2002).

### Potential biases in the review process

This review is subject to patient selection bias. Most of the included studies had a mean participant age over 60 years, and only one focused on perimenopausal women. In all but one of the 24 studies that reported mean participant age, the mean age at enrolment was over 50 years. This does not reflect usual clinical practice with respect to prescribing of hormone therapy, which is most likely to occur for treatment of vasomotor symptoms during perimenopause (Pedersen 2003). Moreover, participants described as 'relatively healthy' in this review were derived largely from WHI 1998. Investigators reported a high frequency of obesity and hypertensive disorders amongst WHI 1998 participants. Only 30% were of normal weight, and 30% were morbidly obese (body mass index (BMI) > 30 kg/m<sup>2</sup>); 36% were receiving treatment for hypertension or had blood pressure exceeding 140/90 mmHg at enrolment.

Despite extensive searching, we may have failed to identify all relevant studies. However, it is unlikely that we missed any study large enough to substantially influence our overall findings, given the dominance of WHI 1998 in the review. Similarly, although our data could conceivably have been organised differently (e.g. by study population categories and hormone therapy doses), again, the dominance of WHI 1998 makes it extremely unlikely that this would have influenced our findings. We chose not to pool studies that used different types of hormone therapy, and this approach is supported by contrasting findings in the two arms (combined

versus oestrogen-only hormone therapy) of WHI 1998 for some outcomes.

The choice of one year's duration of hormone therapy as a cut-off point for inclusion of studies was arbitrary but is unlikely to have introduced bias, as it was a prespecified criterion.

### Agreements and disagreements with other studies or reviews

Our findings are consistent with those of a Cochrane review of hormone therapy for preventing cardiovascular disease in postmenopausal women (Boardman 2015), which concluded that use of hormone therapy in peri- or postmenopausal women has little to no benefit for primary or secondary prevention of cardiovascular disease and causes an increase in the risk of stroke and venous thromboembolic events. Boardman 2015 differed from the current review in that the review authors pooled data related to unopposed oestrogen with data for combined hormone therapy.

Our findings differ from those of some older systematic reviews (Hemminki 1997; Hemminki 2000b), which provided insufficient evidence to exclude potential benefit from hormone therapy. Our review also differs from Beral 2002, which pooled the results of four randomised controlled trials (RCTs) of hormone therapy published between 1998 and 2002 (EVTET 2000; HERS 1998; WEST 2001; WHI 1998). The most notable difference is that we found a clear increase in the risk of coronary heart disease amongst women taking combined continuous hormone therapy, particularly in the first year. Unlike our review, Beral 2002 pooled results from studies in differing participant groups and types of hormone therapy; this appears to explain why our overall findings differ.

Salpeter 2004 meta-analysed 17 RCTs of hormone therapy that reported at least one death. They concluded that, compared to placebo, the risk of death was significantly reduced in women with a mean age under 60 years who were taking hormone therapy, although results showed no difference when older women were compared. This meta-analysis pooled studies that differed widely with respect to the type of hormone therapy used and the clinical status of participants; in several studies, death was not a prespecified outcome. Moreover, women with a poor prognosis for ovarian cancer accounted for 60% of events in the meta-analysis of studies of younger women. In our review, the evidence shows no survival advantage in taking hormone therapy, although only one of the included studies analysed published subgroup analyses for younger women for this outcome (WHI 1998 (oestrogen-only hormone therapy)). Of the 17 studies that Salpeter and colleagues included in their meta-analysis of younger women, only two met the inclusion criteria for our review. The other 15 studies included in Salpeter 2004 were not blinded, did not report mortality as a primary or secondary outcome or were of less than one year's duration.

One meta-analysis of 30 case-control studies, seven cohort studies, one randomised controlled study and four cancer registry studies reported a small increase in the risk of ovarian cancer with the use of oestrogen-only hormone therapy (RR 1.28, 95% CI 1.18 to 1.40) or combined hormone therapy (RR 1.11, 95% CI 1.02 to 1.21) (Greiser 2006). This risk applied to a range of common histological subtypes of ovarian cancer. Review authors noted that their review was limited by reliance on observational data; however, heterogeneity was low or moderate in most analyses.

A study which analysed data from one large population database from Korea included 2,506,271 participants, with 18 years of follow-up (Yuk 2023). Authors reported no clear evidence of a difference in the risk of ovarian cancer for women taking combined hormone therapy (hazard ratio (HR) 1.07, 95% CI 0.94 to 1.22) or oestrogen-only hormone therapy (oral hormone therapy: HR 1.03, 95% CI 0.85 to 1.24; and topical hormone therapy: HR 1.07, 95% CI 0.45 to 2.59).

Another systematic review meta-analysed 28 RCTs of hormone therapy that reported stroke events (Bath 2005). Hormone therapy was associated with an increased risk of stroke, particularly ischaemic stroke. Moreover, participants in the hormone therapy group who had a stroke seemed to have a worse outcome. This review had very broad inclusion criteria and pooled a wide range of studies, which used different types of hormone therapy for a range of indications, some with male participants and some without placebo control. It is unclear to what extent these findings apply to perimenopausal and postmenopausal women.

A Danish study of 1006 recently postmenopausal or perimenopausal women (aged 45 to 58 years) was not eligible for this review owing to lack of blinding (Schierbeck 2012). Investigators reported that after 10 years of treatment, women taking triphasic 17-B oestradiol with norethisterone acetate or (in those who had had a hysterectomy) oestradiol alone, were at lower risk of experiencing their primary outcome – a composite of death and admission to hospital for heart failure or myocardial infarction (HR 0.48, 95% CI 0.26 to 0.87;  $P = 0.015$ ). No increase in risk of cancer, venous thromboembolism or stroke was apparent. Study authors attributed the difference between their findings and those of WHI 1998 to the younger age and proximity to menopause of their participants. This study was limited by the lack of blinding and by the use of a composite primary outcome that was not prespecified in the study protocol (Marjoribanks 2012; Schroll 2012; Shapiro 2003). It has been noted that the cardiovascular outcomes were not ascertained by researchers but were based on data from individual clinicians entered into a national database (Abelin 2012).

We found no randomised studies or systematic reviews that provided evidence about the risks of long-term hormone therapy in perimenopausal women or those younger than 50 years of age.

Current recommendations from the North American Menopause Society (NAMS) state that hormone therapy is the most effective treatment for vasomotor symptoms (VMS) and genitourinary syndrome of menopause (GSM) (NAMS 2022). The NAMS position statement advises that hormone therapy has been shown to prevent bone loss, and that fracture risks differ according to the person, depending on the type, dose, administration route, initiation time, duration of use and whether progestogen is used. The statement advises that hormone therapy should be used for the time needed to manage symptoms, and that no arbitrary age-based stopping rule is clinically appropriate. The New Zealand and Australia current recommendations are similar, recommending caution beyond five to seven years of use, and advising that hormone therapy should be managed based on individual cases (RANZCOG 2020).

The UK's NICE guideline on menopause management has recently been updated (NICE 2024). Similar to our results, the update concludes that oestrogen-only hormone therapy does not increase the risk of coronary events or mortality from coronary events for generally healthy women. The NICE guideline indicates that

oral oestrogen-only hormone therapy (not transdermal) increases the risk of stroke and venous thromboembolism, and states that there is little to no difference in breast cancer incidence with oestrogen-only hormone therapy. For combined hormone therapy, the guideline also concludes that there is no clear evidence of a difference in the risk of coronary disease compared to placebo. The guideline reported that the general risk of stroke does not change with combined hormone therapy. Still, authors report that it increases with combined hormone therapy containing oral oestrogen, treatment lasting longer than five years and is greater with the increase in age at first starting hormone therapy. These details may explain the difference between the NICE guideline and our review on implications for stroke, as the study contributing most of the data for this review used oral oestrogen and participants were older (i.e. more time between menopause and the start of hormone therapy) (WHI 1998). The NICE guideline reports an increased risk of venous thromboembolism with combined hormone therapy. Similar to our results, the NICE guideline reports an increase in the risk of breast cancer for those receiving combined hormone therapy; a risk which rises with duration of use.

One publication used individual participant data from 108,647 participants from prospective studies assessing breast cancer risk and the time and type of menopausal hormone therapy (CGHFBC 2019). It reported an increased risk for women of average weight starting hormone therapy at 50 years and using it for five years, independent of the type of hormone therapy. The risk increased according to the length of use, reduced with cessation, but there was still evidence of a difference 10 years after cessation.

## AUTHORS' CONCLUSIONS

### Implications for practice

Women with menopausal symptoms who wish to use hormone therapy should weigh the benefits and the small absolute risk of harm arising from short-term use of low-dose hormone therapy, provided they do not have specific contraindications.

Hormone therapy is not indicated for primary or secondary prevention of cardiovascular disease or dementia, nor for preventing the deterioration of cognitive function in postmenopausal women.

Hormone therapy is not indicated for use in women with cardiovascular disease or with a history of venous thrombosis or breast cancer. Randomised evidence provides no specific contraindications for its use in those with a history of endometrial cancer or ovarian cancer, although data are scanty.

Although hormone therapy is effective for the reduction of postmenopausal fractures, it is generally considered an option only for women at significant risk, for whom non-oestrogen therapies are unsuitable.

Women at high risk of lung cancer (current smokers or long-term past smokers) should be aware that combined continuous hormone therapy increases the risk of death from non-small cell lung cancer.

The risk of endometrial pathology amongst women with a uterus taking oestrogen-only hormone therapy was beyond the scope of

this review, but should be taken into account when considering hormonal treatment.

## Implications for research

No studies have adequately assessed the safety of hormone therapy used for symptom relief by perimenopausal women, women younger than 50 years or women with temporary or permanent iatrogenic ovarian failure. Not enough is known about factors that may modulate the risks involved, such as clinical characteristics or biomarkers affecting individual women, different oestrogens and progestogens, different time frames for the use of hormone therapy and different doses and routes of administration (e.g. unopposed oestrogen and intrauterine progestogen, or whether the risk of thromboembolism is diminished by the use of patches). Reliable evidence is needed to show the efficacy and safety of alternatives to hormone therapy for control of menopausal symptoms amongst women who may wish to avoid using hormone therapy, or for whom its use is unsuitable.

Data for the risk of breast cancer associated with different types, dosages and routes of hormone therapy are limited, especially for different types of progestogens.

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## 2025 update: editorial and peer-reviewer contributions

Cochrane Gynaecology and Fertility supported the authors in the development of this updated intervention review. The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Dr Sarah Lensen, Gynaecology and Fertility group, University of Melbourne
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Samuel Hinsley, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Jacob Hester, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Faith Armitage, Cochrane Central Production Service
- Peer-reviewers (provided comments and recommended an editorial decision): Jin-Sung Yuk, MD, PhD, Department of Obstetrics and Gynecology, Sanggye Paik Hospital, School of Medicine, Inje University, Seoul, Republic of Korea (clinical/content review); Dr Denny Mathew John, Assistant Professor, Department of Community Medicine Saveetha Medical College and Hospital, Chennai Tamil Nadu, India (consumer review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review). One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

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\* Indicates the major publication for the study

## CHARACTERISTICS OF STUDIES

### Characteristics of included studies [ordered by study ID]

#### Barakat 2006

##### Study characteristics

|         |                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods | Stated purpose: to determine the effect of oestrogen therapy on recurrence rate and survival in women who have undergone surgery for stage I or II endometrial cancer |
|         | Stratification: stratified by stage                                                                                                                                   |
|         | No. of women screened for eligibility: unclear                                                                                                                        |
|         | No. randomised: 1236 (see Notes)                                                                                                                                      |
|         | No. analysed: 1236                                                                                                                                                    |
|         | Losses to follow-up: none stated                                                                                                                                      |
|         | Adherence to treatment: 41% in HT group, 50% in placebo group at trial end                                                                                            |
|         | Analysis by intention to treat: yes                                                                                                                                   |
|         | No. of centres: not stated                                                                                                                                            |
|         | Years of recruitment: June 1997 to January 2003                                                                                                                       |
|         | Design: parallel                                                                                                                                                      |
|         | Funding: National Cancer Institute grant                                                                                                                              |

## Barakat 2006 (Continued)

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b></p> <p>Women post total hysterectomy and bilateral salpingo-oophorectomy (at least) for surgically staged stage I or II endometrial cancer within 20 weeks of study entry, with indication for use of oestrogen therapy including hot flushes, vaginal atrophy, increased risk of CHD or increased risk of osteoporosis. Had to have undergone clinical exam with history, pelvic exam and chest X-ray before study entry. Normal hepatic function and normal mammogram or negative breast biopsy within previous year</p> <p><b>Excluded</b></p> <p>Women with history or suspicion of breast cancer or other malignancy with exception of non-melanoma skin cancer within past 5 years or with history of acute liver disease or thromboembolic disease</p> <p>Median age: 57</p> <p>Age range: 26–91. About half of the participants were over 60 years old</p> <p>Means of recruitment: not stated</p> <p>Baseline equality of treatment groups: well-balanced</p> <p>Country: USA</p> |
| Interventions | <p>HT arm: 0.625 mg CEE (unopposed oestrogen)</p> <p>Control arm: placebo</p> <p>Duration: planned for 3 years with 2 years' additional follow-up. Closed early with median follow-up 35.7 months</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Outcomes      | <p>Total deaths</p> <p>CHD deaths</p> <p>Coronary event deaths</p> <p>Endometrial cancer deaths</p> <p>Endometrial cancer (recurrence)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Notes         | <p>Enrolment decreased after WHI was published in July 2002. Study closed prematurely owing to poor accrual. In addition, preponderance of participants had low risk profile, so low event rate meant power unlikely to be reached with original power calculation. This study planned to enrol 2108 women.</p> <p>Numbers randomised not entirely clear: study refers to 1236 "eligible and assessable women".</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                            |
|---------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Remotely generated                                                                                                               |
| Allocation concealment (selection bias)                                   | Low risk           | Remotely dispensed drugs                                                                                                         |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | High risk          | No losses to follow-up reported, but numbers randomised not entirely clear: study refers to 1236 "eligible and assessable women" |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants and physicians blinded                                                                                              |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Review authors believe risk of bias low owing to 'hard' nature of outcomes                                                       |

## Barakat 2006 (Continued)

|                                      |          |                                                                                                                                                                                                                                                    |
|--------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selective reporting (reporting bias) | Low risk | All expected outcomes reported                                                                                                                                                                                                                     |
| Other bias                           | Low risk | Funding: supported by National Cancer Institute Grant No. CA 27469 to the Gynecologic Oncology Group Administrative Office and Grant No. CA 37517 to the Gynecologic Oncology Group Statistical and Data Center<br><br>Groups balanced at baseline |

## ELITE 2014

### Study characteristics

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>2 × 2 double-blinded placebo-controlled parallel-group RCT</p> <p>Stated purpose: to assess the effect of HT on progression of subclinical atherosclerosis and cognitive effects initiated between early and late postmenopause</p> <p>Stratification: not stated</p> <p>No. of women screened for eligibility: 3061 (n = 2166 via telephone, n = 895 in person)</p> <p>No. of women randomised: 643 (323 to HT, 320 to placebo; subgrouped by time since menopause with respect to initiation of HT)</p> <p>No. of women analysed: 567 underwent cognitive baseline assessment, total of 567 women provided cognitive outcomes at 2.5 years and 455 women provided outcomes at 5 years.</p> <p>Losses to f/u: 2.2% of women were lost to follow-up, and another 10.0% discontinued participation before cognitive outcomes were assessed (14 lost to follow-up, 22 dropouts due to adverse events, 40 discontinued for other reasons before contributing to cognitive outcomes at 2.5 years).</p> <p>Adherence to treatment: mean adherence for oestradiol or placebo was ≥ 98% for early and late group women.</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 1</p> <p>Years of recruitment: July 2005 and September 2008</p> <p>Design: parallel</p> <p>Funding: supported by National Institutes of Health grant for initial and supplemental funding of ELITE and ELITE-Cog. Study drugs and placebo were supplied without charge or restriction by Teva Pharmaceuticals, Watson Pharmaceuticals and Abbott Laboratories.</p> |
| Participants | <p>643 healthy postmenopausal women with clinical evidence of CVD or diabetes, subgrouped by time since menopause (&lt; 6 years since menopause (n = 271) or &gt; 10 years since menopause (n = 372))</p> <p><b>Included</b></p> <p>Women with a serum oestradiol level &lt; 25 picogram/mL and cessation of regular menses &gt; 6 months who are &lt; 6 years and &gt; 10 years postmenopausal</p> <p><b>Excluded</b></p> <p>Clinical signs, symptoms or personal history of cardiovascular disease, indeterminate time since menopause, DM or fasting serum glucose ≥ 140 mg/dL, uncontrolled hypertension (diastolic blood pressure ≥ 110 mmHg), untreated thyroid disease, serum creatinine &gt; 2.0 mg/dL, plasma triglyceride</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**ELITE 2014** (Continued)

levels > 500 mg/dL, life-threatening disease with prognosis < 5 years, cirrhosis or liver disease, hx of deep vein thrombosis or pulmonary embolism, history of breast cancer, current HT

Median age: 53.4 years for early, 63.6 years for late

Age range: 41 to 84

Means of recruitment: telephone and in person

Baseline equality of treatment group: no statistically significant difference in baseline characteristics. Women not contributing to analysis were similar to other women in most but not all comparisons.

Country: USA

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <p>1. Oral 17<math>\beta</math>-oestradiol 1 mg daily with (uterine intact) or without (hysterectomy) vaginal micronised progesterone gel 4% (45 mg) 10 days per month: Study publication does not state how many women were in each group.</p> <p>2. Placebo</p> <p>Originally planned for 5 years, extended to 7.5 years</p>                                                                                                                  |
| Outcomes      | <p><b>Primary study outcome</b></p> <p>Progression of subclinical atherosclerosis - not relevant for current review</p> <p><b>Secondary outcomes</b></p> <p>Cognitive function at 2.5 and 5 years, cardiovascular events (fatal or nonfatal MI, silent MI, sudden death), stroke, venous thromboembolism (DVT or PE), cancer (breast, uterine, ovarian, gastrointestinal, lung), bone fracture, all-cause mortality, non-coronary mortality</p> |
| Notes         | <p>Power calculation: 506 sample size provides power of 80% needed to detect difference in rate of change in carotid artery intima media thickness and effect size of 0.22 in early and late groups combined.</p> <p>Registered in 2005. NCT00114517</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded</p>                                                |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Randomisation sequence generated by computer by study statistician                                                                                                                                                                                                                   |
| Allocation concealment (selection bias)                                   | Low risk           | "Stratified randomization list [was] used to prepare the study products. After determining a participant's eligibility, clinic staff pulled the next study product in sequence from the appropriate stratum, recorded the product identification number, and dispensed the product". |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk       | 567/643 women (88%) analysed for cognitive outcomes at 2.5 years, 455/643 (71%) at 5 years                                                                                                                                                                                           |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Investigators, participants, clinic staff and data monitors were blinded.                                                                                                                                                                                                            |

**ELITE 2014** (Continued)

|                                                                 |          |                                                                                                                                                                                 |
|-----------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blinding of outcome assessment (detection bias)<br>All outcomes | Low risk | Not stated whether outcome assessor blinded, but most probably, as study author mentioned trial was extended before blinding was unmasked after receiving supplementary funding |
| Selective reporting (reporting bias)                            | Low risk | All expected outcomes reported                                                                                                                                                  |
| Other bias                                                      | Low risk | No other potential source of bias identified. Protocol <a href="#">NCT00114517</a>                                                                                              |

**EPAT 2001**
**Study characteristics**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to determine the effect of oestrogen-alone HT on progression of subclinical atherosclerosis in healthy postmenopausal women without pre-existing cardiovascular disease, as measured by changes in thickness of carotid artery wall</p> <p>Stratification: by low-density lipoprotein cholesterol level (threshold &lt; 4.15 mmol/L), previous duration of HT (threshold &lt; 5 years) and diabetes mellitus status</p> <p>Blinding: participants, gynaecologists, clinical staff and image analysts. The data monitor and the data analyst were blinded to treatment assignment until analyses were completed.</p> <p>No. of women screened for eligibility: 1161 pre-screened by phone, 422 screened on site, of whom 52% randomised</p> <p>No. randomised: 222</p> <p>No. analysed: 222 for clinical outcomes</p> <p>Losses to follow-up: 33 women were not evaluable for primary study endpoints, but clinical endpoints were reported for all.</p> <p>Adherence to treatment in evaluable women: During the trial, mean pill adherence was 95% in the oestradiol group and 92% in the placebo group (P = 0.08).</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 1</p> <p>Years of recruitment: 1994 to 1998</p> <p>Design: parallel</p> <p>Funding: National Institute on Aging</p> |
| Participants  | <p><b>Included</b></p> <p>Postmenopausal women aged &gt; 45 years, no pre-existing cardiovascular disease, low-density lipoprotein levels &gt; 3.37 mmol/L</p> <p><b>Excluded</b></p> <p>Women with previous breast or gynaecological cancer, frequent hot flushes, diastolic blood pressure &gt; 110, uncontrolled diabetes or thyroid disease, abnormal bloods, smokers</p> <p>Mean age: 61.15</p> <p>Age range: 51.4 to 69.2</p> <p>Means of recruitment: not stated</p> <p>Baseline equality of treatment groups: no significant differences in demographics or clinical variables</p> <p>Country: USA</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Interventions | <p>HT arm: unopposed micronised 17B-oestradiol 1 mg daily</p> <p>Control arm: placebo</p> <p>Duration: 2 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Outcomes      | <p><b>Primary outcomes</b></p> <p>Carotid artery wall thickness on ultrasound</p> <p>Myocardial infarction</p> <p>Cerebrovascular accident</p> <p>Transient ischaemic attack</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Long-term hormone therapy for perimenopausal and postmenopausal women (Review)**

## EPAT 2001 (Continued)

Deep vein thrombosis  
Pulmonary embolism

Notes

Power calculation: sample size of 200 required to detect treatment effect size (difference in carotid artery wall thickness) of 0.40 or greater with 80% power

Registered on clinicaltrials.gov NCT00115024

Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                      |
|---------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computer-generated random numbers                                                                                                                                          |
| Allocation concealment (selection bias)                                   | Low risk           | Blinded medication packets assigned sequentially and remotely after eligibility confirmed                                                                                  |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | 33 women were not evaluable for primary (physiological) study endpoints, but clinical endpoints were reported for all by ITT analysis.                                     |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants, gynaecologists, clinical staff and image analysts. The data monitor and the data analyst were blinded to treatment assignment until analyses were completed. |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | "Adverse clinical symptoms and bleeding were assessed by the study gynaecologist, who was blinded to treatment assignment".                                                |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                                                                                             |
| Other bias                                                                | Low risk           | No apparent source of other bias                                                                                                                                           |

## EPHT 2006

### Study characteristics

Methods

Stated purpose: to ascertain harms and benefits of combined HT among healthy postmenopausal Estonian women

Stratification: by centre

No. of women screened for eligibility: 39,713 (whole female population aged 50–64 from 2 areas of Estonia)

No. randomised and consented: 777 for clinical outcomes (Veerus 2006 publication), 796 for quality of life (Veerus 2010 publication) (see Notes)

No. analysed: 777 for clinical outcomes (HT 404, placebo 373), 796 for quality of life (HT 415, placebo 381)

Losses to follow-up: none stated

Adherence to treatment: < 40% in HT group and < 30% in placebo group by 3 years (estimated from graph)

Analysis by intention to treat: yes

No. of centres: 3

**EPHT 2006** (Continued)

Years of recruitment: 1999 to 2001  
Design: parallel  
Funding: academic and government grants

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b><br/>Postmenopausal women &gt; 12 months since last period</p> <p><b>Excluded</b><br/>Women who had used hormone therapy during the previous 6 months; with untreated endometrial adenomatosis or atypical hyperplasia of the endometrium; history of breast cancer, endometrial cancer or ovarian cancer; any other cancer treated less than 5 years ago; history of meningioma; myocardial infarction within the past 6 months; history of hepatitis or functional liver disorders in the past 3 months; history of deep vein thrombosis, pulmonary embolism or cerebral infarction; porphyria; hypertension greater than 170/110 mmHg despite medication; laparoscopically or histologically confirmed endometriosis</p> <p>Mean age: 59<br/>Age range: 50 to 70<br/>Means of recruitment: invitation sent to whole female population aged 50 to 64 from 2 areas of Estonia<br/>Baseline equality of treatment groups: more prior use of oral contraceptive in HT group - 9.2% vs 6.4%; HT group older (59 vs 58.5)<br/>Country: Estonia</p> |
| Interventions | <p>HT arm: combined oestrogen and progesterone as 1 daily tablet containing CEE 0.625 mg and medroxyprogesterone acetate 2.5 mg<br/>Control arm: matching placebo<br/>Duration: mean follow up 3.43 years (range 2-5). Planned for 10-year follow-up but closed early</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Outcomes      | <p>Death<br/>CHD<br/>Cancers<br/>Fractures<br/>CVD</p> <p>Quality of life measured with EuroQol-5D questionnaire at 3 years (also measured with Women's Health Questionnaire at 1 year), but no baseline measure, and results pooled for blinded and unblinded HT arms (data not reported in this review)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Notes         | <p>Women randomised before eligibility and consent checked - envelopes opened only once these processes were completed. Additional 1001 women in unblinded trial arms</p> <p>Designed as part of international WISDOM trial</p> <p>Mean follow-up only 3.43 years (range 2-5) for clinical outcomes, 3.6 years for quality of life. Planned for 10-year follow-up but closed early</p> <p><a href="http://www.controlled-trials.com">www.controlled-trials.com</a><br/>ISRCTN35338757/35338757</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Risk of bias**

| Bias                                        | Authors' judgement | Support for judgement                  |
|---------------------------------------------|--------------------|----------------------------------------|
| Random sequence generation (selection bias) | Low risk           | Remotely randomised in permuted blocks |
| Allocation concealment (selection bias)     | Low risk           | Non-transparent sealed envelopes       |

## EPHT 2006 (Continued)

|                                                                           |              |                                                                                                                                                                        |
|---------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk | No stated losses to follow-up or dropouts, analysed by intention to treat. However, stated participation rates differ slightly across trial publications (796 vs 777). |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | Participants and investigators blinded                                                                                                                                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk     | Assessed by entries in cancer registry - review authors believe low risk of bias                                                                                       |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported                                                                                                                                         |
| Other bias                                                                | Unclear risk | Quality of life not measured (with EuroQol-5D) at baseline - possible baseline differences                                                                             |

## ERA 2000

### Study characteristics

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>Stated purpose: to evaluate effects of HT on progression of coronary atherosclerosis</p> <p>Stratification: according to lipid lowering therapy at baseline and hospital where angiogram was performed</p> <p>Blinding: participants, clinic staff and all outcome assessors blinded</p> <p>Unblinding: treatment assignment available to designated member of data management staff. Questions related to adverse effects directed to gynaecology physician and nurse not connected with study</p> <p>No. of women screened for eligibility: not stated</p> <p>No. randomised: 309</p> <p>No. analysed: 309 (for clinical events)</p> <p>Losses to follow-up: none (for clinical events)</p> <p>Adherence to treatment in 248 participants evaluated was as follows: unopposed oestrogen group took 74.5% of their prescribed medication, and combined HT group took 84%; placebo group took 85.8%. 5 women in the placebo group started to take HT.</p> <p>Analysis by intention to treat: although only 248 participants were available for the primary trial endpoint (which was biological), clinical adverse events, including outcomes of interest to this review, were reported for all participants at 3.2 years by intention to treat.</p> <p>No. of centres: 6</p> <p>Years of recruitment: January 1996 to December 1997</p> <p>Design: parallel</p> <p>Funding: grants from National Heart, Lung and Blood Institute and National Center for Research Resources General Clinical Research Center, study medications from Wyeth-Ayerst Research</p> |
| Participants | <p><b>Included</b></p> <p>Postmenopausal women aged 55–80 years (non-natural menses for at least 5 years, or for 1 year and FSH &gt; 40 mu/mL or oophorectomy) with at least 1 stenosis &gt; 30% in any single coronary artery confirmed by coronary angiography within 4 months of randomisation, baseline gynaecological examination normal</p> <p><b>Excluded</b></p> <p>Failure to achieve &gt; 80% compliance during 4-week placebo run-in phase, breast or endometrial cancer, history of DVT or PE, symptomatic gallstones or elevated liver enzymes, fasting plasma triglycerides &gt; 400 mg/dL, MI within 4 weeks, renal insufficiency, dye allergy, &gt; 70% stenosis of coronary</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Long-term hormone therapy for perimenopausal and postmenopausal women (Review)

**ERA 2000** (Continued)

artery, uncontrolled hypertension, uncontrolled diabetes, planned or prior coronary artery bypass graft, revascularisation of only qualifying lesion (for study), inadequate baseline angiogram for study, other non-CHD disease likely to be fatal or to prevent adequate follow-up, participation in other intervention studies, plans to leave area within 3 years

Mean age: 66

Age range: 41.8 to 79.9

Means of recruitment: media announcements, contact through hospital records and admissions, screening logs from other studies

Baseline equality of treatment groups

Country: USA

Follow-up: 3 months, 6 months, then 6-monthly clinic visits; annual smear and mammography, annual endometrial aspiration

|               |                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <b>HT arm: 1 of the following</b><br>1. 0.625 mg CEE (unopposed oestrogen)<br>2. 0.625 mg CEE plus 2.5 mg MPA (combined continuous therapy)<br>Control arm: placebo<br>Duration: 3.2 years mean                                                   |
| Outcomes      | Primary outcome angiographic<br>MI<br>Stroke<br>Death<br>DVT<br>PE                                                                                                                                                                                |
| Notes         | Power calculation: 80% power for primary angiographic outcome<br><br>Clinical trial register: NCT00000549<br><br>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                              |
|---------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computerised in random blocks                                                      |
| Allocation concealment (selection bias)                                   | Low risk           | Computer-displayed treatment assignment after eligible participant details entered |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | No losses to follow-up for clinical adverse events. Analysed by intention to treat |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants and clinicians blinded                                                |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Outcome assessors blinded                                                          |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                     |

**ERA 2000** (Continued)

|            |              |                                                                                                           |
|------------|--------------|-----------------------------------------------------------------------------------------------------------|
| Other bias | Unclear risk | More in unopposed oestrogen group using nitrates at baseline; otherwise prognostic balance between groups |
|------------|--------------|-----------------------------------------------------------------------------------------------------------|

**ESPRIT 2002**
**Study characteristics**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to ascertain whether unopposed oestrogen reduces the risk of further cardiac events in postmenopausal women who survive a first myocardial infarction</p> <p>Stratification: by clinical centre</p> <p>Blinding: participants, clinicians, outcome assessors. Pharmaceutical company dispensed medication/placebo in identical numbered packages.</p> <p>Unblinding: on request of family doctor or if participant withdrew from treatment (in later stages of study, only if withdrawing participant had not had a hysterectomy). Outcome assessors remained blinded throughout.</p> <p>No. of women screened for eligibility: 3121 met inclusion criteria for MI (reasons for non-participation listed in study).</p> <p>No. randomised: 1017</p> <p>No. analysed: 1017</p> <p>Dropouts: dropouts included 43 women in the HT group (8%) and 57 in the placebo group (11%) who did not take any of the trial medication.</p> <p>Losses to follow-up: none</p> <p>Known non-adherence with allocated treatment was as follows: at 1 year, 51% of participants on the HT arm and 31% on the placebo arm were not taking their allocated tablets regularly. At 2 years, 57% of participants on the HT arm and 37% on the placebo arm were not taking their allocated tablets regularly.</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 35</p> <p>Years of recruitment: July 1996 to Feb 2000</p> <p>Design: parallel</p> <p>Funding: Schering AG provided medication</p> |
| Participants  | <p><b>Included</b></p> <p>Postmenopausal women admitted to coronary care units or general medical wards at participating centres, who met diagnostic criteria for myocardial infarction, were discharged alive within 31 days of admission</p> <p><b>Excluded</b></p> <p>Women with previously documented MI who had used HT or had vaginal bleeding in the 12 months before admission, history of breast, ovarian or endometrial cancer, active thrombophlebitis, history of DVT or PE, liver disease, Rotor syndrome, Dubin-Johnson syndrome or severe renal disease</p> <p>Mean age: 62 years (SD 5)</p> <p>Means of recruitment: research nurses checked hospital case notes and approached potentially eligible women if their family doctor agreed to collaborate.</p> <p>Baseline equality of treatment groups: yes</p> <p>Countries: England and Wales</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Interventions | <p>HT arm: unopposed oestradiol valerate 2 mg daily</p> <p>Control arm: placebo</p> <p>Duration: 2 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Outcomes      | <p>Recurrent MI</p> <p>Cardiac death</p> <p>All-cause death</p> <p>Endometrial cancer</p> <p>Breast cancer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**ESPRIT 2002** (Continued)

Stroke  
Thromboembolism

Notes

Power calculation: needed 1700 participants to give 80% power to detect 33.3% decrease in incidence of non-fatal reinfarction or cardiac death (2-sided  $P = 0.05$ )

Accrual lower than anticipated: study closed with only 1017 participants, giving 56% power to detect above-mentioned outcomes, assuming full compliance.

Clinical trial register: not available

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | List of random numbers generated by trial statistician in blocks of 4                |
| Allocation concealment (selection bias)                                   | Low risk           | Women assigned consecutively to numbers kept on list accessible to statistician only |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | No losses to follow-up; analysed by intention to treat                               |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants and clinicians blinded                                                  |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Outcome assessors blinded                                                            |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                       |
| Other bias                                                                | Low risk           | No apparent source of other bias                                                     |

**EVTET 2000**
**Study characteristics**

Methods

Stated purpose: to determine if HT alters risk of venous thromboembolism in high-risk women

Randomisation method: computer-generated 1:1 block randomisation with fixed block sizes of 10

Allocation method: not described

Stratification: by age < 60 years or ≥ 60 years; 37 (23 HT and 14 placebo) women did not attend all visits owing to premature termination of the study

Blinding: double-blind

No. of women screened for eligibility: 328

No. randomised: 140 (71 HT, 69 placebo)

No. analysed: 140

Losses to follow-up: nil, although 37 (23 HT and 14 placebo) women did not attend all visits owing to premature termination of the study

Adherence to treatment: 33 dropouts: 10 in HT group (2 wanted to be sure of being treated with oestrogen for postmenopausal symptoms, 8 had adverse effects), 23 in the placebo group (11 wanted to be sure of being treated with oestrogen for postmenopausal symptoms, 10 had adverse effects, 2 no reason stated)

**EVTET 2000** (Continued)

Analysis by intention to treat: main findings were not reported by intention to treat because dropouts from the placebo group were not included in the denominator for the rate of recurrent thromboembolism. We included all randomised participants in the analysis for this review.

No. of centres: not stated

Years of recruitment: February 1996 to March 1999

Design: stratified double triangular sequential design

Funding: Novo-Nordisk Pharmaceutical and research forum Ulleval University Hospital

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b></p> <p>Postmenopausal women with history of VTE, aged &lt; 70 years, previous VTE verified by objective means (i.e. venography or ultrasonography in cases of DVT; lung scan, helical computed tomography or angiography in cases of PE)</p> <p><b>Excluded</b></p> <p>Current use or use of anticoagulants within past 3 months, familial antithrombin deficiency, any type of malignant disease including known, suspected or past history of carcinoma of the breast; acute or chronic liver disease or history of liver disease in which liver function tests had failed to return to normal; porphyria; known drug abuse or alcoholism; life expectancy less than 2 years; women who had taken part in other clinical trials within 12 weeks before study entry</p> <p>Mean age: 55.8 years</p> <p>Age range: 42-69 years</p> <p>Means of recruitment: letters to family doctors, gynaecologists and hospitals, health bulletins and media</p> <p>Baseline equality of treatment groups: baseline characteristics were similar for HT group and placebo group with regard to previous disease (coronary heart disease, hypertension, stroke, diabetes), smoking habits and serum lipids. All women had previously suffered at least 1 VTE, and the total number of previous VTEs was 75 in the placebo group and 77 in the HT group.</p> <p>Country: Norway</p> |
| Interventions | <p>HT arm: 2 mg oestradiol plus 1 mg norethisterone acetate 1 mg</p> <p>Control arm: placebo</p> <p>Duration: planned 2 years, stopped prematurely at median 1.3 years' follow-up</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Outcomes      | <p>Venous thrombosis</p> <p>Myocardial infarction</p> <p>Transient ischaemic attacks</p> <p>Stroke</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Notes         | <p>Study was terminated early; only 140 women enrolled of 240 planned</p> <p>Power calculation: at a significance level of 5% and a power of 90%, sample size was estimated to a maximum of 240 women</p> <p>After publication of results of the HERS study, which showed as a secondary endpoint increased risk of VTE, recruitment of women was discontinued in September 1998, until reviewed by the safety monitoring committee. The committee was also concerned about a non-significant clustering of endpoints in 1 study group, without knowing treatment allocation. The committee advised on premature termination of the study, even though formal boundaries showing excess risk of VTE were not reached. The final decision on termination of the study was made in February 1999, and by the end of March 1999, all participants had completed a final follow-up visit.</p> <p>Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.</p>                                                                                                                                                                                                                                                                                                                                                   |

**Risk of bias**

| Bias                                        | Authors' judgement | Support for judgement                                                   |
|---------------------------------------------|--------------------|-------------------------------------------------------------------------|
| Random sequence generation (selection bias) | Low risk           | Computer-generated 1:1 block randomisation with fixed block sizes of 10 |

**EVTET 2000** (Continued)

|                                                                           |              |                                                                                                                                                                              |
|---------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Allocation concealment (selection bias)                                   | Unclear risk | Not described                                                                                                                                                                |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk | Main findings were not reported by intention-to-treat because dropouts from the placebo group were not included in the denominator for the rate of recurrent thromboembolism |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | Blinded participants and personnel - "equal-looking" placebo                                                                                                                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk     | Blinded outcome assessment                                                                                                                                                   |
| Selective reporting (reporting bias)                                      | Low risk     | Retrospectively registered protocol on trials register. Reports all expected outcomes                                                                                        |
| Other bias                                                                | Low risk     | No apparent source of other bias                                                                                                                                             |

**Ferenczy 2002**
**Study characteristics**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>Stated purpose: to assess endometrial safety and bleeding patterns of 17B-oestradiol sequentially combined with dydrogesterone</p> <p>Stratification: not mentioned</p> <p>Blinding: double-blinded</p> <p>No. of women screened for eligibility: 844</p> <p>No. randomised: 595 (HT group 1: 117, HT group 2: 114, HT group 3: 117, HT group 4: 118, placebo group: 113 (see Interventions))</p> <p>No. analysed: 442 (for endometrial cancer, which is the only outcome of interest for this review)</p> <p>Losses to follow-up: endometrial status was evaluated by a biopsy, which was available only for women who remained on active treatment for over a year, or who received placebo and completed the 2-year study. This resulted in 153 losses to follow-up for this outcome (87 from active treatment groups (24%) and 50 from the placebo group (44%), plus another 16 who received no study medication).</p> <p>Adherence to treatment: not reported</p> <p>Analysis by intention to treat: no</p> <p>No. of centres: multicentre (number not stated)</p> <p>Years of recruitment: not stated</p> <p>Design: parallel</p> <p>Funding: Solvay Pharmaceutical</p> |
| Participants | <p><b>Included</b></p> <p>Postmenopausal women with a uterus with amenorrhoea of at least 6 months or surgically postmenopausal (following bilateral oophorectomy without hysterectomy, more than 3 months before enrolment), FSH within normal postmenopausal range</p> <p><b>Excluded</b></p> <p>Abnormal (uninvestigated bleeding) vaginal bleeding, use of oestrogens and/or progestogens and/or androgens in the preceding 6 months or more, and any previous use of oestradiol pellet/implant therapy</p> <p>Age range: 45–65</p> <p>Baseline equality of treatment groups: yes</p> <p>Countries: Canada and Netherlands</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Ferenczy 2002 (Continued)

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <b>HT arm</b><br><br>1 mg/d 17B-oestradiol/5 mg dydrogesterone for the last 14 days of each 28-day cycle<br>1 mg/d 17B-oestradiol/10 mg dydrogesterone for the last 14 days of each 28-day cycle<br>2 mg/d 17B-oestradiol/10 mg dydrogesterone for the last 14 days of each 28-day cycle<br>2 mg/d 17B-oestradiol/20 mg dydrogesterone for the last 14 days of each 28-day cycle<br>Control arm: placebo<br>Duration: 26 cycles (104 weeks) |
| Outcomes      | Endometrial cancer                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Notes         | Power calculation: not stated<br><br>Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.                                                                                                                                                                                                                                                                   |

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Unclear risk       | Not described                                                                                                                                                                                                                                               |
| Allocation concealment (selection bias)                                   | Unclear risk       | Not described                                                                                                                                                                                                                                               |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | High risk          | Endometrial status was evaluated by a biopsy, which was available only for women who remained on active treatment for over a year, or who received placebo and completed the 2-year study. This resulted in 153 losses to follow-up (26%) for this outcome. |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | States double-blinded - review authors believe risk of bias low owing to 'hard' nature of outcomes                                                                                                                                                          |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Review authors believe risk of bias low owing to 'hard' nature of outcomes                                                                                                                                                                                  |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                                                                                                                                                                              |
| Other bias                                                                | Low risk           | No apparent source of other bias                                                                                                                                                                                                                            |

## Greenspan 2005

### Study characteristics

|         |                                                                                                                                                                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods | Stated purpose: to determine effects of HT on clinical outcomes, including cognitive function, in elderly women<br>Stratification: no<br>Blinding: double-blinded<br>Unblinding: not described<br>No. of women screened for eligibility: 573<br>No. randomised: 373 (187 to HT, 186 to placebo)<br>No. analysed: 373 |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Long-term hormone therapy for perimenopausal and postmenopausal women (Review)

## Greenspan 2005 (Continued)

Losses to follow-up: 8 (6 in HT group, 2 in placebo group)  
Adherence to treatment: 61% on HT, 67% on placebo  
Analysis by intention to treat: yes  
No of centres: 1  
Years of recruitment: study conducted January 1996 to May 2001  
Design: parallel  
Funding: academic research funding Pharmaceutical companies provided the drugs.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b><br/>Community-dwelling women aged 65 years or older with (n = 243) or without (n = 130) a uterus, with complete medical history, physical examination and lab evaluation; tolerated HT in run-in phase</p> <p><b>Excluded</b><br/>Women with any illnesses or taking medications that could affect bone mineral metabolism within past year, or with known contraindication to HT<br/>Mean age: 71 years<br/>Age range: 65–90<br/>Means of recruitment: advertisements, presentations, physical referrals<br/>Baseline equality of treatment groups: yes<br/>Country: USA; n = 373 healthy women, mean age 71.3 years, 243 with a uterus and 130 without a uterus</p> |
| Interventions | <p>Three-month open run-in phase on HT</p> <p>1. CEE oral (0.625 mg/d) or CEE oral (0.625 mg/d) + medroxyprogesterone acetate (2.5 mg/d) in women with a uterus<br/>2. Placebo</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Outcomes      | MMSE, breast cancer, DVT, clinical fractures, colon cancer (and other outcomes not relevant to this review)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Notes         | <p>Half of participants also took alendronate; all took calcium and vitamin D supplement and a multi-vitamin.</p> <p>Clinical trial register not available</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                        |
|---------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Unclear risk       | Randomised lists "prepared by study statistician"                                                                                            |
| Allocation concealment (selection bias)                                   | Low risk           | Research pharmacist assigned treatment                                                                                                       |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | All participants included in analysis; low losses to follow-up (6/373)                                                                       |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Described as double-blinded                                                                                                                  |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | "Those who assessed the outcomes were blinded to treatment assignment... block sizes randomly determined to enhance blinding of study staff" |

## Greenspan 2005 (Continued)

|                                         |              |                                                                                                           |
|-----------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------|
| Selective reporting (re-reporting bias) | Low risk     | All expected outcomes reported                                                                            |
| Other bias                              | Unclear risk | Possibly some potential for confounding from 3-month run-in phase on HT and concurrent use of alendronate |

## HERS 1998

### Study characteristics

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>Stated purpose: to determine if combined HT alters risk for CHD events in postmenopausal women with established coronary disease</p> <p>Stratification: by clinical centre</p> <p>Blinding: participants, clinical centre staff, outcome assessors, data analysts, funders blinded</p> <p>Unblinding: when required for safety or symptom control, participants reported directly to gynaecology staff, who were located separately from clinical staff, did not communicate with them about breast or gynaecological problems and were not involved in outcome ascertainment.</p> <p>No. of women screened for eligibility: 3463, of whom 43% were excluded (ineligible, declined to participate, did not return for appointment or did not comply with placebo run-in therapy)</p> <p>No. randomised: 2763</p> <p>No. analysed: 2763</p> <p>Losses to follow-up: vital status known for all women at end of trial. 59 women did not complete follow-up (32 in experimental arm, 27 in placebo arm).</p> <p>Adherence to treatment by women evaluated: by self-report: at 1 year: 82% HT arm, 91% control arm; at 3 years: 75% HT arm, 81% control arm; by pill count in HT arm: at 1 year: 79%; at 3 years: 70% HT arm</p> <p>Analysis by intention to treat: yes (also analysed by treatment received, with inclusion limited to women with &gt; 80% compliance)</p> <p>No. of centres: 20</p> <p>Years of recruitment: February 1993 to September 1994</p> <p>Design: parallel</p> <p>Funding: pharmaceutical (Wyeth-Ayerst)</p> <p>UNBLINDED CONTINUATION OF <a href="#">HERS 1998</a>:</p> <p>N.B. Follow-up continued unblinded, as an open-label observational study</p> <p>2321 women (93% of 2510 surviving HERS participants) followed up for a further 2.7 years – originally planned for additional 4 years, but executive committee decided no further useful information likely to emerge</p> <p>No. analysed: 2311 for vital status</p> <p>Losses to follow-up: 10 women (1%) not contacted at final follow-up (2 in HT arm, 8 in control arm); of these, vital status known for 5</p> <p>Adherence to treatment: among women originally assigned to the HT group, 45% reported at least 80% compliance during the sixth year of follow-up. Among women originally assigned to placebo, 8% reported taking HT at 6 years.</p> |
| Participants | <p><b>Included</b></p> <p>Postmenopausal women younger than 80 years, with a uterus, with coronary disease (myocardial infarction, coronary artery bypass surgery, percutaneous coronary revascularisation or angiographic evidence of at least 50% narrowing of 1 or more major arteries, as documented by baseline ECG or hospital discharge summary), likely to be available for follow-up for at least 4 years</p> <p><b>Excluded</b></p> <p>Women whose coronary event occurred within 6 months of randomisation, use of hormone therapy within 3 months of randomisation, serum triglycerides <math>\geq 300</math> mg/dL, history or baseline findings suggestive of venous thromboembolism, breast cancer, endometrial cancer, cervical cancer, uncontrolled hypertension, uncontrolled diabetes, severe congestive heart failure, other life-threatening disease, alcoholism, drug abuse, history of intolerance of HT, any pre-existing condition indicating unsuitability for long-term HT or placebo therapy, &gt; 80% compliance with placebo medication during run-in phase</p> <p>Mean age: 67 years (SD 7)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**HERS 1998** (Continued)

Age range: 44–79  
Means of recruitment: lists of cardiac patients, mass mailing, direct advertising  
Baseline equality of treatment groups: more women in control arm on statins at randomisation (67% vs 54%). When adjusted in analyses, made no statistically significant difference  
Country: USA

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <p>HT arm: conjugated equine oestrogen 0.625 mg with medroxyprogesterone acetate 2.5 mg<br/>Control arm: placebo identical in appearance<br/>Continuous oral regimen<br/>Adherence to treatment defined as &gt; 80% compliance with medication or placebo<br/>Duration: 4.2 years, mean</p> <p>FOR UNBLINDED CONTINUATION OF <a href="#">HERS 1998</a>:<br/>continuation planned for an additional 4 years but stopped after mean additional 2.7 years, as no additional useful data anticipated</p> |
| Outcomes      | <p>Coronary events (MI or coronary death)<br/>Venous thromboembolism<br/>Fracture<br/>Gallbladder disease<br/>Endometrial, breast or ovarian cancer<br/>Death</p>                                                                                                                                                                                                                                                                                                                                    |
| Notes         | <p>Power calculation: 90% power to observe 24% reduction in coronary events at an average of 4.2 years' (P = 0.05) follow-up<br/>Further unblinded follow-up 2.7 years (HERS II) – see below</p> <p>Clinical trial register: not available</p> <p>Trial authors contacted for additional information; response received but no details recorded.</p>                                                                                                                                                 |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                             |
|---------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computer-generated random numbers in blocks of 4                                                                                                                  |
| Allocation concealment (selection bias)                                   | Low risk           | Computer displayed after participant details entered                                                                                                              |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | Vital status known for all women at end of trial. 59 women did not complete follow-up (32 in experimental arm, 27 in placebo arm). Analysed by intention to treat |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants, clinical centre staff, data analysts and funders blinded                                                                                            |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Outcome assessors blinded                                                                                                                                         |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                                                                                    |
| Other bias                                                                | Low risk           | More women in control arm on statins at randomisation (67% vs 54%). When adjusted in analyses, made no statistically significant difference                       |

## KEEPS 2012

### Study characteristics

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to test whether menopausal HT initiated within 3 years of menopause can delay progression of atherosclerosis</p> <p>Stratification: by study centre</p> <p>No. of women screened for eligibility: 4532</p> <p>No. randomised: 727 (oral HT: 230, transdermal HT: 222, placebo: 275)</p> <p>No. analysed: 580</p> <p>Losses to follow-up: in oral HT, transdermal HT and placebo groups, respectively, for the following reasons: withdrawals for AEs = 16/230, 9/222, 12/275; personal reasons 11/230, 8/222, 24/275; non-adherence 1/230, 4/222, 3/275; unknown reasons 15/230, 22/222, 18/275</p> <p>Dropouts/adherence to treatment: 11% of women non-adherent to treatment but included in analysis (116/580)</p> <p>For KEEPS-COG study, sample sizes were oral HT: 220, transdermal HT: 211, placebo: 262.</p> <p>Analysis by intention to treat: no - data not imputed for women lost to follow-up</p> <p>No. of centres: 9</p> <p>Years of recruitment: 2005-2008</p> <p>Design: double-blinded parallel-group RCT</p> <p>Funding: Aurora Foundation (not-for-profit) and other academic grants. Study medications provided in part by pharmaceutical companies</p> |
| Participants  | <p><b>Included</b></p> <p>Women aged 42–58 within 6–36 months of final menses</p> <p><b>Excluded</b></p> <p>Women post hysterectomy, BMI &gt; 35 kg/m<sup>2</sup>, low-density lipoprotein cholesterol &gt; 160 mg/dL, coronary artery calcium over 50 Agaston units at baseline, smoking over 10 cigarettes per day, history of diabetes, myocardial infarction, stroke, thromboembolic disease or cancer</p> <p>Mean age: 52.7 years, mean 1.8 years since menopause</p> <p>Age range: 42–58</p> <p>Means of recruitment: mass mailings, posters, print and online advertising, Internet web page</p> <p>Baseline equality of treatment groups: high-density lipoprotein cholesterol lower in placebo group, otherwise no statistically significant difference in baseline characteristics</p> <p>Country: USA</p>                                                                                                                                                                                                                                                                                                                                                                           |
| Interventions | <p><b>HT arm</b></p> <p>0.45 mg/d oral CEE + cyclic oral micronised progesterone 200 mg/d × 12 days per month</p> <p>0.05 mg/d transdermal oestradiol + cyclic oral micronised progesterone 200 mg/d × 12 days per month</p> <p><b>Control:</b> placebo</p> <p>Duration: 4 years (original protocol was for 5 years, shortened during first year after reconsideration of study design)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Outcomes      | <p>Primary outcome: carotid artery intima media thickness</p> <p>Outcomes relevant to this review: quality of life, clinical CVD events (including MI, stroke) reported as adverse events</p> <p>Cognition: KEEPS-COG ancillary study enrolled 93% of women in <a href="#">KEEPS 2012</a> (participation by invitation). Measured with MMSE</p> <p>Global cognitive function also reported in a subset of participants (CEE 29, combined HT 59, placebo 36) in conjunction with magnetic resonance imaging monitoring of brain structure: data not included in this review (Kantarci 2015)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Notes         | Registered NCT00154180 2005 Kronos Early Estrogen Prevention Study (KEEPS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**KEEPS 2012** (Continued)

NCT03718494

Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computer-generated randomly sequenced blocks of 13 in a ratio 4:4:5 (oral CEE/transdermal CEE/placebo)                                                                                                                                         |
| Allocation concealment (selection bias)                                   | Low risk           | Remotely generated sequence; database key not accessible to study personnel                                                                                                                                                                    |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | High risk          | 80% of women included in analysis for primary clinical endpoint at 4 years (580/727): 43 withdrew in each HT group, 57 withdrew in placebo group<br><br>619/693 (89%) women in KEEPS-COG were included in analysis (85% of total KEEPS sample) |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Study pharmacist provided blinded packets of study drugs for each participant.                                                                                                                                                                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Outcome assessors blinded to study allocation                                                                                                                                                                                                  |
| Selective reporting (reporting bias)                                      | Low risk           | Reports all expected outcomes and all outcomes planned in protocol. Reporting of AEs actively solicited                                                                                                                                        |
| Other bias                                                                | Low risk           | No other source of potential bias identified                                                                                                                                                                                                   |

**Mulnard 2000**
**Study characteristics**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>Stated purpose: to determine whether oestrogen-only HT affects global, cognitive or functional decline in women with mild to moderate Alzheimer's disease</p> <p>Stratification: not mentioned</p> <p>No. of women screened for eligibility: 153</p> <p>No. randomised: 120 (CEE 0.625 mg: 42, CEE 1.25 mg: 39, placebo: 39)</p> <p>No. analysed: 120</p> <p>Losses to follow-up: nil</p> <p>Adherence to treatment: 23 dropouts (7 in placebo group, 7 in CEE 0.625 mg group, 9 in CEE 1.25 mg/d).</p> <p>Adherence to treatment was measured and was defined as the proportion of individuals who ingested at least 80% of the study medication but was not reported in the trial publication.</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 32</p> <p>Years of recruitment: not stated</p> <p>Design: parallel placebo-controlled</p> <p>Funding: National Institute on Aging, Wyeth Ayerst</p> |
| Participants | <b>Included</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Mulnard 2000 (Continued)

Women with a diagnosis of probable Alzheimer's disease according to National Institute of Neurological and Communicative Disorders and Stroke-Alzheimer Disease and Related Disorders Association Criteria in mild or moderate stage (study protocol specified MMSE score of 14–28; several exceptions were made by the project director to allow for participants with MMSE scores as low as 12); female sex; previous hysterectomy (oophorectomy not required); older than 60 years; absence of major clinical depressive disorder (as measured by score < 17 on the Hamilton Depression Rating Scale (Ham D); normal gynaecological, breast and mammography results

### Excluded

Myocardial infarction within 1 year, history of thromboembolic disease or hypercoagulable state, hyperlipidaemia, use of excluded medications (i.e. oestrogens within 3 months); current use of antipsychotics, anticonvulsants, anticoagulants, beta-blockers, narcotics, methyldopa, clonidine or prescription cognitive-enhancing or antiparkinson medications, including experimental medications within 60 days before baseline. Stable doses of neuroleptics, antidepressants, anxiolytics, sedatives and hypnotics were allowed. At initiation of the protocol, individuals treated with donepezil or tacrine were excluded, but a protocol amendment after 20 months of enrolment allowed stable use (minimum of 4 weeks) of these medications before screening for the study

Mean age: 75

Age range: 56–91

Means of recruitment: not stated

Baseline equality of treatment groups: no significant differences between the 3 groups in terms of baseline and demographic characteristics

Country: USA

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <b>HT arm</b><br>CEE oral 0.625 mg/d<br>CEE 1.25 mg/d<br>Control: placebo<br>Duration: 1 year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Outcomes      | <b>Primary outcome</b><br><br>Progression of Alzheimer's disease (Alzheimer's Disease Co-operative Study version of the Clinical Global Impression of Change Scale)                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Notes         | Power calculation: 81% to detect a 29% difference in the proportion of participants who worsened in the 2 groups (60% worse in the placebo group vs 31% worse in the oestrogen group) using a 2-tailed ( $\alpha$ ) = 0.05 (based on data from a similar trial, with 40 participants receiving placebo and 80 receiving oestrogen)<br>* Inclusion criteria state > 60 years, but age range at baseline was 56–91<br><br>Clinical trial register: not available<br><br>Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded. |

### Risk of bias

| Bias                                                     | Authors' judgement | Support for judgement                                         |
|----------------------------------------------------------|--------------------|---------------------------------------------------------------|
| Random sequence generation (selection bias)              | Low risk           | Computer-generated in blocks of 6                             |
| Allocation concealment (selection bias)                  | Unclear risk       | Not described                                                 |
| Incomplete outcome data (attrition bias)<br>All outcomes | Low risk           | No losses to follow-up stated. Analysed by intention to treat |

## Mulnard 2000 (Continued)

|                                                                           |              |                                                                             |
|---------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------|
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | States double-blinded; used "identically appearing tablets"                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Unclear risk | States double-blinded; no further details                                   |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported                                              |
| Other bias                                                                | Unclear risk | Inclusion criteria state > 60 years, but age range at baseline was 56 to 91 |

## Nachtigall 1979

### Study characteristics

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to evaluate effects of HT</p> <p>Stratification: not mentioned</p> <p>Unblinding: code broken if a major medical complication or death occurred (13 times in HT group, 17 times in control group)</p> <p>No. of women screened for eligibility: 403 (235 excluded: 74 ineligible, 31 refused, 130 no match for pair found)</p> <p>No. randomised: 168</p> <p>No. analysed: 168</p> <p>Losses to follow-up: none</p> <p>Adherence to treatment: not mentioned</p> <p>Analysis by intention to treat: yes, although any events occurring after unblinding were not recorded</p> <p>No. of centres: 1</p> <p>Years of recruitment: unclear - study lasted 10 years and was complete by 1976</p> <p>Design: parallel</p> <p>Funding: not stated</p>                                                                                                                                                                                             |
| Participants  | <p><b>Included</b></p> <p>Postmenopausal inpatients with chronic disease (last menstrual period &gt; 2 years previously, FSH &gt; 105.5 mU, total urinary oestrogen &lt; 10 micrograms/dL), never taken HT. All hospitalised for entire study period; screened with history, physical examination, medical record review; matched on the basis of chronic disease diagnosis, as follows: diabetes mellitus (14 pairs), custodial care (20 pairs), arteriosclerosis (9 pairs). Other pairs matched on the basis of chronic neurological disorders</p> <p><b>Excluded</b></p> <p>Acute heart disease, hypertension (blood pressure &gt; 160/94), apparent malignancy, hysterectomy</p> <p>Mean age: 55</p> <p>Baseline equality of treatment groups: correlation for diagnosis was identical. Correlation for some other risk factors was low between individual pairs, but group means were similar.</p> <p>Country: New York Hospital for Chronic Diseases</p> |
| Interventions | <p>HT arm: CEE 2.5 mg daily, plus MPA 10 mg for 7 days each month</p> <p>Control arm: placebo</p> <p>Duration: 10 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Outcomes      | Death, myocardial infarction, "serious embolism" (pulmonary embolus), breast cancer, colon cancer, endometrial cancer, gallstones                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Notes         | Power calculation: not mentioned                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Long-term hormone therapy for perimenopausal and postmenopausal women (Review)

## Nachtigall 1979 (Continued)

Re generalisability: study authors point out that almost all women had long-term chronic disease, were hospitalised for the entire study period, had much lower than normal overall parity and had more prolonged bed rest than the average woman.

Clinical trial register: not available

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                    |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Unclear risk       | Women matched for diagnosis of chronic disease. From matched pairs, research nurse randomly selected which member would be assigned to which group. Method not described |
| Allocation concealment (selection bias)                                   | Unclear risk       | Not described                                                                                                                                                            |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | No losses to follow-up described. Analysed by intention to treat, but any events occurring after unblinding not recorded                                                 |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | States participants and research physicians blinded                                                                                                                      |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | States participants and research physicians blinded                                                                                                                      |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                                                                                           |
| Other bias                                                                | Unclear risk       | Correlation for some baseline prognostic factors was low between individual pairs, but group means were similar.                                                         |

## Notelovitz 2002

### Study characteristics

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods | <p>Stated purpose: to determine the lowest effective dose of an oestradiol transdermal delivery system for preventing bone loss in postmenopausal women</p> <p>Stratification: not described</p> <p>Unblinding: not described</p> <p>No. of women screened for eligibility: not stated</p> <p>No. randomised: 355 (0.025 mg dose: 89; 0.05 mg dose: 90; 0.075 mg dose: 89; placebo: 87)</p> <p>No. analysed: 355 (data imputed for losses to follow-up)</p> <p>Losses to follow-up: 34 (9.6%)</p> <p>Adherence to treatment: 125 dropouts: 125 (35%) did not complete 2 years' treatment (88 in active treatment arms, 37 in placebo arm). One participant was withdrawn for failure to adhere to the treatment schedule. Overall level of adherence to treatment in women who continued with their allocated treatment is not described.</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 22</p> <p>Years of recruitment: not stated</p> <p>Design: parallel</p> <p>Funding: Proctor and Gamble Pharmaceuticals</p> |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Notelovitz 2002 (Continued)

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b></p> <p>Postmenopausal, non-osteoporotic, ambulatory women younger than 70 years of age who had had a hysterectomy, with or without bilateral oophorectomy, at least 12 months earlier. Postmenopausal status documented by serum oestrogen &lt; 23 picograms/mL and FSH serum levels &gt; 40 mIU/mL. Non-osteoporotic status defined by dual energy x-ray absorptiometry (DXA) minimum T-score of -2.5</p> <p><b>Excluded</b></p> <p>Participants who had received oral oestrogens within 2 months of enrolment, or who had contraindications to oestrogen therapy or history of oestrogen intolerance, women with clinically significant systemic or psychiatric disorders; history of cancer (other than basal cell carcinoma in remission or uterine cancer treated by hysterectomy); history of osteomalacia, hyperparathyroidism or untreated hyperthyroidism, abnormal serum lipids, creatinine or liver enzymes; use of medications within 3 months of enrolment that could modify BMD, radiographic abnormalities of the lumbar spine on anterior/posterior or lateral view, which would preclude precise DXA [dual-energy X-ray absorptiometry] measurements</p> <p>Mean age: not stated<br/>Age range: not stated<br/>Means of recruitment: not stated<br/>Baseline equality of treatment groups: yes<br/>Country: USA</p> |
| Interventions | <p>HT arm: 2 patches, delivering daily dose of oestradiol: 0.025 mg, 0.05 mg or 0.075 mg</p> <p>Control arm: 2 placebo patches</p> <p>Duration: 2 years (26 cycles)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Outcomes      | <p>Breast cancer (regular mammograms)</p> <p>Fractures</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Notes         | <p>Power calculation: not mentioned</p> <p>Clinical trial register: not available</p> <p>Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                         |
|---------------------------------------------------------------------------|--------------------|---------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Unclear risk       | Not described                                                 |
| Allocation concealment (selection bias)                                   | Unclear risk       | Not described                                                 |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | 34 (9.6%) losses to follow-up. Analysed by intention to treat |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | States double-blinded, double-dummy                           |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | States double-blinded, double-dummy; "hard" outcomes          |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                |

## Notelovitz 2002 (Continued)

|            |          |                                  |
|------------|----------|----------------------------------|
| Other bias | Low risk | No apparent source of other bias |
|------------|----------|----------------------------------|

## Obel 1993

### Study characteristics

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to compare combined and sequential therapy with respect to relief of climacteric symptoms, effects on the endometrium and on vaginal cellular maturation, steroid metabolism and side effects</p> <p>Stratification: not mentioned</p> <p>Unblinding: not described</p> <p>No. of women screened for eligibility: 176, of whom 21 unwilling to take placebo, 2 found not post-menopausal, 2 excluded for private reasons</p> <p>No. randomised: 151 (combined HT: 50, sequential HT: 50, placebo: 51)</p> <p>No. analysed: 129 (in the groups to which they were allocated)</p> <p>Losses to follow-up: 22 (11 from combined group, 5 from sequential group, 6 from placebo group)</p> <p>Adherence to treatment: not described</p> <p>Analysis by intention to treat: no</p> <p>No. of centres: 1</p> <p>Years of recruitment: not stated</p> <p>Design: parallel</p> <p>Funding: Pharmaceutical Division, Novo Nordisk</p> |
| Participants  | <p><b>Included</b></p> <p>Women in early menopause (last spontaneous vaginal bleeding &gt; 6 and &lt; 24 months earlier), no HT within preceding 24 months</p> <p><b>Excluded</b></p> <p>Women with previous or current oestrogen-dependant neoplasia, thromboembolic disease, liver or pancreatic disease, diabetes mellitus, severe obesity, disease with high or low bone turnover and medication known to influence bone metabolism or provoke induction of liver enzymes</p> <p>Mean age: not stated</p> <p>Age range: not stated</p> <p>Means of recruitment: 11 5800 women born between 1930 and 1933 in Frederiksborg County, Denmark, were invited to participate.</p> <p>Baseline equality of treatment groups: yes</p> <p>Country: Denmark</p>                                                                                                                                                                                       |
| Interventions | <p><b>HT arm</b></p> <p>Oral oestradiol 2 mg + norethisterone 1 mg</p> <p>Oral oestradiol 2 mg days 1-22 + norethisterone acetate days 13-22, then oestradiol 1 mg days 22-28</p> <p><b>Control arm:</b> placebo</p> <p>Duration: 2 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Outcomes      | Only outcomes of interest to this review: endometrial cancer, quality of life                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Notes         | <p>Power calculation: not mentioned</p> <p>Clinical trial register: not available</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Risk of bias

| Bias | Authors' judgement | Support for judgement |
|------|--------------------|-----------------------|
|------|--------------------|-----------------------|

**Obel 1993** (Continued)

|                                                                           |              |                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Unclear risk | Not described                                                                                                                                                                                                                                                  |
| Allocation concealment (selection bias)                                   | Unclear risk | Not described                                                                                                                                                                                                                                                  |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk | 22 (15%) losses to follow-up for endometrial cancer (11 from combined group, 5 from sequential group, 6 from placebo group), analysed by ITT. However, only 70% of women included for quality of life outcomes (loss to follow-up rates similar across groups) |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | Participants and personnel blinded - identical placebo                                                                                                                                                                                                         |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Unclear risk | States double-blinded - no further details                                                                                                                                                                                                                     |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported                                                                                                                                                                                                                                 |
| Other bias                                                                | Unclear risk | Baseline quality of life scores on several measures appear substantially lower for placebo group.                                                                                                                                                              |

**PEPI 1995**
**Study characteristics**

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods | <p>Stated purpose: to investigate effects of oestrogen-only and combined therapies on cardiovascular disease risk factors, as well as on endometrial status, breast changes, bone density, menopausal symptoms and quality of life factors</p> <p>Stratification: by clinical centre and hysterectomy status</p> <p>Blinding: participants, clinical and laboratory personnel blinded; medication packages visually indistinguishable</p> <p>Unblinding: unblinding officer at each trial centre or by phone call to co-ordinating centre; referral gynaecologist at each centre not directly involved with data collection or patient care; able to access treatment assignment for management of safety issues</p> <p>No. of women screened for eligibility: approximately 1460 (states that 60% of women screened were randomised)</p> <p>No. randomised: 875</p> <p>No. analysed: 847 (97%)</p> <p>Losses to follow-up: 28 (CEE-only group: 5/170, CEE + MPA sequential group: 5/174, CEE + MPA continuous group: 4/174, CEE + MPA sequential group: 5/178, placebo group: 9/174)</p> <p>Adherence to treatment: dropout rate disproportionately high in women with a uterus assigned unopposed oestrogen: 55% had to discontinue assigned therapy, largely owing to endometrial hyperplasia. Of 847 women who attended 3-year follow-up, 75% with a uterus and 80% without a uterus had at least 80% adherence to treatment. (Note: 55% of women with a uterus assigned unopposed oestrogen were required to discontinue assigned therapy owing to endometrial hyperplasia.)</p> <p>Analysis by intention to treat: no - but 97% of women analysed by ITT</p> <p>No. of centres: 7</p> <p>Years of recruitment: December 1989-February 1990</p> <p>Design: parallel</p> <p>Funding: research grants from National Heart, Lung and Blood Institute, National Institute of Child Health and Human Development, National Institute of Health and Human Development, National In-</p> |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**PEPI 1995** (Continued)

stitute of Arthritis and Musculoskeletal and Skin Diseases, National Institute of Diabetes and Digestive and Kidney Diseases, National Institute on Aging, USA

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants  | <p><b>Included</b><br/>Healthy postmenopausal women 45–65 years of age, with or without a uterus; ceased menstruation 12 months before entry or had hysterectomy at least 2 months before entry and FSH levels &lt; 40 mU/mL</p> <p><b>Excluded</b><br/>Women who had used hormones within past 3 months, women treated with thyroid hormone unless stabilised on treatment, serious illness including heart or thromboembolic disease, previous endometrial or breast cancer, contraindications to oestrogen</p> <p>Mean age: 56 years (SD 4)<br/>Age range: 45–64<br/>Means of recruitment: through mass media and community efforts<br/>Baseline equality of treatment groups: women assigned to placebo had higher mean levels of fibrinogen and low density lipoprotein-C at baseline.<br/>Country: USA</p> |
| Interventions | <p><b>HT arm: 1 of the following regimens</b><br/>CEE 0.625 mg daily (unopposed oestrogen)<br/>CEE 0.625 mg daily plus MPA 10 mg daily for first 10 days (combined sequential treatment)<br/>CEE 0.625 mg plus MPA 2.5 mg daily (combined continuous treatment)<br/>CEE 0.625 mg plus MP 200 mg daily for first 12 days (combined sequential treatment)<br/>Control arm: placebo<br/>Duration: 3 years</p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| Outcomes      | <p>Primary endpoints: biological markers, not relevant to this review; however, the following prespecified outcomes were also measured.</p> <p>Breast cancer<br/>Endometrial cancer<br/>Cardiovascular disease<br/>Thromboembolism<br/>Gallbladder disease</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Notes         | <p>Power calculation: based on primary (biological) outcome: a sample of 840 women was projected to provide minimum power of 0.92 to detect differences of 5 mg/dL in HDL cholesterol for any pair-wise comparison of treatment arms at 3 years.</p> <p>55% of women with a uterus assigned unopposed oestrogen were required to discontinue assigned therapy owing to endometrial hyperplasia.</p> <p>Clinical trial register: not available</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded.</p>                                                                                                                                                                                                                       |

**Risk of bias**

| Bias                                        | Authors' judgement | Support for judgement                                                                                                                                                                  |
|---------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias) | Low risk           | Computer-generated blocks of variable length                                                                                                                                           |
| Allocation concealment (selection bias)     | Low risk           | Allocation assignments on encrypted file loaded on computer at clinical centre and issued once eligibility confirmed (or by phone to co-ordinating centre in case of computer failure) |
| Incomplete outcome data (attrition bias)    | Low risk           | 28 (3%) lost to follow-up; 97% of women analysed by intention to treat                                                                                                                 |

**PEPI 1995** (Continued)

## All outcomes

|                                                                           |              |                                                                                                                                    |
|---------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | Participants, clinical and laboratory personnel blinded; medication packages visually indistinguishable                            |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk     | Clinical and laboratory personnel blinded                                                                                          |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported                                                                                                     |
| Other bias                                                                | Unclear risk | Placebo group had higher levels of fibrinogen and low-density lipoprotein C at baseline; otherwise, groups prognostically balanced |

**Ran 2017**
**Study characteristics**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | Double-blind, randomised, parallel-group, placebo-controlled study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Participants  | <p><b>Inclusion</b><br/>Women 40 to 55 years, with obvious menopause symptoms or irregular cycles.<br/>All the participants were in good physical and mental health and were informed of the possible risk of breast neoplasia.</p> <p><b>Exclusion</b><br/>History or presence of any type of malignancy, cardiovascular disease, severe chronic disease, or endocrine disease.<br/>Diagnosed with secondary osteoporosis.<br/>Previous use of oestrogen or calcitonin, or had used bisphosphonate within the past 6 months.<br/>Evidence of alcohol or drug abuse</p> <p>Groups were divided into early and late menopause. The inclusion criteria for participants in early menopause were as follows: women aged between 45 and 60 years with spontaneous amenorrhoea for 6 months to 5 years.</p> |
| Interventions | <p>Intervention: oestradiol valerate 1 mg/day + MPA 5 mg daily for 12 days per month</p> <p>Control: blank starch tablets (same method of administration as used for the HT group)</p> <p>All participants received twice-daily calcium and vitamin D supplements of 500 mg and 200 IU/day, respectively, and participated in a standard exercise program (fast walking for 10 minutes two times/day).</p>                                                                                                                                                                                                                                                                                                                                                                                             |
| Outcomes      | <p>Breast cancer</p> <p>Cardiovascular event (not included: atrial fibrillation and angina)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Notes         | Clinical trial register: not available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Risk of bias**

| Bias | Authors' judgement | Support for judgement |
|------|--------------------|-----------------------|
|------|--------------------|-----------------------|

## Ran 2017 (Continued)

|                                                                           |              |                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk     | "The two groups of Chinese women were assigned to the MHT group and the placebo group according to their recruitment order using a random number table". "The menopausal transition and early menopause groups had specific inclusion criteria with limited overlap" |
| Allocation concealment (selection bias)                                   | Unclear risk | Not stated                                                                                                                                                                                                                                                           |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk | 30% dropout for unclear reasons                                                                                                                                                                                                                                      |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | "Blank starch tablets (which copied the appearance of E2V and MPA tablets) were used as the placebo"                                                                                                                                                                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Unclear risk | "Double-blind" in description but did not specify which was the second blind-ed group, and the methods used                                                                                                                                                          |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported<br>Study protocol is available<br>"An intention-to-treat analysis was conducted"                                                                                                                                                      |
| Other bias                                                                | Low risk     | Groups were similar at baseline, The study appears to be free of other sources of bias.<br><br>The work was supported by the 'Chinese Medical Board'<br><br>Protocol NCT00860964                                                                                     |

## REPLENISH 2018

### Study characteristics

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | Prospective, randomised, double-blind, placebo-controlled, parallel-group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Participants | <p><b>Inclusion criteria</b></p> <p>Women aged 40 to 65 years old with intact uterus<br/> Postmenopausal (serum estradiol, <math>\leq 50</math> pg/mL)<br/> Generally healthy per prespecified criteria<br/> BMI <math>\leq 34</math> kg/m<sup>2</sup><br/> Seeking treatment for menopause-related vasomotor symptoms<br/> Willing to abstain from non-study hormone products<br/> Use of no more than 2 antihypertensive drugs<br/> Negative screening mammogram; normal breast exam and endometrial biopsy</p> <p><b>Exclusion criteria</b></p> <p>Contraindications to hormone use<br/> Heavy smoker (<math>\geq 15</math> cigarettes/day)<br/> History of endometrial hyperplasia or of undiagnosed vaginal bleeding<br/> History of melanoma or of breast, uterine, or ovarian cancer<br/> History of clinically significant, relevant physical or mental illness, including but not limited to thromboembolic disorder or other vascular disease, clotting or malabsorption disorder, oestrogen-dependent neoplasia, or chronic kidney or liver disease<br/> Recent use of a CYP3A4 inhibitor, certain hormones, or an intrauterine device</p> |

**REPLENISH 2018** (Continued)

|               |                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <p>For the outcomes relevant to this review, five groups:</p> <p>Oestradiol (1 mg OR 0.5 mg) AND progesterone (Pg) 100 mg daily for 53 weeks</p> <p>OR oestradiol (0.5 mg OR 0.25 mg) AND progesterone 50 mg</p> <p>Placebo</p> |
| Outcomes      | <p>Has over 200 symptom-related outcomes not relevant to this review.</p> <p>Endometrial hyperplasia or malignancy assessed at 12 months of treatment.</p> <p>Adverse events: breast cancer and DVT</p>                         |
| Notes         | <p>Setting USA (80 sites)</p> <p>Protocol 2013, NCT01942668</p> <p>Funding: the REPLENISH study and preparation of the article was funded by TherapeuticsMD, Inc</p>                                                            |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                                                                    |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | "Randomization was performed by the clinical research organization using a reproducible, computer-generated, block schedule with a block size of five for the vasomotor symptoms substudy and four for the non-substudy" |
| Allocation concealment (selection bias)                                   | Unclear risk       | Not stated in the main publication or the protocol.                                                                                                                                                                      |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk       | Reported only 1275 women at 12 months' follow-up, out of 1845 randomised women. Reasons for withdrawal were not stated                                                                                                   |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | All investigators, involved staff, and participants were blinded using a double-dummy technique because different doses were different-sized capsules.                                                                   |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | All investigators, involved staff, and participants were blinded using a double-dummy technique because different doses were different-sized capsules.                                                                   |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                                                                                                                                           |
| Other bias                                                                | Low risk           | <p>Baseline characteristics similar. No other bias detected</p> <p>Protocol 2013, NCT01942668</p> <p>Funding: the REPLENISH Study and preparation of the article was funded by TherapeuticsMD, Inc</p>                   |

**Tierney 2009**
**Study characteristics**
**Long-term hormone therapy for perimenopausal and postmenopausal women (Review)**

**Tierney 2009** (Continued)

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to determine whether oestradiol and norethindrone HT prevents decline in delayed verbal recall in older women with normal to mildly impaired memory functioning</p> <p>Stratification: none reported</p> <p>Unblinding: adverse effects managed by data safety monitoring board, which did not have access to study participants</p> <p>No. of women screened for eligibility: 987</p> <p>No. randomised: 142</p> <p>No. analysed: 128 at 2 years</p> <p>Losses to follow-up: 14 (8 in HT group, 6 in placebo group)</p> <p>Adherence to treatment: 26 discontinued intervention in HT group, 16 discontinued in placebo group</p> <p>Analysis by intention to treat: no, but 128/142 analysed by ITT (90%)</p> <p>No. of centres: 1</p> <p>Years of recruitment: 2000 to 2004</p> <p>Design: parallel</p> <p>Funding: Canadian Institutes of Health Research, Insitute of Neurosciences Mental Health and Addic-tion, Shire Biochemistry. Pharmaceutical companies provided tablets.</p> |
| Participants  | <p><b>Included</b></p> <p>Women at least 60 years of age, at least 12 months since last menstrual cycle, normal to below-normal scores on screening for short-delay verbal recall, fluent in English and with normal reading and hearing abilities</p> <p><b>Excluded</b></p> <p>Women with dementia or history of a condition that would affect cognition; women with conditions considered to be exacerbated by oestrogen, or taking specific medications (listed in the publication) in-cluding HT within past 2 years; women who received neuropsychological testing to rule out dementia</p> <p>Mean age: not stated</p> <p>Age range: 61 to 87</p> <p>Means of recruitment: advertisements, display booths (e.g. in hospitals, seniors' clubs), family physi-cian referrals</p> <p>Baseline equality of treatment groups: similar baseline scores</p> <p>Country: Canada</p>                                                                                                                           |
| Interventions | <p>HT group: 1 mg 17-B oestradiol daily for 4 days a week followed by 1 combined oestrogen/progestin ampoule (1 mg 17-B oestradiol and 0.35 mg norethindrone) per day for 3 days a week</p> <p>Control: placebo</p> <p>Duration: 2 years</p> <p>All women were given the same intervention, whether or not they had a uterus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Outcomes      | <p>Short-delay verbal recall of the California Verbal Learning Test</p> <p>Adverse events, including cardiovascular events, cancer, fractures</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Notes         | Clinical trial register: not available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**Risk of bias**

| Bias                                         | Authors' judgement | Support for judgement |
|----------------------------------------------|--------------------|-----------------------|
| Random sequence genera-tion (selection bias) | Low risk           | Computer-generated    |
| Allocation concealment (selection bias)      | Low risk           | Pharmacy allocation   |

### Tierney 2009 (Continued)

|                                                                           |              |                                                                                                                                                                                |
|---------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Unclear risk | 10% losses to follow-up by 2 years; those lost to follow-up did not differ on baseline short-delay recall scores from those who stayed in the study                            |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk     | All study personnel and participants were blinded to treatment assignment for the duration of the study; placebo capsules were identical in appearance to the active capsule.  |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Unclear risk | States that study personnel and participants were blinded – no specific statement about outcome assessment                                                                     |
| Selective reporting (reporting bias)                                      | Low risk     | All expected outcomes reported                                                                                                                                                 |
| Other bias                                                                | Unclear risk | More women in HT group than in placebo group reported breast tenderness, vaginal bleeding and discharge, suggesting that they may have been aware that they were receiving HT. |

### WAVE 2002

#### Study characteristics

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods      | <p>Stated purpose: to determine whether HT or antioxidant vitamin supplements, alone or in combination, influence the progress of coronary artery disease in postmenopausal women, as measured by angiography</p> <p>Stratification: clinical centre, hysterectomy status</p> <p>Unblinding: adverse effects managed by gynaecologist not involved in outcome assessment who had access to treatment assignment if necessary, with permission of co-ordinating centre</p> <p>No. randomised: 211</p> <p>No. analysed: 206 for clinical status at end of study</p> <p>Losses to follow-up: 5 (3 in HT group, 2 in placebo group)</p> <p>Adherence to treatment: evaluated for 159/211 who had angiographic follow-up: HT group took 67% of medication, placebo group took 70%; 9/108 women in placebo group crossed to open-label oestrogen</p> <p>Analysis by intention to treat: no – but 98% of women analysed by ITT</p> <p>No. of centres: 7</p> <p>Years of recruitment: July 1997 to August 1999</p> <p>Design: parallel</p> <p>Funding: National Heart, Lung and Blood Institute contract, General Clinical Research Center grant, USA</p> |
| Participants | <p><b>Included</b></p> <p>Postmenopausal women with 1 or more 15% to 75% coronary stenoses in an artery not subjected to intervention, seen on angiogram within 4 months of study entry. Postmenopausal defined as post bilateral oophorectomy, younger than 55 years of age with an FSH of 40 Mu/mL or higher, or older than 55 years</p> <p><b>Excluded</b></p> <p>HT use within 3 months, concurrent use of more than 60 mg/d of vitamin C or 30 IU daily of vitamin E and unwilling to stop taking them; suspected breast, uterine or cervical cancer; uncontrolled diabetes or hypertension, MI within 4 weeks, elevated triglycerides or creatinine levels, symptomatic gallstones, heart failure, history of haemorrhagic stroke, bleeding diathesis, PE, DVT or untreated osteoporosis</p> <p>Mean age: 65</p> <p>Age range: 56 to 74</p> <p>Means of recruitment: recruited at clinical sites in USA and Canada</p>                                                                                                                                                                                                                      |

**WAVE 2002** (Continued)

Baseline equality of treatment groups: higher prevalence of diabetes and higher fasting blood glucose levels in the HT group  
Countries: USA and Canada

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions | <p><b>HT arm: 1 of the following regimens</b><br/>CEE 0.625 (oestrogen-only therapy) - for women who had had a hysterectomy<br/>CEE 0.625 and MPA 2.5 mg daily (continuous combined therapy) - for women who had not had a hysterectomy<br/>Control arm: placebo</p> <p>Duration: 3 years</p> <p>In addition, this study included women who were prescribed a regimen of vitamins E and C or placebo vitamins. The only comparison considered in this review was HT/placebo vitamins vs placebo HT/placebo vitamins.</p>                                 |
| Outcomes      | <p>Primary outcome was biological: change in minimum lumen diameter of qualifying coronary lesions<br/><b>Outcomes of interest to review</b><br/>All-cause death<br/>Total mortality<br/>Cardiovascular events<br/>Venous thromboembolism<br/>Stroke<br/>Breast cancer<br/>Quality of life</p>                                                                                                                                                                                                                                                           |
| Notes         | <p>Study publication pools results for women receiving unopposed and combined therapies.<br/>Power calculation: based on primary (biological) outcome: 423 women provide 90% power to detect an effect size of at least 0.33 (corresponding to a change in minimum lumen diameter of 0.1 mm and assuming 20% of women would not undergo a follow-up angiogram)</p> <p>Clinical trial register not available</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded.</p> |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                              |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computer-randomised, permuted block design with random blocks of 2 and 4                                           |
| Allocation concealment (selection bias)                                   | Low risk           | Remotely by phone call to study co-ordinating centre                                                               |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | Losses to follow-up: 5 (3 in HT group, 2 in placebo group); 98% of women analysed by intention to treat            |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants, investigators and staff at clinical centres blinded, except (when necessary) the study gynaecologist |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Participants, investigators and staff at clinical centres blinded - main outcomes "hard"                           |

**WAVE 2002** (Continued)

|                                         |              |                                                                                                                                         |
|-----------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Selective reporting (re-reporting bias) | Low risk     | All expected outcomes reported                                                                                                          |
| Other bias                              | Unclear risk | Active group had higher prevalence of diabetes and higher fasting blood glucose levels; otherwise, groups were prognostically balanced. |

**WEST 2001**
**Study characteristics**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to determine whether 17B-oestradiol reduces risk of recurrent stroke or death among postmenopausal women who have experienced a transient ischaemic attack or a non-disabling ischaemic stroke</p> <p>Stratification: by trial centre and risk level (3 levels)</p> <p>Unblinding: study internist unblinded in the case of overriding concern about a woman's clinical care</p> <p>No. of women screened for eligibility: 5296 (2772 ineligible, 1843 declined to participate, 17 unable to be randomised within protocol time frame)</p> <p>No. randomised: 664 (HT: 337, placebo: 327)</p> <p>No. analysed: 664</p> <p>Losses to follow-up: nil</p> <p>Adherence to treatment: 34% of the oestradiol group and 24% of the placebo group dropped out. Non-adherence to allocated treatment: overall mean: HT group: 44%; placebo: 36%. Among women who continued with treatment, adherence to treatment was 90% in both groups.</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 21 (single recruitment hub)</p> <p>Years of recruitment: December 1993 to May 1998</p> <p>Design: parallel</p> <p>Funding: National Institute of Neurological Disorders and Stroke grant, Medical Research Council of Canada grant. Mead Johnson Laboratories provided support and study drug.</p> |
| Participants  | <p><b>Included</b></p> <p>Postmenopausal women (i.e. amenorrhoea for at least 12 months, or having undergone hysterectomy and &gt; 55 years of age) over 44 years of age within 90 days of a qualifying ischaemic stroke or transient ischaemic attack</p> <p><b>Excluded</b></p> <p>Women whose index event was disabling or occurred while taking oestrogen; women with history of breast or endometrial cancer, who had had a venous thromboembolic event while receiving oestrogen replacement therapy, had had a neurological or psychiatric disease that could complicate evaluation of endpoints or had a coexisting condition that limited life expectancy</p> <p>Mean age: 71</p> <p>Age range: 46–91</p> <p>Means of recruitment: admissions to 20 largest regional hospitals in Connecticut and Massachusetts; also via contact with selected neurology groups and direct referral from physicians</p> <p>Baseline equality of treatment groups: yes</p> <p>Country: USA</p>                                                                                                                                                                                                                                                                                                                                   |
| Interventions | <p>HT arm: 17-beta oestradiol 1 mg daily plus, for women with a uterus, a course of medroxyprogesterone acetate once a year, 5 mg daily for 12 days</p> <p>Control arm: placebo</p> <p>Duration: 2.8 years</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Outcomes      | <p>Death or recurrent stroke</p> <p>Myocardial infarction</p> <p>Cognitive function</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Notes         | Study publication pools results for women receiving unopposed and combined therapies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## WEST 2001 (Continued)

Power calculation: 652 women required to give 80% power to detect a reduction in the rate of death or non-fatal stroke from 25% in the placebo group to 15% in the HT group (2-tailed  $P = 0.05$ )

Clinical trial register not available

Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.

### Risk of bias

| Bias                                                                      | Authors' judgement | Support for judgement                                  |
|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Computer-generated at pharmacy, in blocks of 4         |
| Allocation concealment (selection bias)                                   | Low risk           | By remote contact with trial pharmacy                  |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | No losses to follow-up; analysed by intention to treat |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | Participants and investigators blinded                 |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | Endpoint assessors blinded                             |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                         |
| Other bias                                                                | Low risk           | No apparent source of other bias                       |

## WHI 1998

### Study characteristics

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods | <p>Stated purpose: to test the hypothesis that women taking HT will have lower rates of coronary heart disease and osteoporosis-related fractures</p> <p>COMBINED HT ARM:<br/>Stratification: by clinical centre site and age group<br/>Re blinding: all participants, clinic staff and outcome assessors blinded, with the exception of 331 participants who were unblinded from the unopposed oestrogen arm and were reassigned to the combined HT arm owing to change in the protocol (see Notes). A further 432 women (248 in the experimental arm and 183 in the placebo arm) had a hysterectomy after randomisation (for reasons other than cancer) and were switched to unopposed oestrogen or corresponding placebo in the unopposed oestrogen study arm.<br/>Unblinding: when required for safety or symptom management, unblinding officer, unblinded clinical gynaecologist, who was not involved with outcomes assessment. At average 5.2-year follow-up, 3444 women in experimental group and 548 women in placebo group had been unblinded, mainly to manage persistent vaginal bleeding.<br/>No. randomised: 16,608 (8506 to experimental group, 8102 to placebo group)<br/>No. analysed: 16,608</p> |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**WHI 1998** (Continued)

Losses to follow-up: 583 participants (3.5%) - i.e. no outcome data for > 18 months: 307 in HT arm (3%), 276 in control group (3.5%). Vital status known for 96.5%

Dropouts/Non-adherence to allocated treatment: women with adherence to treatment less than 80% (by pill count) were counted as dropouts. Dropout rates at 5.6 years were 42% in the experimental arm and 38% in the placebo group. In addition, 10.7% of women in the placebo group crossed to receive active treatment.

Analysis by intention to treat: yes (analysed with and without unblinded group in experimental arm)

No. of centres: 40

Power calculation: Sample gives 80%-95% power for primary endpoint comparisons at 5% significance, assuming an intervention effect of 20% for CHD and 21% for combined fractures at 6-9 year follow-up, and an intervention effect of 22% for breast cancer at 14-year follow-up (risk ratio of 1.3 assumed for increased risk of breast cancer in intervention group).

Years of recruitment: 1993 to 1998

Note: planned 8.5 years' follow-up. Trial was stopped after mean of 5.6 years, as test statistic for breast cancer exceeded predetermined stopping boundary, and global risk index indicated risks exceeding benefits.

This study continued follow-up for breast cancer outcomes beyond the planned trial completion date for women who consented to continue follow-up (n = 12,788: 83% of those eligible, of whom 2.7% dropped out (Manson 2013)). Seventeen percent of surviving women declined to be re-consented, and their data were censored for the additional follow-up period. Baseline characteristics were evenly distributed between the 2 groups.

**WHI 1998 UNOPPOSED OESTROGEN ARM:**

Stratification: as above

Blinding: as above

Unblinding: as above

No. randomised: 10,739 (including 248 in experimental arm, 183 in placebo arm) joined this study after randomisation to corresponding arms in WHI 2002 and having subsequently had hysterectomy (for reasons other than cancer).

No. analysed: 10,739

Losses to follow-up: 563

Dropouts/Non-adherence to allocated treatment: women with adherence to treatment of under 80% by pill count were counted as dropouts. The dropout rate was 53.8% by the end of the study (6.8 years) and did not vary significantly between study arms. In addition, 9.1% of women in the placebo arm and 5.7% in the active treatment group initiated hormone use outside of the study through their own physician.

Analysis by intention to treat: yes

No. of centres: 40

Power calculation: 12,375 participants needed to detect a 21% reduction in CHD rates over projected 9-year average follow-up

Years of recruitment: 1993-1998

N.B. This arm of WHI 1998 was stopped early after a mean follow-up of 7.1 years (planned for 9), when it was determined that the prospect of obtaining more precise evidence about effects of the intervention was unlikely to outweigh potential harms, although no predefined safety boundaries had been crossed. This study continued follow-up for all outcomes beyond the planned trial completion date for women who consented to continue follow-up (n = 7645; 78% of those eligible, of whom 3% dropped out). Twenty-two percent of surviving women declined to be re-consented, and their data were censored for the additional follow-up period. Baseline characteristics were similar in the 2 groups, and imputation analyses suggested that this loss to follow-up did not significantly influence study findings.

**WHIMS ANCILLARY STUDY:**

WHI 1998 enrolled 7479 participants who were free of probable dementia and were 56-79 years of age. Of these, 4532 were from the combined HT arm of WHI 1998 [WHI 1998 (WHIMS:combined arm) and 2947 were from the unopposed oestrogen arm [WHI 1998 (WHIMS:unopposed oestrogen arm)]. Overall, 92.4% of eligible women participated.

Years of recruitment: May 1996 to December 1999

Analysis by intention to treat: all analysed by ITT for planned primary and secondary outcomes. For a third outcome (global cognitive function), which was not formally preplanned, 178 participants (3.9%) were excluded from the combined arm (151 because relevant follow-up data were missing, and 27 because they consented to join WHIMS more than 6 months after WHI treatment assignment, by which

**WHI 1998** (Continued)

time treatment effects may already have been under way), and 139 (4.7%) were excluded from the unopposed oestrogen arm (109 owing to missing follow-up data and 30 as the result of enrolment 6 months or more after randomisation).

Adherence to allocated treatment (i.e. proportion taking > 80% of study medication): unopposed oestrogen arm: year 1: 77.2% in HT group vs 84.1% in placebo group; year 6: 42% in HT group vs 47.8% in placebo group; combined HT arm: year 1: 71% in HT group vs 83% in the placebo group; year 4: 49% in HT group vs 61% in placebo group.

Power: designed to provide > 80% power to detect an observed 40% relative reduction in the incidence rate of clinically diagnosed all-cause dementia

Duration: Mean time from randomisation to [WHI 1998](#) to the last WHIMS cognitive screening examination was 4.05 years for women in the combined HT arm and 5.21 years for women in the unopposed oestrogen arm.

**WHISCA ANCILLARY STUDY**

Randomised in 1999

Enrolled 2304 women who had been enrolled in WHIMS for a mean of 3 years (1106 women from WHIMS: combined arm; 886 from WHIMS: unopposed oestrogen arm)

Re-randomised in 2004-2005 for further follow-up to September 2007; 84% of the original cohort agreed to continue (n = 1933). Those who participated were more likely than those who did not to be younger, non-smokers, free of diabetes and cardiovascular disease and prior users of oral contraceptives, and to have higher MMSE scores. Among ongoing participants, active and placebo groups had similar characteristics.

**Participants**
**COMBINED HT ARM**
**Included**

Postmenopausal women (no vaginal bleeding for 6 months, or for 12 months for 50-54 year olds; any use of postmenopausal hormones), with a uterus, aged 50-79 at initial screening, likely to reside in area for 3 years, provision of written informed consent

**Excluded**

Medical condition predictive of survival time < 3 years, invasive cancer in past 10 years (except non-melanoma skin cancer), breast cancer at any time or suspicion of breast cancer at baseline screening, acute myocardial infarction, stroke, transient ischaemic attack in previous 6 months, known chronic active hepatitis or severe cirrhosis, blood counts indicative of disease, severe hypertension or current use of oral corticosteroids, femoral neck bone mineral density more than 3 standard deviations below the corresponding age-specific mean, endometrial cancer or endometrial hyperplasia at baseline, malignant melanoma, pulmonary embolism or deep vein thrombosis that was non-traumatic or had occurred in the previous 6 months, bleeding disorder, lipaemic serum and hypertriglyceridaemia diagnosis, current use of anticoagulants or tamoxifen, PAP smear or pelvic abnormalities, unwillingness or inability to complete baseline study requirements, alcoholism, drug dependency, mental illness, dementia, severe menopausal symptoms inconsistent with assignment to placebo, inability or unwillingness to discontinue current HT use or oral testosterone use, inadequate adherence with placebo run-in, unwillingness to have baseline or follow-up endometrial aspirations, active participant in another randomised clinical trial

Mean age: 63 years (SD 7)

Age range: 50-79. Age ratio of 33%:45%:21% for baseline age categories of 50-59, 60-69 and 70-79, respectively (enrolment targeted to achieve ratio of 30:45:25)

Recruitment: letter of invitation in conjunction with media awareness programme. Sampling method gave women from minority groups 6-fold higher odds of selection than Caucasian women and resulted in a sample with 84% racially/ethnically designated "white", and 16% non-"white"

Screening: interested women were screened by phone or mail for eligibility, then attended 3 screening visits for history, clinical exam and tests. Three-month washout period before baseline evaluation of women using postmenopausal hormones at baseline screening. Lead-in placebo pills given for at least 4 weeks during screening process to establish compliance with pill taking

Baseline equality of treatment groups: no substantive differences between study groups at baseline

Country: USA

**WHI 1998** (Continued)

**UNOPPOSED OESTROGEN ARM**
**Included**

Postmenopausal women who had undergone hysterectomy (therefore considered postmenopausal for enrolment purposes), aged 50-79 at initial screening, likely to reside in area for 3 years, provision of written informed consent

**Excluded**

As above

Mean age: 64

Age range: 50 to 79.

Age ratio of 33%:45%:21% for baseline age categories of 50-59, 60-69 and 70-79, respectively (enrolment targeted to achieve ratio of 30:45:25)

Recruitment: as above

Screening: as above

Baseline equality of treatment groups: no substantive differences between study groups at baseline

Country: USA

WHIMS ancillary study

**Included**

Participants in either arm of [WHI 1998](#), at least 65 years of age and free of probable dementia

**WHISCA ANCILLARY STUDY**

Women from 14 of the WHIMS clinical sites

**Interventions**
**COMBINED HT ARM**

Experimental group: combined oestrogen and progesterone as 1 daily tablet containing conjugated equine oestrogen 0.625 mg and medroxyprogesterone acetate 2.5 mg

Control group: matching placebo

Duration: 5.6 years (mean duration of treatment)

Permanent discontinuation of medication: women who developed breast cancer, endometrial hyperplasia not responsive to treatment, endometrial atypia, endometrial cancer, deep vein thrombosis, pulmonary embolus, malignant melanoma, meningioma, triglyceride level over 1000 mg/dL, prescription of oestrogen, testosterone or selective oestrogen-receptor modulators by their personal physician

Temporary discontinuation of medication: women who had acute MI, stroke, fracture, major injury involving hospitalisation, surgery involving anaesthesia, illness resulting in immobilisation for longer than 1 week, other severe illness for which hormone use is temporarily inappropriate

N.B. [WHI 1998](#) (WHIMS) investigators reported outcomes according to study arm (unopposed oestrogen or combined HT therapy) and also (as per protocol) reported results pooled across the 2 arms. However, results showed significant baseline prognostic differences between the 2 arms (see Quality Table). We have not pooled the results in this review.

**UNOPPOSED OESTROGEN ARM**

Experimental group: 0.635 mg CEE daily

Control arm: placebo

Permanent discontinuation of medication: as above

**WHIMS ANCILLARY STUDY**

As for either arm of [WHI 1998](#) above

**Outcomes**
**COMBINED HT ARM**

Cardiovascular disease: acute MI, silent MI, coronary death, stroke, pulmonary embolus

Cancer: breast, colorectal, endometrial, other cancers

Fractures: hip, vertebral, osteoporotic

**UNOPPOSED OESTROGEN ARM**

As above (with the exception of endometrial cancer)

**WHIMS ANCILLARY STUDY**

**WHI 1998** (Continued)

Cognitive function  
Mild cognitive impairment  
Dementia  
For assessment of outcomes, women in WHIMS underwent up to 4 phases of testing as follows.  
1. Participants underwent cognitive screening with the Modified MMSE at baseline and annually.  
2. Women who scored below an education-adjusted cut-off point proceeded to a battery of psychoneurological tests and standardised interviews, plus interviews with a designated informant (friend or relative).  
3. Clinical assessments by local physicians.  
4. CT and blood tests to rule out reversible pathology.  
All cases judged locally as probable dementia were independently evaluated by 2 adjudicators blinded to the diagnosis, as were 50% of cases of mild cognitive impairment and 10% of all cases without dementia.  
Mild cognitive impairment defined as per current DSM IV criteria - operationally defined as follows: poor performance (< 10th percentile) on a battery of neuropsychological tests, report of mild functional impairment from designated informant, no evidence of a psychiatric or medical explanation for the cognitive decline, absence of dementia  
Dementia defined as per DSM-IV criteria

**Notes**

N.B. The original WHI protocol allowed women with a uterus to be randomised to receive unopposed oestrogen. As evidence emerged (from the PEPI trial) that this could be unsafe, 331 participants with a uterus in the intervention group in the unopposed oestrogen arm were reassigned to the intervention group in the combined HT arm. Both arms closed early: combined arm at 5.6 years (8.5 planned); oestrogen-only arm at 6.8 years (9 planned)

Clinical trial register: NCT00000611

Trial authors contacted for additional information in previous versions of the review; no response received and no details recorded.

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Centrally randomised by permuted block algorithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Allocation concealment (selection bias)                                   | Low risk           | By local access to remote study database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | <p>Combined HT arm: 583 participants (3.5%) lost to follow-up. Vital status known for 96.5%</p> <p>For 11-year breast cancer outcomes in the combined HT arm, 17% of women had withdrawn. Imputation analyses and comparison of baseline characteristics suggested that this did not significantly influence effect estimates.</p> <p>Oestrogen-only arm: 563 (5%) - analysed by intention to treat</p> <p>WHISCA: for ongoing follow-up (2004-2007), 16% of women withdrew. Clinical and demographic characteristics of those continuing differed from those discontinuing, but active and placebo groups did not differ significantly.</p> |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | All participants and clinic staff were blinded, with the exception of 331 participants who were unblinded from the unopposed oestrogen arm and reassigned to the combined HT arm owing to a change in protocol.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Blinding of outcome assessment (detection bias)                           | Low risk           | Outcome assessors blinded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**WHI 1998** (Continued)

## All outcomes

|                                      |              |                                                                                                                                                                                                                                                          |
|--------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selective reporting (reporting bias) | Low risk     | All expected outcomes reported                                                                                                                                                                                                                           |
| Other bias                           | Unclear risk | In the unopposed oestrogen arm, greater use of aspirin at bedtime in the placebo group at baseline.<br>In the combined HT arm, lower prevalence of stroke and higher percentage of participants using statins in the active treatment group at baseline. |

**WISDOM 2007**
**Study characteristics**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to assess long-term benefits and risks of HT</p> <p>Stratification: by hysterectomy status and intended use of HT: women with no uterus and unwilling to take placebo randomised to CEE or combined HT. Equal probability of any treatment within each stratum</p> <p>No. of women screened for eligibility: 14,203</p> <p>No. randomised: 4385; 2196 on combined therapy, 2189 on placebo (see Notes)</p> <p>No. analysed: 4385</p> <p>Losses to follow-up: 5</p> <p>Adherence to treatment: 615 (14%) had dropped out from randomised treatment by trial closure. Trial treatment delivered 73% of time to women in combined HT arm and 86% of time to women on placebo</p> <p>Analysis by intention to treat: yes</p> <p>No. of centres: 384 UK, 91 Australian and 24 New Zealand general practitioner practices</p> <p>Years of recruitment: 1999-2002</p> <p>Design: parallel</p> <p>Funding: non-commercial medical research funding</p> |
| Participants  | <p><b>Included</b></p> <p>Postmenopausal women 50–69 years of age</p> <p><b>Excluded</b></p> <p>History of breast cancer, any other cancer in past 10 years except basal or squamous cell skin cancer, endometriosis or endometrial hyperplasia, venous thromboembolism, gallbladder disease, MI, unstable angina, cardiovascular accident, subarachnoid haemorrhage, transient ischaemic attack, use of HT within past 6 months, unlikely to be able to give informed consent</p> <p>Mean age: 63</p> <p>Age range: 50–69</p> <p>Means of recruitment: general practitioner practice registers</p> <p>Countries: UK, Australia, New Zealand</p>                                                                                                                                                                                                                                                                                                                  |
| Interventions | <p>HT arm: daily CEE 0.625 mg plus medroxyprogesterone acetate 2.5 mg (for women with or without a uterus), or daily CEE 0.625 (for women without a uterus)</p> <p>Control arm: placebo</p> <p>Duration: planned for median 10 years, but prematurely closed after median 11.9 months (range 7.1-19.6)</p> <p>Women with a uterus within 3 years of last period, those aged 50–53 and older women with unacceptable breakthrough bleeding took medroxyprogesterone acetate 5.0 mg.</p> <p>Women with a uterus who experienced unacceptable spotting or bleeding on the above therapy were offered open-label CEE 0.625 mg plus medroxyprogesterone acetate 10.0 mg daily for the last 14 days of a 28-day cycle.</p> <p>All women took placebo medication during run-in: those who achieved 80% compliance were randomised.</p>                                                                                                                                   |

**WISDOM 2007** (Continued)

A further 1307 women were randomised to a comparison of oestrogen-only vs combined HT: these results are not reported here.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outcomes | <p>Major cardiovascular disease (primary)</p> <p>Osteoporotic fractures (primary)</p> <p>Breast cancer</p> <p>Mortality</p> <p>VTE</p> <p>CVD</p> <p>Dementia (no follow-up data collected)</p> <p>Adverse events</p> <p>Quality of life (reported among 3721 women with an intact uterus or subtotal hysterectomy, among whom 1862 were randomised to combined HT and 1859 to placebo). A variety of overall and symptom-specific measures were used, including EuroQoL-5D (which measures health-related quality of life) and a generic VAS scale (which measures quality of all aspects of life) – only these 2 measures are included in this review.</p> |
| Notes    | <p>Powered in protocol to detect 25% reduction in CHD over 10 years - this assumed an 18,000 sample size, but trial stopped early with 26% of target</p> <p>A further 1307 women were included in comparison of combined therapy vs oestrogen only and were not included in this review.</p> <p>Clinical trial register not available</p> <p>Trial authors contacted for additional information in previous versions of the review; response received but no details recorded.</p>                                                                                                                                                                           |

**Risk of bias**

| Bias                                                                      | Authors' judgement | Support for judgement                                                                              |
|---------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------|
| Random sequence generation (selection bias)                               | Low risk           | Remote computer-generated                                                                          |
| Allocation concealment (selection bias)                                   | Low risk           | Remote computer-generated                                                                          |
| Incomplete outcome data (attrition bias)<br>All outcomes                  | Low risk           | 615 (14%) had withdrawn from randomised treatment by trial closure; analysed by intention to treat |
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk           | All participants and clinic staff blinded except when vaginal bleeding triggered a code break      |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk           | All outcome assessors blinded except when vaginal bleeding triggered a code break                  |
| Selective reporting (reporting bias)                                      | Low risk           | All expected outcomes reported                                                                     |
| Other bias                                                                | Low risk           | No apparent source of other bias                                                                   |

Yaffe 2006

### Study characteristics

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | <p>Stated purpose: to investigate effects of unopposed ultra-low-dose transdermal oestradiol on cognition and health-related quality of life in postmenopausal women</p> <p>Stratification: by clinical centre</p> <p>No. of women screened for eligibility: 1509 (of whom 605 had 1-week run-in phase. 10/605 were non-compliant and 1678 were found ineligible or refused to continue screening)</p> <p>No. randomised: 417 (treatment group: 208; placebo group: 209)</p> <p>No. analysed: using a time × treatment interaction. 388 at year 1, 376 at year 2</p> <p>Losses to follow-up: 40</p> <p>Adherence to treatment: dropouts: 41. Among those who completed treatment, 84% used at least 75% of study drug during the entire 2 years.</p> <p>No. of centres: 9</p> <p>Years of recruitment: 1999-2000</p> <p>Design: parallel</p> <p>Funding: industry funded</p>          |
| Participants  | <p><b>Included</b></p> <p>Women 60–80 years of age with intact uterus, at least 5 years post menopause, normal bone density</p> <p><b>Excluded</b></p> <p>Women with unexplained uterine bleeding, endometrial hyperplasia, endometrium &gt; mm double thickness on ultrasonography, abnormal mammogram suggestive of breast cancer, history of metabolic bone disease, cancer, coronary disease, stroke, transient ischaemic attack, VTE, uncontrolled hypertension, uncontrolled thyroid disease, liver disease, abnormal fasting triglyceride or fasting glucose, ever taken fluoride, calcitonin or bisphosphates, oestrogen or progestin within past 3 months</p> <p>Median age: 67</p> <p>Means of recruitment: not stated</p> <p>Baseline equality of treatment groups: mean MMSE scores slightly higher in intervention group (<math>P = 0.04</math>)</p> <p>Country: USA</p> |
| Interventions | <p>HT arm: oestradiol patch delivering approx 0.014 mg oestradiol daily, applied to abdomen weekly</p> <p>Control arm: identical placebo patch</p> <p>Duration: 2 years</p> <p>All participants also received 400 mg calcium twice daily and 400 IU vitamin D daily.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Outcomes      | <p>Preplanned secondary outcomes</p> <p>Changes in global cognition (MMSE)</p> <p>Short-Form Health Survey (SF-36): Physical Component Scale and Mental Component Scale</p> <p>Bone mineral density was primary outcome (not reported here).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Notes         | Clinical trial register not available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### Risk of bias

| Bias                                                     | Authors' judgement | Support for judgement                                                 |
|----------------------------------------------------------|--------------------|-----------------------------------------------------------------------|
| Random sequence generation (selection bias)              | Low risk           | Remotely by computer                                                  |
| Allocation concealment (selection bias)                  | Low risk           | By pharmacy                                                           |
| Incomplete outcome data (attrition bias)<br>All outcomes | Low risk           | 40/417 (9.5%) women lost to follow-up; analysed by intention to treat |

**Yaffe 2006** (Continued)

|                                                                           |          |                                        |
|---------------------------------------------------------------------------|----------|----------------------------------------|
| Blinding of participants and personnel (performance bias)<br>All outcomes | Low risk | Participants and investigators blinded |
| Blinding of outcome assessment (detection bias)<br>All outcomes           | Low risk | Outcome assessors blinded              |
| Selective reporting (reporting bias)                                      | Low risk | All expected outcomes reported         |
| Other bias                                                                | Low risk | No apparent source of other bias       |

**AE:** adverse events; **BMD:** bone mineral density; **BMI:** body mass index; **CEE:** conjugated equine oestrogen; **CHD:** coronary heart disease; **CVD:** cardiovascular disease; **DM:** diabetes mellitus; **DSM:** Diagnostic and Statistical Manual of Mental Disorders; **DVT:** deep vein thrombosis; **ECG:** electrocardiogram; **FSH:** follicle-stimulating hormone; **f/u:** follow-up; **HT:** hormone therapy; **hx:** history of; **ITT:** intention-to-treat; **MI:** myocardial infarction; **MMSE:** Mini Mental State Examination; **MPA:** medroxyprogesterone acetate; **PE:** pulmonary embolism; **RCT:** randomised controlled trial; **TIA:** transient ischaemic attack; **VAS:** visual analogue scale; **VTE:** venous thromboembolism

**Characteristics of excluded studies** [ordered by study ID]

| Study                              | Reason for exclusion                         |
|------------------------------------|----------------------------------------------|
| <a href="#">AHT 2015</a>           | Not blinded                                  |
| <a href="#">Aitken 1971</a>        | No outcomes of interest measured             |
| <a href="#">Aitken 1973</a>        | No outcomes of interest measured             |
| <a href="#">Angerer 2001</a>       | Duration less than 1 year                    |
| <a href="#">Bloch Thomsen 2002</a> | No outcomes of interest measured             |
| <a href="#">Chen 2001</a>          | No placebo, no outcomes of interest measured |
| <a href="#">Christiansen 1981</a>  | No outcomes of interest measured             |
| <a href="#">Corrado 2002</a>       | No placebo, no outcomes of interest measured |
| <a href="#">Corson 1999</a>        | No placebo, no outcomes of interest measured |
| <a href="#">De Roo 1999</a>        | No outcomes of interest measured             |
| <a href="#">Dowling 2017</a>       | No outcomes of interest measured             |
| <a href="#">Eiken 1996</a>         | No outcomes of interest measured             |
| <a href="#">ESTRATAB 1997</a>      | No outcomes of interest measured             |
| <a href="#">EWA 2000</a>           | No placebo, no outcomes of interest measured |
| <a href="#">Genant 1990</a>        | No outcomes of interest measured             |
| <a href="#">Gordon 2018</a>        | No outcomes of interest measured             |

| Study                 | Reason for exclusion                                               |
|-----------------------|--------------------------------------------------------------------|
| Graser 2001           | Duration less than 1 year, no outcomes of interest measured        |
| HABITS 2004           | Not double-blinded                                                 |
| Haines 2003           | No outcomes of interest measured                                   |
| Hall 1998             | Not double-blinded                                                 |
| Jensen 1985           | No outcomes of interest measured                                   |
| Komulainen 1998       | Not blinded                                                        |
| Lufkin 1992           | No outcomes of interest measured                                   |
| Maki 2004             | No outcomes of interest measured                                   |
| Mirkin 2016           | Co-intervention (bazedoxifene) in the HT group                     |
| Mizunuma 2010         | No comparisons of interest (i.e. hormone therapy to placebo only)  |
| NCT00006399           | Duration less than 1 year                                          |
| Ng 1992               | No outcomes of interest measured                                   |
| Nielsen 2006          | No outcomes of interest measured                                   |
| Ory 1998              | No outcomes of interest measured                                   |
| Os 2002               | No placebo, no outcomes of interest measured                       |
| Paoletti 2015         | No outcomes of interest measured                                   |
| Papworth 2002         | No placebo                                                         |
| Pefanco 2007          | No outcomes of interest measured                                   |
| Post 2001             | No outcomes of interest measured                                   |
| Rasgon 2014           | No placebo group, interim outcomes measured                        |
| Saitta A 2001         | Duration less than 1 year                                          |
| Sanchez-Guerrero 2007 | No outcomes of interest measured                                   |
| Schierbeck 2012       | Not blinded                                                        |
| SMART 5               | No outcomes of interest measured                                   |
| Steiner 2007          | Combines EPAT and WELL-HART data. No outcomes of interest measured |
| Stevenson 2016        | No outcomes of interest measured                                   |
| Teede 2002            | No outcomes of interest measured                                   |
| ULTRA 2005            | No outcomes of interest measured                                   |

| Study                         | Reason for exclusion                                                                            |
|-------------------------------|-------------------------------------------------------------------------------------------------|
| <a href="#">Virtanen 1999</a> | Duration less than 1 year, no outcomes of interest measured                                     |
| <a href="#">Wharton 2011</a>  | Planned duration of 1 year. Owing to high dropout rate, results were reported only at 3 months. |
| <a href="#">Yoon 2018</a>     | The study was designed using hormone therapy as a co-intervention with a drug for dementia.     |

## Characteristics of studies awaiting classification [ordered by study ID]

### ChiCTR-IPR-16008754

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | Parallel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Participants  | <p>Inclusion criteria</p> <p>"1. Aged 40 ~ 60 years; 2. taking no estrogen therapy in the past three months; 3. FSH &gt; 40 IU/L; 4. without pregnancy; 5. comply with the plan and follow-up on time; 6. understanding test content, signed informed consent; 7. normal or mild menopausal syndrome, mood disorders KMI scale, depression scale, PSQI scale score; 8. normal or mildly bone loss by bone mineral density; 9. normal blood fat; 10. without serious heart and kidney disease; 11. without contraindications to estrogen therapy, such as breast cancer."</p> <p>Exclusion criteria</p> <p>"Taking estrogen or other steroid hormone drugs the recent three months; 2. with malignant tumor; 3. serious mental disorders; 4. difficult understanding the experiment content and sign the informed consent; 5. pregnancy; 6. serious nerve dysfunction; 7. the expectant survival &lt; 2 years; 8. joining other studies at the experimental period; 9. hypertension, hyperglycemia, hyperlipidemia, cardiovascular disease, ovarian disease."</p> |
| Interventions | <p>Oestrogen</p> <p>Placebo</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Outcomes      | Outcome of interest: stroke                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Notes         | To 2023, no publications. The only reference we found was in <i>Hormone Therapy for the Primary Prevention of Chronic Conditions in Postmenopausal Persons: An Evidence Review</i> for the U.S. Preventive Services Task Force, saying intervention lasted less than one year.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Hodis 2003

|               |                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methods       | Randomised controlled trial                                                                                                                                                                                                                                                                                                                                                  |
| Participants  | Women postmenopausal, with or without a uterus, 75 years of age or younger, with low-density lipoprotein (LDL) cholesterol level of 100 to 250 mg per deciliter (2.59 to 6.46 mmol per liter) and a total triglyceride level of less than 400 mg per deciliter (4.52 mmol per liter), and at least one coronary-artery lesion occluding 30% or more of the luminal diameter. |
| Interventions | <p>- 1 mg of oral micronized 17b-estradiol (Estrace, Mead Johnson) daily, plus a placebo tablet matching medroxyprogesterone acetate for 12 consecutive days of every month.</p> <p>- 1 mg of oral micronized 17b-estradiol daily, plus 5 mg of medroxyprogesterone acetate (Provera, Upjohn) daily for 12 consecutive days of every month.</p>                              |

**Hodis 2003** (Continued)

- two placebo tablets, one matching the 17b-estradiol and taken daily and the other matching the medroxyprogesterone acetate and taken for 12 consecutive days of every month.

|          |                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------|
| Outcomes | Adverse events                                                                                                            |
| Notes    | Identified after finalising the review. Will be considered for inclusion in the next version. Data do not change results. |

**Characteristics of ongoing studies** [ordered by study ID]

**NCT04090957**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study name   | Estetrol for the Treatment of Moderate to Severe Vasomotor Symptoms in Postmenopausal Women (E4Comfort)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Methods      | Randomised parallel trial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Participants | <p>40 to 65 years. Body mass index &gt;18 and &lt; 38</p> <p>If uterus in situ, endometrial thickness &lt; 4 mm at randomisation. Normal mammogram.</p> <p><b>Exclusion criteria</b></p> <p>"History of malignancy or any clinically significant findings at the breast examination and/or on mammography suspicious of breast malignancy that would require additional clinical testing to rule out breast cancer. Papanicolaou (PAP) test with ASC-US or higher.</p> <p>For uterus in situ history or presence of uterine cancer, endometrial hyperplasia, or disordered proliferative endometrium; Presence of endometrial polyp; Undiagnosed vaginal bleeding or undiagnosed abnormal uterine bleeding; Endometrial ablation; Any uterine/endometrial abnormality that in the judgment of the investigator contraindicates the use of estrogen and/or progestin therapy. This includes presence or history of adenomyosis or significant myoma; HBP, VTE, coagulopathy,</p> <ol style="list-style-type: none"> <li>1. Laboratory values of fasting glucose above 125 mg/dL and/or glycated hemoglobin above 7%;</li> <li>2. (LDL &gt;190 mg/dL and/or triglycerides &gt;300 mg/dL);</li> <li>3. Subjects smoking &gt;15 cigarettes per day;</li> <li>4. Presence or history of gallbladder disease, unless cholecystectomy has been performed;</li> <li>5. Systemic lupus erythematosus;</li> <li>6. Any malabsorption disorders including gastric bypass surgery;</li> <li>7. History of acute liver disease in the preceding 12 months before the start of screening or presence or history of chronic or severe liver disease [alanine transaminase (ALT) or aspartate transaminase (AST) &gt;2 x upper limit of normal (ULN), bilirubin &gt;1.5 ULN], or liver tumors;</li> <li>8. Chronic or current acute renal impairment (estimated glomerular filtration rate &lt;60 ml/min);</li> <li>9. Porphyria;</li> <li>10. Diagnosis or treatment of major psychiatric disorder (e.g., schizophrenia, bipolar disorder, etc.) in the judgement of the Investigator;</li> <li>11. Use of estrogen/progestin containing drug(s) up to: 1 week before screening start for vaginal non-systemic hormonal products (rings, creams, gels); 4 weeks before screening start for vaginal or transdermal estrogen or estrogen/progestin products; 8 weeks before screening start for oral estrogen and/or progestin products and/or selective estrogen receptor modulator therapy; 8 weeks before screening start for intrauterine progestin therapy; 3 months before screening start for progestin implants or estrogen alone injectable drug therapy; 6 months before screening start for estrogen pellet therapy or progestin injectable drug therapy;</li> <li>12. Use of androgen/dehydroepiandrosterone (DHEA) containing drugs: 8 weeks before screening start for oral, topical, vaginal or transdermal androgen; 6 months before screening start for implantable or injectable androgen therapy;</li> </ol> |

## NCT04090957 (Continued)

13. Use of phytoestrogens or black cohosh for treatment of VMS up to 2 weeks before the start of screening;
14. For the women participating in the Efficacy Study part: use of prescription or over-the-counter products used for the treatment of VMS, e.g., anti-depressants: paroxetine, escitalopram, methyl-dopa, opioid and clonidine up to 4 weeks before the start of screening, and venlafaxine and desvenlafaxine up to 3 months before the start of screening, and not willing to stop these during their participation in the trial;
15. Not willing to stop any hormonal products as described in exclusion criteria 18, 19 and 20 during their participation in the trial;
16. Inadequately treated hyperthyroidism with abnormal TSH and free T4 at screening. Subjects with low or high TSH are allowed if free T4 at screening is within normal range;
17. History or presence of allergy/intolerance to the investigational product or drugs of this class or any component of it, or history of drug or other allergy that, in the opinion of the Investigator contraindicates subject participation;
18. For non-hysterectomized subjects: history or presence of allergy to peanuts;
19. History of alcohol or substance abuse (including marijuana, even if legally allowed) or dependence in the previous 12 months before the start of screening as determined by the Investigator, based on reported observations;
20. Sponsor or contract research organization (CRO) employees or employees under the direct supervision of the Investigator and/or involved directly in the trial;
21. Subjects with known or suspected history of a clinically significant systemic disease, unstable medical disorders, life-threatening disease or current malignancies that would pose a risk to the subject in the opinion of the Investigator;
22. Participation in another investigational drug clinical trial within 1 month (30 days) or having received an investigational drug within the last month (30 days) before the start of screening;
23. Is judged by the Investigator to be unsuitable for any reason."

|                     |                                                 |
|---------------------|-------------------------------------------------|
| Interventions       | Estretol 15 to 20 mg daily versus placebo       |
| Outcomes            | Endometrial cancer (Bp at 13 months)            |
| Starting date       | 2019                                            |
| Contact information | Estetra                                         |
| Notes               | Initial protocol of <a href="#">NCT04090957</a> |

## NCT05586724

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study name   | PROBES study                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Methods      | Micronized Progesterone Versus Norethisterone Acetate in Combination With Estrogen as Menopausal Hormone Therapy                                                                                                                                                                                                                                                                                                                                                        |
| Participants | <p><b>Inclusion criteria</b></p> <ul style="list-style-type: none"> <li>• Age 45–60 years</li> <li>• Body mass index (BMI) &gt; 19 kg/m<sup>2</sup> and ≤ 32 kg/m<sup>2</sup></li> <li>• Intact uterus</li> <li>• In case of previous menopausal hormone therapy (MHT) use, washout 8 weeks for oral MHT and 4 weeks for transdermal MHT or local oestrogen treatment before screening</li> <li>• Written informed consent</li> </ul> <p><b>"Exclusion criteria</b></p> |

## NCT05586724 (Continued)

- Previous history or risk factors for breast cancer, breast cancer in situ or abnormal mammogram at baseline as assessed clinically by a radiology expert
- Previous history or risk factors for endometrial cancer or hyperplasia or abnormal/proliferative endometrial biopsy at baseline
- Vaginal bleeding
- Any concomitant medical treatment except for well-controlled hypertension, non-insulin treated type 2 diabetes, asthma and hypothyroidism
- History or presence of or risk factor for cardiovascular disease including thromboembolic disorder or cerebrovascular disease
- History or presence of liver and gallbladder disease, familial hyperlipidemia, epilepsy or classical migraine with aura
- History or presence of clinically significant depression or other psychiatric disorder that might in anyway compromise the performance of the trial or undermine its scientific validity
- Porphyria, Systemic lupus erythematosus and otosclerosis
- Current use of MHT or local estrogen treatment
- Alcohol and/or drug abuse
- Clinically significant findings on physical and/or gynecological examination at baseline
- Hypersensitivity to any of the study treatments"

|                     |                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interventions       | Micronised progesterone in continuous combination with oral oestrogen<br><br>Norethisterone acetate in continuous combination with oral oestrogen<br><br>Placebo |
| Outcomes            | Outcomes of interest: endometrial cancer                                                                                                                         |
| Starting date       | 2022                                                                                                                                                             |
| Contact information | Karolinska University Hospital. Angelica Hirschberg: angelica.linden-hirschberg@regionstockholm.se                                                               |
| Notes               |                                                                                                                                                                  |

## DATA AND ANALYSES

### Comparison 1. Peri or postmenopausal people without major health problems

| Outcome or subgroup title                                               | No. of studies | No. of participants | Statistical method              | Effect size       |
|-------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 1.1 Death from any cause: estrogen-only HT                              | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 1.1.1 Estradiol 1 mg (low dose) for 2 years                             | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.10] |
| 1.1.2 CEE 0.625 mg (mod dose) for 7.1 years                             | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.03 [0.88, 1.20] |
| 1.1.3 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up) | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.02 [0.91, 1.13] |

| Outcome or subgroup title                                                                            | No. of studies | No. of participants | Statistical method              | Effect size        |
|------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| 1.1.4 CEE 0.625 mg (mod dose) for 18 years (cumulative follow-up)                                    | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.94 [0.89, 1.00]  |
| 1.2 Death from any cause: combined HT                                                                | 5              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.2.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                    | 2              | 20993               | Risk Ratio (M-H, Fixed, 95% CI) | 1.31 [0.76, 2.27]  |
| 1.2.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                               | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.05 [0.71, 1.56]  |
| 1.2.3 CEE 0.625 mg (mod dose) + P (as per footnotes) for 3 years                                     | 3              | 18075               | Risk Ratio (M-H, Fixed, 95% CI) | 1.09 [0.81, 1.46]  |
| 1.2.4 CEE 0.045 mg (lowish dose) + 200 mg sequential progesterone for 4 years                        | 1              | 505                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.58 [0.15, 87.57] |
| 1.2.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 5.6 years                                             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.00 [0.84, 1.19]  |
| 1.2.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 7.9 years                                             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.06 [0.93, 1.20]  |
| 1.2.7 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow-up)            | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.00 [0.92, 1.08]  |
| 1.2.8 CEE 0.625 mg (mod dose) + MPA 2.5 mg 18 years (cumulative follow-up)                           | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.01 [0.96, 1.07]  |
| 1.3 Death from any cause: estrogen with or without sequential progesterone vaginal gel               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.3.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after median 5 years | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.99 [0.06, 15.77] |
| 1.4 Death from coronary heart disease: estrogen-only HT                                              | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.4.1 Oestradiol 1 mg (low dose) daily for 2 years                                                   | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.10]  |
| 1.4.2 CEE 0.625 mg (mod dose) for 7.1 years                                                          | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.98 [0.69, 1.38]  |
| 1.4.3 CEE 0.625 mg (mod dose) after 10.7 years (includes extra follow-up)                            | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.92 [0.71, 1.19]  |
| 1.4.4 CEE 0.625 mg (mod dose) for 18 years (cumulative follow-up)                                    | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.89 [0.75, 1.05]  |
| 1.5 Death from coronary heart disease: combined continuous HT                                        | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |

| Outcome or subgroup title                                                                          | No. of studies | No. of participants | Statistical method              | Effect size        |
|----------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| 1.5.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.06 [0.68, 1.66]  |
| 1.5.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 7.9 years                                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.99 [0.72, 1.38]  |
| 1.5.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 18 years (cumulative follow-up)                   | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.04 [0.88, 1.21]  |
| <b>1.6 Death from coronary heart disease: combined sequential HT</b>                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.6.1 1 mg 17-B-oestradiol (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.34 [0.01, 8.27]  |
| <b>1.7 Death from stroke: estrogen-only HT</b>                                                     | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.7.1 CEE 0.625 mg (low dose) for 7.1 years                                                        | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.16 [0.58, 2.32]  |
| 1.7.2 CEE 0.625 mg (low dose) for 18 years                                                         | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.98 [0.77, 1.24]  |
| <b>1.8 Death from stroke: combined continuous HT</b>                                               | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.8.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for median 1 year                                       | 1              | 4385                | Risk Ratio (M-H, Fixed, 95% CI) | 2.99 [0.12, 73.37] |
| 1.8.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.04 [0.46, 2.35]  |
| 1.8.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 18 years (cumulative follow up)                     | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.11 [0.90, 1.37]  |
| <b>1.9 Death from stroke: combined sequential HT</b>                                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.9.1 1 mg 17-B-oestradiol (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.08 [0.13, 74.46] |
| <b>1.10 Death from colorectal cancer: estrogen-only HT</b>                                         | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.10.1 CEE 0.625 mg (mod dose) for 7.1 years                                                       | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.28 [0.66, 2.46]  |
| 1.10.2 CCE 0.625 mg (mod dose) for 18 years (cumulative follow up)                                 | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.20 [0.79, 1.83]  |
| <b>1.11 Death from colorectal cancer: combined continuous HT</b>                                   | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |

| Outcome or subgroup title                                                                                  | No. of studies | No. of participants | Statistical method              | Effect size       |
|------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 1.11.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.40, 2.29] |
| 1.11.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 7.1 years                                                  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.01 [0.52, 1.96] |
| 1.11.3 CEE 0.0625 mg (mod dose) + MPA 2.5 mg after 11.6 years                                              | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.31 [0.80, 2.14] |
| 1.11.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 18 years (cumulative follow up)                            | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.01 [0.69, 1.48] |
| <b>1.12 Death from breast cancer: estrogen-only HT</b>                                                     | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 1.12.1 CEE 0.625 mg (mod dose) after a median 10.8 years (includes extra follow-up)                        | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.58 [0.32, 1.02] |
| 1.12.2 CEE 0.625 mg (mod dose) for 18 years (cumulative follow up)                                         | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.55 [0.33, 0.92] |
| 1.12.3 CEE 0.625 mg (mod dose) 20 years (cumulative follow up)                                             | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.67 [0.42, 1.05] |
| <b>1.13 Death from breast cancer: combined continuous HT</b>                                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 1.13.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.48 [0.98, 2.25] |
| 1.13.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 12.5 yrs                                              | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.48 [0.98, 2.25] |
| 1.13.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 18 years (cumulative follow up)                            | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.45 [0.98, 2.16] |
| 1.13.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 20 years (cumulative follow up)                            | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.28 [0.90, 1.82] |
| <b>1.14 Death from lung cancer: estrogen-only HT (moderate dose)</b>                                       | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 1.14.1 Death from lung cancer (non-small cell or small cell): CEE 0.625 mg (mod dose) after mean 7.9 years | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.05 [0.65, 1.70] |
| 1.14.2 Death from non-small cell lung cancer: CEE 0.625 mg (mod dose) after mean 7.9 years                 | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.88 [0.52, 1.50] |
| 1.14.3 Death from small cell lung cancer: CEE 0.625 mg (mod dose) after mean 7.9 years                     | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 2.04 [0.62, 6.79] |

| Outcome or subgroup title                                                                           | No. of studies | No. of participants | Statistical method              | Effect size        |
|-----------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| <b>1.15 Death from lung cancer: combined continuous HT (moderate dose)</b>                          | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.15.1 Death from lung cancer (non-small cell or small cell) at mean 7.9 years                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.74 [1.18, 2.55]  |
| 1.15.2 Death from non-small cell lung cancer at mean 7.9 years                                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.91 [1.24, 2.93]  |
| 1.15.3 Death from small cell lung cancer at mean 7.9 years                                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.16 [0.48, 2.81]  |
| 1.15.4 Death from lung cancer (any type) at median 14 years                                         | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.10 [0.88, 1.39]  |
| <b>1.16 Death from lung cancer: combined sequential HT (low dose oestrogen)</b>                     | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.16.1 1 mg 17-B-oestradiol (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.08 [0.13, 74.46] |
| <b>1.17 Death from any cancer: combined continuous HT</b>                                           | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.17.1 CEE 0.625 mg daily (mod dose) + MPA 2.5 mg for 3 years                                       | 1              | 777                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.77 [0.11, 67.80] |
| 1.17.2 CEE 0.625 mg daily (mod dose) + MPA 2.5 mg for mean 5.2 years                                | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.16 [0.87, 1.53]  |
| <b>1.18 Coronary events (MI or cardiac death): estrogen-only HT</b>                                 | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.18.1 Oestradiol 1 mg (low dose) for 2 years                                                       | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.50 [0.05, 5.43]  |
| 1.18.2 CEE 0.625 mg (mod dose) for 3 years                                                          | 1              | 349                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.98 [0.12, 72.72] |
| 1.18.3 CEE 0.625 mg (mod dose) for mean 7.1 years                                                   | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.94 [0.78, 1.13]  |
| 1.18.4 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up)                            | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.82, 1.10]  |
| <b>1.19 Coronary events (MI or cardiac death): combined continuous HT</b>                           | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.19.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                  | 2              | 20993               | Risk Ratio (M-H, Fixed, 95% CI) | 1.89 [1.15, 3.10]  |
| 1.19.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.49 [1.05, 2.12]  |

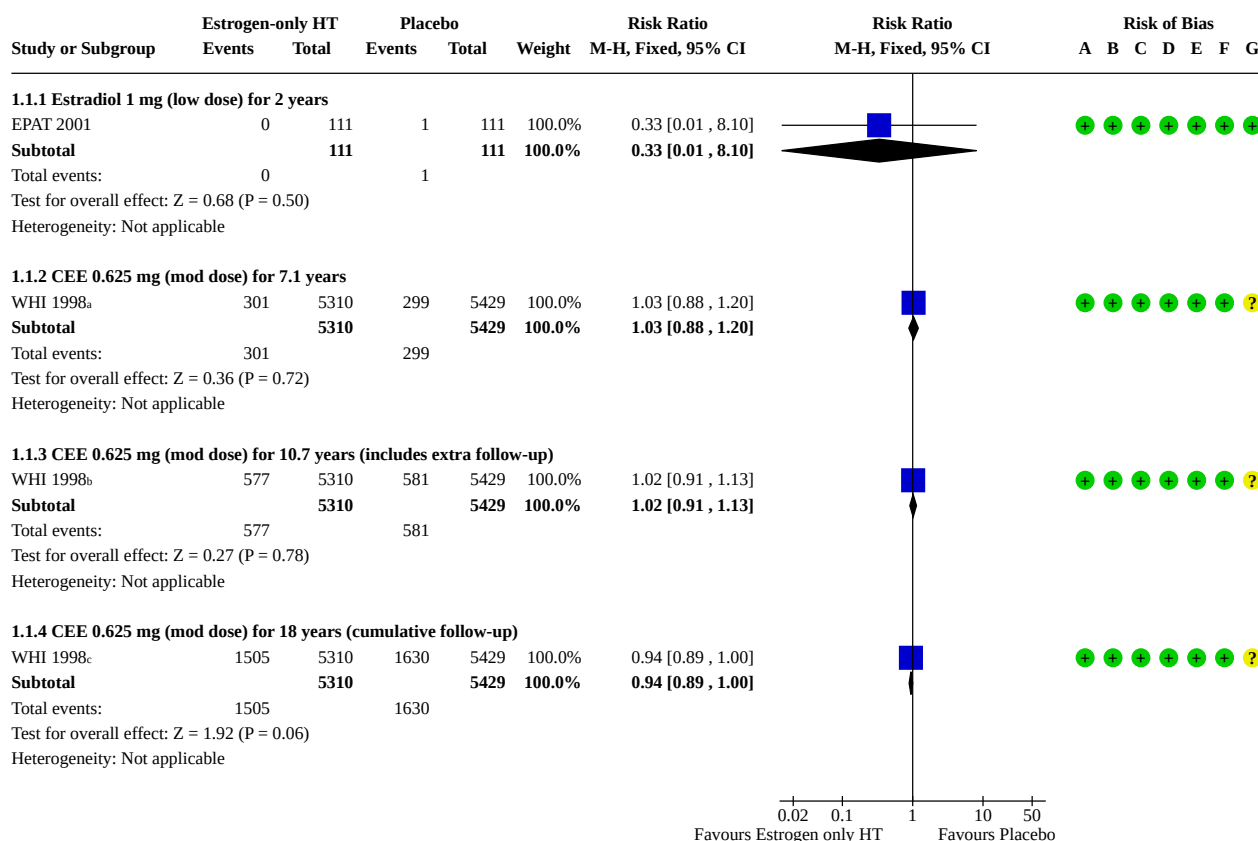
| Outcome or subgroup title                                                                                       | No. of studies | No. of participants | Statistical method              | Effect size         |
|-----------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 1.19.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years                                                         | 2              | 17385               | Risk Ratio (M-H, Fixed, 95% CI) | 1.45 [1.07, 1.98]   |
| 1.19.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                                  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.17 [0.95, 1.44]   |
| 1.19.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow-up)                      | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.08 [0.95, 1.22]   |
| 1.19.6 Estradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily                                                 | 1              | 1835                | Risk Ratio (M-H, Fixed, 95% CI) | 0.45 [0.02, 9.35]   |
| <b>1.20 Coronary events (MI or cardiac death): combined sequential HT</b>                                       | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.20.1 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years                     | 1              | 352                 | Risk Ratio (M-H, Fixed, 95% CI) | 4.89 [0.24, 101.09] |
| 1.20.2 1 mg (low dose) 17-B-oestradiol daily plus (3 days weekly) 0.35 mg norethindrone for 2 years             | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.34 [0.01, 8.27]   |
| 1.20.3 Oestradiol patch 0.05 mg (mod dose) + 200 mg sequential progesterone for 4 years                         | 1              | 497                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.71 [0.15, 90.70]  |
| <b>1.21 Coronary events (MI or cardiac death): estrogen with or without sequential progesterone vaginal gel</b> | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.21.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after median 5 years           | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.03, 3.16]   |
| <b>1.22 Stroke: Estrogen only HT</b>                                                                            | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.22.1 Oestradiol 1 mg (low dose) for 2 years                                                                   | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.00 [0.12, 72.86]  |
| 1.22.2 CEE 0.625 mg (mod dose) for mean 7.1 years                                                               | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.33 [1.06, 1.67]   |
| 1.22.3 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up)                                        | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.17 [0.97, 1.40]   |
| <b>1.23 Stroke: combined continuous HT</b>                                                                      | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.23.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 1 year                                                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.49, 1.86]   |

| Outcome or subgroup title                                                                            | No. of studies | No. of participants | Statistical method              | Effect size        |
|------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| 1.23.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                              | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.31 [0.83, 2.06]  |
| 1.23.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years                                              | 2              | 17385               | Risk Ratio (M-H, Fixed, 95% CI) | 1.46 [1.02, 2.09]  |
| 1.23.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                       | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.39 [1.09, 1.77]  |
| 1.23.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 7.9 years                                       | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.29 [1.06, 1.56]  |
| 1.23.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow-up)           | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.15 [0.99, 1.33]  |
| <b>1.24 Stroke: combined sequential HT</b>                                                           | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.24.1 1 mg (low dose) 17-B-oestradiol daily plus (3 days weekly) 0.35 mg norethindrone for 2 years  | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.08 [0.13, 74.46] |
| 1.24.2 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                               | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.00 [0.12, 73.14] |
| 1.24.3 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years          | 1              | 352                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.93 [0.12, 71.51] |
| 1.24.4 CEE 0.45 mg daily + micronized progesterone 200 mg daily for 12 days/month at 4 years         | 1              | 482                 | Risk Ratio (M-H, Fixed, 95% CI) | Not estimable      |
| <b>1.25 Transient ischaemic attack: estrogen-only HT</b>                                             | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.25.1 Oestradiol 1 mg (low dose) for 2 years                                                        | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.00 [0.12, 72.86] |
| <b>1.26 Transient ischaemic attack: combined sequential HT</b>                                       | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 1.26.1 1 mg 17-B-oestradiol (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years  | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.03 [0.07, 16.13] |
| 1.26.2 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                               | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.00 [0.12, 73.14] |
| <b>1.27 Transient ischaemic attack: estrogen with or without sequential progesterone vaginal gel</b> | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |

| Outcome or subgroup title                                                                             | No. of studies | No. of participants | Statistical method              | Effect size         |
|-------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 1.27.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after median 5 years | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.50 [0.05, 5.44]   |
| <b>1.28 Stroke or transient ischaemic attack: combined continuous HT (moderate dose)</b>              | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.28.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                    | 1              | 4385                | Risk Ratio (M-H, Fixed, 95% CI) | 0.73 [0.37, 1.46]   |
| <b>1.29 Venous thromboembolism (DVT or PE): estrogen-only HT</b>                                      | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.29.1 CEE 0.625 mg (mod dose) for up to 2 years                                                      | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 2.22 [1.12, 4.39]   |
| 1.29.2 CEE 0.625 mg (mod dose) for 3 years                                                            | 1              | 349                 | Risk Ratio (M-H, Fixed, 95% CI) | 6.96 [0.36, 133.75] |
| 1.29.3 CEE 0.625 mg (mod dose) for 7.1 years                                                          | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.32 [1.00, 1.74]   |
| 1.29.4 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up)                              | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.04 [0.84, 1.29]   |
| <b>1.30 Venous thromboembolism (DVT or PE): combined continuous HT</b>                                | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.30.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                    | 2              | 20993               | Risk Ratio (M-H, Fixed, 95% CI) | 4.28 [2.49, 7.34]   |
| 1.30.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                               | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 2.98 [1.88, 4.71]   |
| 1.30.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years                                               | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 2.54 [1.73, 3.72]   |
| 1.30.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                        | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 2.03 [1.55, 2.64]   |
| 1.30.5 CEE 0.625 mg (mod dose) + 2.5 mg MPA for mean 7.9 years                                        | 1              | 16707               | Risk Ratio (M-H, Fixed, 95% CI) | 1.65 [1.32, 2.05]   |
| <b>1.31 Venous thromboembolism (DVT or PE): combined sequential HT</b>                                | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 1.31.1 1 mg 17-B-oestradiol (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years   | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.08 [0.13, 74.46]  |
| 1.31.2 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                                | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.00 [0.12, 73.14]  |
| 1.31.3 CEE 0.045 mg (lowish dose) + 200 mg sequential progesterone for 4 years                        | 1              | 505                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.40 [0.02, 9.73]   |

| Outcome or subgroup title                                                                                    | No. of studies | No. of participants | Statistical method                  | Effect size         |
|--------------------------------------------------------------------------------------------------------------|----------------|---------------------|-------------------------------------|---------------------|
| 1.31.4 Oestradiol patch 0.05 mg (mod dose) + 200 mg sequential progesterone for 4 years                      | 1              | 497                 | Risk Ratio (M-H, Fixed, 95% CI)     | 1.24 [0.08, 19.69]  |
| 1.31.5 Estradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily                                              | 1              | 1835                | Risk Ratio (M-H, Fixed, 95% CI)     | 0.27 [0.01, 6.61]   |
| <b>1.32 Venous thromboembolism (DVT or PE): estrogen with or without sequential progesterone vaginal gel</b> | 1              |                     | Risk Ratio (M-H, Random, 95% CI)    | Subtotals only      |
| 1.32.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after median 5 years        | 1              | 643                 | Risk Ratio (M-H, Random, 95% CI)    | 1.49 [0.25, 8.83]   |
| <b>1.33 Global cognitive function</b>                                                                        | 4              |                     | Mean Difference (IV, Fixed, 95% CI) | Subtotals only      |
| 1.33.1 Transdermal estradiol 0.014 mg (low dose) for 2 years: MMSE scores (baseline MMSE ≤ 90)               | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -1.21 [-5.05, 2.63] |
| 1.33.2 Transdermal estradiol 0.014 mg (low dose) for 2 years: MMSE scores (baseline MMSE > 90)               | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -0.30 [-0.73, 0.13] |
| 1.33.3 CEE 0.625 mg (mod dose) for mean 5.2 years: MMSE scores                                               | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -0.26 [-0.52, 0.00] |
| 1.33.4 CEE 0.625 mg (mod dose) with or without 2.5 mg MPA for 3 years: MMSE scores                           | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -0.10 [-0.35, 0.15] |
| 1.33.5 Combined continuous CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 4.2 years: MMSE scores              | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -0.18 [-0.36, 0.00] |
| 1.33.6 Oestrogen with or without sequential progesterone vaginal gel                                         | 1              |                     | Mean Difference (IV, Fixed, 95% CI) | -0.03 [-0.21, 0.15] |
| <b>1.34 Probable dementia: Estrogen only</b>                                                                 | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI)     | Subtotals only      |
| 1.34.1 CEE 0.625 mg (mod dose) for 5.2 years                                                                 | 1              | 2947                | Risk Ratio (M-H, Fixed, 95% CI)     | 1.52 [0.89, 2.59]   |
| <b>1.35 Probable dementia: Combined continuous HT</b>                                                        | 1              | 4532                | Odds Ratio (M-H, Fixed, 95% CI)     | 1.99 [1.17, 3.38]   |
| 1.35.1 Combined continuous CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 4.05 years                          | 1              | 4532                | Odds Ratio (M-H, Fixed, 95% CI)     | 1.99 [1.17, 3.38]   |

### Analysis 1.1. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 1: Death from any cause: estrogen-only HT



#### Footnotes

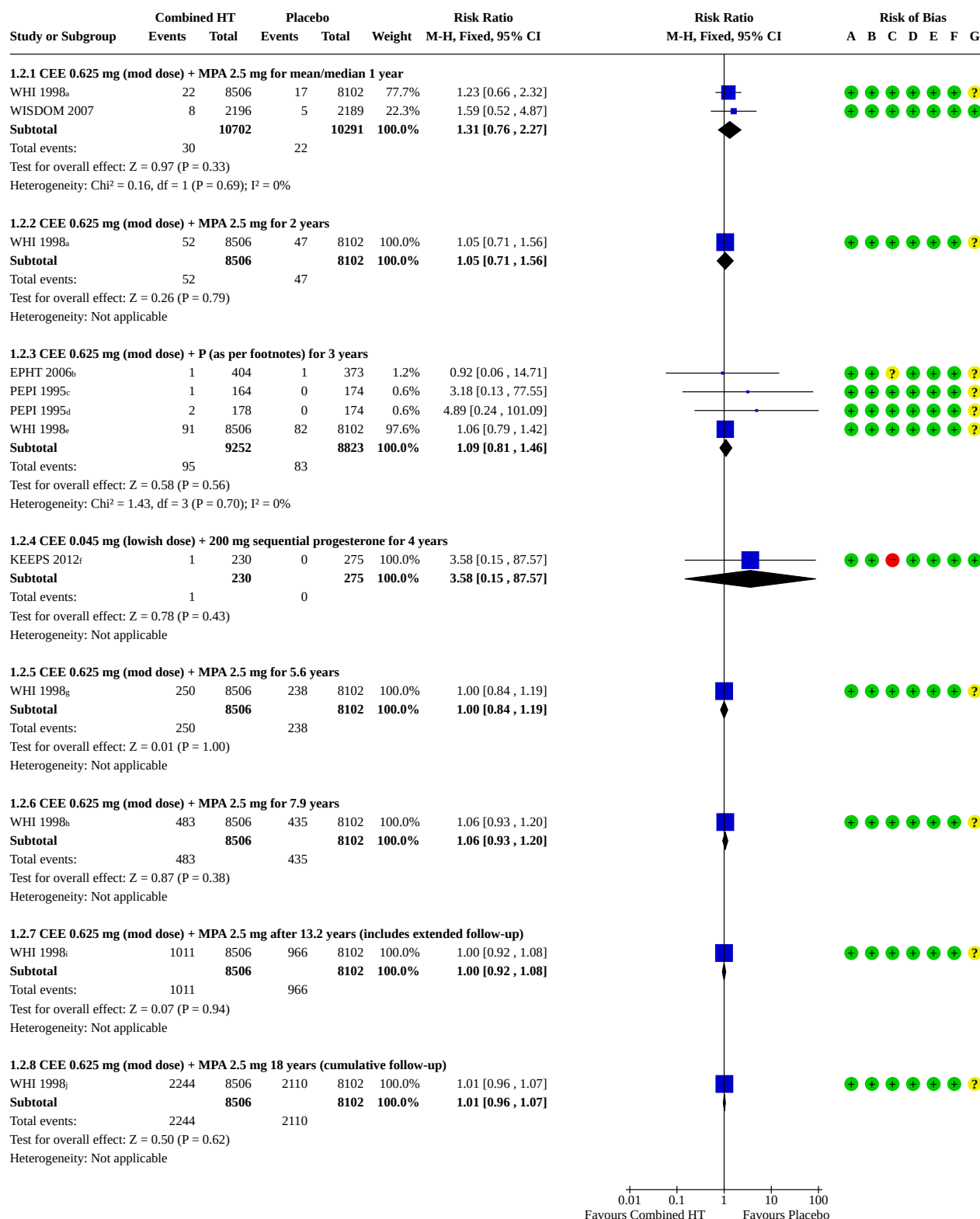
<sup>a</sup>Manson 2013

<sup>b</sup>LaCroix 2011: mean 7.1 yrs planned follow-up + additional follow-up in 78% of participants: mean total follow-up 10.7 yrs

<sup>c</sup>(3) Manson 2017. CCE alone for a median of 7.2 years, and follow up for 18 years

#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

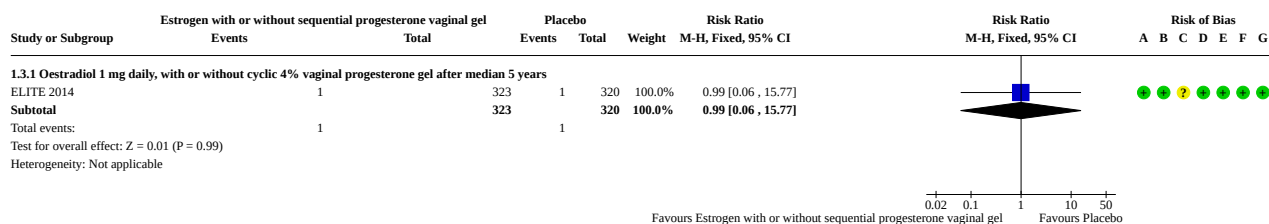
**Analysis 1.2. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 2: Death from any cause: combined HT****Footnotes**<sup>a</sup>Rossouw 2002<sup>b</sup>MPA 2.5 mg<sup>c</sup>MPA 2.5 mg

## Analysis 1.2. (Continued)

<sup>a</sup>Rossouw 2002  
<sup>b</sup>MPA 2.5 mg  
<sup>c</sup>MPA 10 mg days 1-12  
<sup>d</sup>Micronised progesterone 200 mg days 1-12  
<sup>e</sup>MPA 2.5 mg Rossouw 2002  
<sup>f</sup>Placebo group halved  
<sup>g</sup>Manson 2013  
<sup>h</sup>Heiss 2008  
<sup>i</sup>Manson 2013: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women  
<sup>j</sup>Manson 2017. CCE +MPA for a median of 5.6 years, and follow up for 18 years

## Risk of bias legend

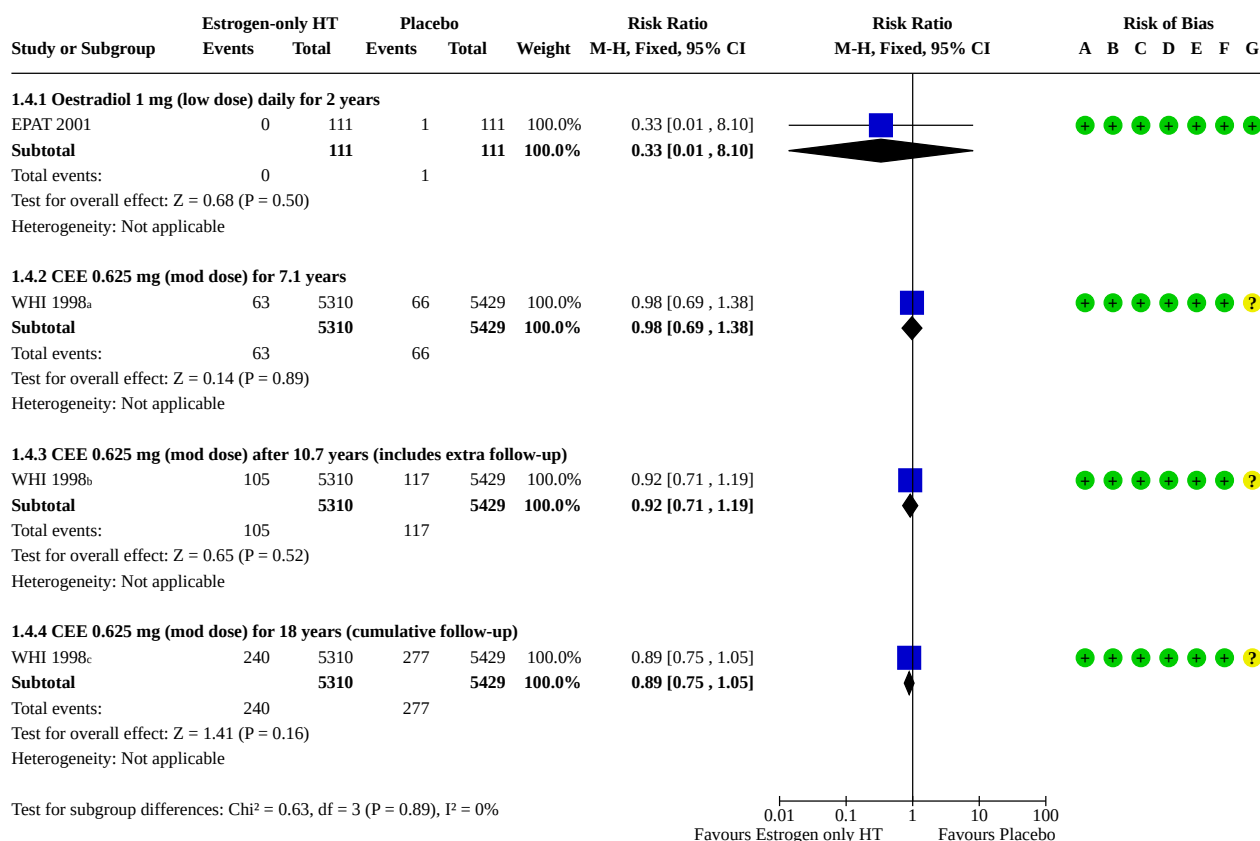
- (A) Random sequence generation (selection bias)  
(B) Allocation concealment (selection bias)  
(C) Incomplete outcome data (attrition bias)  
(D) Blinding of participants and personnel (performance bias)  
(E) Blinding of outcome assessment (detection bias)  
(F) Selective reporting (reporting bias)  
(G) Other bias

**Analysis 1.3. Comparison 1: Peri or postmenopausal people without major health problems ,  
Outcome 3: Death from any cause: estrogen with or without sequential progesterone vaginal gel**

## Risk of bias legend

- (A) Random sequence generation (selection bias)  
(B) Allocation concealment (selection bias)  
(C) Incomplete outcome data (attrition bias)  
(D) Blinding of participants and personnel (performance bias)  
(E) Blinding of outcome assessment (detection bias)  
(F) Selective reporting (reporting bias)  
(G) Other bias

### Analysis 1.4. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 4: Death from coronary heart disease: estrogen-only HT



#### Footnotes

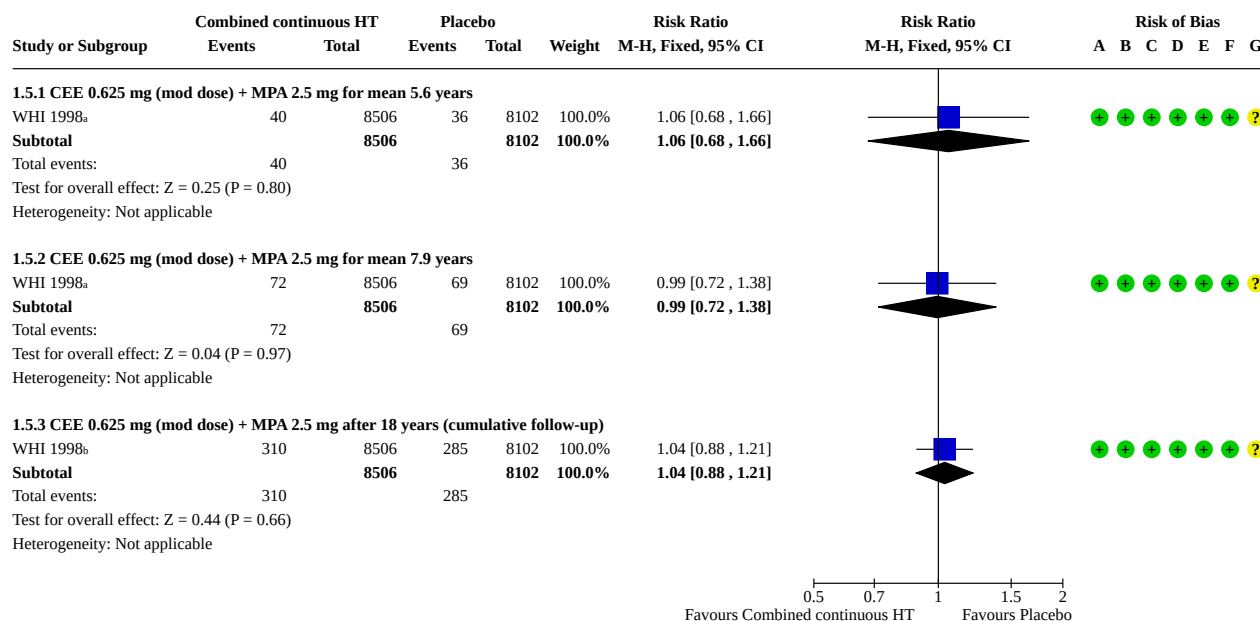
<sup>a</sup>LaCroix 2011

<sup>b</sup>7.1 yrs planned follow-up + additional follow-up in 78% of participants: mean total follow-up 10.7 yrs ( LaCroix 2011).

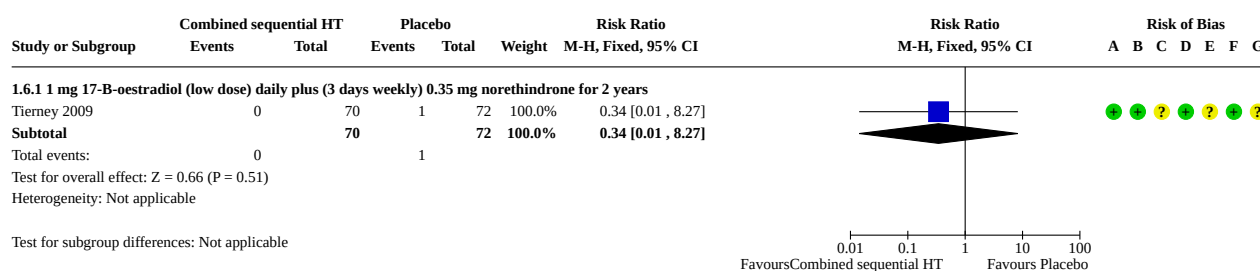
<sup>c</sup>Manson 2017. CCE alone for a median of 7.2 years, and follow up for 18 years

#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

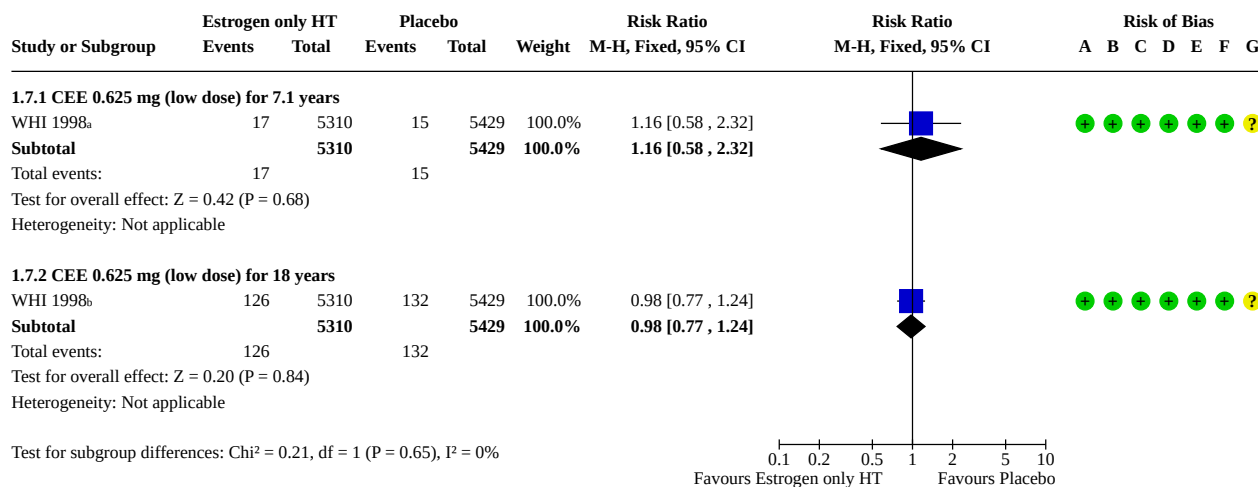
**Analysis 1.5. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 5: Death from coronary heart disease: combined continuous HT****Footnotes**<sup>a</sup>Heiss 2008<sup>b</sup>Manson 2017**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.6. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 6: Death from coronary heart disease: combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

### Analysis 1.7. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 7: Death from stroke: estrogen-only HT



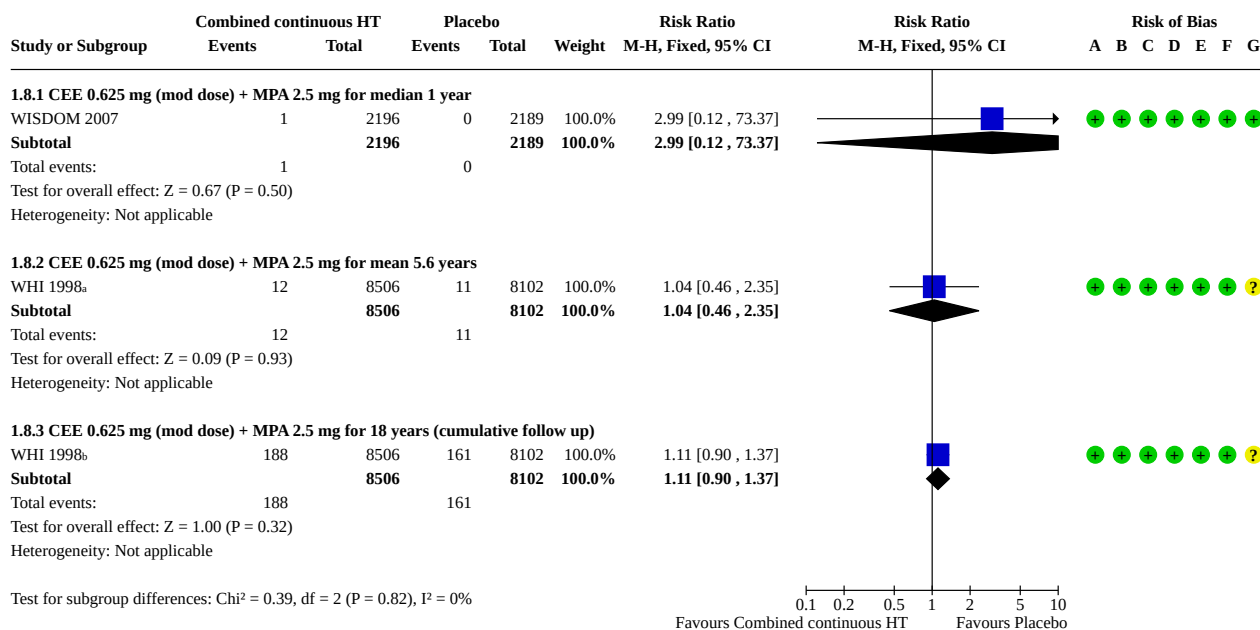
#### Footnotes

<sup>a</sup>Hendrix 2006

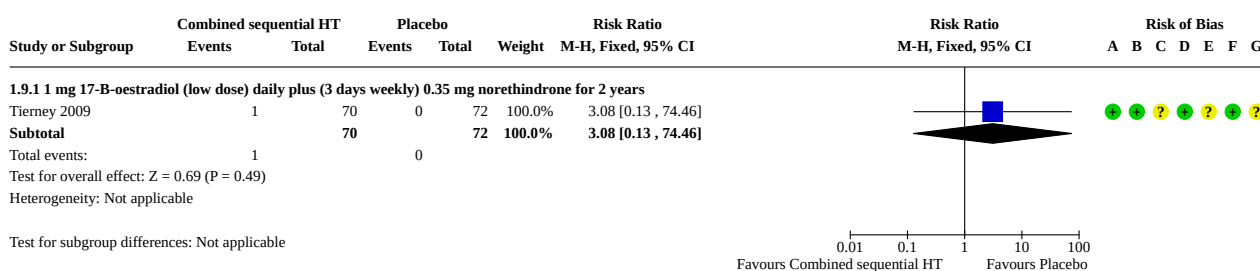
<sup>b</sup>Manson 2017

#### Risk of bias legend

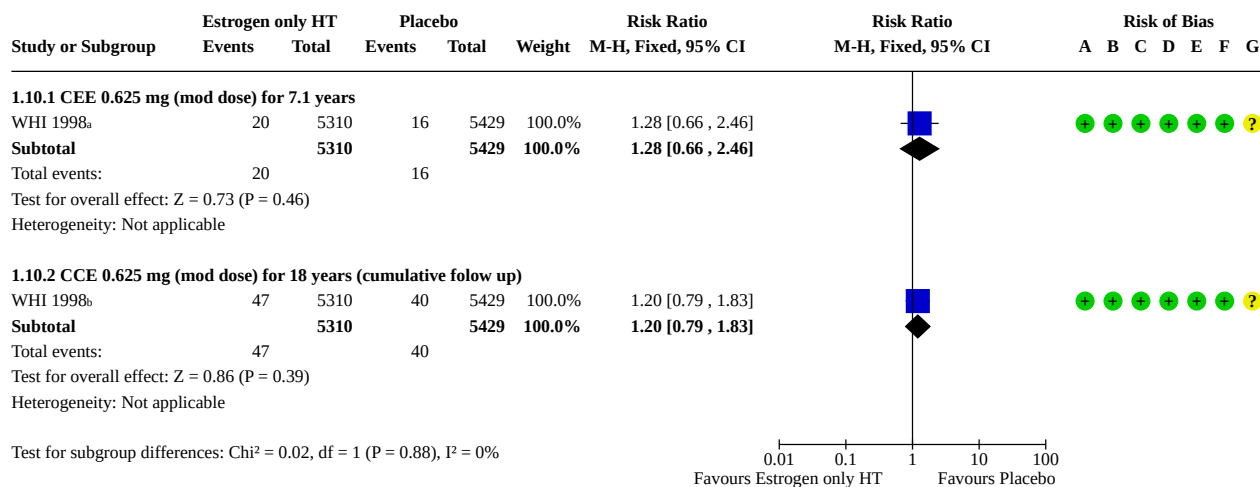
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.8. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 8: Death from stroke: combined continuous HT****Footnotes**<sup>a</sup>Wassertheil-Smoller 2006<sup>b</sup>Manson 2017. CCE +MPA for a median of 5.6 years, and follow up for 18 years**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

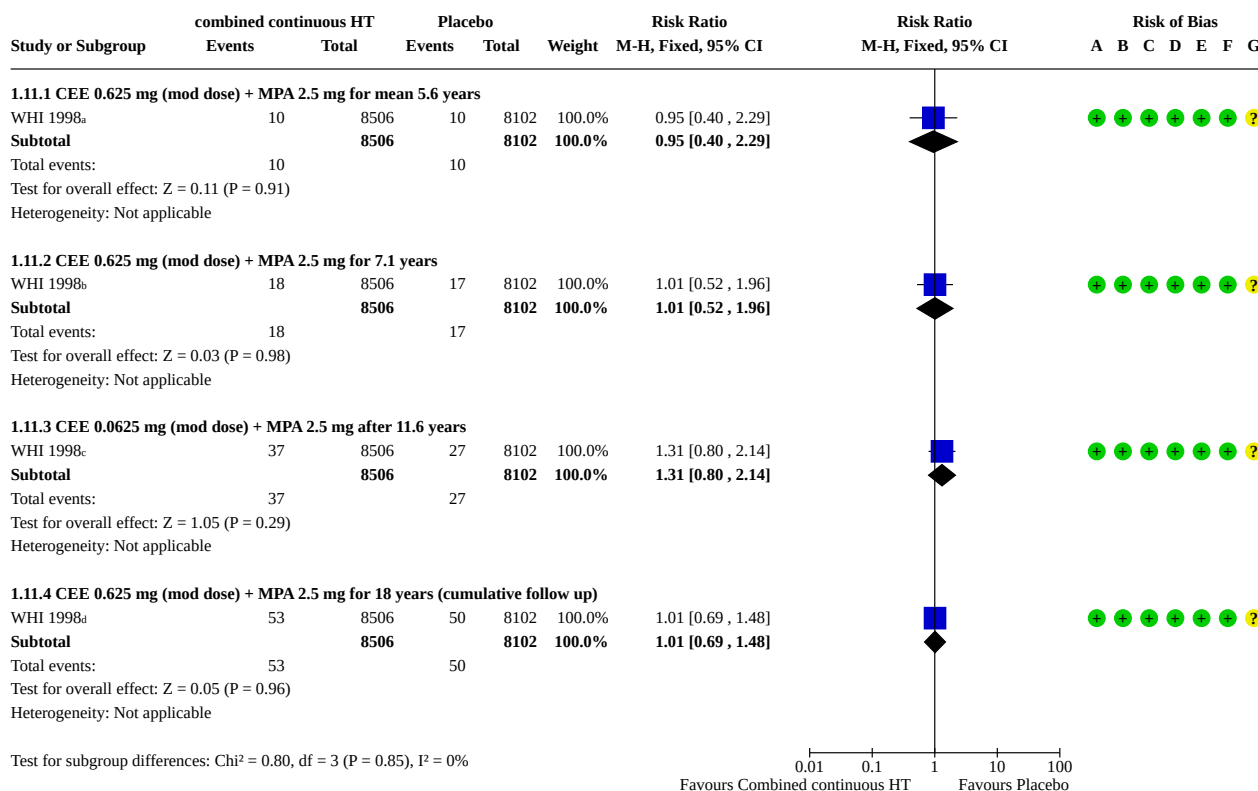
**Analysis 1.9. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 9: Death from stroke: combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.10. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 10: Death from colorectal cancer: estrogen-only HT****Footnotes**<sup>a</sup>Ritenbaugh 2008<sup>b</sup>Mason 2017. CCE alone for a median of 7.2 years, and follow up for 18 years**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# **Analysis 1.11. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 11: Death from colorectal cancer: combined continuous HT**



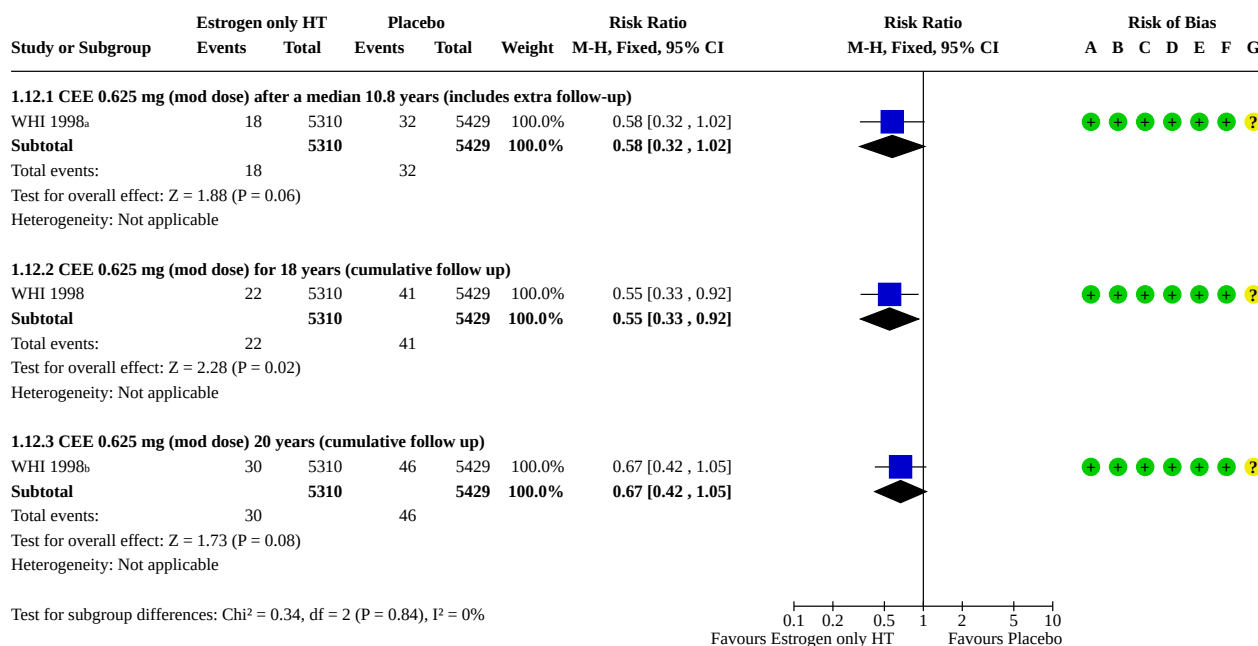
## **Footnotes**

- <sup>a</sup>Prentice 2009
- <sup>b</sup>Prentice 2009: includes 5.6 years active intervention plus follow up to 7.1 years
- <sup>c</sup>Simon 2012: mean 5.6 years' intervention plus postintervention follow up to 11.6 years in 83% participants
- <sup>d</sup>Manson 2017. CCE +MPA for a median of 5.6 years, and follow up for 18 years

## **Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# **Analysis 1.12. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 12: Death from breast cancer: estrogen-only HT**

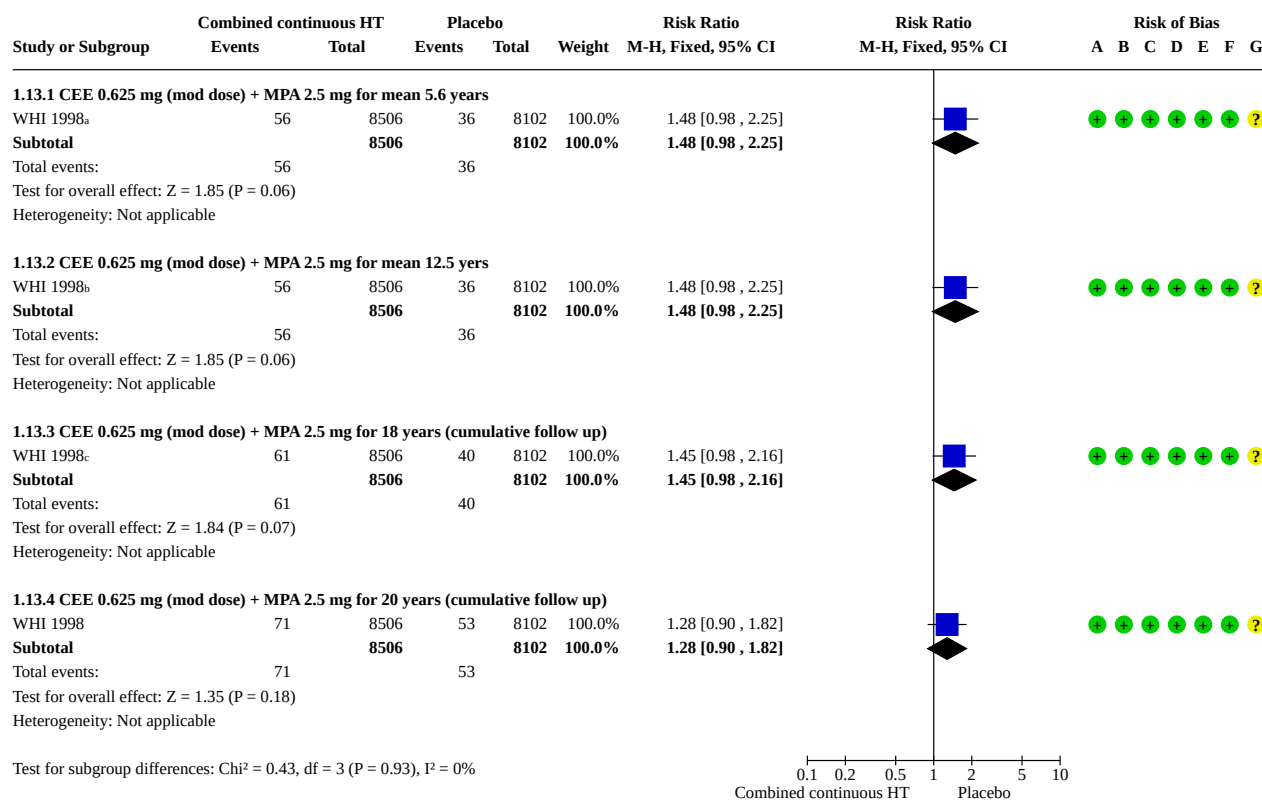


## **Footnotes**

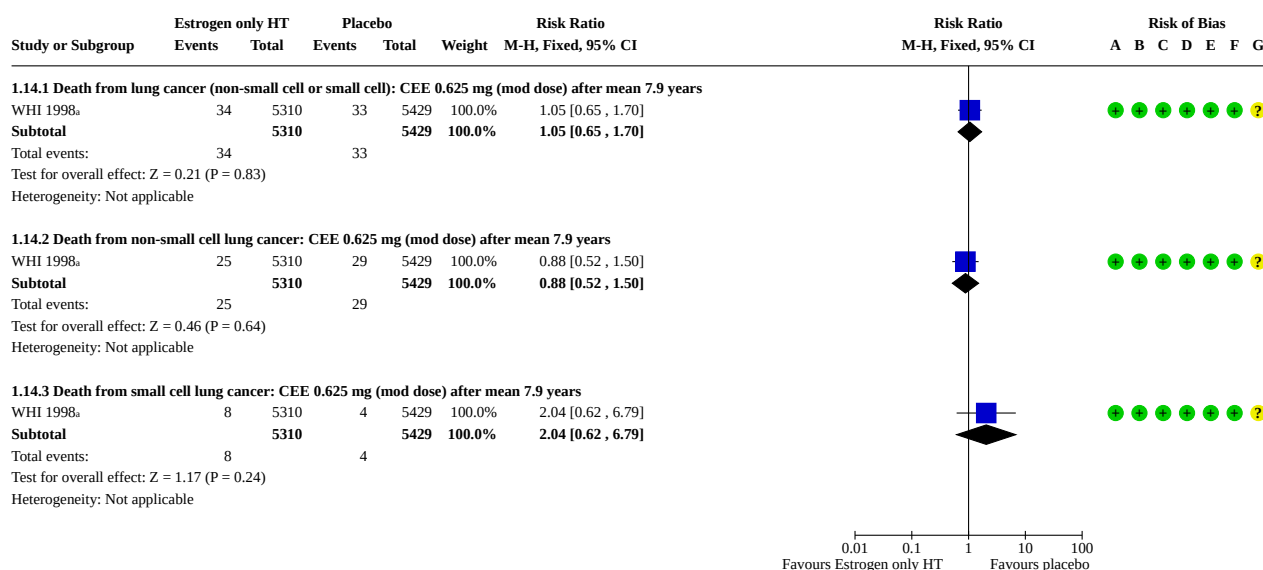
<sup>a</sup>Anderson 2012: overall median follow-up 11.8 yrs: 7.2 yrs median planned follow up + median 4.7 yrs extended follow-up in 78% of women  
<sup>b</sup>Chlebowski 2020.

## **Risk of bias legend**

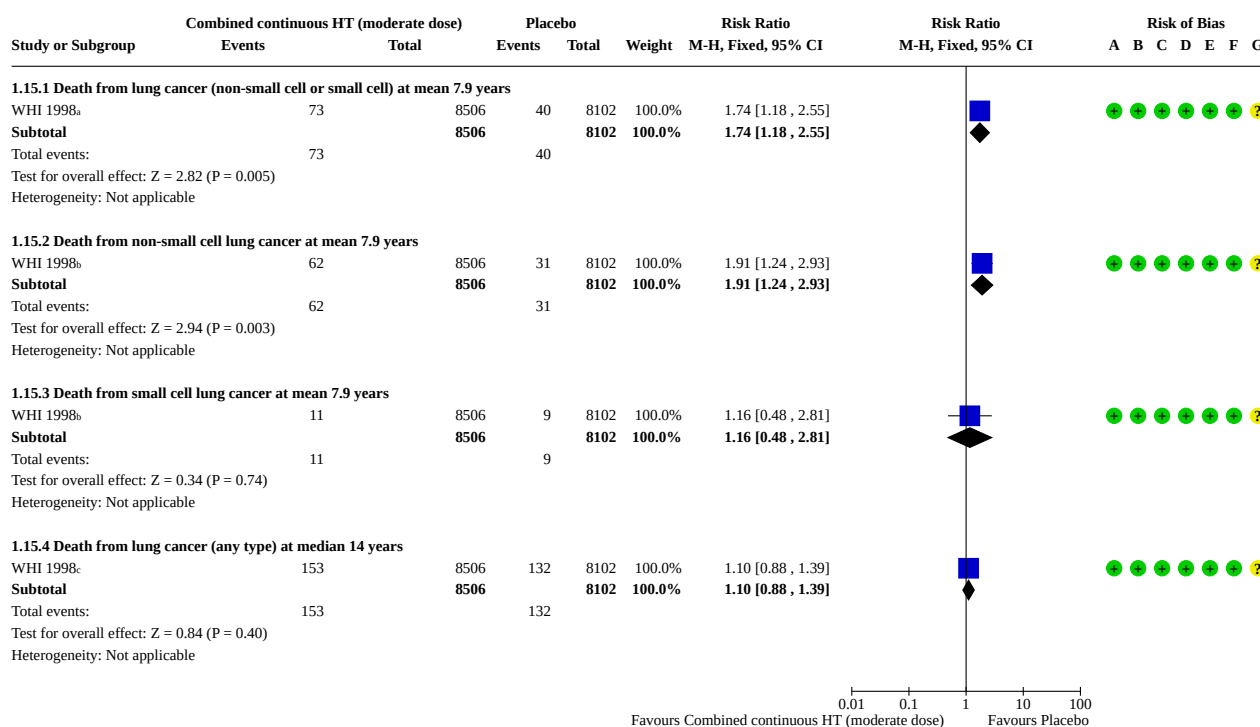
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.13. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 13: Death from breast cancer: combined continuous HT****Footnotes**<sup>a</sup>Chlebowski 2003<sup>b</sup>Chlebowski 2010: Mean follow-up 11 yrs: 7.1 yrs follow-up to original planned completion date (including intervention phase of 5.6 years) + additional follow up in 83% of surviving women<sup>c</sup>Manson 2017. CEE +MPA for a median of 5.6 years, and follow up for 18 years**Risk of bias legend**

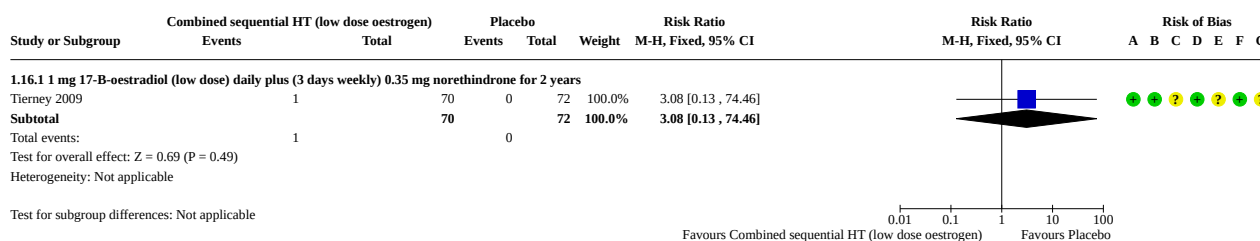
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.14. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 14: Death from lung cancer: estrogen-only HT (moderate dose)****Footnotes**<sup>a</sup>Chlebowski 2010b: Post hoc analysis. Mean follow-up 7.9 years.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

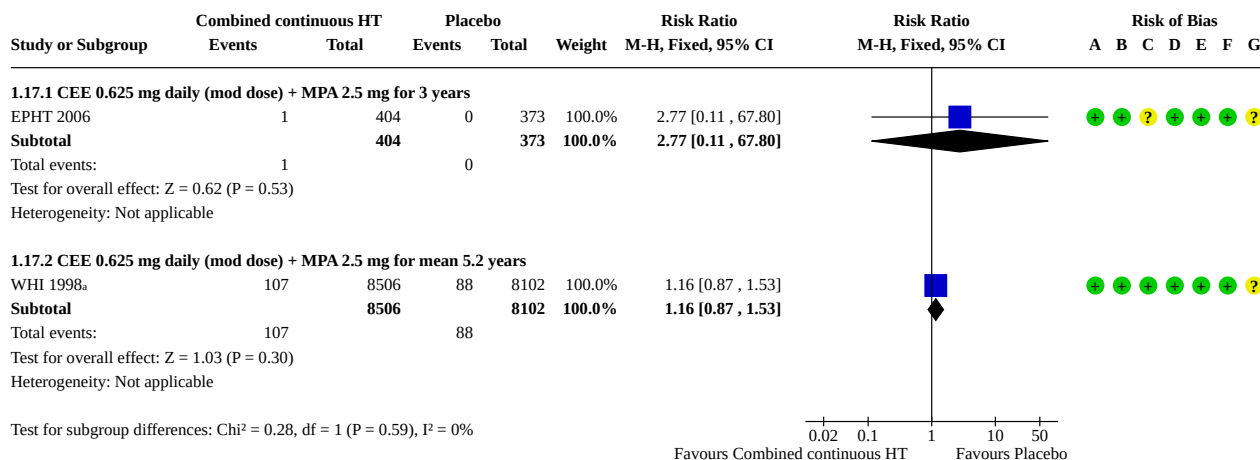
**Analysis 1.15. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 15: Death from lung cancer: combined continuous HT (moderate dose)****Footnotes**<sup>a</sup>Chlebowski 2009 and 2016a: Post hoc analysis. Mean 7.9 yrs follow-up (median 8 years)<sup>b</sup>Chlebowski 2009: Post hoc analysis. Mean 7.9 yrs follow-up.<sup>c</sup>Chlebowski 2016a: Post hoc analysis. Median 14 yrs follow-up**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.16. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 16: Death from lung cancer: combined sequential HT (low dose oestrogen)****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

### Analysis 1.17. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 17: Death from any cancer: combined continuous HT



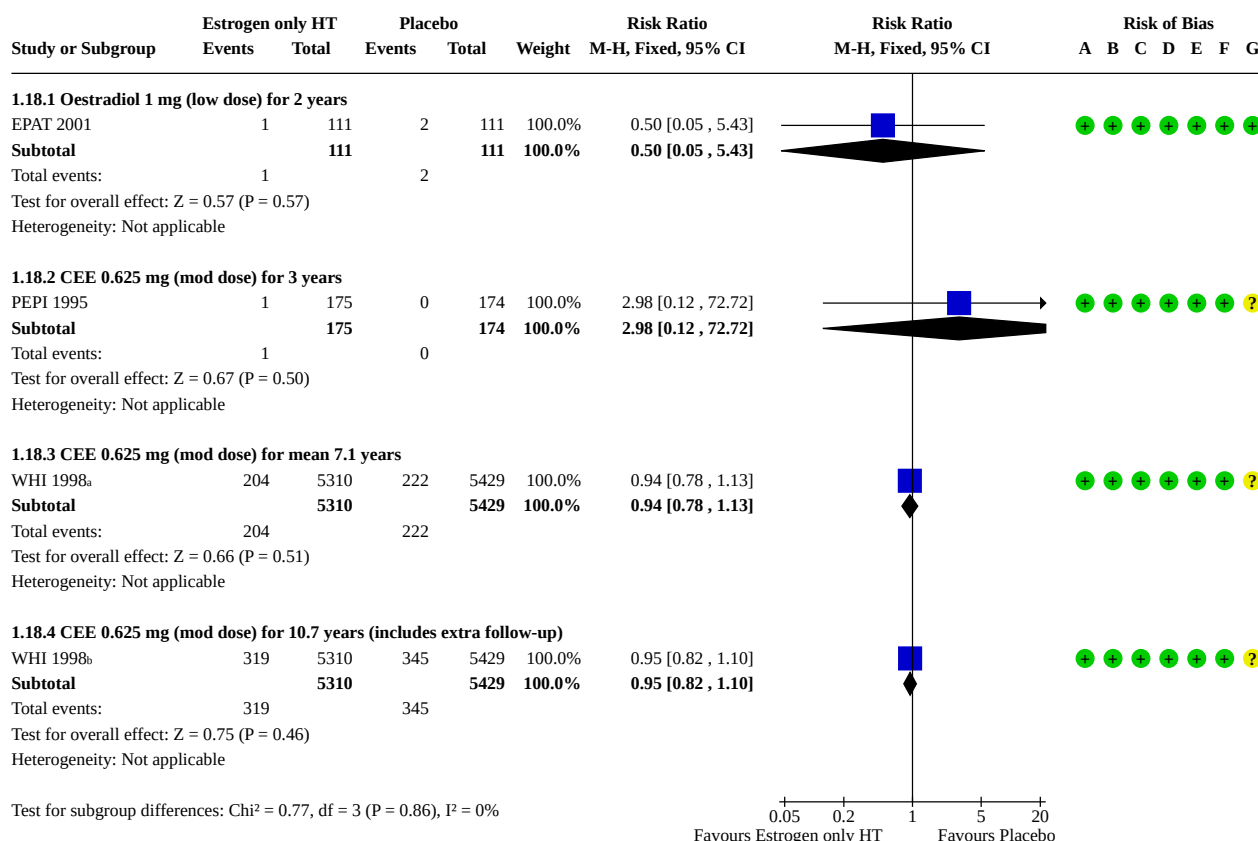
#### Footnotes

<sup>a</sup>Rossouw 2002

#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# **Analysis 1.18. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 18: Coronary events (MI or cardiac death): estrogen-only HT**



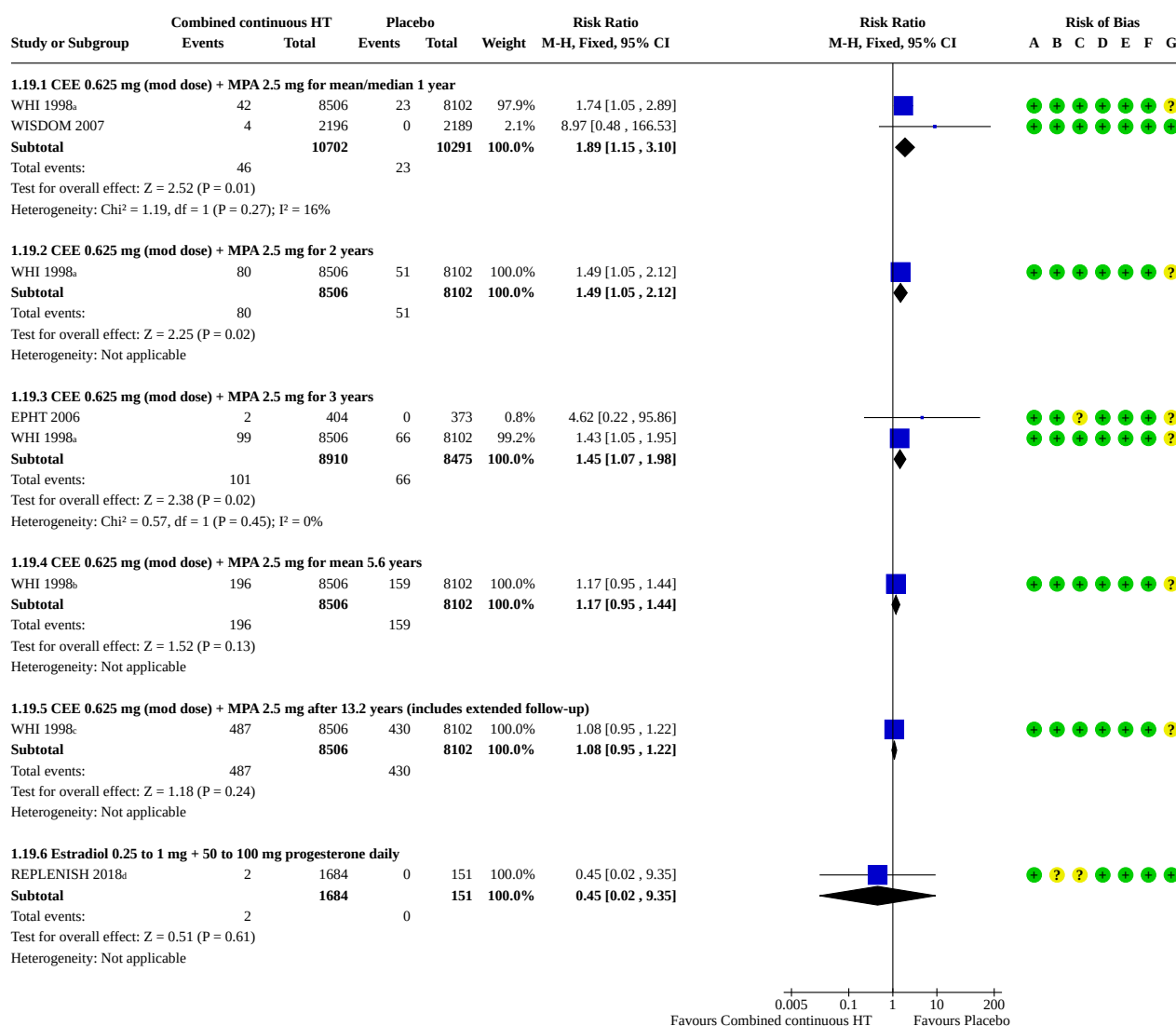
## **Footnotes**

<sup>a</sup>Manson 2013. Mean follow-up time 7.1 years, median 7.2 years

<sup>b</sup>7.1 yrs planned follow-up + additional follow-up in 78% of participants: mean total follow-up 10.7 yrs (LaCroix 2011).

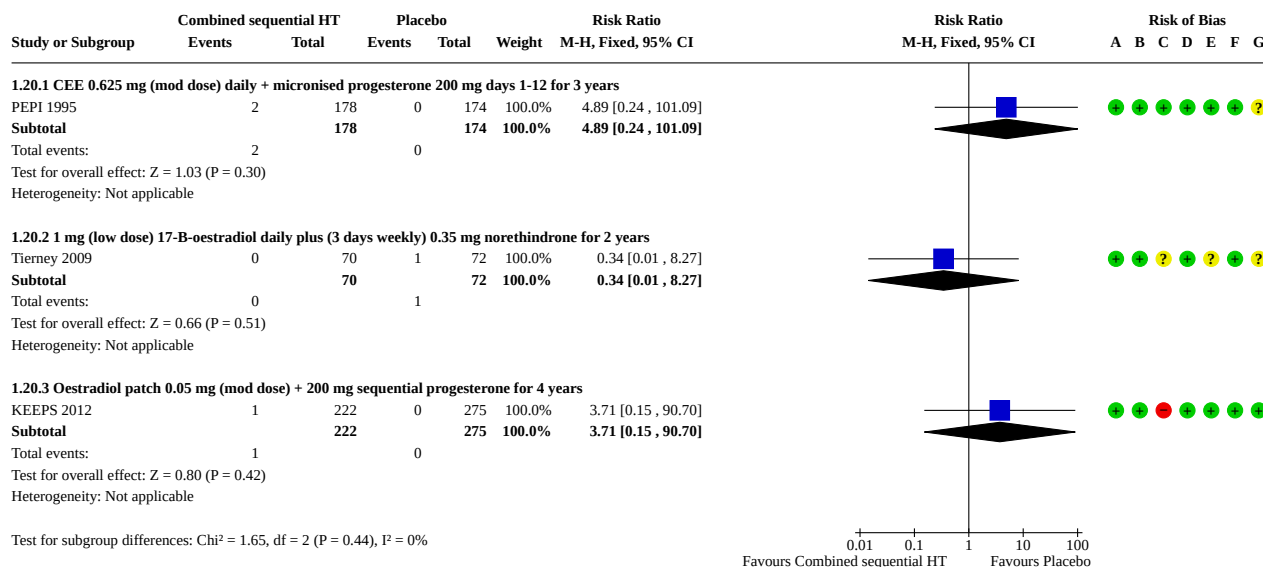
## **Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.19. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 19: Coronary events (MI or cardiac death): combined continuous HT****Footnotes**<sup>a</sup>Manson 2003<sup>b</sup>Manson 2013<sup>c</sup>Manson 2013: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women<sup>d</sup>Daily estradiol(mg)-progesterone (mg) (1/100, 0.5/100, 0.5/50, or 0.25/50 all combined**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

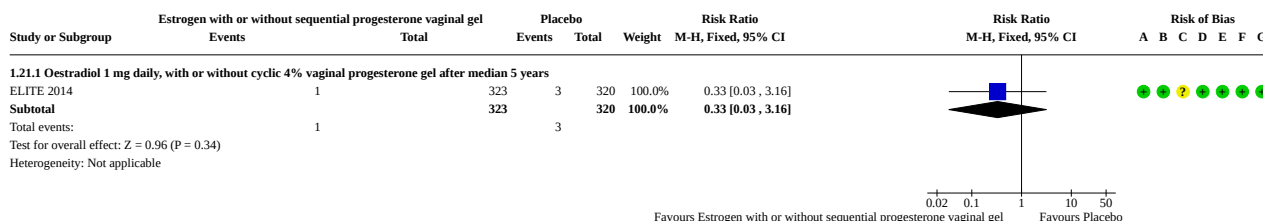
### Analysis 1.20. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 20: Coronary events (MI or cardiac death): combined sequential HT



#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

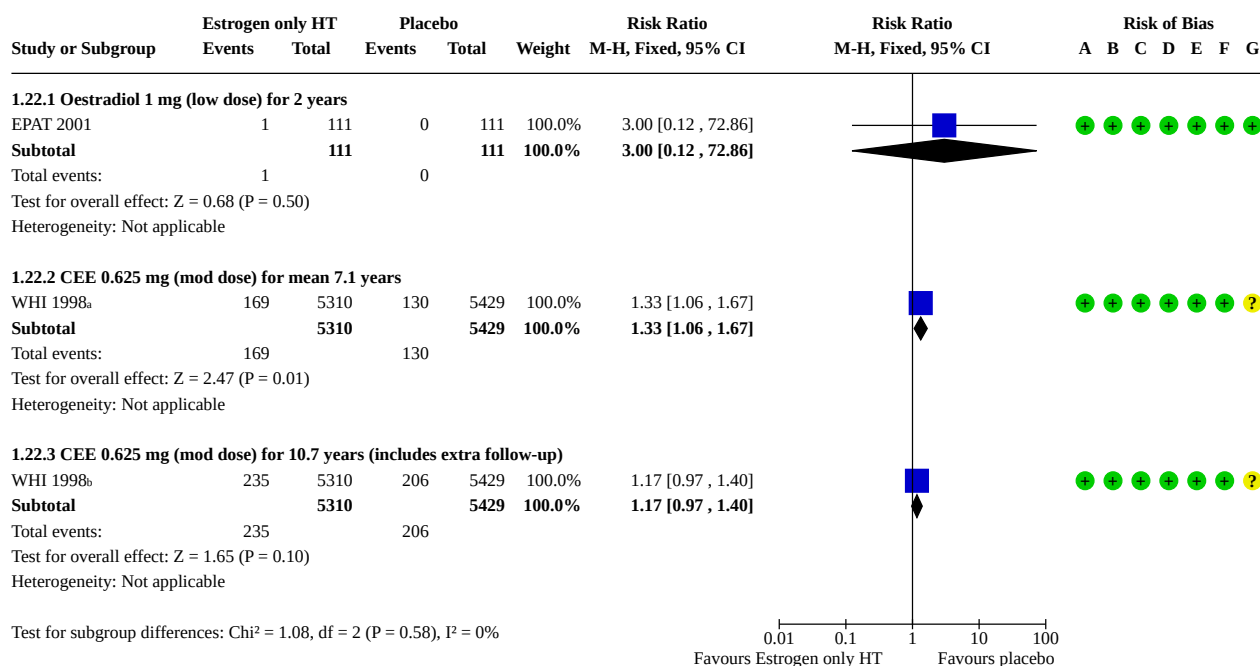
### Analysis 1.21. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 21: Coronary events (MI or cardiac death): estrogen with or without sequential progesterone vaginal gel



#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 1.22. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 22: Stroke: Estrogen only HT



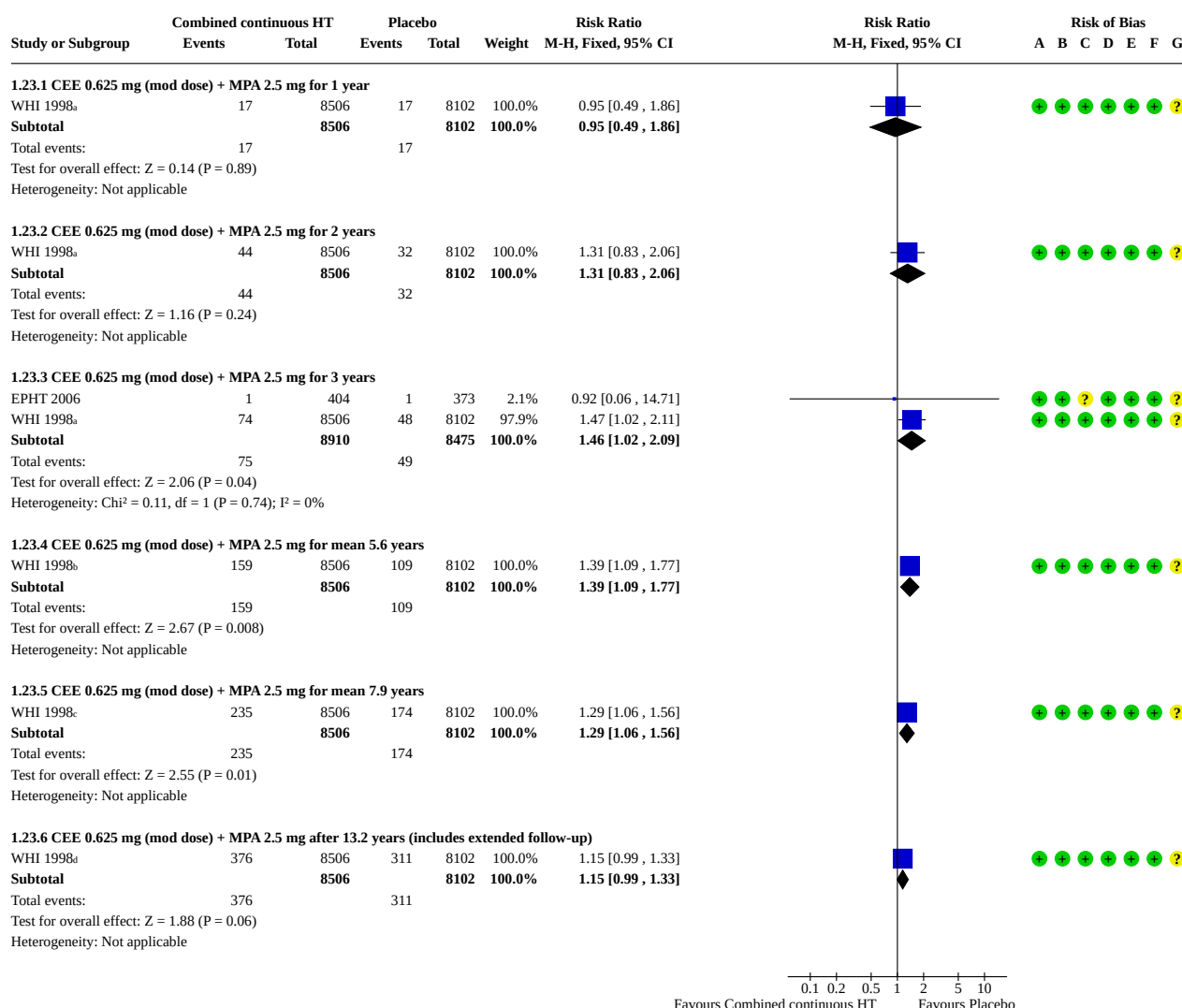
### Footnotes

<sup>a</sup>Manson 2013

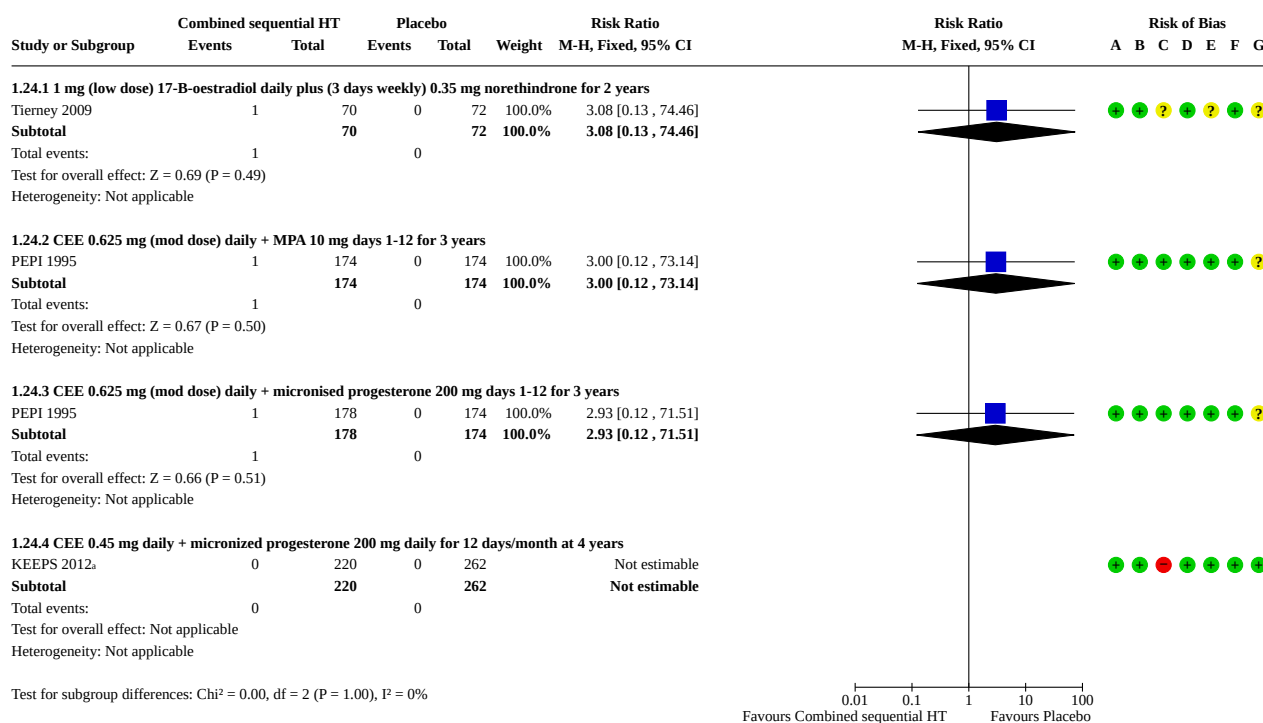
<sup>b</sup>LaCroix 2011: 7.1 yrs planned follow-up + extended follow-up in 78% of women

### Risk of bias legend

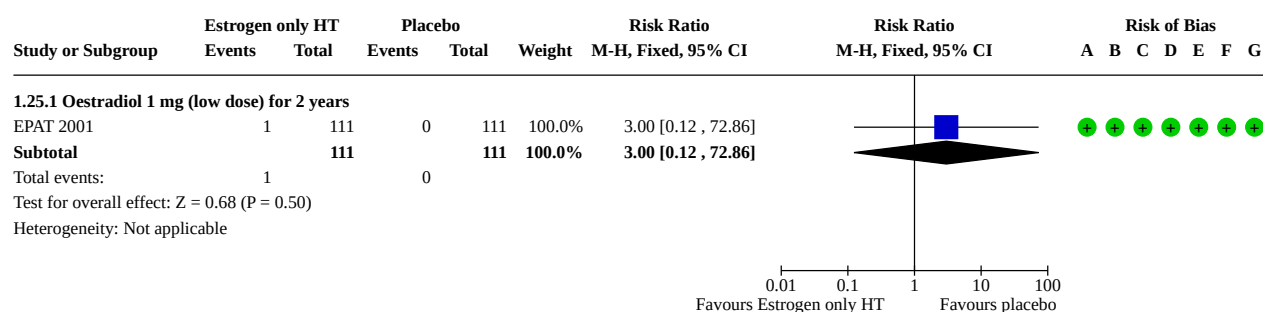
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.23. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 23: Stroke: combined continuous HT****Footnotes**<sup>a</sup>Rossouw 2002<sup>b</sup>Manson 2013<sup>c</sup>Heiss 2008<sup>d</sup>Manson 2013: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women**Risk of bias legend**

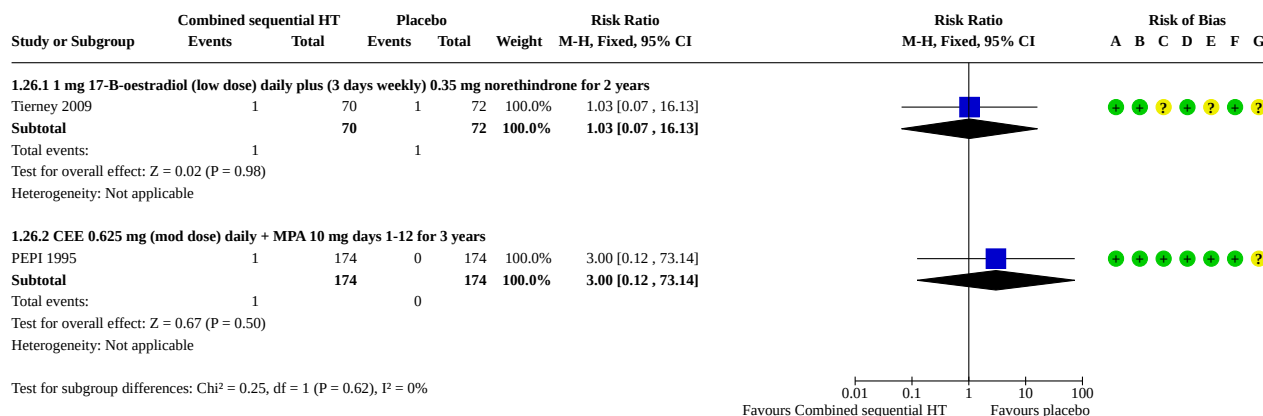
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.24. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 24: Stroke: combined sequential HT****Footnotes**<sup>a</sup>Gleason 2015**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

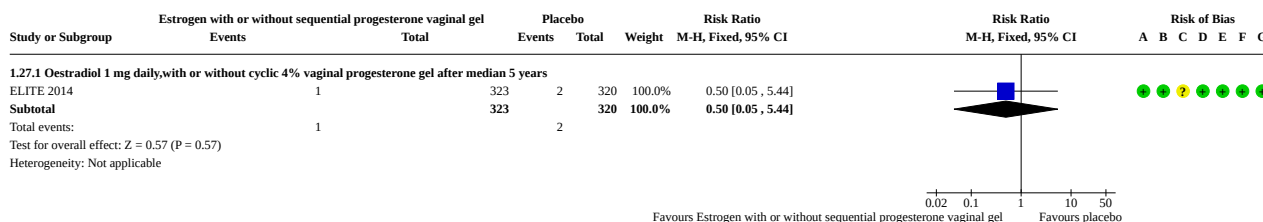
**Analysis 1.25. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 25: Transient ischaemic attack: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.26. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 26: Transient ischaemic attack: combined sequential HT**

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.27. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 27: Transient ischaemic attack: estrogen with or without sequential progesterone vaginal gel**

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

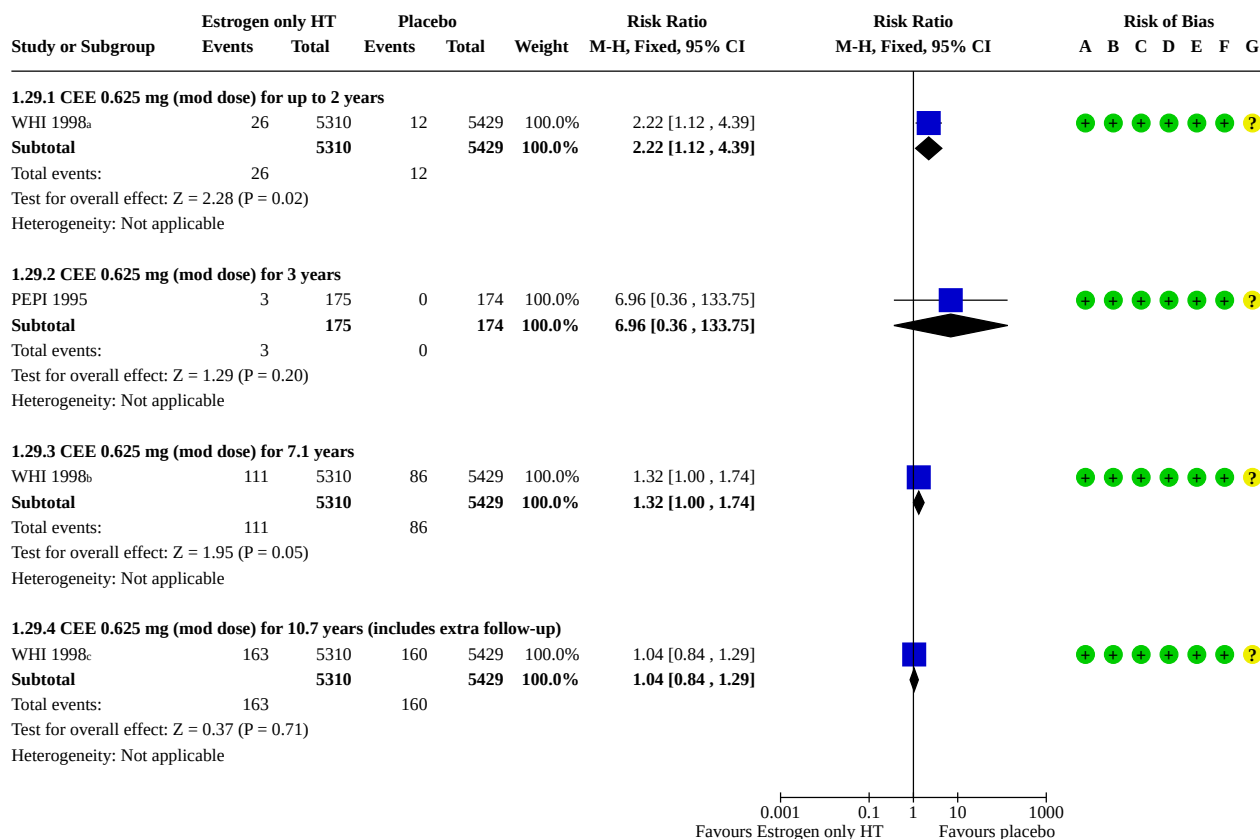
## Analysis 1.28. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 28: Stroke or transient ischaemic attack: combined continuous HT (moderate dose)



### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 1.29. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 29: Venous thromboembolism (DVT or PE): estrogen-only HT



### Footnotes

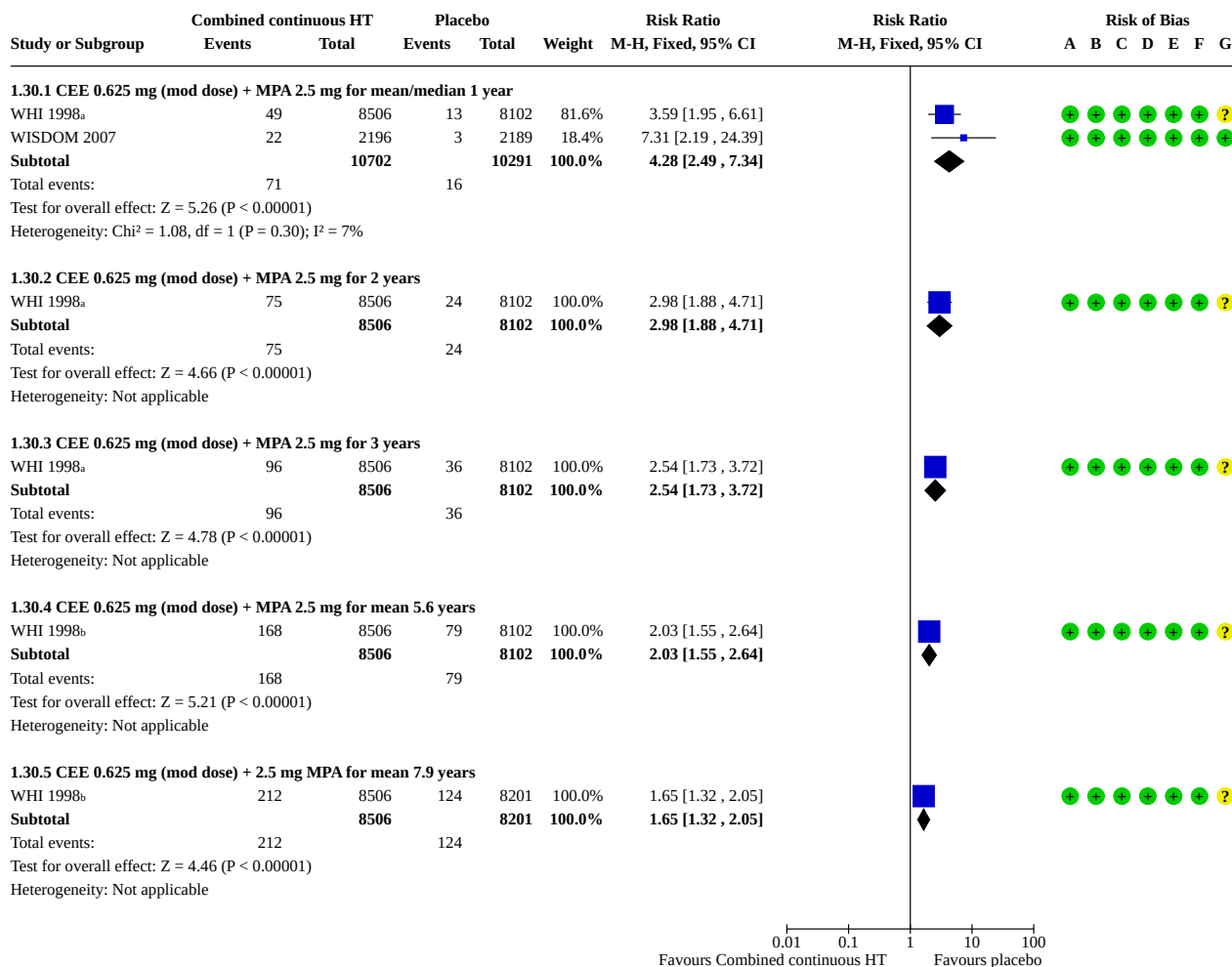
<sup>a</sup>Curb 2006

<sup>b</sup>LaCroix 2011

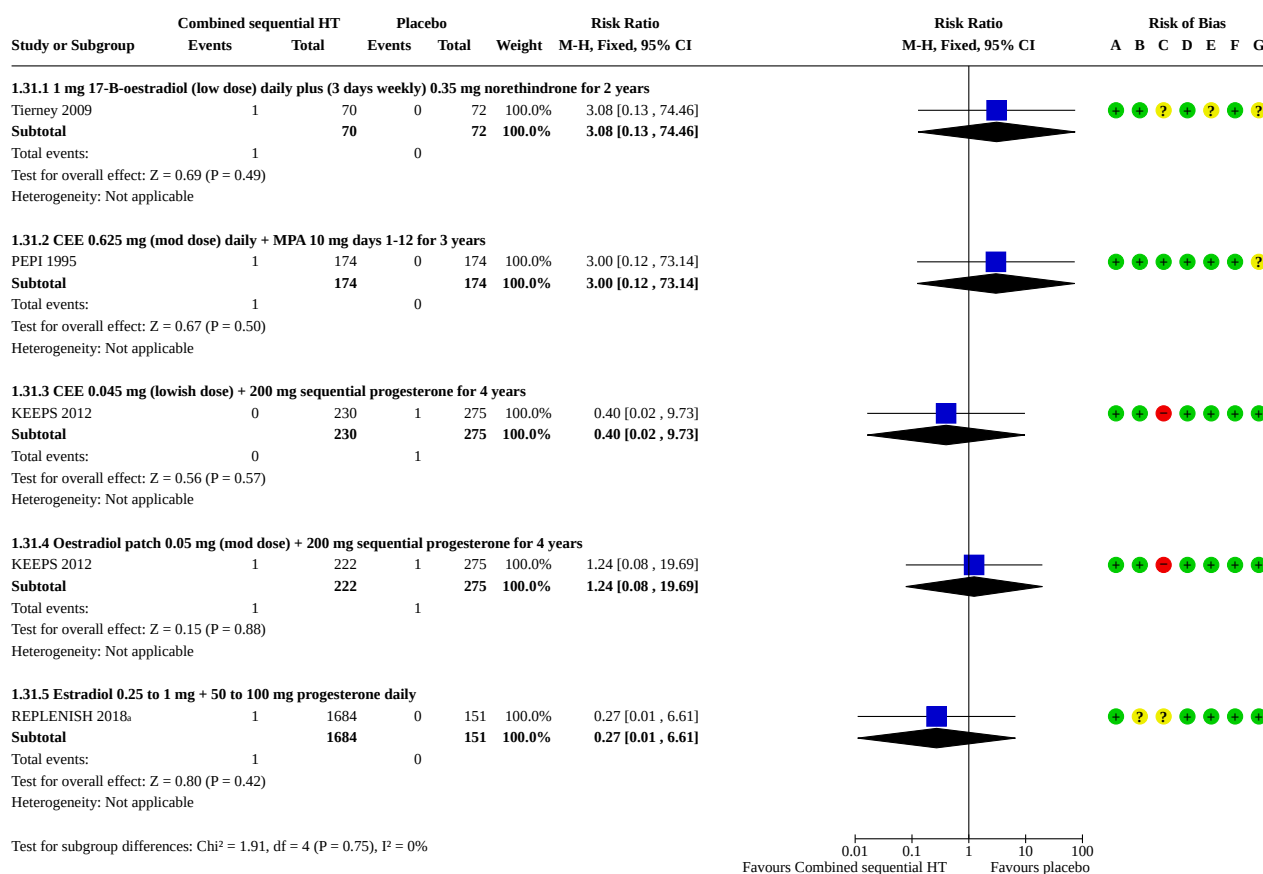
<sup>c</sup>LaCroix 2011: 7.1 yrs planned follow-up + extended follow-up in 78% of women.

### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.30. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 30: Venous thromboembolism (DVT or PE): combined continuous HT****Footnotes**<sup>a</sup>Rossouw 2002<sup>b</sup>Heiss 2008**Risk of bias legend**

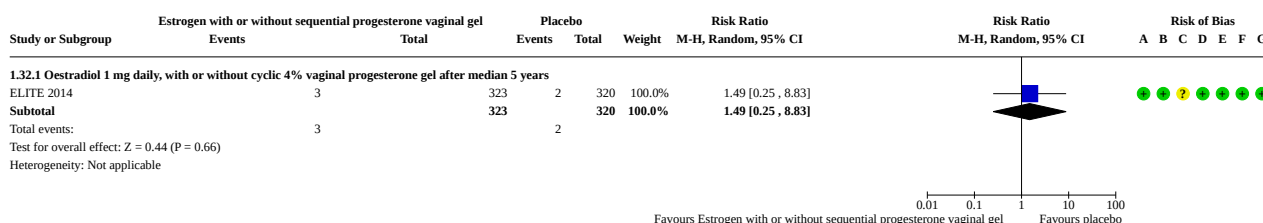
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.31. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 31: Venous thromboembolism (DVT or PE): combined sequential HT****Footnotes**

aDaily estradiol (mg) and progesterone (mg) (1/100, 0.5/100, 0.5/50, or 0.25/50 all combined)

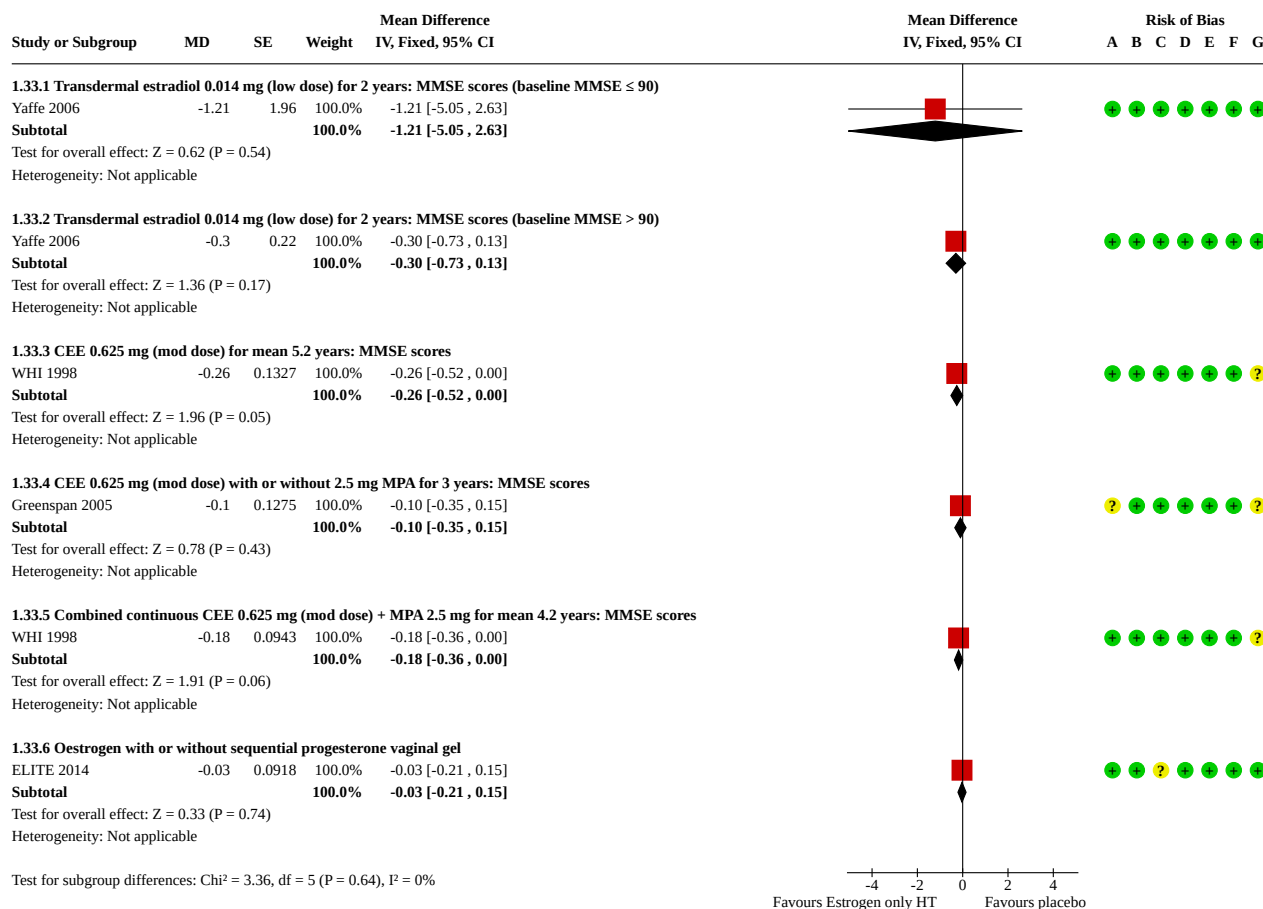
**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.32. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 32: Venous thromboembolism (DVT or PE): estrogen with or without sequential progesterone vaginal gel****Risk of bias legend**

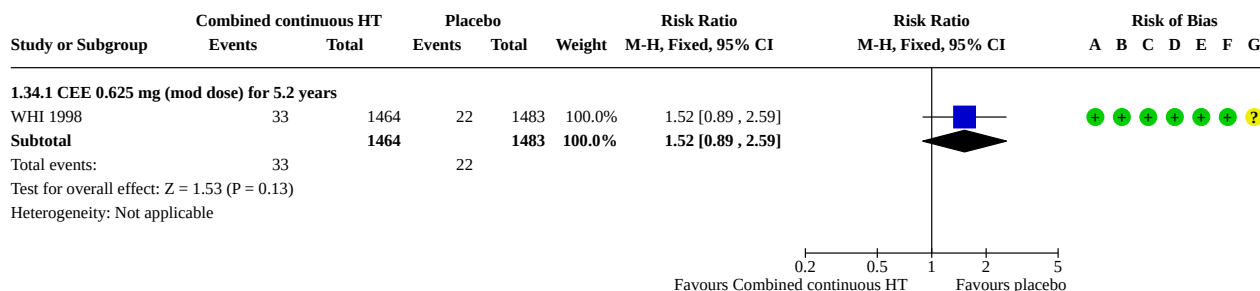
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# Analysis 1.33. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 33: Global cognitive function

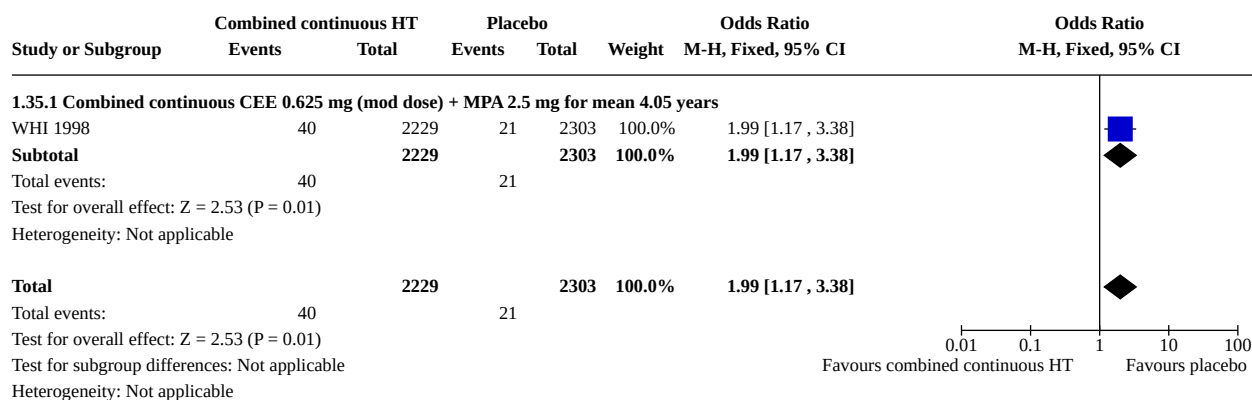


## Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.34. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 34: Probable dementia: Estrogen only****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.35. Comparison 1: Peri or postmenopausal people without major health problems , Outcome 35: Probable dementia: Combined continuous HT****Comparison 2. Peri or postmenopausal people with cardiovascular disease**

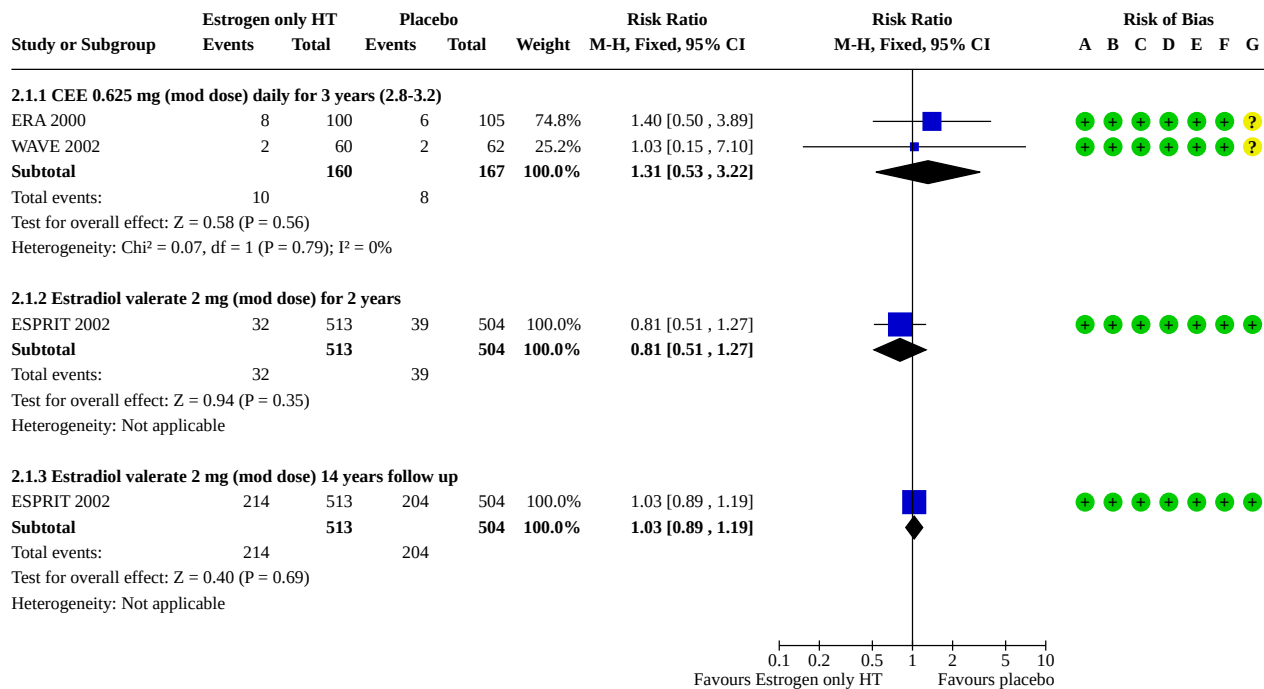
| Outcome or subgroup title                                   | No. of studies | No. of participants | Statistical method              | Effect size       |
|-------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 2.1 Death from any cause: estrogen-only HT                  | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.1.1 CEE 0.625 mg (mod dose) daily for 3 years (2.8-3.2)   | 2              | 327                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.31 [0.53, 3.22] |
| 2.1.2 Estradiol valerate 2 mg (mod dose) for 2 years        | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 0.81 [0.51, 1.27] |
| 2.1.3 Estradiol valerate 2 mg (mod dose) 14 years follow up | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.03 [0.89, 1.19] |

| Outcome or subgroup title                                                                                               | No. of studies | No. of participants | Statistical method              | Effect size       |
|-------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| <a href="#">2.2 Death from any cause: estrogen-only or combined sequential HT</a>                                       | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.2.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.14 [0.77, 1.67] |
| <a href="#">2.3 Death from any cause: combined continuous HT</a>                                                        | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.3.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2.8-3.2 years                                                            | 2              | 297                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.86 [0.28, 2.62] |
| 2.3.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                                                  | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.06 [0.84, 1.34] |
| 2.3.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-7 years UNBLINDED                                                      | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.14 [0.90, 1.44] |
| <a href="#">2.4 Death from coronary heart disease: estrogen-only HT</a>                                                 | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.4.1 CEE 0.625 mg (mod dose) daily for 2.8-3.2 years                                                                   | 2              | 327                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.31 [0.36, 4.77] |
| 2.4.2 Oestradiol valerate 2 mg (mod dose) for 2 years                                                                   | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 0.69 [0.40, 1.18] |
| <a href="#">2.5 Death from coronary heart disease: estrogen-only or combined sequential HT</a>                          | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.5.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.82 [0.37, 1.81] |
| <a href="#">2.6 Death from coronary heart disease: combined continuous HT</a>                                           | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.6.1 CEE 0.625 mg (mod dose) daily + MPA 2.5 mg for 1 year                                                             | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.55 [0.73, 3.29] |
| 2.6.2 CEE 0.625 mg (mod dose) daily + MPA 2.5 mg for 2 years                                                            | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.50 [0.90, 2.51] |
| 2.6.3 CEE 0.625 mg (mod dose) daily + MPA 2.5 mg for 3 years (2.8-3.2)                                                  | 3              | 3060                | Risk Ratio (M-H, Fixed, 95% CI) | 1.30 [0.88, 1.90] |
| 2.6.4 CEE 0.625 mg (mod dose) daily + MPA 2.5 mg for 4+ years (median 4.1)                                              | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.19 [0.85, 1.67] |
| 2.6.5 CEE 0.625 mg (mod dose) daily + MPA 2.5 mg for 4-6.8 years                                                        | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 0.99 [0.71, 1.39] |
| <a href="#">2.7 Death from stroke: estrogen-only or combined sequential HT</a>                                          | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |

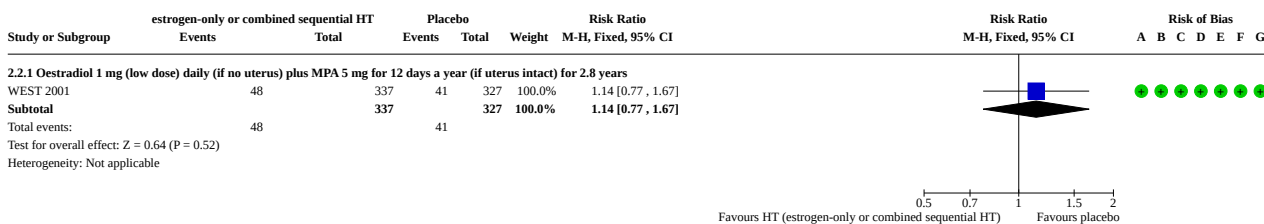
| Outcome or subgroup title                                                                                                | No. of studies | No. of participants | Statistical method              | Effect size       |
|--------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 2.7.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years  | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.91 [0.95, 8.93] |
| <b>2.8 Death from cancer: combined continuous HT</b>                                                                     | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.8.1 CEE 0.625 mg daily (mod dose) + MPA 2.5 mg for 4+ years (median 4.1)                                               | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.88 [0.49, 1.57] |
| 2.8.2 CEE 0.625 mg daily (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                               | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.51 [0.86, 2.65] |
| <b>2.9 Coronary event (MI or cardiac death): estrogen-only HT</b>                                                        | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.9.1 CEE 0.625 (mod dose) daily for 2.8-3.2 years                                                                       | 2              | 327                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.13 [0.54, 2.40] |
| 2.9.2 Oestradiol valerate 2 mg (mod dose) for 2 years                                                                    | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.00 [0.72, 1.39] |
| 2.9.3 Oestradiol valerate 2 mg (mod dose) 14 years follow up                                                             | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.21 [0.91, 1.62] |
| <b>2.10 Coronary event (MI or cardiac death): estrogen-only or combined sequential HT</b>                                | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.10.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.97 [0.57, 1.65] |
| <b>2.11 Coronary event (MI or cardiac death): combined continuous HT</b>                                                 | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.11.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 1 year                                                                   | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.50 [1.00, 2.25] |
| 2.11.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                                                  | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.20 [0.91, 1.58] |
| 2.11.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years (2.8-3.2)                                                        | 3              | 3060                | Risk Ratio (M-H, Fixed, 95% CI) | 1.07 [0.86, 1.33] |
| 2.11.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for median 4.1 years                                                         | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.99 [0.81, 1.19] |
| 2.11.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                                    | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.01 [0.78, 1.29] |
| <b>2.12 Stroke (first or recurrent): estrogen-only HT (mod dose)</b>                                                     | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 2.12.1 CEE 0.625 mg (mod dose) daily for 2.8 years                                                                       | 1              | 122                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.69 [0.12, 3.98] |

| Outcome or subgroup title                                                                                                | No. of studies | No. of participants | Statistical method              | Effect size         |
|--------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 2.12.2 Oestradiol valerate 2 mg (mod dose) for 2 years                                                                   | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.64 [0.60, 4.47]   |
| 2.12.3 Oestradiol valerate 2 mg (mod dose) 14 years follow up                                                            | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.13 [0.54, 2.36]   |
| <a href="#">2.13 Stroke (first or recurrent): estrogen-only HT or combined sequential</a>                                | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 2.13.1 Oestradiol 1 mg daily (low dose) (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.09 [0.79, 1.51]   |
| <a href="#">2.14 Stroke (first or recurrent): combined continuous HT (mod dose oestrogen)</a>                            | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 2.14.1 Continuous oestradiol 2 mg (mod dose) + norethisterone acetate 1 mg for 1.3 years                                 | 1              | 140                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.32 [0.01, 7.82]   |
| 2.14.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2.8 years                                                                | 1              | 88                  | Risk Ratio (M-H, Fixed, 95% CI) | 5.23 [0.26, 105.85] |
| 2.14.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for median 4.1 years                                                         | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.23 [0.90, 1.68]   |
| 2.14.4 CEE 0.625 mg (mod dose) + MPA for 4-6.8 years UNBLINDED                                                           | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.05 [0.71, 1.57]   |
| <a href="#">2.15 Transient ischaemic attack: estrogen-only HT (mod dose)</a>                                             | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 2.15.1 Oestradiol valerate 2 mg (mod dose) for 2 years                                                                   | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.13 [0.54, 2.36]   |
| <a href="#">2.16 Transient ischaemic attack: estrogen-only or combined sequential HT</a>                                 | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 2.16.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.16 [0.70, 1.94]   |
| <a href="#">2.17 Transient ischaemic attack: combined continuous HT</a>                                                  | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 2.17.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                                                  | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.80 [0.51, 1.23]   |
| 2.17.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                                    | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.49, 1.84]   |
| <a href="#">2.18 Stroke or transient ischaemic attack: estrogen-only HT</a>                                              | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |

| Outcome or subgroup title                                                                          | No. of studies | No. of participants | Statistical method              | Effect size        |
|----------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| 2.18.1 CEE 0.625 mg (mod dose) daily for 3.2 years                                                 | 1              | 205                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.88 [0.28, 2.78]  |
| <a href="#">2.19 Stroke or transient ischaemic attack: combined continuous HT</a>                  | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 2.19.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3.2 years                                          | 1              | 209                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.01 [0.34, 3.03]  |
| <a href="#">2.20 VTE (first or recurrent PE or DVT): estrogen-only HT</a>                          | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 2.20.1 Oestradiol valerate 2 mg (mod dose) for 2 years                                             | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 1.23 [0.33, 4.55]  |
| 2.20.2 CEE 0.625 mg (mod dose) daily for 2.8-3.2 years                                             | 2              | 327                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.64 [0.44, 6.17]  |
| <a href="#">2.21 Venous thromboembolism (first or recurrent PE or DVT): combined continuous HT</a> | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 2.21.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 1 year                                             | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 3.26 [1.06, 9.96]  |
| 2.21.2 Continuous oestradiol 2 mg (mod dose) + norethisterone acetate 1 mg for 1.3 years           | 1              | 140                 | Risk Ratio (M-H, Fixed, 95% CI) | 6.80 [0.86, 53.85] |
| 2.21.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                            | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 3.51 [1.42, 8.66]  |
| 2.21.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years (2.8-3.2)                                  | 3              | 3060                | Risk Ratio (M-H, Fixed, 95% CI) | 3.01 [1.50, 6.04]  |
| 2.21.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for median 4.1 years                                   | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 2.62 [1.39, 4.94]  |
| 2.21.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-7 years UNBLINDED                                | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.37 [0.63, 2.98]  |

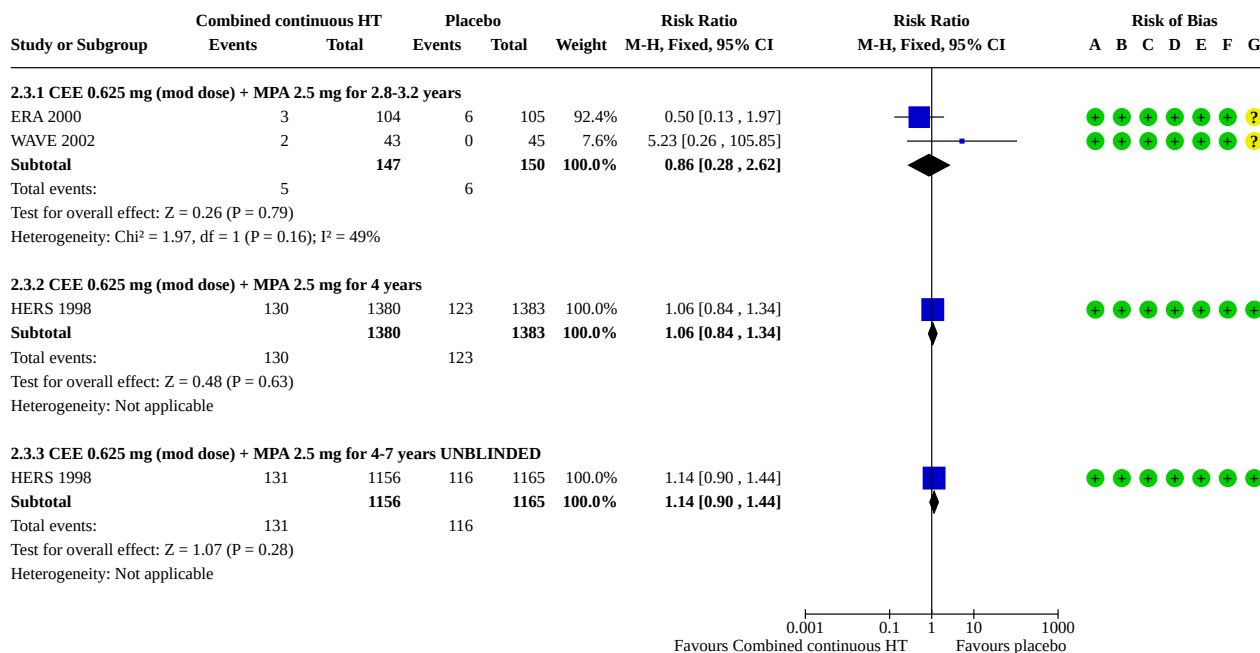
**Analysis 2.1. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 1: Death from any cause: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.2. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 2: Death from any cause: estrogen-only or combined sequential HT****Risk of bias legend**

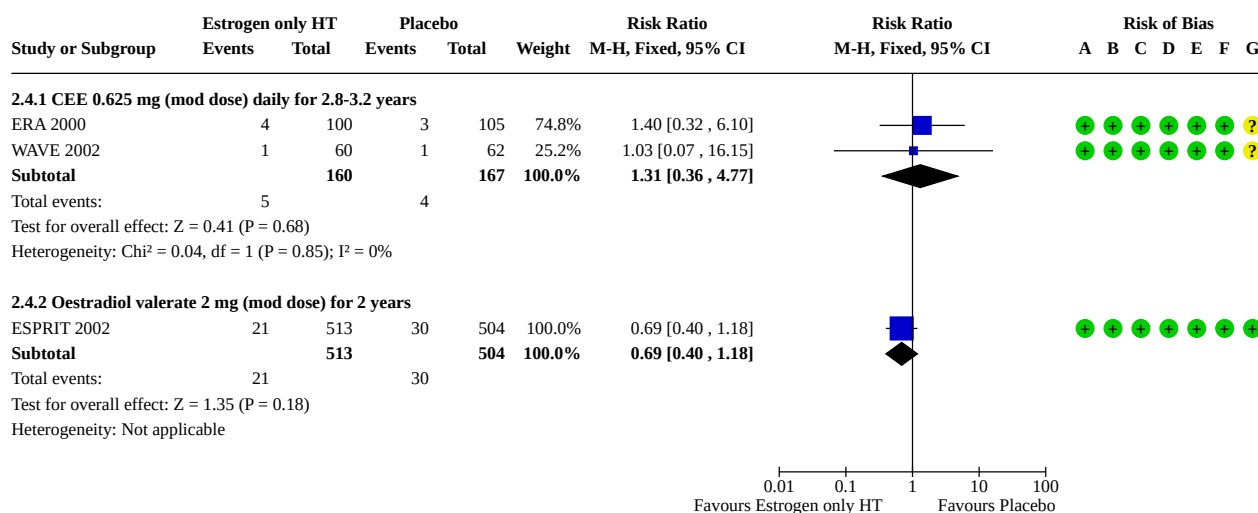
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

### Analysis 2.3. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 3: Death from any cause: combined continuous HT

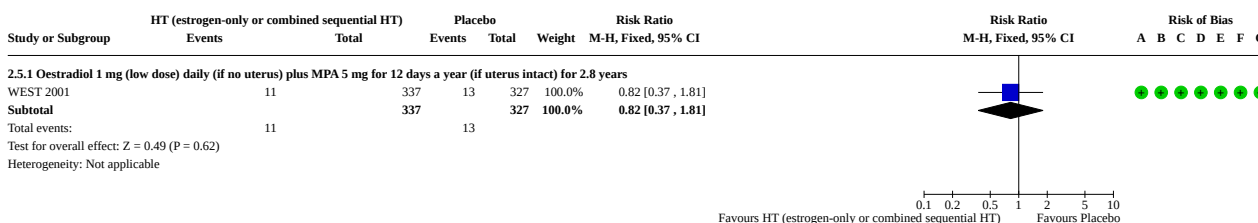


#### Risk of bias legend

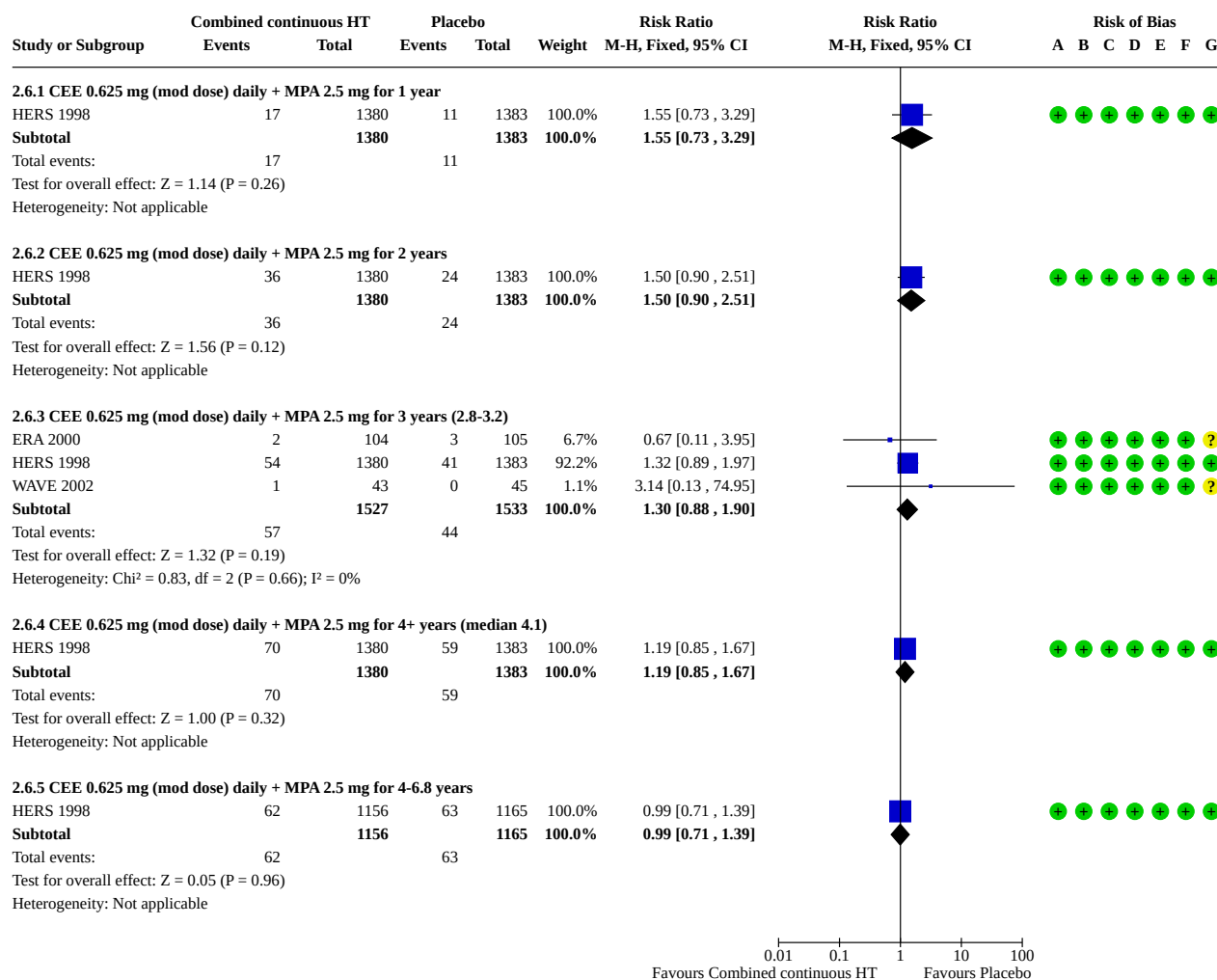
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.4. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 4: Death from coronary heart disease: estrogen-only HT****Risk of bias legend**

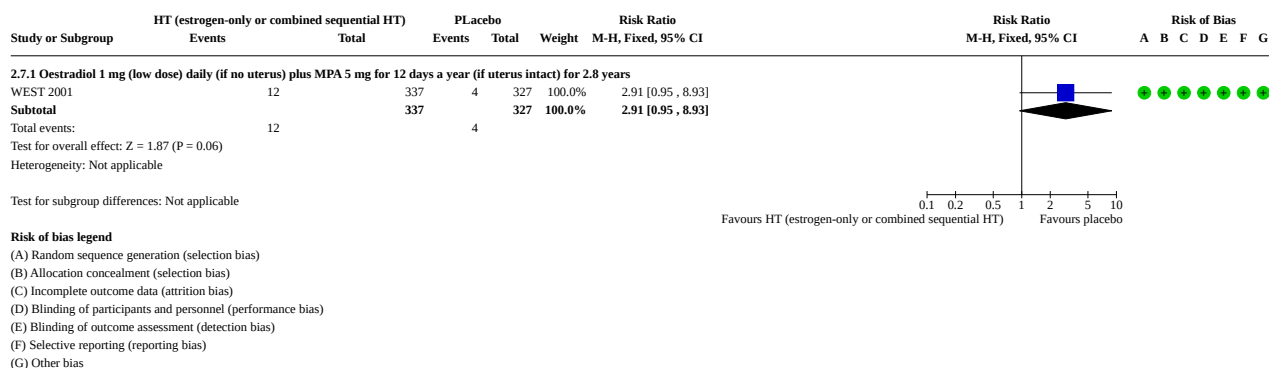
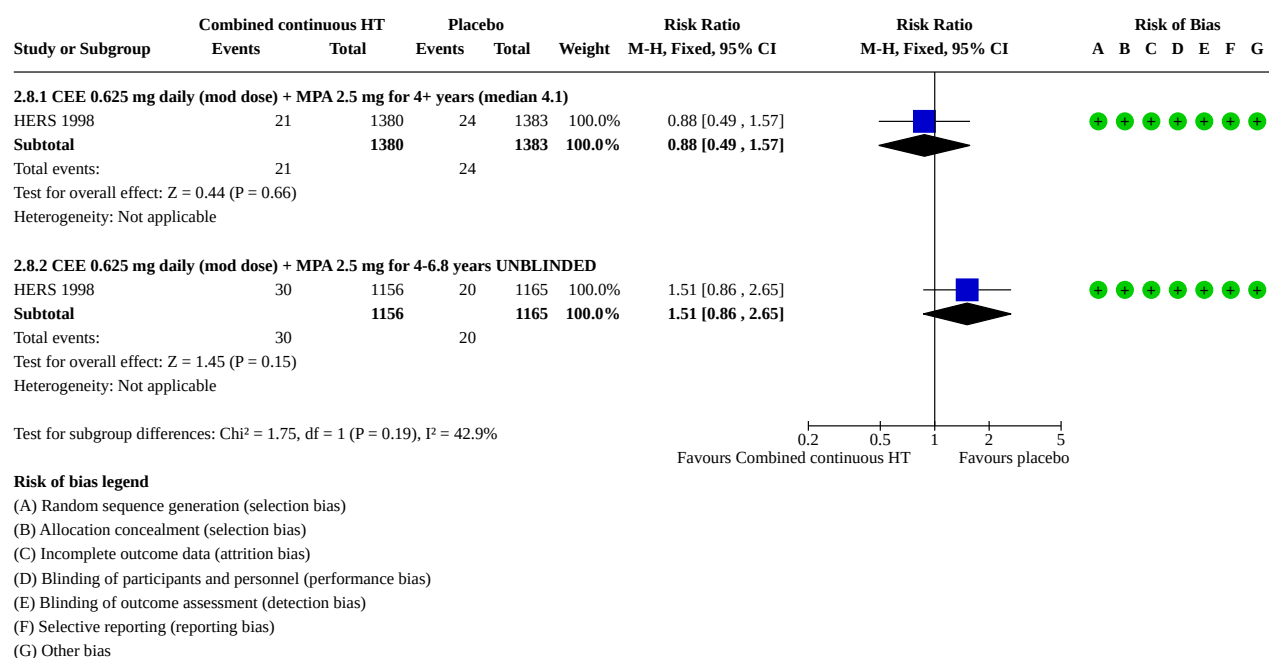
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

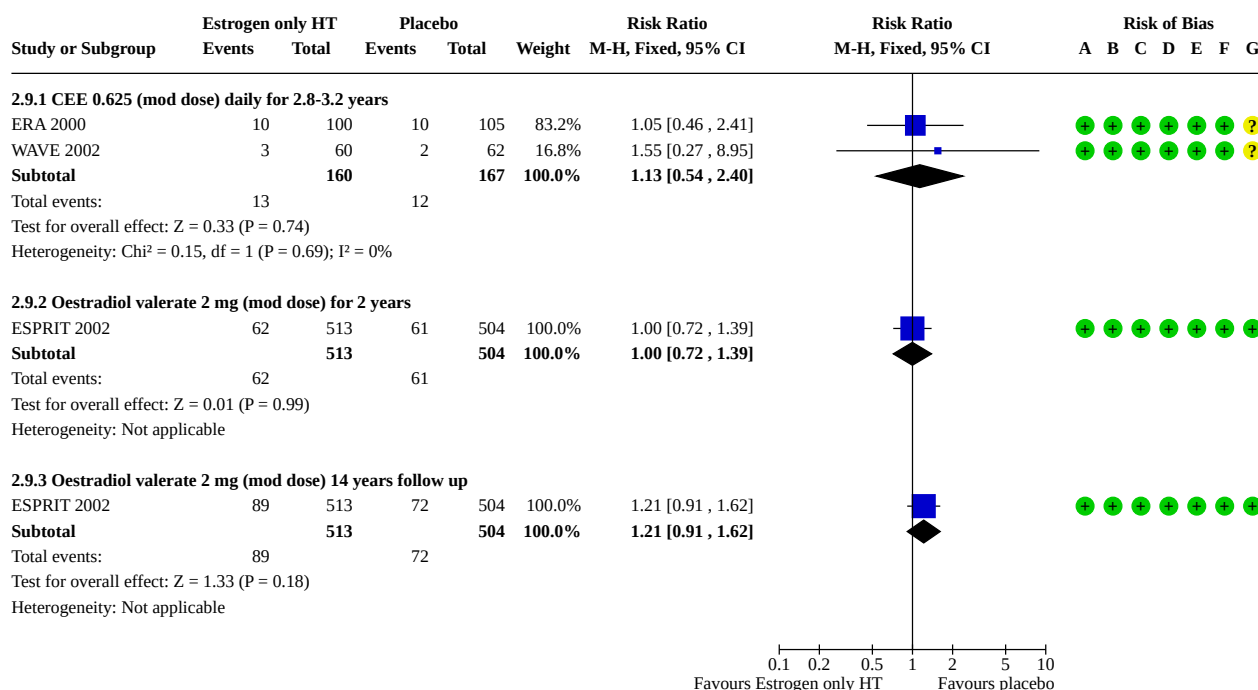
**Analysis 2.5. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 5: Death from coronary heart disease: estrogen-only or combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

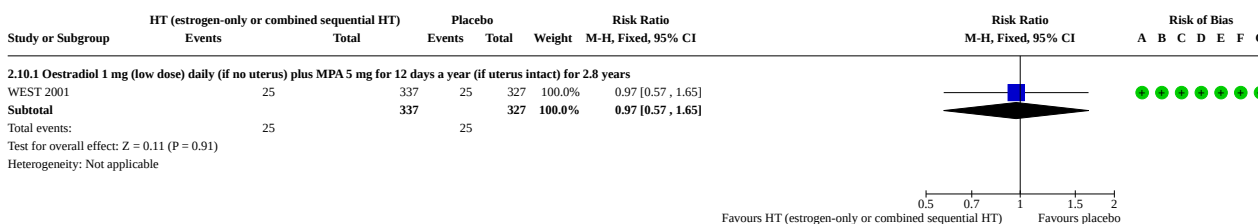
**Analysis 2.6. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 6: Death from coronary heart disease: combined continuous HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.7. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 7: Death from stroke: estrogen-only or combined sequential HT****Analysis 2.8. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 8: Death from cancer: combined continuous HT**

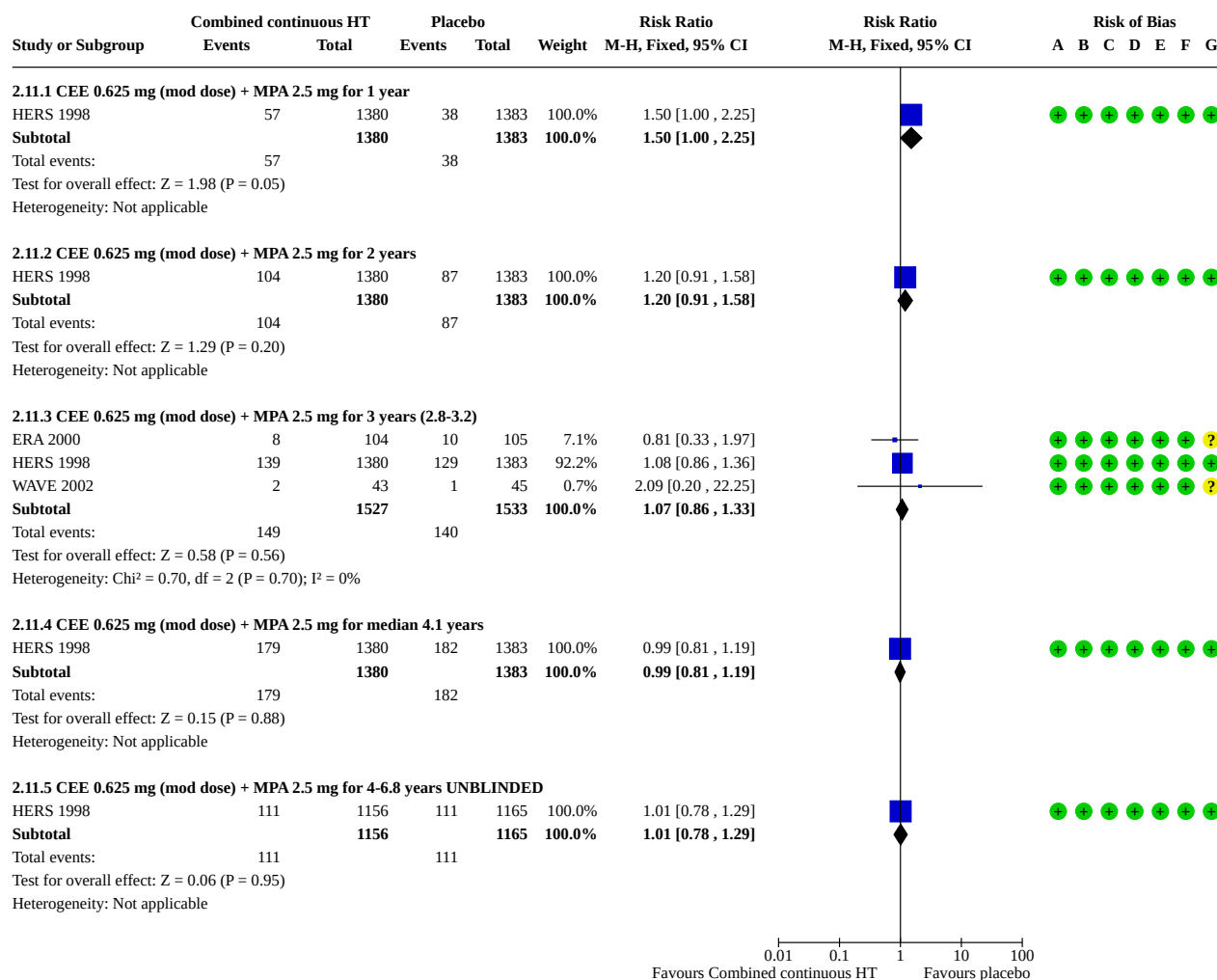
**Analysis 2.9. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 9: Coronary event (MI or cardiac death): estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.10. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 10: Coronary event (MI or cardiac death): estrogen-only or combined sequential HT****Risk of bias legend**

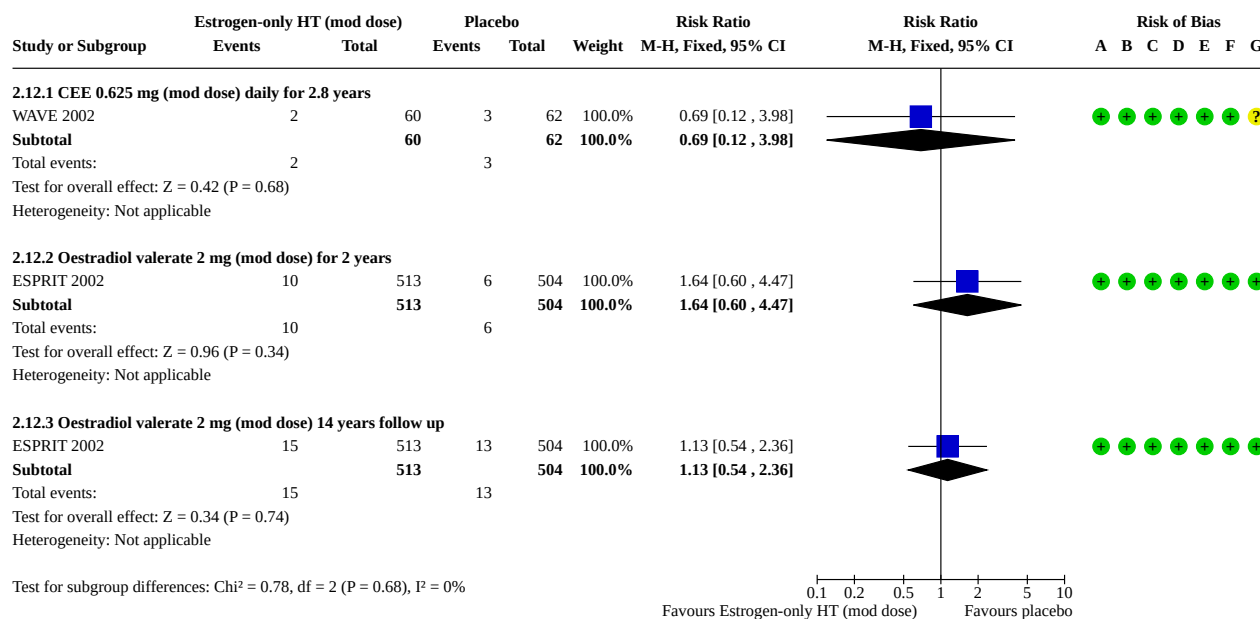
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 2.11. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 11: Coronary event (MI or cardiac death): combined continuous HT

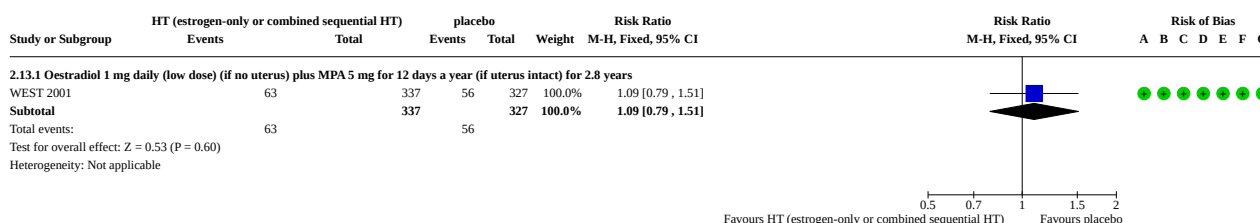


### Risk of bias legend

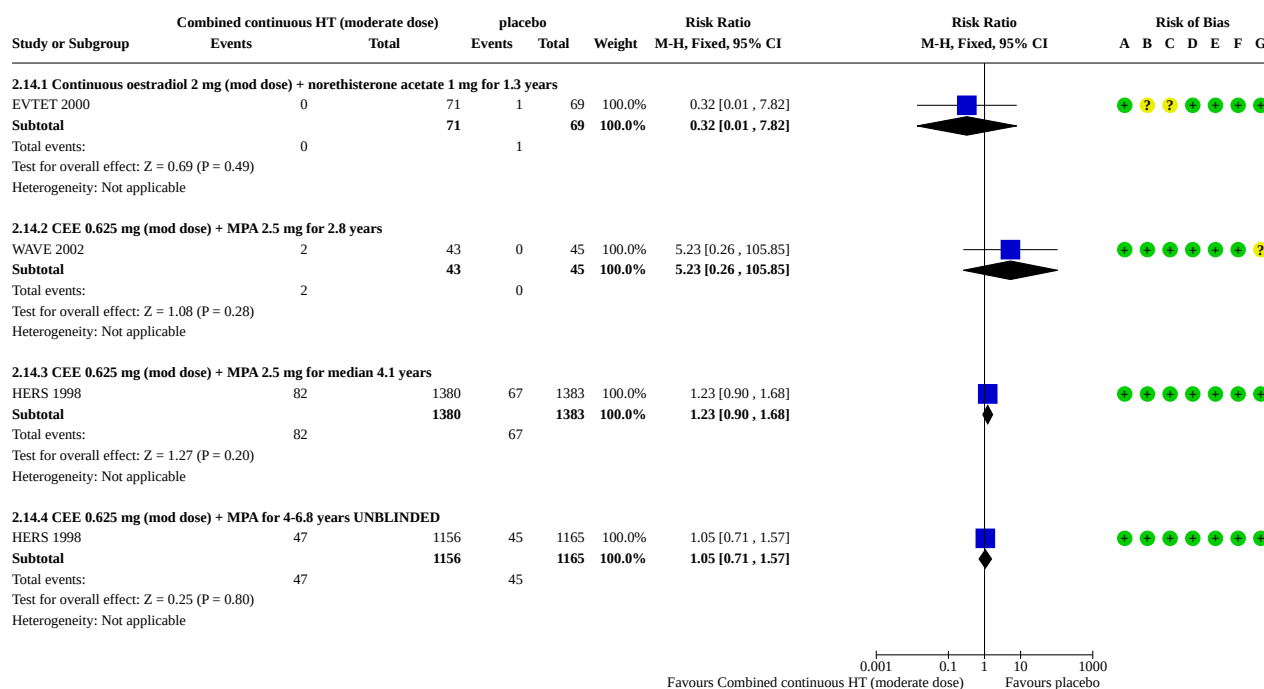
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.12. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 12: Stroke (first or recurrent): estrogen-only HT (mod dose)****Risk of bias legend**

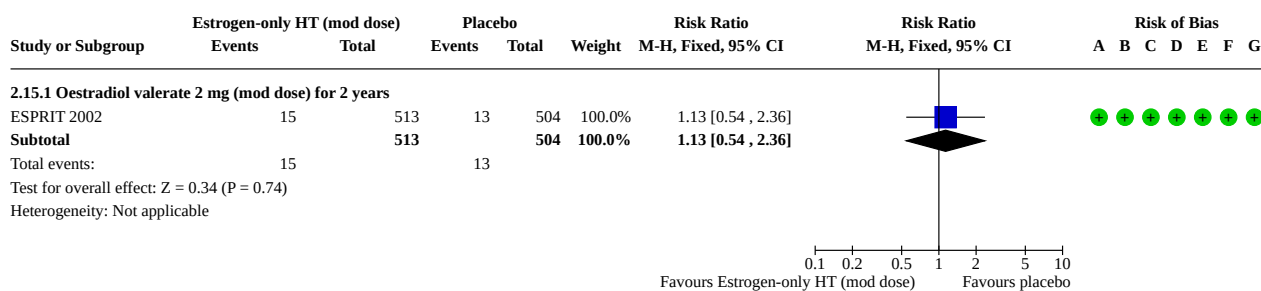
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.13. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 13: Stroke (first or recurrent): estrogen-only HT or combined sequential****Risk of bias legend**

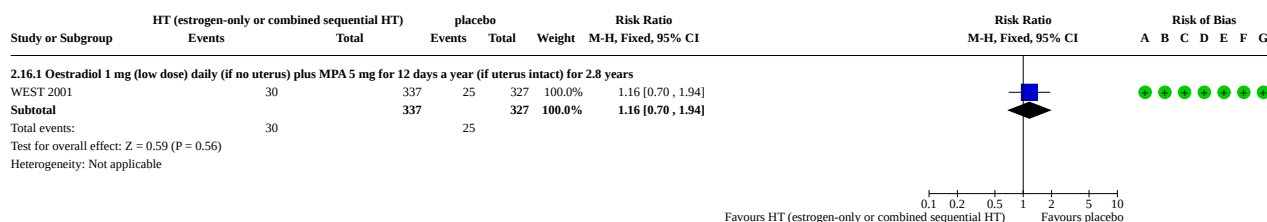
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.14. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 14: Stroke (first or recurrent): combined continuous HT (mod dose oestrogen)****Risk of bias legend**

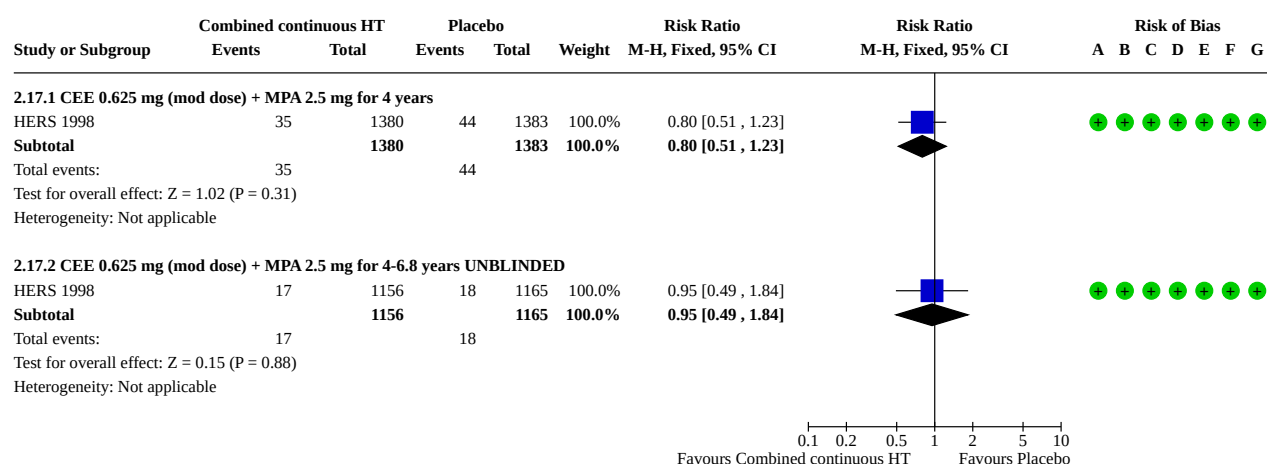
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.15. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 15: Transient ischaemic attack: estrogen-only HT (mod dose)****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

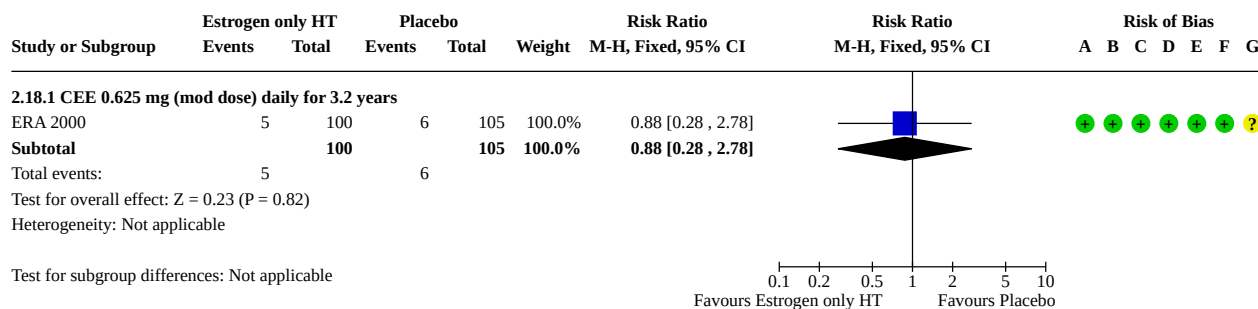
**Analysis 2.16. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 16: Transient ischaemic attack: estrogen-only or combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.17. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 17: Transient ischaemic attack: combined continuous HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

### Analysis 2.18. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 18: Stroke or transient ischaemic attack: estrogen-only HT



#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

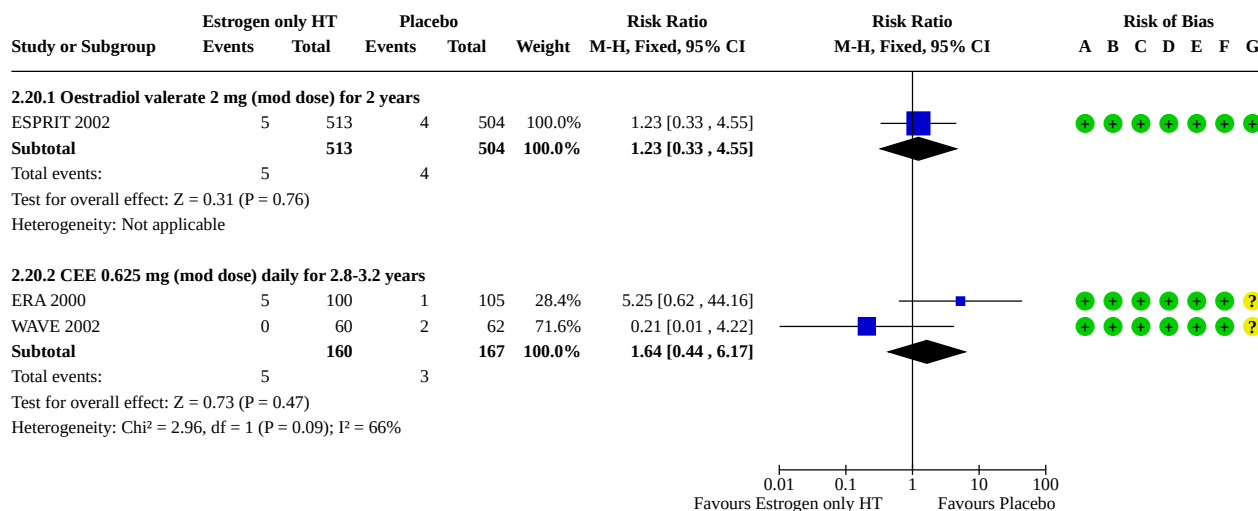
### Analysis 2.19. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 19: Stroke or transient ischaemic attack: combined continuous HT



#### Risk of bias legend

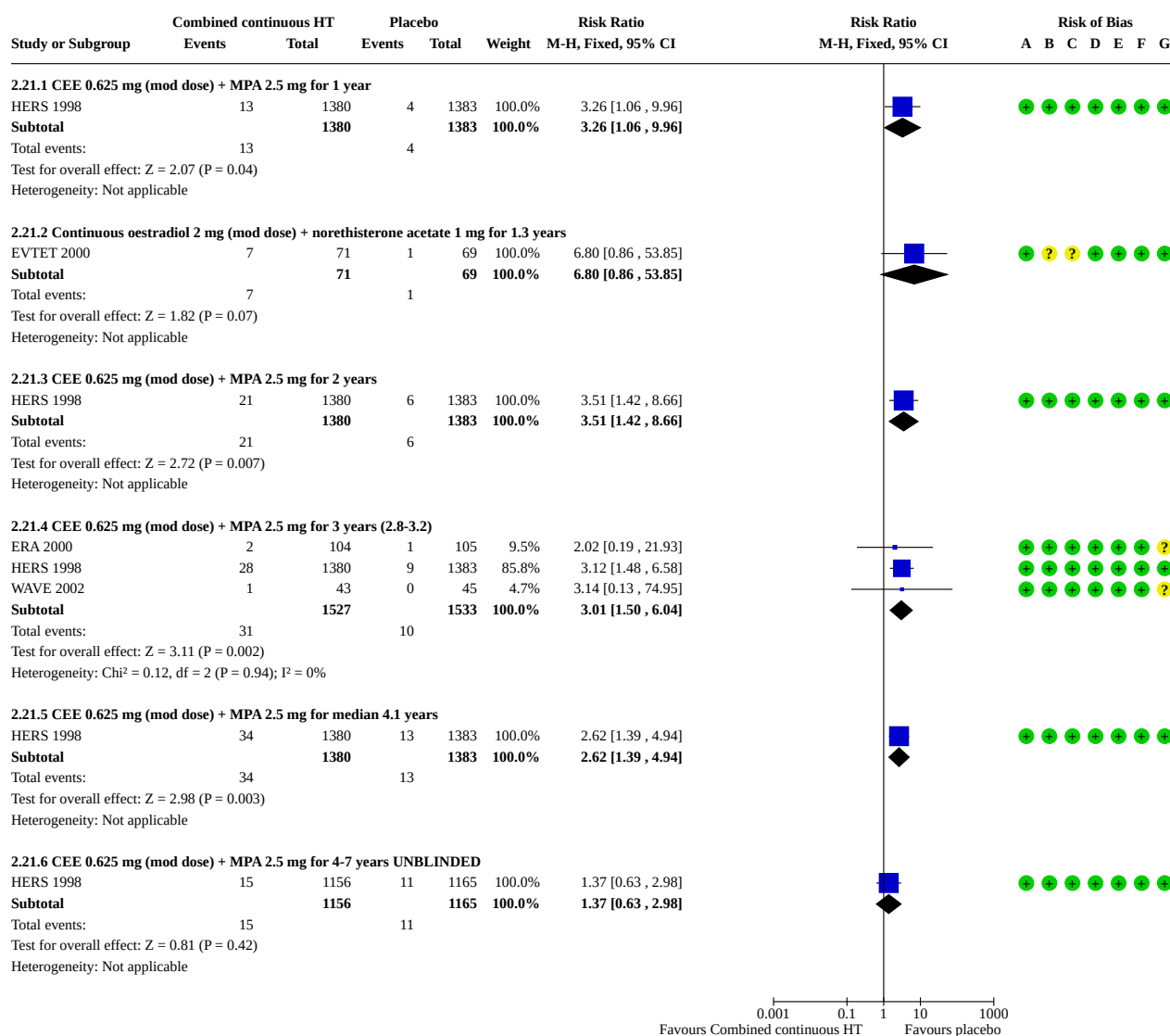
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# **Analysis 2.20. Comparison 2: Peri or postmenopausal people with cardiovascular disease , Outcome 20: VTE (first or recurrent PE or DVT): estrogen-only HT**



## **Risk of bias legend**

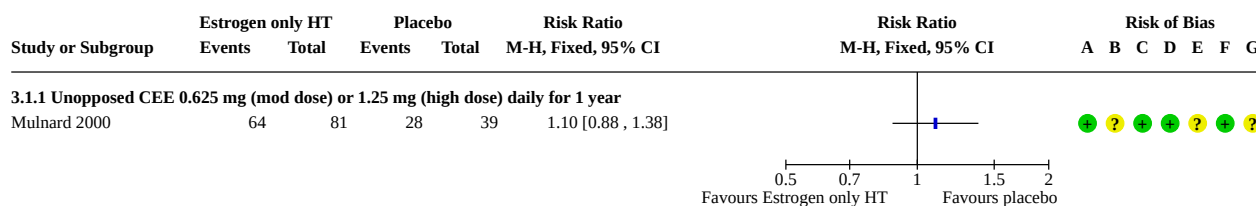
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.21. Comparison 2: Peri or postmenopausal people with cardiovascular disease ,  
Outcome 21: Venous thromboembolism (first or recurrent PE or DVT): combined continuous HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Comparison 3. Peri or postmenopausal people with dementia**

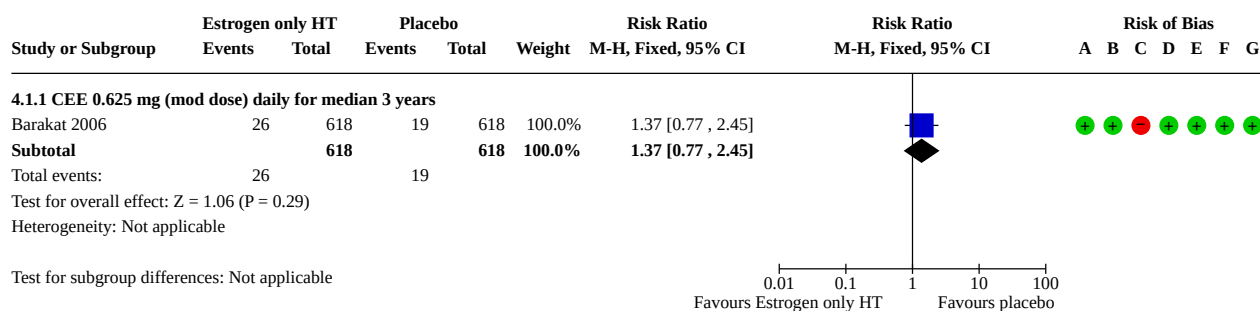
| Outcome or subgroup title                                                       | No. of studies | No. of participants | Statistical method              | Effect size         |
|---------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 3.1 Worsening of dementia on treatment (by ADCS-CGIC score): estrogen-only HT   | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| 3.1.1 Unopposed CEE 0.625 mg (mod dose) or 1.25 mg (high dose) daily for 1 year | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Totals not selected |

**Analysis 3.1. Comparison 3: Peri or postmenopausal people with dementia, Outcome 1: Worsening of dementia on treatment (by ADCS-CGIC score): estrogen-only HT****Risk of bias legend**

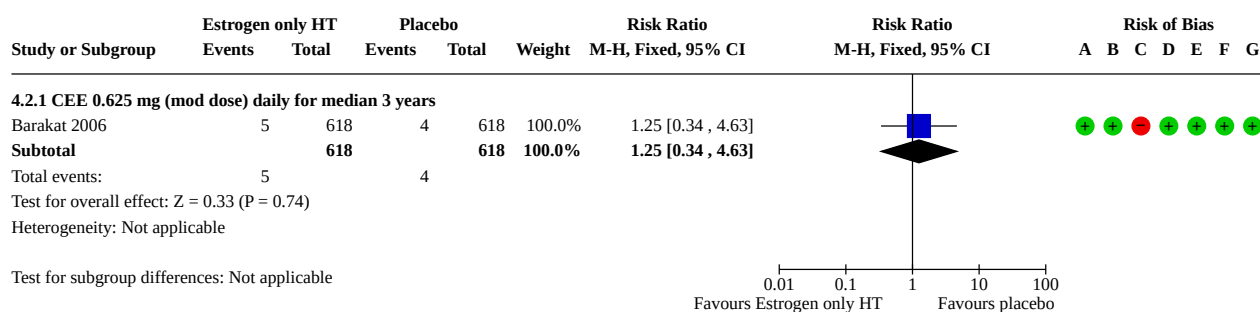
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Comparison 4. Peri or post menopausal people post surgery for early-stage endometrial cancer**

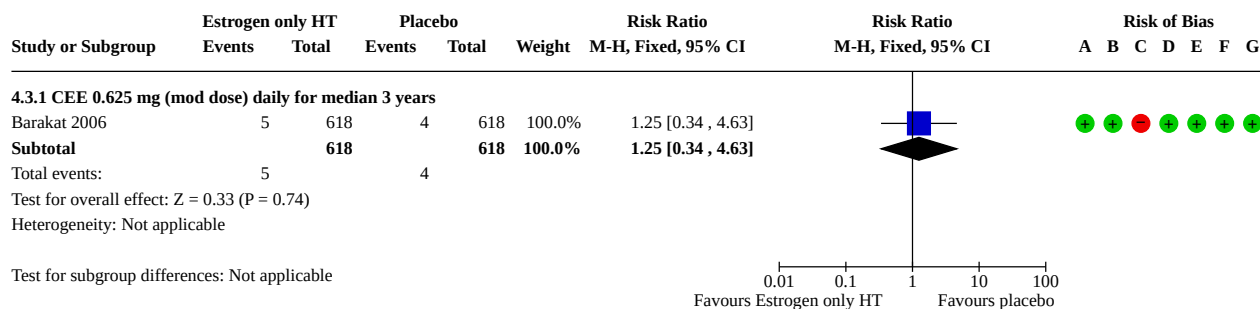
| Outcome or subgroup title                                  | No. of studies | No. of participants | Statistical method              | Effect size       |
|------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| <b>4.1 Death from any cause: estrogen-only HT</b>          | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 4.1.1 CEE 0.625 mg (mod dose) daily for median 3 years     | 1              | 1236                | Risk Ratio (M-H, Fixed, 95% CI) | 1.37 [0.77, 2.45] |
| <b>4.2 Death from endometrial cancer: estrogen-only HT</b> | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 4.2.1 CEE 0.625 mg (mod dose) daily for median 3 years     | 1              | 1236                | Risk Ratio (M-H, Fixed, 95% CI) | 1.25 [0.34, 4.63] |
| <b>4.3 Death from CHD: estrogen-only HT</b>                | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 4.3.1 CEE 0.625 mg (mod dose) daily for median 3 years     | 1              | 1236                | Risk Ratio (M-H, Fixed, 95% CI) | 1.25 [0.34, 4.63] |

**Analysis 4.1. Comparison 4: Peri or post menopausal people post surgery for early-stage endometrial cancer , Outcome 1: Death from any cause: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 4.2. Comparison 4: Peri or post menopausal people post surgery for early-stage endometrial cancer , Outcome 2: Death from endometrial cancer: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

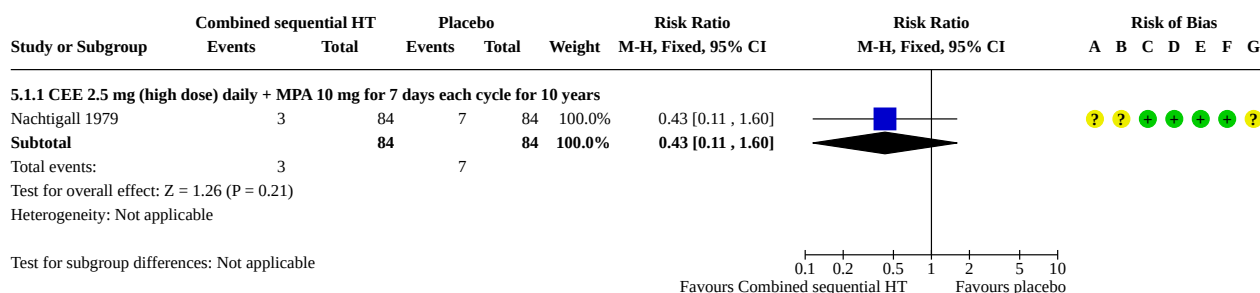
**Analysis 4.3. Comparison 4: Peri or post menopausal people post surgery for early-stage endometrial cancer , Outcome 3: Death from CHD: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Comparison 5. Peri or postmenopausal people hospitalised with chronic illness**

| Outcome or subgroup title                                                         | No. of studies | No. of participants | Statistical method              | Effect size       |
|-----------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 5.1 Death from any cause: combined sequential HT                                  | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 5.1.1 CEE 2.5 mg (high dose) daily + MPA 10 mg for 7 days each cycle for 10 years | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.43 [0.11, 1.60] |
| 5.2 Coronary events (myocardial infarction): combined sequential HT               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 5.2.1 CEE 2.5 mg (high dose) daily + MPA 10 mg for 7 days each cycle for 10 years | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.04, 3.14] |
| 5.3 Venous thromboembolism (DVT or PE): combined sequential HT                    | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 5.3.1 CEE 2.5 mg (high dose) daily + MPA 10 mg for 7 days each cycle for 10 years | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.07] |

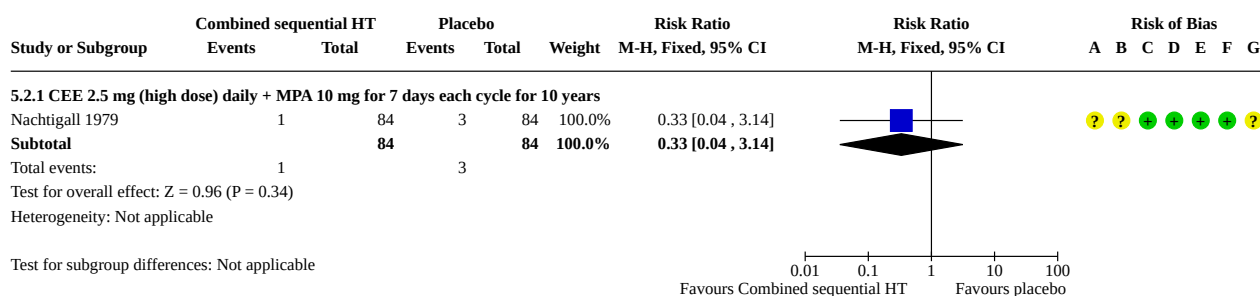
### Analysis 5.1. Comparison 5: Peri or postmenopausal people hospitalised with chronic illness , Outcome 1: Death from any cause: combined sequential HT



#### Risk of bias legend

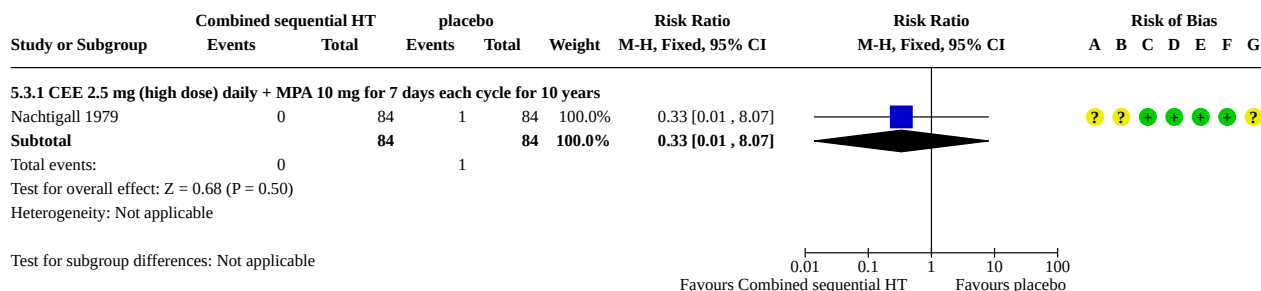
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

### Analysis 5.2. Comparison 5: Peri or postmenopausal people hospitalised with chronic illness , Outcome 2: Coronary events (myocardial infarction): combined sequential HT



#### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 5.3. Comparison 5: Peri or postmenopausal people hospitalised with chronic illness , Outcome 3: Venous thromboembolism (DVT or PE): combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Comparison 6. All peri or postmenopausal people**

| Outcome or subgroup title                                                 | No. of studies | No. of participants | Statistical method              | Effect size         |
|---------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 6.1 Breast cancer: estrogen-only HT                                       | 7              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.1.1 Oestrogen only HRT patch 0.025 (low dose) mg daily for 2 years      | 1              | 176                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.93 [0.12, 71.04]  |
| 6.1.2 Oestradiol 1 mg (low dose) for 2 years                              | 1              | 222                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.10]   |
| 6.1.3 Oestradiol valerate 2 mg (mod dose) for 2 years                     | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 0.98 [0.25, 3.91]   |
| 6.1.4 Oestradiol patch 0.075 mg (high dose) for 2 years                   | 1              | 176                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.93 [0.12, 71.04]  |
| 6.1.5 CEE 0.625 mg (mod dose) for 2.8-3.2 years                           | 3              | 676                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.05 [0.38, 11.04]  |
| 6.1.6 CEE 0.625 mg (mod dose) for 7.1 years                               | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.79 [0.61, 1.01]   |
| 6.1.7 CEE 0.625 mg (mod dose) after 10.7 years (includes extra follow-up) | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.78 [0.63, 0.96]   |
| 6.1.8 CEE 0.625 mg (mod dose) after 13 years (includes extra follow-up)   | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.80 [0.65, 0.97]   |
| 6.1.9 CEE 0.625 mg (mod dose) 20 years cumulative follow up               | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.82 [0.70, 0.97]   |
| 6.2 Breast cancer: estrogen-only or combined HT                           | 1              |                     | Odds Ratio (M-H, Fixed, 95% CI) | Totals not selected |

| Outcome or subgroup title                                                                  | No. of studies | No. of participants | Statistical method              | Effect size         |
|--------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 6.2.1 CEE 0.625 mg (mod dose) with or without 2.5 mg MPA for 3 years                       | 1              |                     | Odds Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| <b>6.3 Breast cancer: combined continuous HT</b>                                           | 6              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.3.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                          | 2              | 23182               | Risk Ratio (M-H, Fixed, 95% CI) | 0.52 [0.28, 0.96]   |
| 6.3.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                     | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.71 [0.47, 1.08]   |
| 6.3.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2.8-3.4 years                               | 3              | 17733               | Risk Ratio (M-H, Fixed, 95% CI) | 0.85 [0.62, 1.18]   |
| 6.3.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                     | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.36 [0.82, 2.27]   |
| 6.3.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 5.6 years                                   | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.27 [1.03, 1.56]   |
| 6.3.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-7 years unblinded                         | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.08 [0.52, 2.23]   |
| 6.3.7 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 7.9 years                                   | 1              | 16607               | Risk Ratio (M-H, Fixed, 95% CI) | 1.27 [1.07, 1.52]   |
| 6.3.8 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 11 years (includes extra follow-up)       | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.25 [1.08, 1.45]   |
| 6.3.9 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow up)  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.28 [1.11, 1.47]   |
| 6.3.10 CEE 0.625 mg (mod dose) + MPA 2.5 mg 20 years (cumulative follow up)                | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.24 [1.10, 1.40]   |
| 6.3.11 Estradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily for one year               | 1              | 1835                | Risk Ratio (M-H, Fixed, 95% CI) | 1.17 [0.07, 20.72]  |
| <b>6.4 Breast cancer: combined sequential HT</b>                                           | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.4.1 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                      | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.00 [0.18, 21.85]  |
| 6.4.2 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years | 1              | 352                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.91 [0.44, 34.64]  |
| 6.4.3 CEE 0.045 mg (lowish dose) + 200 mg sequential progesterone for 4 years              | 1              | 505                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.79 [0.30, 10.64]  |
| 6.4.4 Oestradiol patch 0.05 mg (mod dose) + 200 mg sequential progesterone for 4 years     | 1              | 497                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.86 [0.31, 11.02]  |

| Outcome or subgroup title                                                                            | No. of studies | No. of participants | Statistical method              | Effect size         |
|------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 6.4.5 CEE 2.5 mg daily (high dose) + MPA 10 mg for 7 days each cycle for 10 years                    | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.11 [0.01, 2.03]   |
| 6.4.6 Estradiol valerate 1mg =MPA 5 mg 12 days 1-12 at 5 years                                       | 1              | 153                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.48 [0.04, 5.19]   |
| <b>6.5 Breast cancer: estrogen with or without sequential progesterone vaginal gel</b>               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.5.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel after median 5 years | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.24 [0.50, 3.10]   |
| <b>6.6 Colorectal cancer: estrogen-only HT</b>                                                       | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.6.1 CEE 0.625 mg (mod dose) for 3 years                                                            | 1              | 349                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.08]   |
| 6.6.2 CEE 0.625 (mod dose) for 7.1 years                                                             | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.15 [0.81, 1.63]   |
| 6.6.3 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up)                              | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.11 [0.82, 1.49]   |
| <b>6.7 Colorectal cancer: estrogen-only or combined HT</b>                                           | 1              |                     | Odds Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| 6.7.1 CEE 0.625 mg (mod dose) with or without 2.5 mg MPA for 3 years                                 | 1              |                     | Odds Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| <b>6.8 Colorectal cancer: combined continuous HT</b>                                                 | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.8.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                    | 2              | 20993               | Risk Ratio (M-H, Fixed, 95% CI) | 0.68 [0.32, 1.42]   |
| 6.8.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                               | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.83 [0.46, 1.50]   |
| 6.8.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years                                               | 2              | 16956               | Risk Ratio (M-H, Fixed, 95% CI) | 0.81 [0.49, 1.34]   |
| 6.8.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                               | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.69 [0.32, 1.48]   |
| 6.8.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                        | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.64 [0.44, 0.91]   |
| 6.8.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                 | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 0.81 [0.46, 1.44]   |
| 6.8.7 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 7.9 years (includes extended follow-up)             | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.76 [0.57, 1.01]   |

| Outcome or subgroup title                                                                           | No. of studies | No. of participants | Statistical method              | Effect size        |
|-----------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| 6.8.8 CEE 0.0625 mg (mod dose) + MPA 2.5 mg after 11.6 years (includes extended follow-up)          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.78 [0.61, 0.99]  |
| 6.8.9 CEE 0.0625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow-up)          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.80 [0.63, 1.01]  |
| <b>6.9 Colorectal cancer: combined sequential HT</b>                                                | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.9.1 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                               | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.13]  |
| 6.9.2 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years          | 1              | 352                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 7.95]  |
| 6.9.3 CEE 2.5 mg (high dose) daily + MPA 10 mg for 7 days each cycle for 10 years                   | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.00 [0.06, 15.73] |
| <b>6.10 Colorectal cancer: estrogen with or without sequential progesterone vaginal gel</b>         | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.10.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel                    | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.49 [0.25, 8.83]  |
| <b>6.11 Lung cancer: estrogen-only HT (moderate dose)</b>                                           | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.11.1 Any lung cancer (non-small cell or small cell) at 7.1 years                                  | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 1.04 [0.73, 1.48]  |
| <b>6.12 Lung cancer: combined continuous HT (mod dose oestrogen)</b>                                | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.12.1 Any lung cancer (non-small cell or small cell) at 5.6 years                                  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.06 [0.77, 1.46]  |
| 6.12.2 Any lung cancer at 7.9 years                                                                 | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.22 [0.92, 1.62]  |
| 6.12.3 Any lung cancer after median 14 years (includes extended follow-up)                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.13 [0.93, 1.38]  |
| <b>6.13 Lung cancer: combined sequential HT</b>                                                     | 1              |                     | Odds Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.13.1 17-B-oestradiol 1 mg (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years | 1              | 142                 | Odds Ratio (M-H, Fixed, 95% CI) | 3.13 [0.13, 78.13] |
| <b>6.14 Endometrial cancer: estrogen-only HT</b>                                                    | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |

| Outcome or subgroup title                                                                   | No. of studies | No. of participants | Statistical method              | Effect size         |
|---------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 6.14.1 CEE 0.625 mg (mod dose) for 3-3.2 years                                              | 1              | 238                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.10]   |
| <a href="#">6.15 Endometrial cancer: combined continuous HT (moderate dose estrogen)</a>    | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.15.1 CEE 0.625 mg + MPA 2.5 mg for 1 year                                                 | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.13, 6.76]   |
| 6.15.2 CEE 0.625 mg + MPA 2.5 mg for 2 years                                                | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.31, 2.95]   |
| 6.15.3 CEE 0.625 mg + MPA 2.5 mg for 3-3.2 years                                            | 2              | 16847               | Risk Ratio (M-H, Fixed, 95% CI) | 0.80 [0.35, 1.82]   |
| 6.15.4 CEE 0.625 mg + MPA 2.5 mg for 4 years                                                | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.40 [0.08, 2.06]   |
| 6.15.5 CEE 0.625 mg + MPA 2.5 mg for 5.6 years                                              | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.86 [0.51, 1.44]   |
| 6.15.6 CEE 0.625 mg + MPA 2.5 mg for 4-6.8 years UNBLINDED                                  | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 0.14 [0.01, 2.78]   |
| 6.15.7 CEE 0.625 + MPS 2.5 mg for 7.9 years                                                 | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.81 [0.54, 1.20]   |
| 6.15.8 CEE 0.625 mg + MPA 2.5 mg after median 13 years (includes extended follow-up)        | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.66 [0.48, 0.90]   |
| 6.15.9 Estradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily for one year                | 1              | 1835                | Risk Ratio (M-H, Fixed, 95% CI) | Not estimable       |
| <a href="#">6.16 Endometrial cancer: combined sequential HT</a>                             | 4              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.16.1 17-B-oestradiol 1 mg (low dose) + dydrogesterone 5 mg days 14-28 for 2 years         | 1              | 163                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.90 [0.08, 45.95]  |
| 6.16.2 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years | 1              | 239                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.03]   |
| 6.16.3 Oestradiol 2 mg (mod dose) + dihydrogesterone 20 mg for 2 years                      | 1              | 159                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.30 [0.16, 67.59]  |
| 6.16.4 CEE 0.045 mg (lowish dose) + 200 mg sequential progesterone for 4 years              | 1              | 505                 | Risk Ratio (M-H, Fixed, 95% CI) | 5.97 [0.29, 123.81] |
| 6.16.5 Oestradiol patch 0.05 mg (mod dose) + 200 mg sequential progesterone for 4 years     | 1              | 497                 | Risk Ratio (M-H, Fixed, 95% CI) | 3.71 [0.15, 90.70]  |
| 6.16.6 CEE 2.5 mg (high dose) daily + MPA 10 mg for 7 days each cycle for 10 years          | 1              | 168                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.07]   |

| Outcome or subgroup title                                                                         | No. of studies | No. of participants | Statistical method              | Effect size        |
|---------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|--------------------|
| <a href="#">6.17 Recurrent endometrial cancer: estrogen-only HT</a>                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.17.1 CEE 0.625 mg (mod dose) for median 3 years                                                 | 1              | 1236                | Risk Ratio (M-H, Fixed, 95% CI) | 1.17 [0.54, 2.50]  |
| <a href="#">6.18 Ovarian cancer: combined continuous HT</a>                                       | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.18.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 5.6 years                                         | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.43 [0.76, 2.69]  |
| 6.18.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg after 13.2 years (includes extended follow-up)        | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 1.23 [0.82, 1.85]  |
| <a href="#">6.19 Ovarian cancer: estrogen with or without sequential progesterone vaginal gel</a> | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.19.1 Oestradiol 1 mg daily, with or without cyclic 4% vaginal progesterone gel                  | 1              | 643                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.33 [0.01, 8.08]  |
| <a href="#">6.20 Gallbladder disease requiring surgery: estrogen-only HT</a>                      | 3              | 8930                | Risk Ratio (M-H, Fixed, 95% CI) | 1.75 [1.40, 2.19]  |
| 6.20.1 CEE 0.625 mg (mod dose) for 3-3.2 years                                                    | 2              | 554                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.77 [0.17, 3.39]  |
| 6.20.2 CEE 0.625 mg (mod dose) for 7.1 years                                                      | 1              | 8376                | Risk Ratio (M-H, Fixed, 95% CI) | 1.78 [1.42, 2.24]  |
| <a href="#">6.21 Gallbladder disease requiring surgery: combined continuous HT</a>                | 4              | 17013               | Risk Ratio (M-H, Fixed, 95% CI) | 1.55 [1.29, 1.86]  |
| <a href="#">6.22 Gallbladder disease requiring surgery: combined sequential HT</a>                | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.22.1 CEE 0.625 mg (mod dose) daily + MPA 10 mg days 1-12 for 3 years                            | 1              | 348                 | Risk Ratio (M-H, Fixed, 95% CI) | 2.00 [0.37, 10.78] |
| 6.22.2 CEE 0.625 mg (mod dose) daily + micronised progesterone 200 mg days 1-12 for 3 years       | 1              | 352                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.47 [0.25, 8.67]  |
| <a href="#">6.23 Hip fractures: estrogen-only HT</a>                                              | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only     |
| 6.23.1 CEE 0.625 mg (mod dose) for 7.1 years                                                      | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.66 [0.46, 0.95]  |
| 6.23.2 CEE 0.625 mg (mod dose) for 10.7 years (includes extra follow-up)                          | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.92 [0.71, 1.18]  |
| 6.23.3 CEE 0.625 mg (mod dose) after 13.2 years (includes extended follow-up)                     | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.93 [0.74, 1.17]  |

| Outcome or subgroup title                                                                                                | No. of studies | No. of participants | Statistical method              | Effect size       |
|--------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| <a href="#">6.24 Hip fractures: estrogen-only or combined sequential HT</a>                                              | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.24.1 Oestradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.62 [0.27, 1.42] |
| <a href="#">6.25 Hip fractures: combined continuous HT</a>                                                               | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.25.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean/median 1 year                                                       | 2              | 20993               | Risk Ratio (M-H, Fixed, 95% CI) | 0.64 [0.26, 1.57] |
| 6.25.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 2 years                                                                  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.61 [0.31, 1.18] |
| 6.25.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3 years                                                                  | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.70 [0.42, 1.17] |
| 6.25.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                                                  | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 1.16 [0.55, 2.42] |
| 6.25.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 5.6 years                                                                | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.67 [0.47, 0.96] |
| 6.25.6 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                                    | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 2.10 [1.06, 4.16] |
| 6.25.7 CEE 0.625 mg (mod dose) + 2.5 mg MPA for 7.9 years                                                                | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.77 [0.60, 0.99] |
| 6.25.8 CEE 0.625 mg (mod dose) + 2.5 mg MPA after 13.2 years (includes extended follow-up)                               | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.82 [0.69, 0.97] |
| <a href="#">6.26 Hip fractures: combined sequential HT</a>                                                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.26.1 17-B-oestradiol 1 mg (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years                      | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.34 [0.01, 8.27] |
| <a href="#">6.27 Vertebral fractures: estrogen-only HT</a>                                                               | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.27.1 CEE 0.625 mg (mod dose) for 7.1 years                                                                             | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.64 [0.44, 0.94] |
| <a href="#">6.28 Vertebral fractures: combined continuous HT</a>                                                         | 2              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.28.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                                                  | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.74 [0.37, 1.47] |

| Outcome or subgroup title                                                                                               | No. of studies | No. of participants | Statistical method              | Effect size         |
|-------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|---------------------|
| 6.28.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.68 [0.49, 0.96]   |
| 6.28.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                                   | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.10 [0.49, 2.48]   |
| 6.28.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 7.9 years                                                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.78 [0.60, 1.01]   |
| <b>6.29 All clinical fractures: estrogen-only HT (moderate dose)</b>                                                    | 3              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.29.1 Oestradiol valerate 2 mg (mod dose) for 2 years                                                                  | 1              | 1017                | Risk Ratio (M-H, Fixed, 95% CI) | 0.60 [0.29, 1.26]   |
| 6.29.2 CEE 0.625 mg (mod dose) daily for 3.2 years                                                                      | 1              | 205                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.42 [0.17, 1.04]   |
| 6.29.3 CEE 0.625 mg (mod dose) for 7.1 years                                                                            | 1              | 10739               | Risk Ratio (M-H, Fixed, 95% CI) | 0.73 [0.65, 0.80]   |
| <b>6.30 All clinical fractures: estrogen-only or combined HT</b>                                                        | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| 6.30.1 CEE 0.625 mg (mod dose) with or without 2.5 mg MPA for 3 years                                                   | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Totals not selected |
| <b>6.31 All clinical fractures: estrogen-only or combined sequential HT</b>                                             | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.31.1 Estradiol 1 mg (low dose) daily (if no uterus) plus MPA 5 mg for 12 days a year (if uterus intact) for 2.8 years | 1              | 664                 | Risk Ratio (M-H, Fixed, 95% CI) | 1.23 [0.68, 2.19]   |
| <b>6.32 All clinical fractures: combined continuous HT</b>                                                              | 5              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only      |
| 6.32.1 CEE 0.625 mg (mod dose) + MPA 2.5 mg for median 1 year                                                           | 1              | 4385                | Risk Ratio (M-H, Fixed, 95% CI) | 0.69 [0.46, 1.02]   |
| 6.32.2 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 3.2-3.4 years                                                           | 2              | 986                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.52 [0.32, 0.87]   |
| 6.32.3 CEE 0.625 mg (mod dose) + MPA 2.5 mg for mean 5.6 years                                                          | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.78 [0.71, 0.86]   |
| 6.32.4 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4 years                                                                 | 1              | 2763                | Risk Ratio (M-H, Fixed, 95% CI) | 0.95 [0.76, 1.18]   |
| 6.32.5 CEE 0.625 mg (mod dose) + MPA 2.5 mg for 4-6.8 years UNBLINDED                                                   | 1              | 2321                | Risk Ratio (M-H, Fixed, 95% CI) | 1.23 [0.91, 1.65]   |
| 6.32.6 CEE 0.0625 mg (mod dose) + MPA 2.5 mg for mean 7.9 years                                                         | 1              | 16608               | Risk Ratio (M-H, Fixed, 95% CI) | 0.82 [0.76, 0.89]   |

| Outcome or subgroup title                                                                           | No. of studies | No. of participants | Statistical method              | Effect size       |
|-----------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------------------|-------------------|
| 6.33 All clinical fractures: combined sequential HT                                                 | 1              |                     | Risk Ratio (M-H, Fixed, 95% CI) | Subtotals only    |
| 6.33.1 17-B-oestradiol 1 mg (low dose) daily plus (3 days weekly) 0.35 mg norethindrone for 2 years | 1              | 142                 | Risk Ratio (M-H, Fixed, 95% CI) | 0.44 [0.12, 1.64] |

## Analysis 6.1. Comparison 6: All peri or postmenopausal people , Outcome 1: Breast cancer: estrogen-only HT

| Study or Subgroup                                                         | Estrogen only HT |       | Placebo |       | Weight | Risk Ratio          | Risk Ratio<br>M-H, Fixed, 95% CI                                                      | Risk of Bias                                                                          |   |   |   |   |   |   |
|---------------------------------------------------------------------------|------------------|-------|---------|-------|--------|---------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---|---|---|---|---|---|
|                                                                           | Events           | Total | Events  | Total |        | M-H, Fixed, 95% CI  |                                                                                       | A                                                                                     | B | C | D | E | F | G |
| 6.1.1 Oestrogen only HRT patch 0.025 (low dose) mg daily for 2 years      |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Notelovitz 2002                                                           | 1                | 89    | 0       | 87    | 100.0% | 2.93 [0.12 , 71.04] |    |    |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 89    |         | 87    | 100.0% | 2.93 [0.12 , 71.04] |    |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 1                |       | 0       |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 0.66 (P = 0.51)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.2 Oestradiol 1 mg (low dose) for 2 years                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| EPAT 2001                                                                 | 0                | 111   | 1       | 111   | 100.0% | 0.33 [0.01 , 8.10]  |     |    |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 111   |         | 111   | 100.0% | 0.33 [0.01 , 8.10]  |     |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 0                |       | 1       |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 0.68 (P = 0.50)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.3 Oestradiol valerate 2 mg (mod dose) for 2 years                     |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| ESPRIT 2002                                                               | 4                | 513   | 4       | 504   | 100.0% | 0.98 [0.25 , 3.91]  |    |    |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 513   |         | 504   | 100.0% | 0.98 [0.25 , 3.91]  |    |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 4                |       | 4       |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 0.03 (P = 0.98)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.4 Oestradiol patch 0.075 mg (high dose) for 2 years                   |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Notelovitz 2002                                                           | 1                | 89    | 0       | 87    | 100.0% | 2.93 [0.12 , 71.04] |    |    |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 89    |         | 87    | 100.0% | 2.93 [0.12 , 71.04] |    |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 1                |       | 0       |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 0.66 (P = 0.51)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.5 CEE 0.625 mg (mod dose) for 2.8-3.2 years                           |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| ERA 2000                                                                  | 1                | 100   | 0       | 105   | 24.6%  | 3.15 [0.13 , 76.39] |  |  |   |   |   |   |   |   |
| PEPI 1995                                                                 | 1                | 175   | 1       | 174   | 50.6%  | 0.99 [0.06 , 15.77] |  |  |   |   |   |   |   |   |
| WAVE 2002                                                                 | 1                | 60    | 0       | 62    | 24.8%  | 3.10 [0.13 , 74.59] |  |  |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 335   |         | 341   | 100.0% | 2.05 [0.38 , 11.04] |  |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 3                |       | 1       |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 0.83 (P = 0.40)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Chi² = 0.40, df = 2 (P = 0.82); I² = 0%                    |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.6 CEE 0.625 mg (mod dose) for 7.1 years                               |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| WHI 1998 <sup>a</sup>                                                     | 104              | 5310  | 135     | 5429  | 100.0% | 0.79 [0.61 , 1.01]  |  |  |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 5310  |         | 5429  | 100.0% | 0.79 [0.61 , 1.01]  |  |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 104              |       | 135     |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 1.85 (P = 0.06)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.7 CEE 0.625 mg (mod dose) after 10.7 years (includes extra follow-up) |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| WHI 1998 <sup>b</sup>                                                     | 151              | 5310  | 199     | 5429  | 100.0% | 0.78 [0.63 , 0.96]  |  |  |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 5310  |         | 5429  | 100.0% | 0.78 [0.63 , 0.96]  |  |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 151              |       | 199     |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 2.39 (P = 0.02)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.8 CEE 0.625 mg (mod dose) after 13 years (includes extra follow-up)   |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| WHI 1998 <sup>c</sup>                                                     | 168              | 5310  | 216     | 5429  | 100.0% | 0.80 [0.65 , 0.97]  |  |  |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 5310  |         | 5429  | 100.0% | 0.80 [0.65 , 0.97]  |  |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 168              |       | 216     |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Test for overall effect: Z = 2.27 (P = 0.02)                              |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| Heterogeneity: Not applicable                                             |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| 6.1.9 CEE 0.625 mg (mod dose) 20 years cumulative follow up               |                  |       |         |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |
| WHI 1998 <sup>d</sup>                                                     | 238              | 5310  | 296     | 5429  | 100.0% | 0.82 [0.70 , 0.97]  |  |  |   |   |   |   |   |   |
| Subtotal                                                                  |                  | 5310  |         | 5429  | 100.0% | 0.82 [0.70 , 0.97]  |  |                                                                                       |   |   |   |   |   |   |
| Total events:                                                             | 238              |       | 296     |       |        |                     |                                                                                       |                                                                                       |   |   |   |   |   |   |

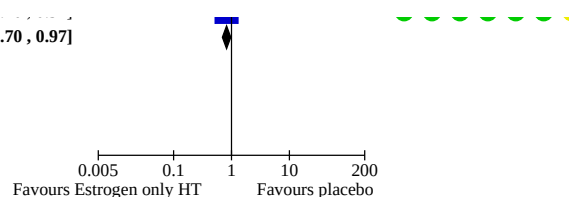
## Analysis 6.1. (Continued)

Subtotal 5310 5429 100.0% 0.82 [0.70, 0.97]

Total events: 238 296

Test for overall effect:  $Z = 2.31$  ( $P = 0.02$ )

Heterogeneity: Not applicable



## Footnotes

<sup>a</sup>LaCroix 2011

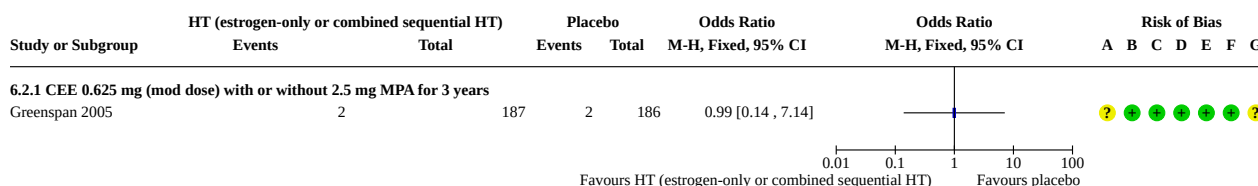
<sup>b</sup>LaCroix 2011: mean 7.1 yrs planned follow-up + extended follow-up in 78% of women

<sup>c</sup>Chlebowski 2015a (supplemental data): 7.1 yrs intervention plus follow up to median 13 years in 78% of eligible participants

<sup>d</sup>Chlebowski 2020.

## Risk of bias legend

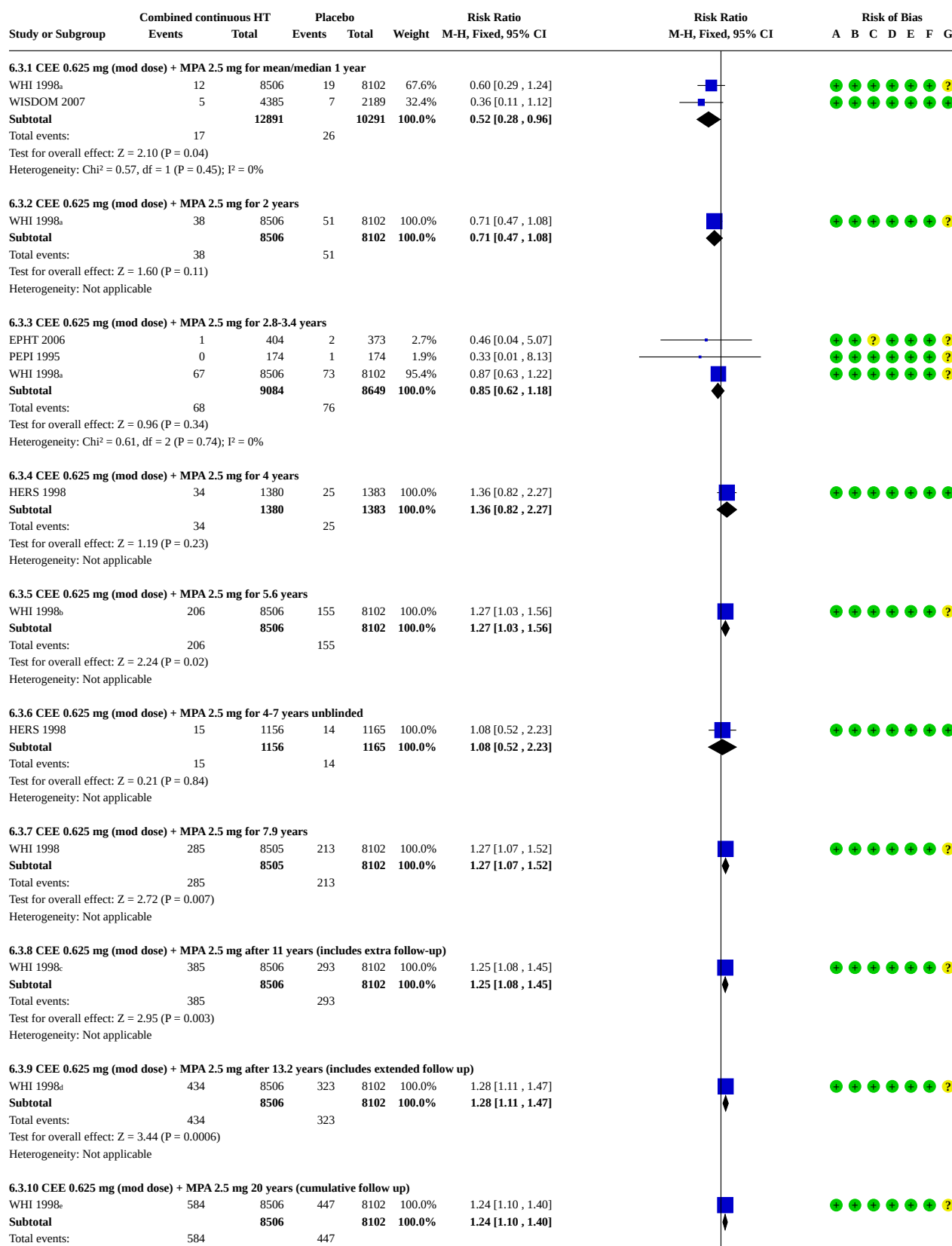
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 6.2. Comparison 6: All peri or postmenopausal people ,  
Outcome 2: Breast cancer: estrogen-only or combined HT

## Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 6.3. Comparison 6: All peri or postmenopausal people , Outcome 3: Breast cancer: combined continuous HT

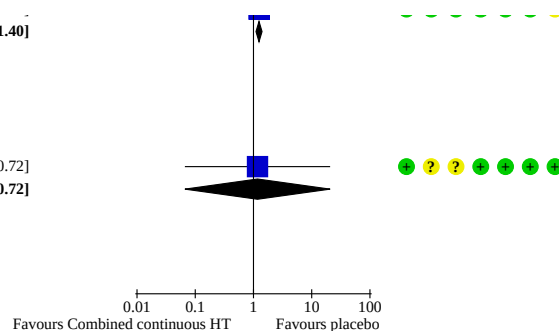


## Analysis 6.3. (Continued)

|                                                      |      |      |        |                   |
|------------------------------------------------------|------|------|--------|-------------------|
| Subtotal                                             | 8506 | 8102 | 100.0% | 1.24 [1.10, 1.40] |
| Total events:                                        | 584  | 447  |        |                   |
| Test for overall effect: $Z = 3.59$ ( $P = 0.0003$ ) |      |      |        |                   |
| Heterogeneity: Not applicable                        |      |      |        |                   |

## 6.3.11 Estradiol 0.25 to 1 mg + 50 to 100 mg progesterone daily for one year

|                                                    |   |      |   |     |        |                    |
|----------------------------------------------------|---|------|---|-----|--------|--------------------|
| REPLENISH 2018 <sup>a</sup>                        | 6 | 1684 | 0 | 151 | 100.0% | 1.17 [0.07, 20.72] |
| Subtotal                                           |   | 1684 |   | 151 | 100.0% | 1.17 [0.07, 20.72] |
| Total events:                                      | 6 |      | 0 |     |        |                    |
| Test for overall effect: $Z = 0.11$ ( $P = 0.91$ ) |   |      |   |     |        |                    |
| Heterogeneity: Not applicable                      |   |      |   |     |        |                    |



## Footnotes

<sup>a</sup>Chlebowski 2003<sup>b</sup>Manson 2013<sup>c</sup>Chlebowski 2010: Mean follow-up 11 yrs: 7.1 yrs follow-up to original planned completion date (including intervention phase of 5.6 years) + additional follow up in 83% of surviving women<sup>a</sup>Manson 2013: median 13.2 yrs: 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women<sup>c</sup>Chlebowski 2020<sup>d</sup>Daily estradiol(mg)–progesterone (mg) (1/100, 0.5/100, 0.5/50, or 0.25/50 all combined)

## Risk of bias legend

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

(C) Incomplete outcome data (attrition bias)

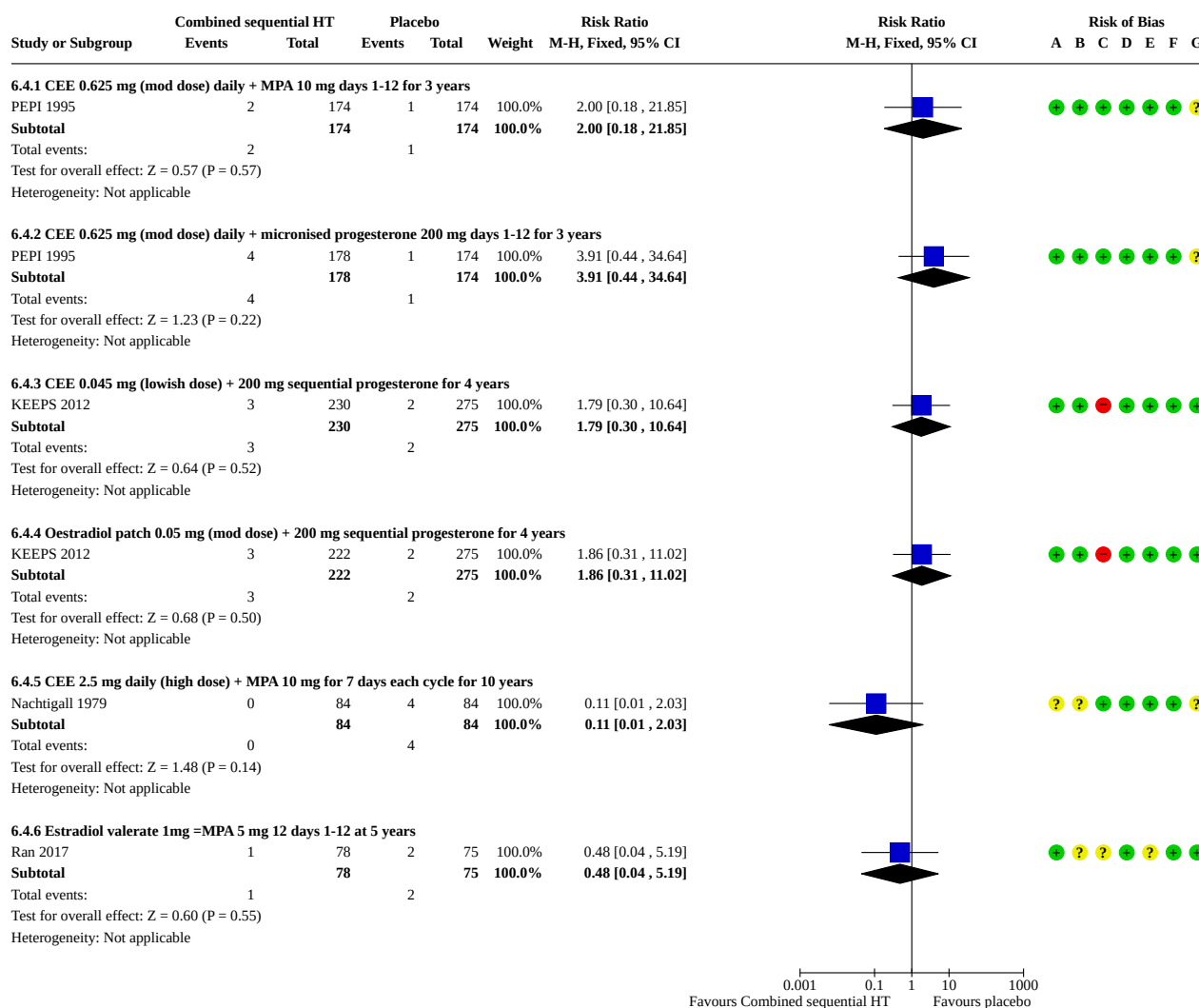
(D) Blinding of participants and personnel (performance bias)

(E) Blinding of outcome assessment (detection bias)

(F) Selective reporting (reporting bias)

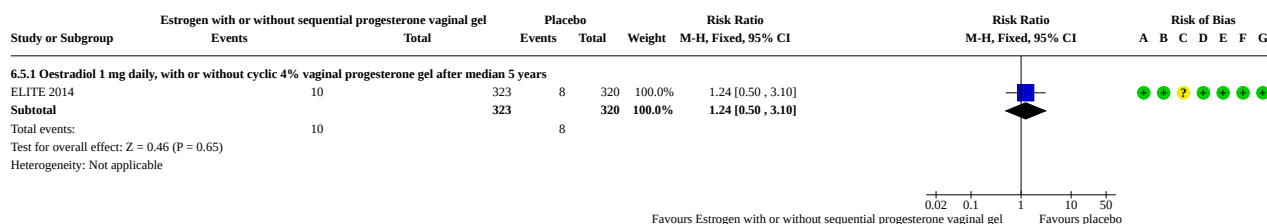
(G) Other bias

## Analysis 6.4. Comparison 6: All peri or postmenopausal people , Outcome 4: Breast cancer: combined sequential HT

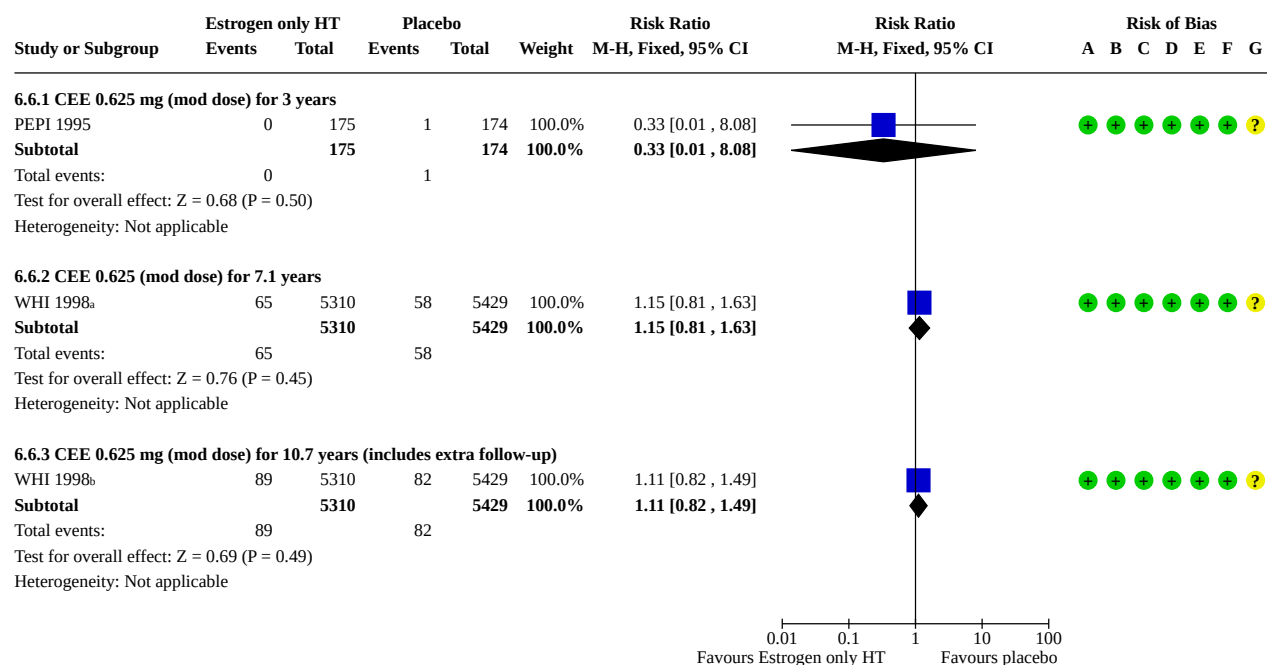


### Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

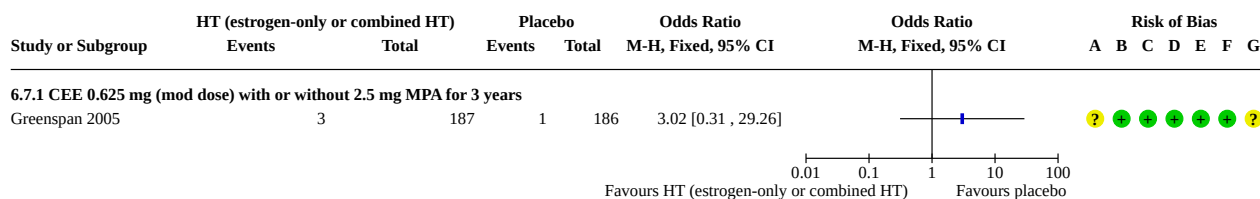
**Analysis 6.5. Comparison 6: All peri or postmenopausal people , Outcome 5: Breast cancer: estrogen with or without sequential progesterone vaginal gel****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.6. Comparison 6: All peri or postmenopausal people , Outcome 6: Colorectal cancer: estrogen-only HT****Footnotes**<sup>a</sup>LaCroix 2011<sup>b</sup>LaCroix 2011: 7.1 yrs planned follow-up + extended follow-up in 78% of women**Risk of bias legend**

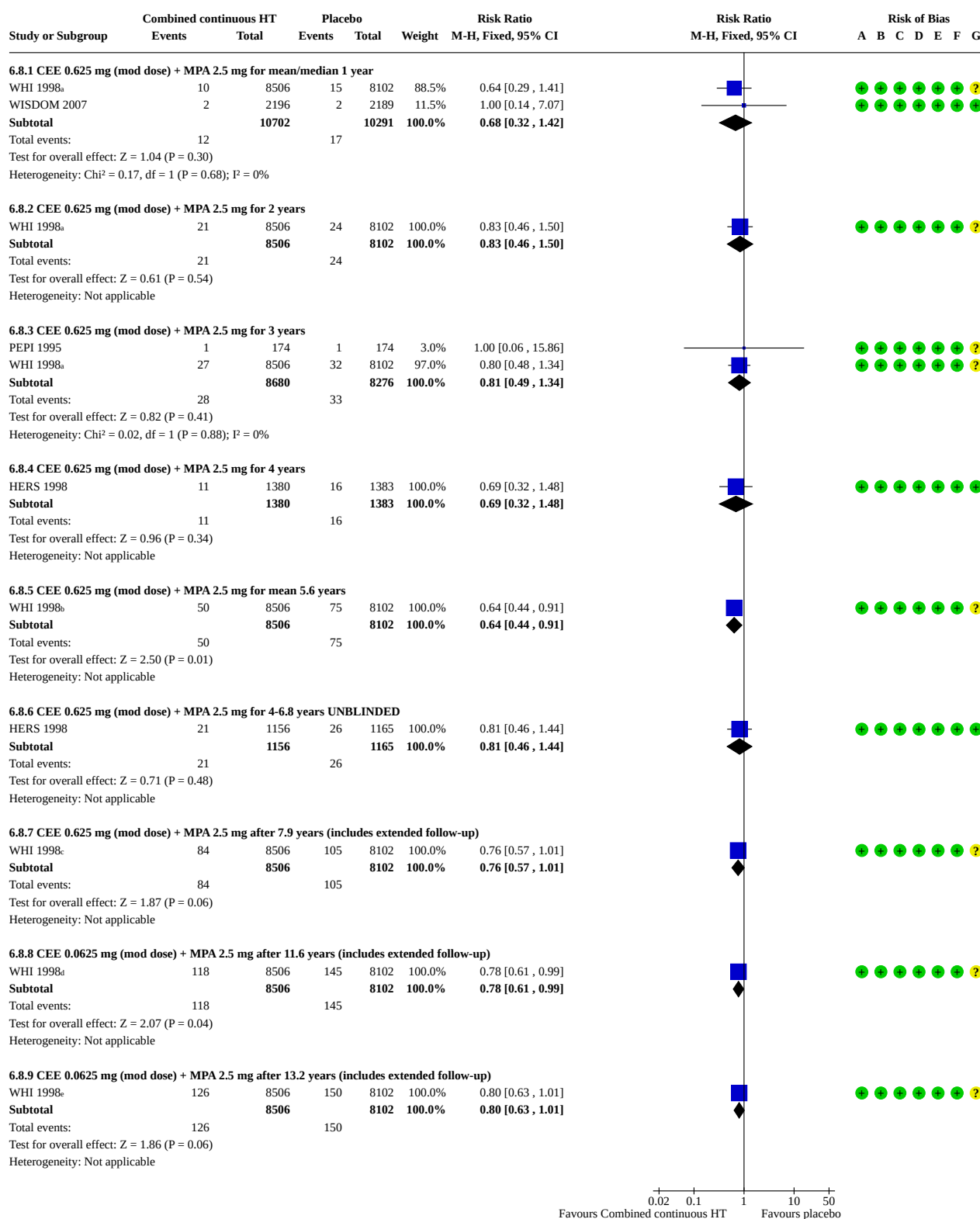
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

# **Analysis 6.7. Comparison 6: All peri or postmenopausal people , Outcome 7: Colorectal cancer: estrogen-only or combined HT**



## **Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.8. Comparison 6: All peri or postmenopausal people , Outcome 8: Colorectal cancer: combined continuous HT****Footnotes**<sup>a</sup>Rossouw 2002<sup>b</sup>Heiss 2008<sup>c</sup>Heiss 2008: Postintervention follow-up to mean 7.9 years total

## Analysis 6.8. (Continued)

<sup>a</sup>Kossouw 2002

<sup>b</sup>Heiss 2008

<sup>c</sup>Heiss 2008: Postintervention follow-up to mean 7.9 years total

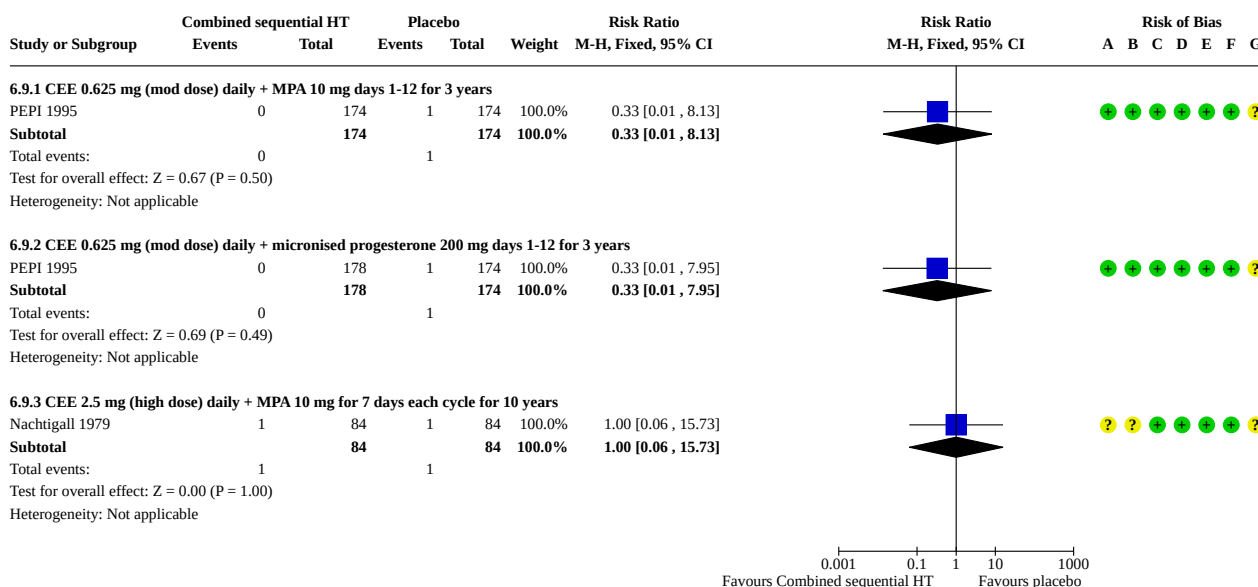
<sup>d</sup>Simon 2012: mean 5.6 years' intervention plus postintervention follow up in 83% participants

<sup>e</sup>Manson 2013: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women

### Risk of bias legend

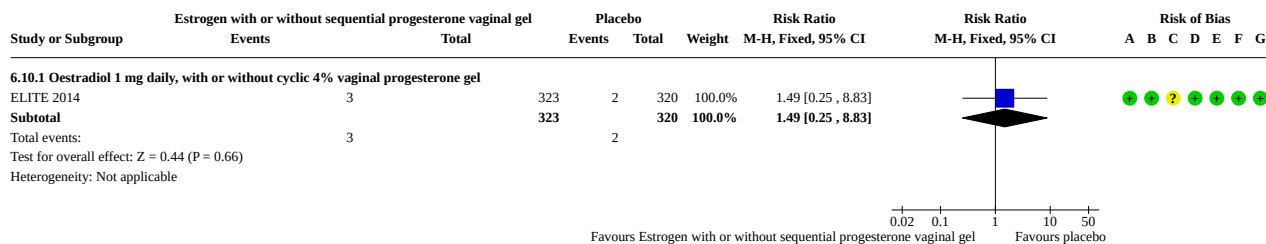
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 6.9. Comparison 6: All peri or postmenopausal people , Outcome 9: Colorectal cancer: combined sequential HT

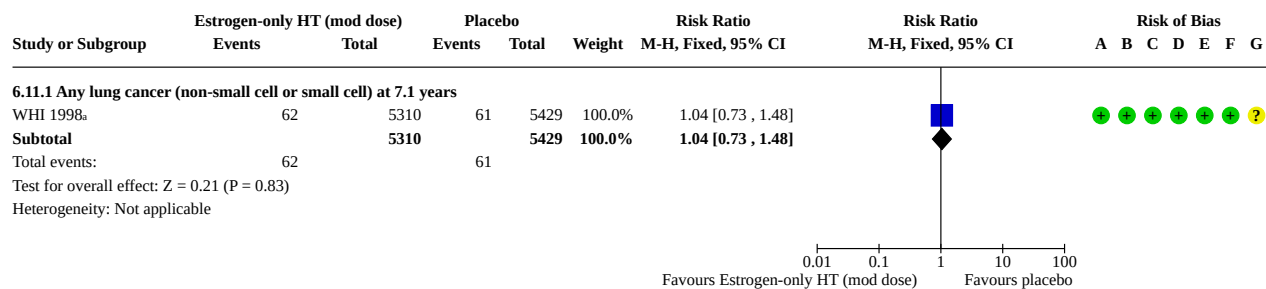


### Risk of bias legend

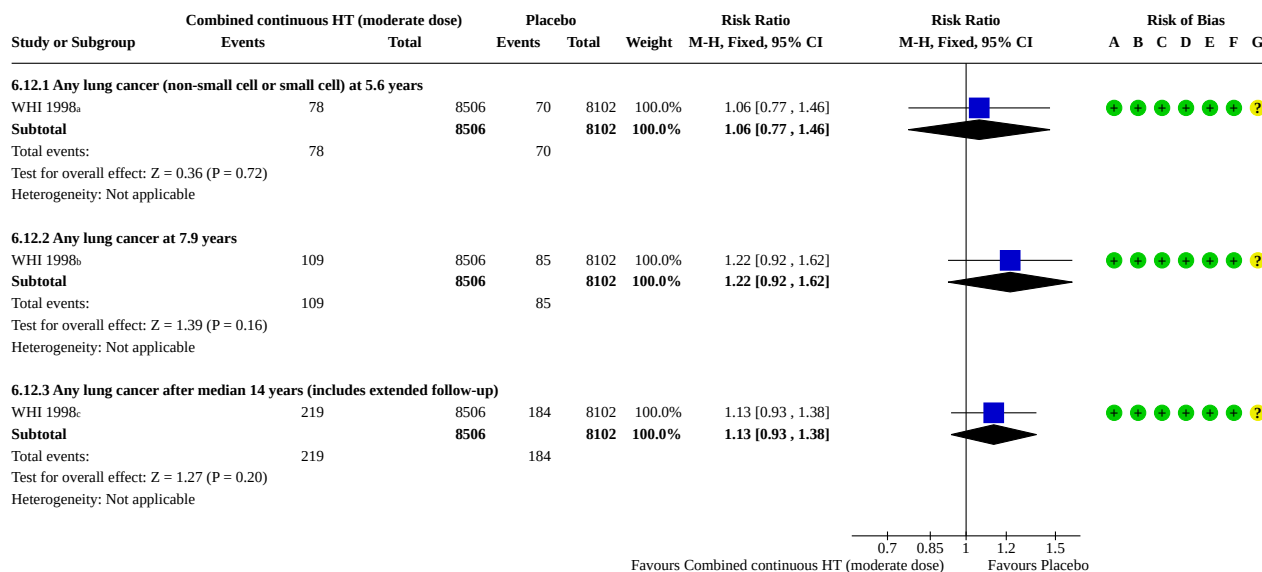
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.10. Comparison 6: All peri or postmenopausal people , Outcome 10: Colorectal cancer: estrogen with or without sequential progesterone vaginal gel****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.11. Comparison 6: All peri or postmenopausal people , Outcome 11: Lung cancer: estrogen-only HT (moderate dose)****Footnotes**<sup>a</sup>Manson 2013**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.12. Comparison 6: All peri or postmenopausal people ,  
Outcome 12: Lung cancer: combined continuous HT (mod dose oestrogen)****Footnotes**<sup>a</sup>Manson 2013<sup>b</sup>Chlebowski 2009: Post hoc analysis. Mean follow-up 7.9 years.<sup>c</sup>Chlebowski 2016a: median 14 years extended follow up in 83% of eligible women (median intervention 5.6 yrs)**Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

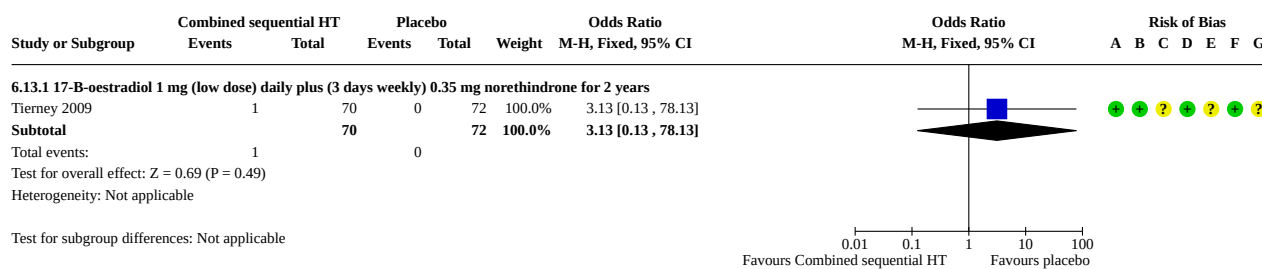
(C) Incomplete outcome data (attrition bias)

(D) Blinding of participants and personnel (performance bias)

(E) Blinding of outcome assessment (detection bias)

(F) Selective reporting (reporting bias)

(G) Other bias

**Analysis 6.13. Comparison 6: All peri or postmenopausal  
people , Outcome 13: Lung cancer: combined sequential HT****Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

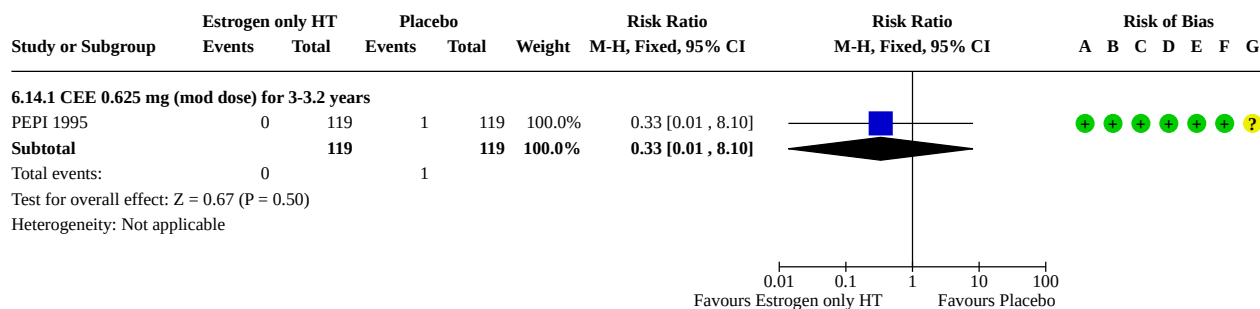
(C) Incomplete outcome data (attrition bias)

(D) Blinding of participants and personnel (performance bias)

(E) Blinding of outcome assessment (detection bias)

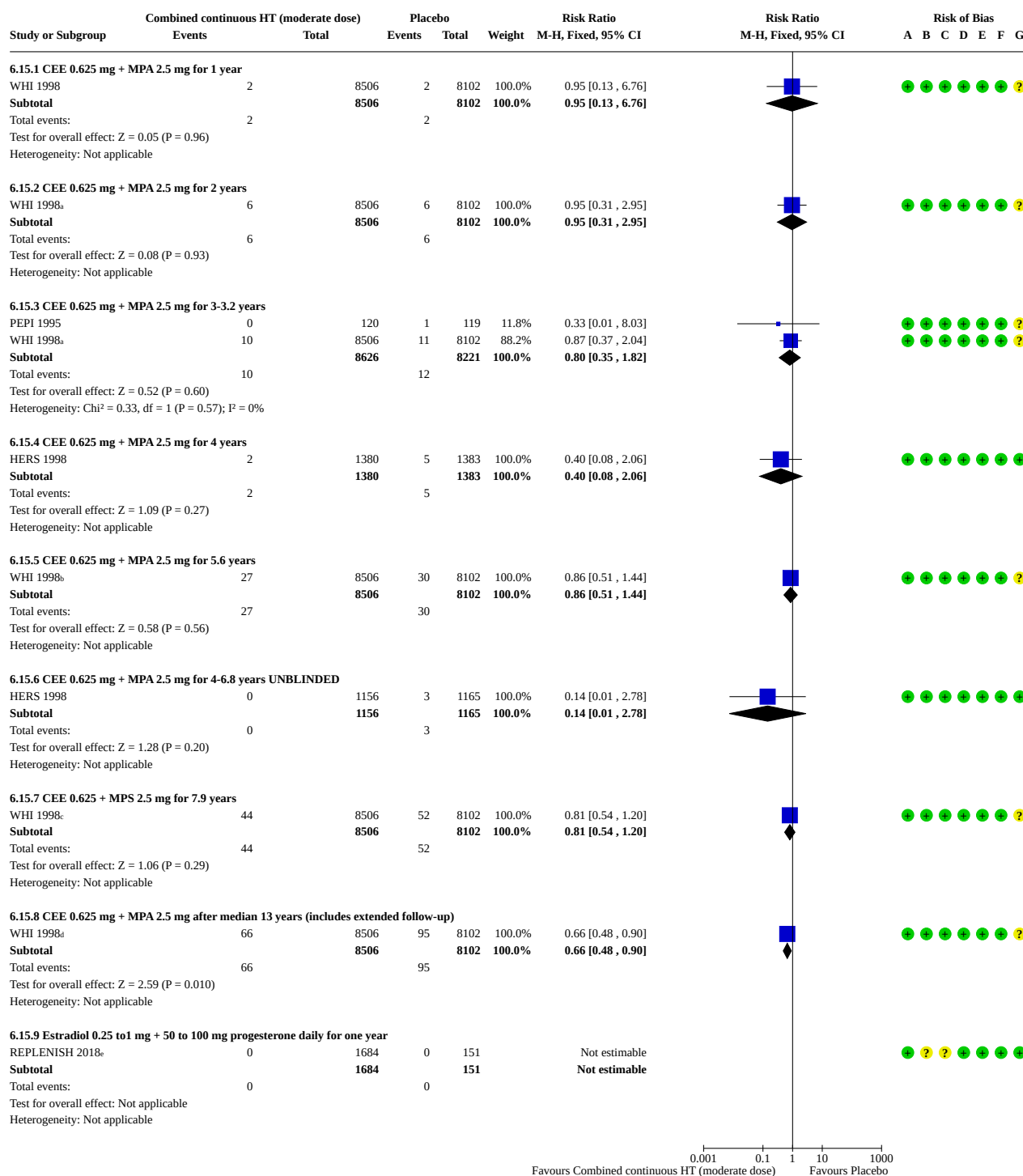
(F) Selective reporting (reporting bias)

(G) Other bias

**Analysis 6.14. Comparison 6: All peri or postmenopausal people , Outcome 14: Endometrial cancer: estrogen-only HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 6.15. Comparison 6: All peri or postmenopausal people , Outcome 15: Endometrial cancer: combined continuous HT (mode dose estrogen)



### Footnotes

Ⓐ Rossouw 2002

Ⓑ Manson 2013

Ⓒ Heiss 2008

Ⓓ Chlebowski 2016: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women

Ⓔ Daily estradiol(mg)-progesterone (mg) (1/100, 0.5/100, 0.5/50, or 0.25/50 all combined

### Risk of bias legend

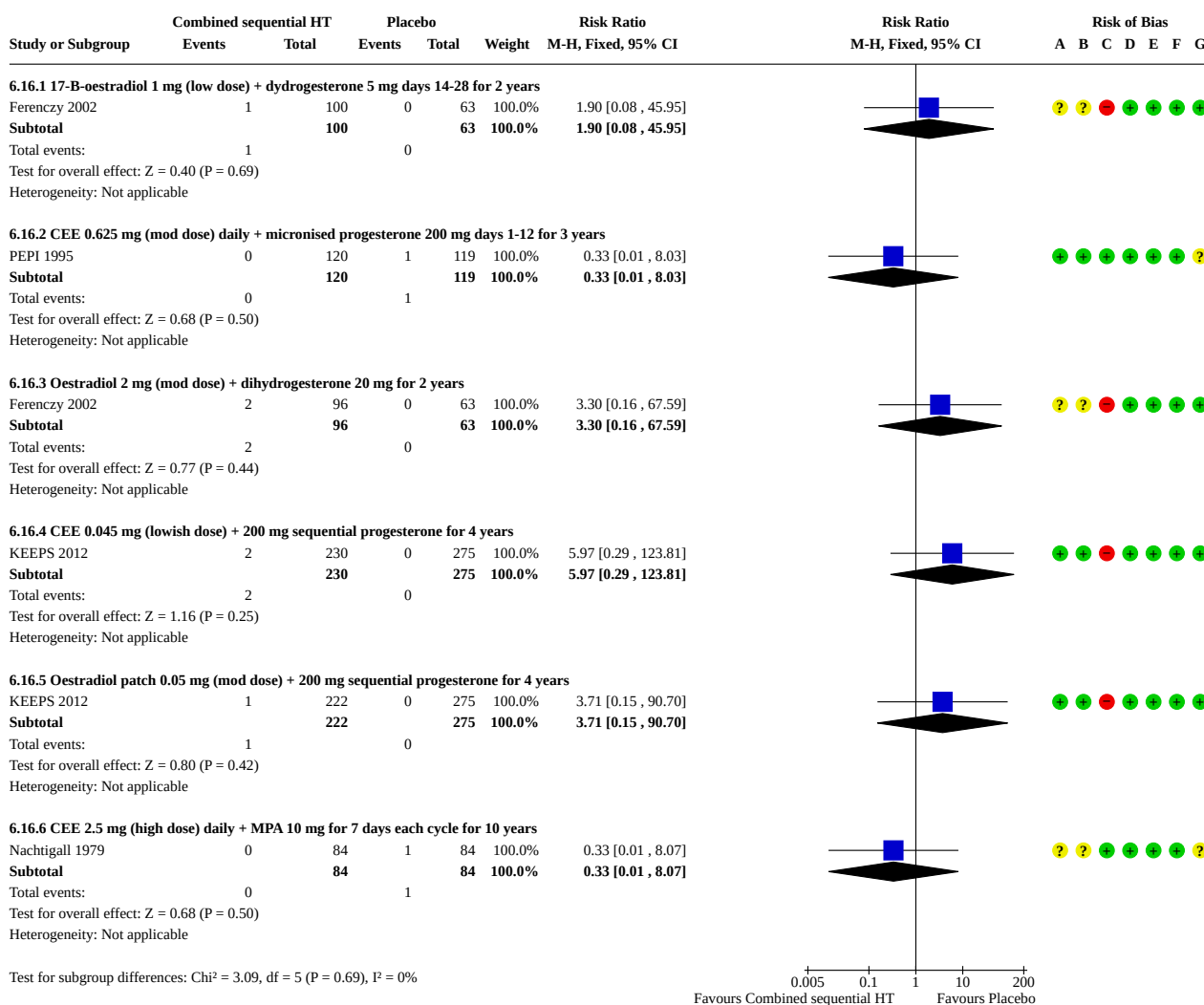
(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

## Analysis 6.15. (Continued)

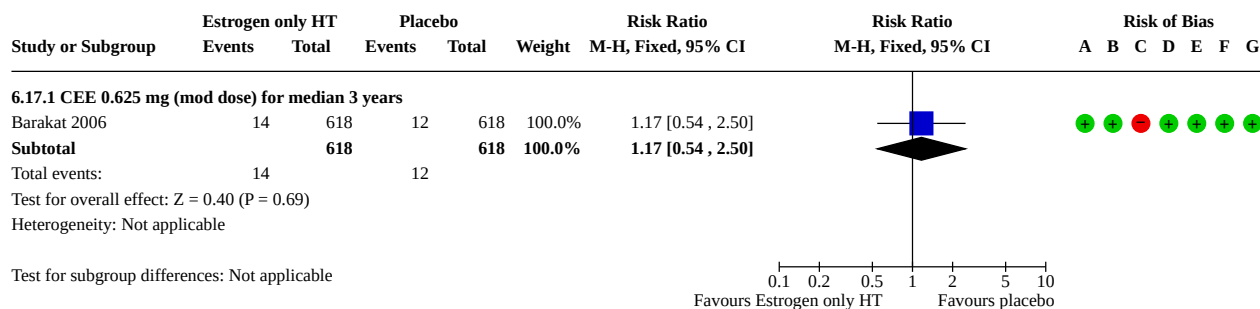
## Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

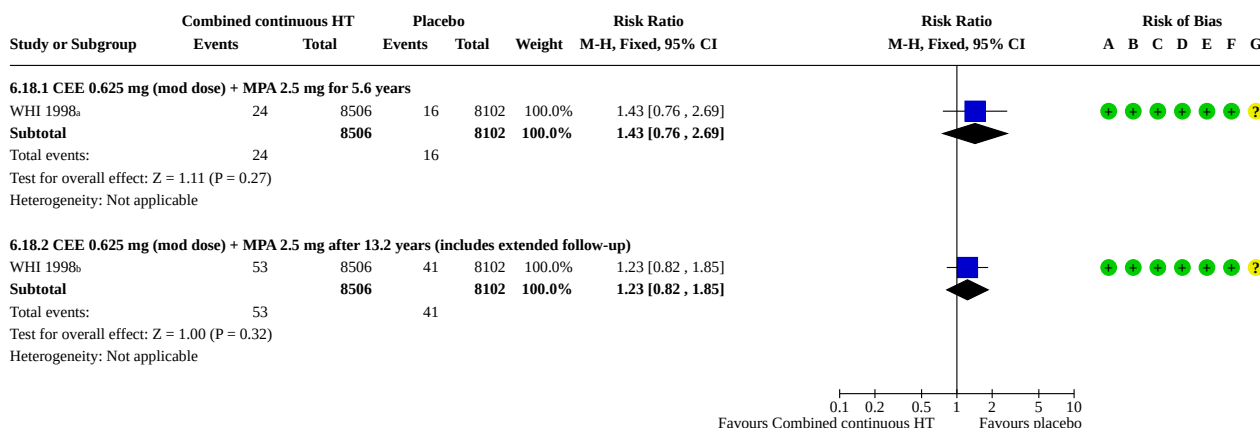
Analysis 6.16. Comparison 6: All peri or postmenopausal people ,  
Outcome 16: Endometrial cancer: combined sequential HT

## Risk of bias legend

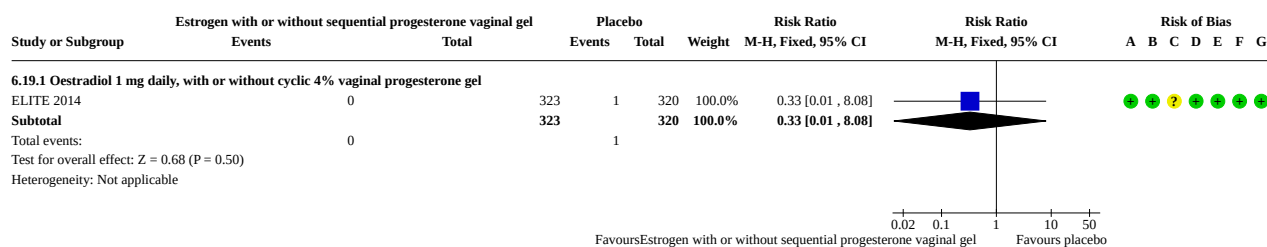
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.17. Comparison 6: All peri or postmenopausal people ,  
Outcome 17: Recurrent endometrial cancer: estrogen-only HT****Risk of bias legend**

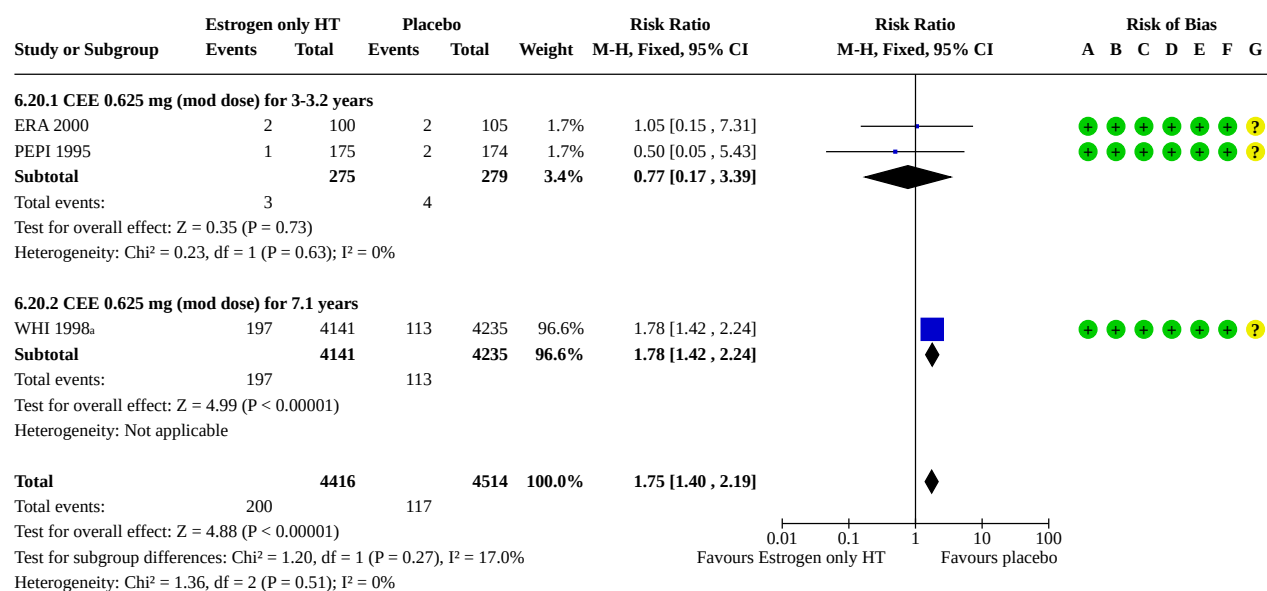
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.18. Comparison 6: All peri or postmenopausal  
people , Outcome 18: Ovarian cancer: combined continuous HT****Footnotes**<sup>a</sup>Manson 2013<sup>b</sup>Manson 2013; median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.19. Comparison 6: All peri or postmenopausal people , Outcome 19: Ovarian cancer: estrogen with or without sequential progesterone vaginal gel****Risk of bias legend**

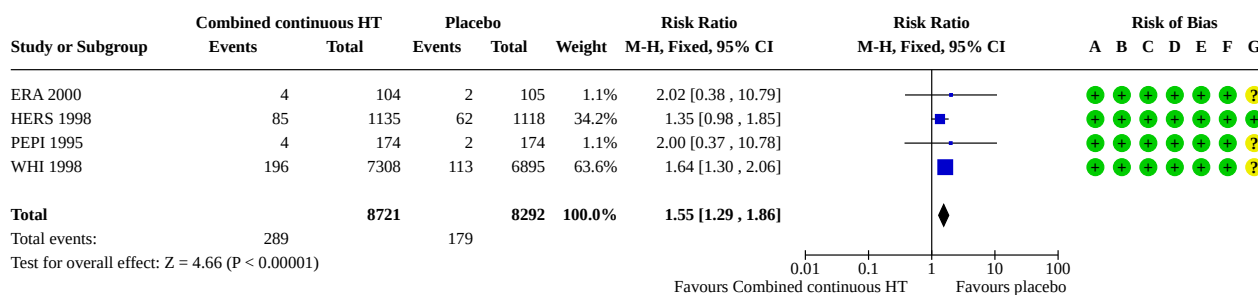
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.20. Comparison 6: All peri or postmenopausal people , Outcome 20: Gallbladder disease requiring surgery: estrogen-only HT****Footnotes**

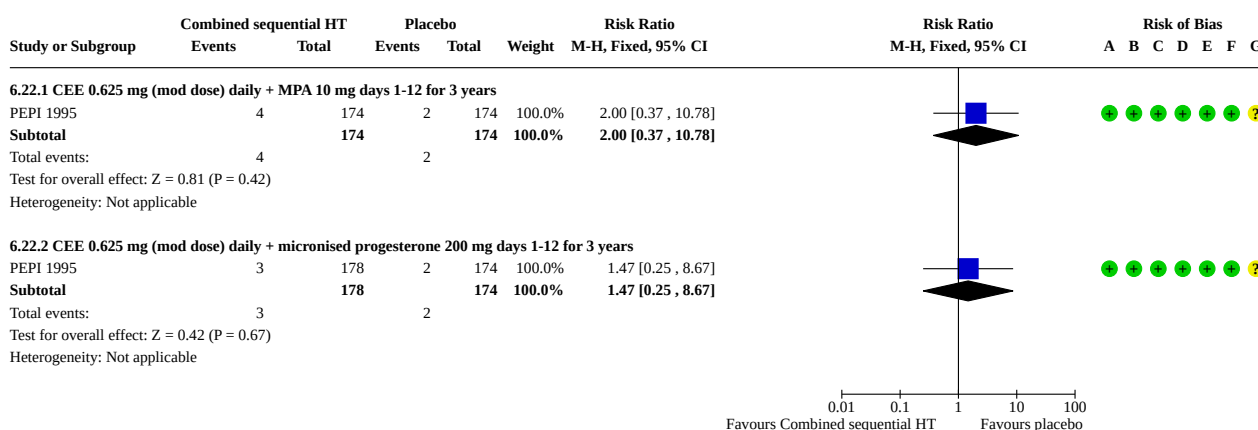
aCirillo 2005

**Risk of bias legend**

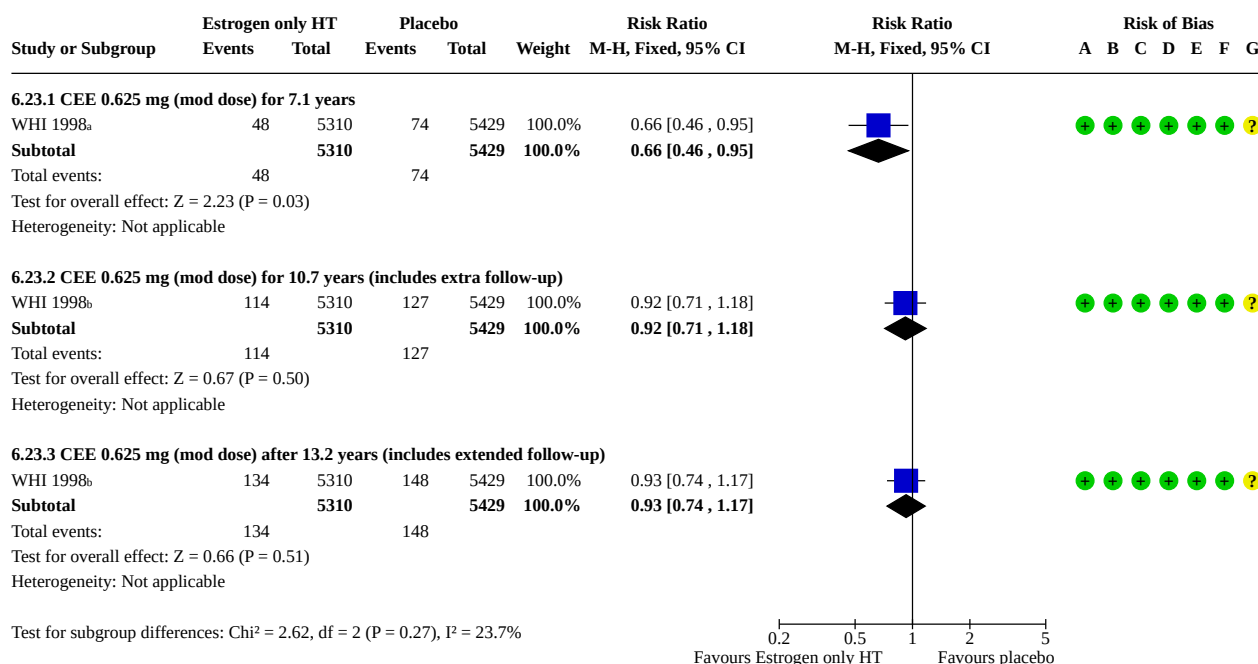
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.21. Comparison 6: All peri or postmenopausal people , Outcome 21: Gallbladder disease requiring surgery: combined continuous HT****Risk of bias legend**

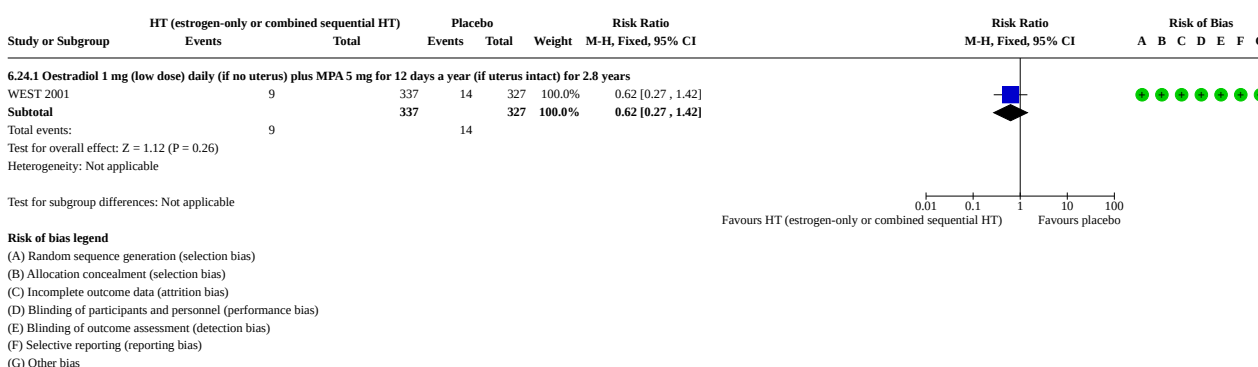
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.22. Comparison 6: All peri or postmenopausal people , Outcome 22: Gallbladder disease requiring surgery: combined sequential HT****Risk of bias legend**

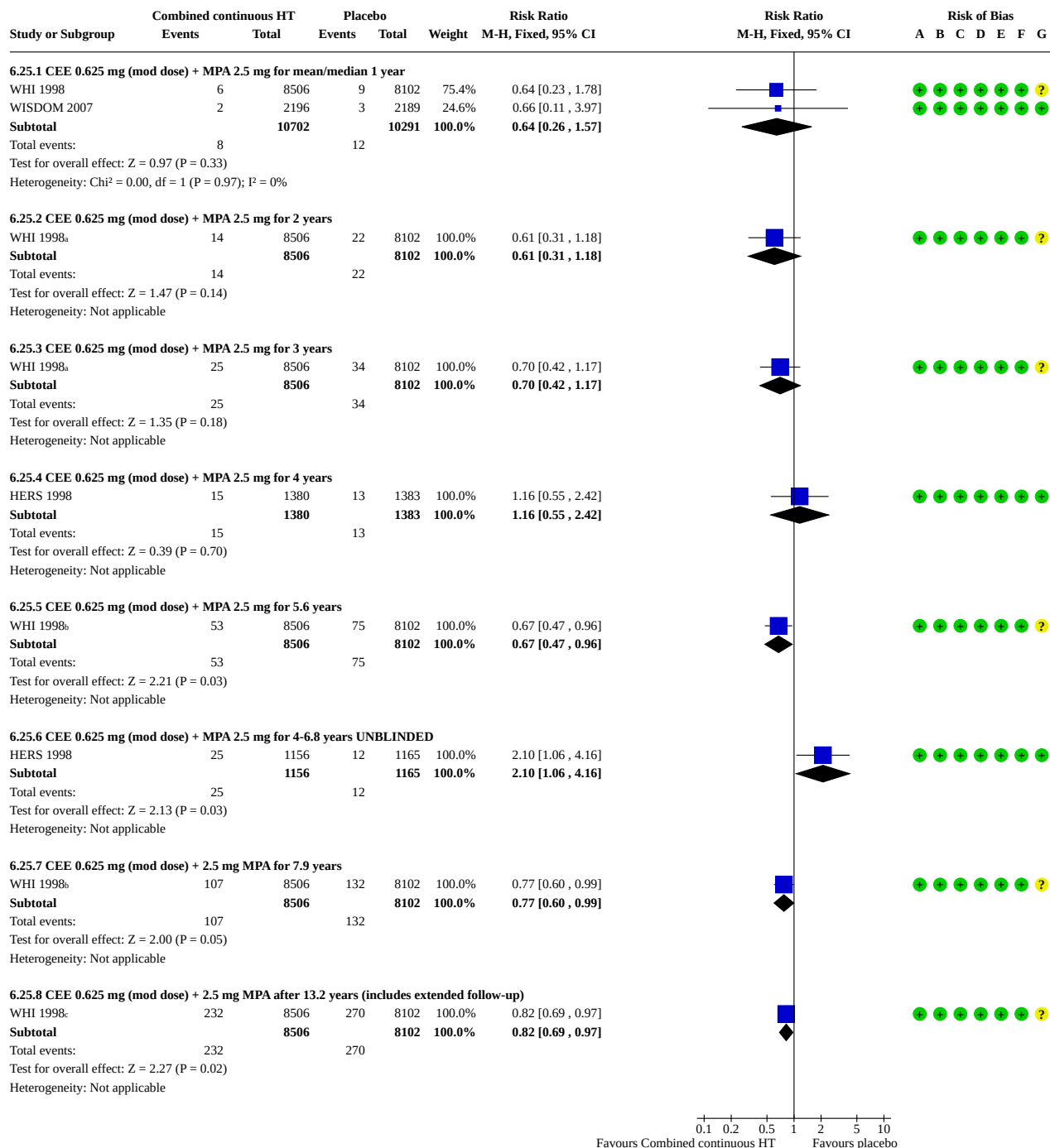
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.23. Comparison 6: All peri or postmenopausal people , Outcome 23: Hip fractures: estrogen-only HT****Footnotes**<sup>a</sup>LaCroix 2011<sup>b</sup>LaCroix 2011: 7.1 yrs planned follow-up + additional follow-up in 78% of women.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.24. Comparison 6: All peri or postmenopausal people , Outcome 24: Hip fractures: estrogen-only or combined sequential HT****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

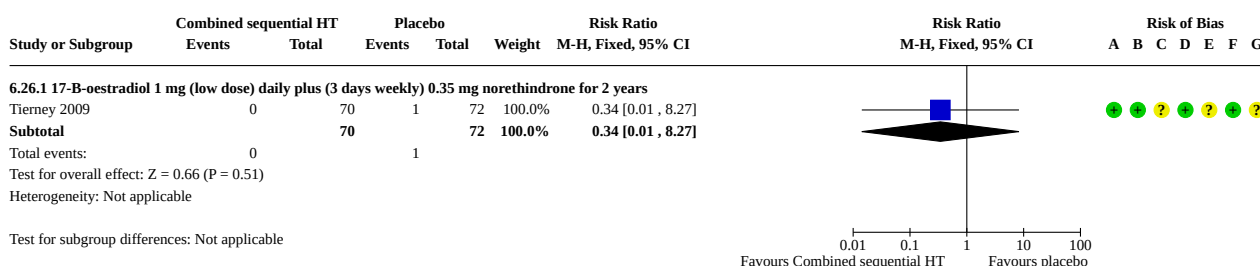
**Analysis 6.25. Comparison 6: All peri or postmenopausal people , Outcome 25: Hip fractures: combined continuous HT****Footnotes**<sup>a</sup>Rossouw 2002<sup>b</sup>Heiss 2008<sup>c</sup>Manson 2013: median 5.6 yrs follow up plus median 8.2 yrs extended follow up in 83% of eligible women**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)

## Analysis 6.25. (Continued)

- (U) Blinding of participants and personnel (performance bias)  
(E) Blinding of outcome assessment (detection bias)  
(F) Selective reporting (reporting bias)  
(G) Other bias

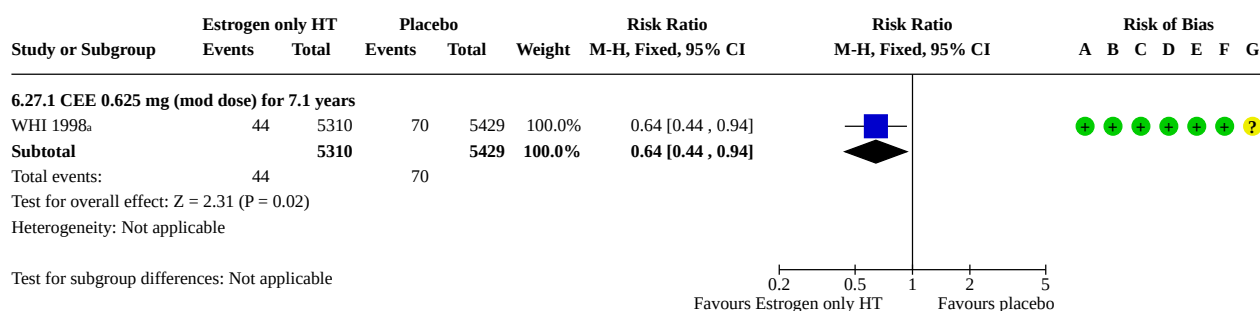
## Analysis 6.26. Comparison 6: All peri or postmenopausal people , Outcome 26: Hip fractures: combined sequential HT



## Risk of bias legend

- (A) Random sequence generation (selection bias)  
(B) Allocation concealment (selection bias)  
(C) Incomplete outcome data (attrition bias)  
(D) Blinding of participants and personnel (performance bias)  
(E) Blinding of outcome assessment (detection bias)  
(F) Selective reporting (reporting bias)  
(G) Other bias

## Analysis 6.27. Comparison 6: All peri or postmenopausal people , Outcome 27: Vertebral fractures: estrogen-only HT



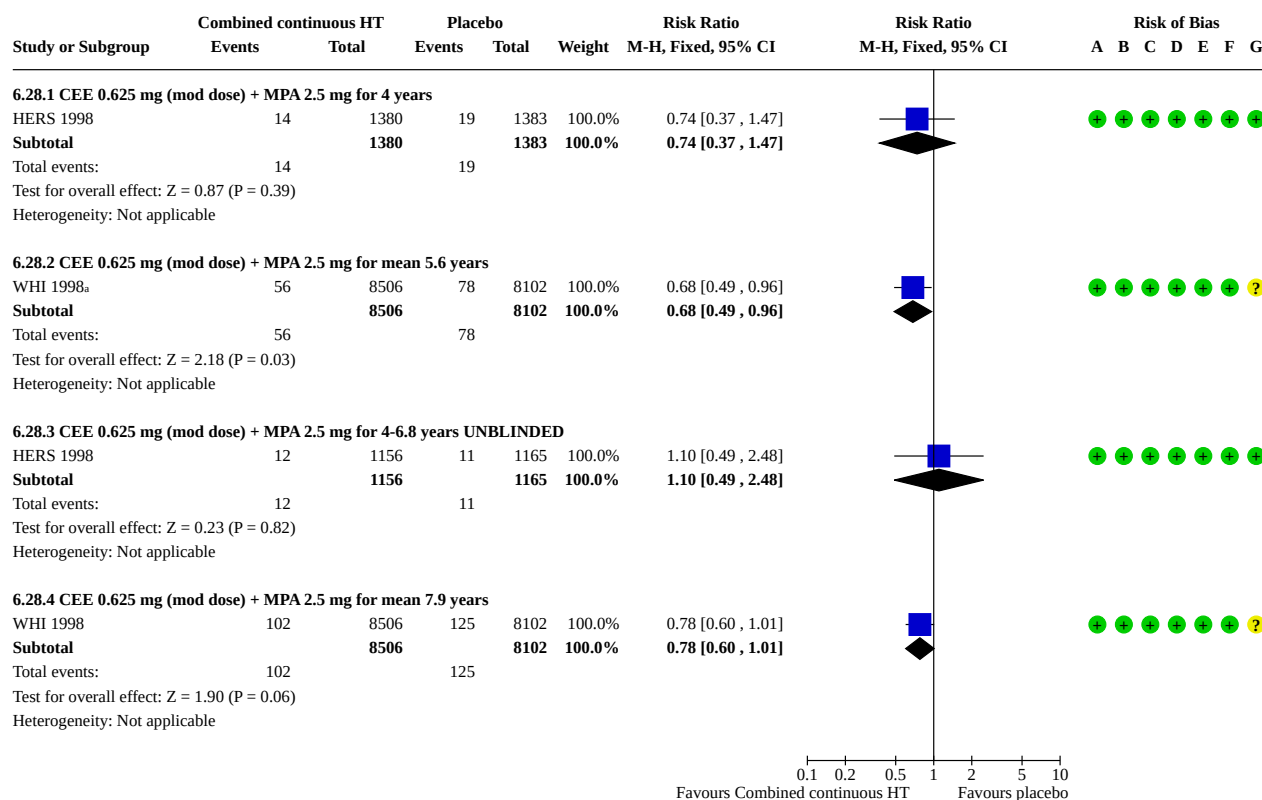
## Footnotes

<sup>a</sup>Jackson 2006

## Risk of bias legend

- (A) Random sequence generation (selection bias)  
(B) Allocation concealment (selection bias)  
(C) Incomplete outcome data (attrition bias)  
(D) Blinding of participants and personnel (performance bias)  
(E) Blinding of outcome assessment (detection bias)  
(F) Selective reporting (reporting bias)  
(G) Other bias

**Analysis 6.28. Comparison 6: All peri or postmenopausal people ,  
Outcome 28: Vertebral fractures: combined continuous HT**

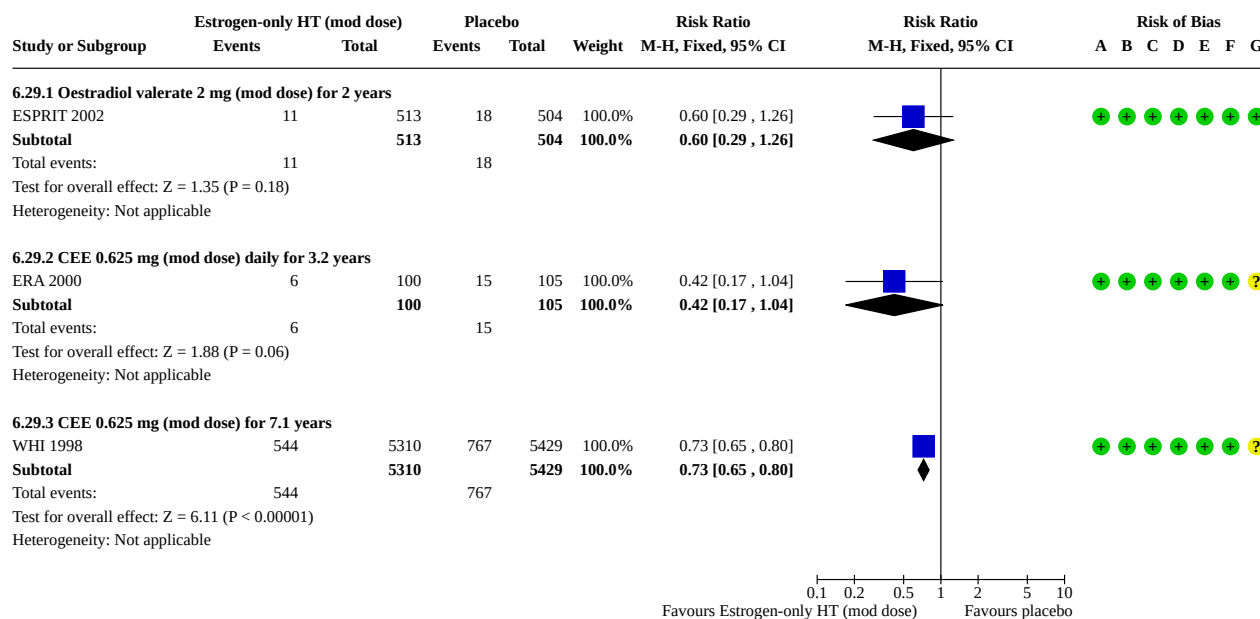


**Footnotes**

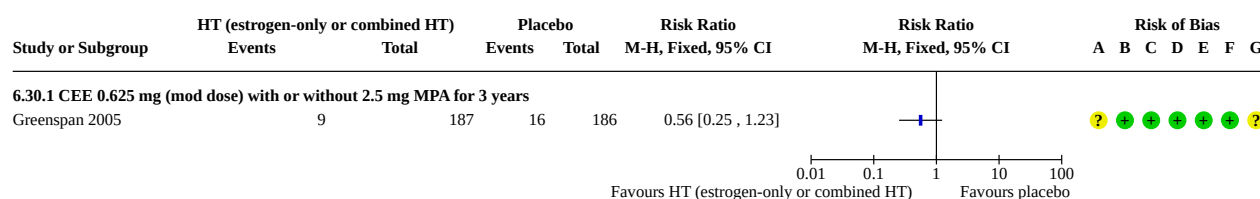
<sup>a</sup>Heiss 2008

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

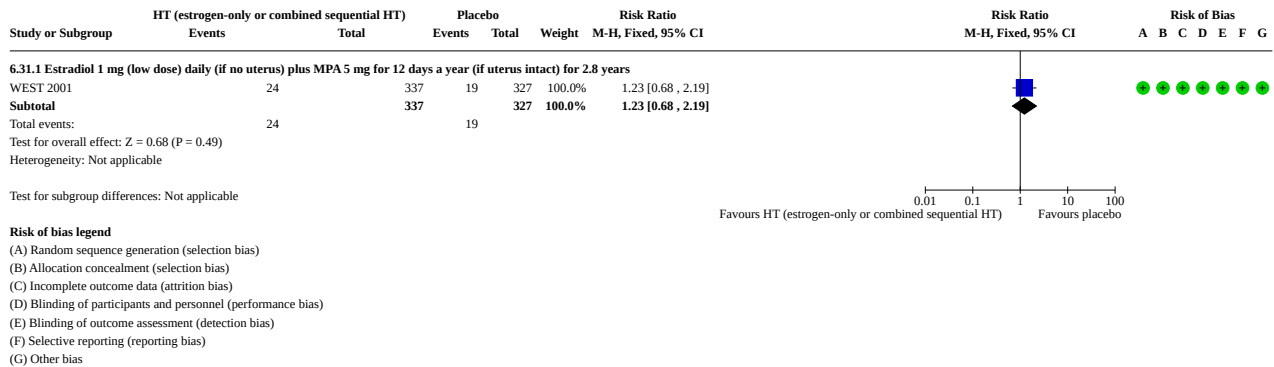
**Analysis 6.29. Comparison 6: All peri or postmenopausal people ,  
Outcome 29: All clinical fractures: estrogen-only HT (moderate dose)****Risk of bias legend**

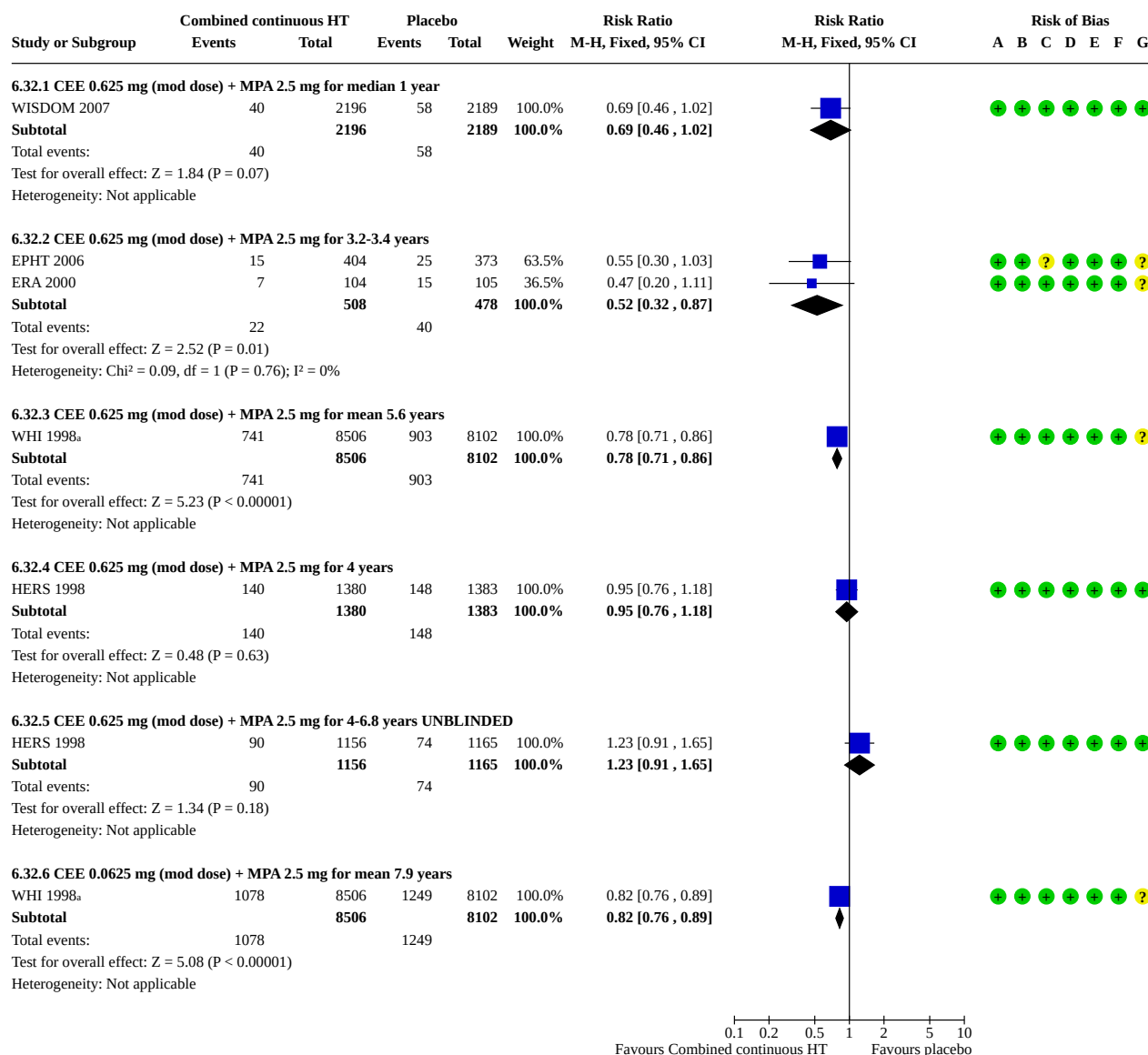
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.30. Comparison 6: All peri or postmenopausal people ,  
Outcome 30: All clinical fractures: estrogen-only or combined HT****Risk of bias legend**

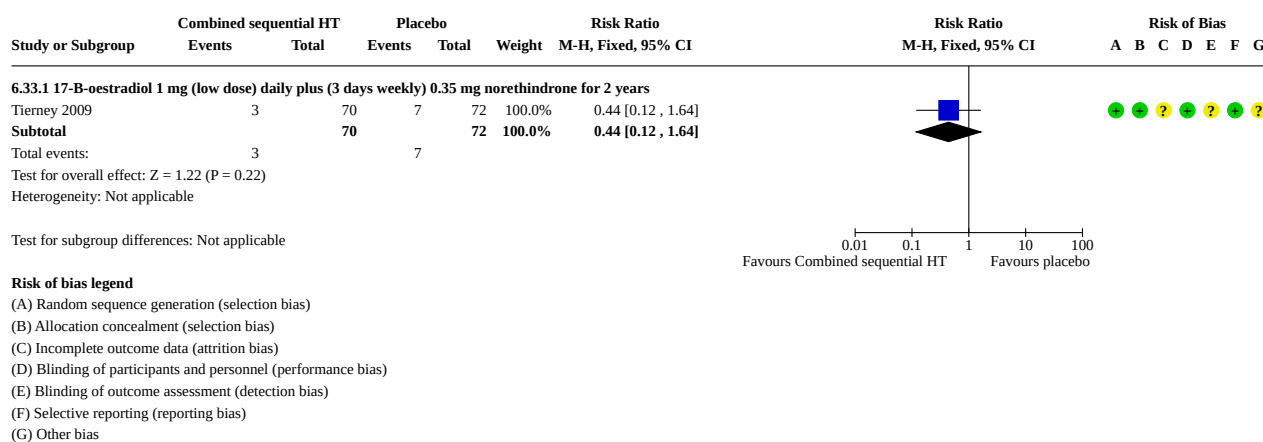
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

## Analysis 6.31. Comparison 6: All peri or postmenopausal people , Outcome 31: All clinical fractures: estrogen-only or combined sequential HT



**Analysis 6.32. Comparison 6: All peri or postmenopausal people ,  
Outcome 32: All clinical fractures: combined continuous HT****Footnotes**<sup>a</sup>Heiss 2008**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Incomplete outcome data (attrition bias)
- (D) Blinding of participants and personnel (performance bias)
- (E) Blinding of outcome assessment (detection bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 6.33. Comparison 6: All peri or postmenopausal people ,  
Outcome 33: All clinical fractures: combined sequential HT****ADDITIONAL TABLES****Table 1. Other data**

| Study      | Comparison                                                                                                                                     | Instrument                                    | Measure                                                                              | Outcome          | Intervention                               | Effect    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------|------------------|--------------------------------------------|-----------|
| KEEPS 2012 | Oestrogen (CEE or oestradiol) + cyclic oral micronised progesterone 200 mg/d × 12 days per month<br><br>versus placebo (n = 275) for 48 months | Modified Mini Mental State Examination (MMSE) | Differences between intervention and placebo groups in mean rate of change over time | Global cognition | 0.45 mg/d oral CEE (n = 230)               | P = 0.178 |
|            |                                                                                                                                                |                                               |                                                                                      |                  | 0.05 mg/d transdermal oestradiol (n = 222) | P = 0.840 |

CEE: conjugated equine oestrogens

**Table 2. Adherence to treatment**

| Study        | How defined                                                                        | Assessment                             | Hormone therapy group                                       | Placebo group                                               | Notes |
|--------------|------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|-------|
| Barakat 2006 | Discontinuation of therapy for longer than a month (or use of HT in placebo group) | Not stated                             | 41.1% compliant for whole follow-up period (median 3 years) | 50.1% compliant for whole follow-up period (median 3 years) |       |
| ELITE 2014   | > 80% of prescribed treatment taken                                                | Pill counts                            | Median > 98% over median of 5 years                         | Median > 98% over median of 5 years                         |       |
| EPAT 2001    | Percentage of study medication consumed                                            | Pill counts                            | Level of adherence 95% in the 87% of participants evaluated | Level of adherence 92% in the 92% of participants evaluated |       |
| EPHT 2006    | > 80% of prescribed treatment taken                                                | Number of collected and returned drugs | < 40% compliant at 3 years (estimated from graph)           | < 30% compliant at 3 years (estimated from graph)           |       |

**Table 2. Adherence to treatment** *(Continued)*  
and clinic re-  
ports

|                |                                                                             |                                                                                                              |                                                                                                                                                                                     |                                                                                                                                |                                                                                                                                        |
|----------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| ERA 2000       | Percentage of study medication taken                                        | Pill counts                                                                                                  | Level of adherence at 3.2 years:<br>Women on unopposed oestrogen, measured in 79% of participants only: 74%<br><br>Women on combined HRT, measured in 82% of participants only: 84% | Level of adherence at 3.2 years:<br>Measured in 80% of participants only: 86%<br><br>5 women initiated treatment outside study |                                                                                                                                        |
| ESPRIT 2002    | "Regular tablet use"                                                        | Self-report to family doctor. Self-report to study nurse at 6 weeks and whenever in contact with trial staff | Number non-adherent: 51% at 12 months 57% at 24 months                                                                                                                              | Number non-adherent: 31% at 12 months 337% at 24 months                                                                        | Trialists attribute higher non-adherence in HRT group to prevalence of vaginal bleeding (reported by 56% in HRT group, 7% in controls) |
| EVTET 2000     | Adherence not described                                                     |                                                                                                              |                                                                                                                                                                                     |                                                                                                                                |                                                                                                                                        |
| Ferenczy 2002  | Adherence not described                                                     |                                                                                                              |                                                                                                                                                                                     |                                                                                                                                |                                                                                                                                        |
| Greenspan 2005 | "Taking at least 80% of medication for at least 80% of entire study period" | Pill counts 6-monthly                                                                                        | 90% adherent at 3 years                                                                                                                                                             | 94% adherent at 3 years                                                                                                        |                                                                                                                                        |
| HERS 1998      | Taking at least 80% of study medication                                     | Pill counts                                                                                                  | 79% adherent at 1 year<br>70% adherent at 3 years<br>3% initiated treatment outside study<br>About 50% continued to use open-label HRT during unblinded follow-up (4.2-6.8 years)   | 91% adherent at 1 year<br>81% non-adherent at 3 years<br>Less than 10% used HRT during unblinded follow-up (4.2-6.8 years)     | Proportion of women who reported taking study medication at 1 year: HRT group: 82% Placebo group: 91%                                  |
| KEEPS 2012     | Pill or patch counts, percentage used                                       | Pill counts or weights                                                                                       | 94%-95% in all groups, among women who completed trial at 4 years                                                                                                                   |                                                                                                                                |                                                                                                                                        |
| Mulnard 2000   | Taking at least 80% of study medication                                     | Plasma oestradiol level evaluation at each visit<br>Pill counts at each visit                                | No information given in publication                                                                                                                                                 |                                                                                                                                |                                                                                                                                        |

**Table 2. Adherence to treatment** (Continued)

|                                    |                                                                                                    |                                                                                                                         |                                                                                                                                                                                                 |                                                                                                                                                                                                 |                                                                               |
|------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Nachtigall 1979                    | Adherence not described                                                                            |                                                                                                                         |                                                                                                                                                                                                 |                                                                                                                                                                                                 |                                                                               |
| Notelovitz 2002                    | Adherence not described                                                                            |                                                                                                                         |                                                                                                                                                                                                 |                                                                                                                                                                                                 |                                                                               |
| Obel 1993                          | Adherence not described                                                                            |                                                                                                                         |                                                                                                                                                                                                 |                                                                                                                                                                                                 |                                                                               |
| PEPI 1995                          | Taking at least 80% of study medication                                                            | Study diary reviewed at clinic visits<br>Pill counts                                                                    | Number adherent at 36 months:<br>Women without uterus: 80%-89% at 36 months<br><br>Women with uterus:<br>1. On unopposed CEE: 44%<br>2. On combined therapy: 80%                                | Number adherent at 36 months:<br>Women without uterus: 67%<br><br>Women with uterus: 76%                                                                                                        |                                                                               |
| Tierney 2009                       | Taking at least 80% of study medication                                                            | Pill counts weekly                                                                                                      | No information given in publication                                                                                                                                                             |                                                                                                                                                                                                 |                                                                               |
| WAVE 2002                          | Percentage of study medication taken                                                               | Pill counts                                                                                                             | At 2.8 years:<br>Adherence 67% in the 78% of women analysed                                                                                                                                     | At 2.8 years:<br>Adherence 70% in the 81% of women analysed                                                                                                                                     |                                                                               |
| WEST 2001                          | Percentage of study medication taken                                                               | Self-report to study nurse 3-monthly<br>Computer chip in medication bottle records opening date and time<br>Pill counts | At 2.8 years:<br>Mean adherence including dropouts: 70%<br><br>Mean adherence excluding dropouts: 90%<br>35% discontinued medication by 2.8 years, of whom 1% initiated treatment outside study | At 2.8 years:<br>Mean adherence including dropouts: 74% over 2.8 years<br><br>Mean adherence excluding dropouts: 90%<br><br>24% discontinued medication<br>2% initiated treatment outside study |                                                                               |
| WHI 1998 (unopposed oestrogen arm) | Taking at least 80% of study medication. Temporary discontinuation (e.g. during surgery) permitted | Weighing of returned medication bottles                                                                                 | At 6.8 years, about 53.8% of women were non-adherent<br>In addition, 5.7% of women had initiated hormone use through their own physician                                                        | At 6.8 years, about 53.8% of women were non-adherent<br>In addition, 9.1% of women had initiated hormone use through their own physician                                                        |                                                                               |
| WHI 1998 (combined arm)            | Taking at least 80% of study medication. Temporary discontinuation (e.g. during surgery) permitted | Weighing of returned medication bottles                                                                                 | 42% non-adherent by 5.2 years<br>Of these, 6.2% initiated HRT outside study                                                                                                                     | 10.7% crossed to active treatment by 5.2 years                                                                                                                                                  | Analyses censoring events 6 months after non-adherence increased effect sizes |
| WISDOM 2007                        | Supply of study medication                                                                         | Time at risk minus temporary interruptions                                                                              | 73% of time                                                                                                                                                                                     | 86% of time                                                                                                                                                                                     | Women had a 3-month run-in period on                                          |

**Table 2. Adherence to treatment** *(Continued)*

|            |                               | and time af-<br>ter withdrawal<br>from treatment                  |     |             | placebo. Only<br>women who<br>took 80% of<br>tablets were<br>randomised                  |
|------------|-------------------------------|-------------------------------------------------------------------|-----|-------------|------------------------------------------------------------------------------------------|
| Yaffe 2006 | Supply of study<br>medication | Patch counts:<br>75% use over 2<br>years counted<br>as compliance | 84% | 84% of time | Women had a<br>1-week run-in<br>period.<br><br>Only adherent<br>women were<br>randomised |

CEE: conjugated equine oestrogens; HT/HRT: hormone (replacement) therapy

## APPENDICES

### Appendix 1. Cochrane Gynaecology and Fertility (CGF) Group Specialised Register search strategy

ProCite platform

Searched on 26 September 2024 for trials from the inception of the database to 23 November 2023 as the CGF Specialised Register has not been updated since 23 November 2023

Keywords CONTAINS "menopausal" or "Menopause" or "menopause-surgical" or "perimenopausal" or "perimenopause" or "postmenopausal" or "postmenopause" or "climacteric" or Title CONTAINS "menopausal" or "menopause-surgical" or "perimenopausal" or "perimenopause" or "postmenopausal" or "postmenopause" or "climacteric" or "Menopause"

AND

Keywords CONTAINS "HRT" or "HT" or "hormone replacement therapy" or "Hormone Substitution" or "hormone therapy" or "Estradiol" or "Estriol" or "estrogen" or "Estrogens" or "oestrodiol" or "oestrogen" or "CEE" or "CEE + MPA" or "conjugated equine" or "premarin" or "17-beta estradiol" or "17beta-estradiol + norethisterone acetate" or "SERM" or "selective estrogen receptor modulator" or "bazedoxifene acetate" or "Evista" or "Clomiphene" or "clomiphene citrate" or "Tamoxifen" or "tamoxifen citrate" or "toremifen" or "toremifene" or "raloxifene" or "ospemifene" or "Levormeloxifene" or "tibolone" or Title CONTAINS "HRT" or "HT" or "hormone replacement therapy" or "Hormone Substitution" or "hormone therapy" or "Estradiol" or "Estriol" or "estrogen" or "Estrogens" or "oestrodiol" or "oestrogen" or "CEE" or "CEE + MPA" or "conjugated equine" or "premarin" or "17-beta estradiol" or "17beta-estradiol + norethisterone acetate" or "selective estrogen receptor modulator" or "Tamoxifen" or "tibolone"

(4056 records)

### Appendix 2. Cochrane Central Register of Controlled Trials (CENTRAL) search strategy

Ovid Platform

Searched on 26 September 2024 (Issue 8)

- 1 menopause/ or menopause, premature/ or perimenopause/ or postmenopause/ (8348)
- 2 (postmenopaus\* or menopaus\* or perimenopaus\*).tw. (30646)
- 3 exp Ovariectomy/ (419)
- 4 Ovariectom\*.tw. (343)
- 5 (oophorectom\* or hysterectom\*).tw. (9131)
- 6 (climacteri\* or vasomotor).tw. (3291)
- 7 (cognition or cognitive).tw. (102457)
- 8 or/1-7 (140816)
- 9 exp Hormone Replacement Therapy/ (3777)
- 10 (hormon\* adj1 (therap\* or substitut\* or replace\*)).tw. (9187)
- 11 replace\* therap\*.tw. (10852)
- 12 (Oestrogen\* or estrogen\*).tw. (13738)
- 13 (estrone\* or mestranol\* or quinestrol\*).tw. (735)

**Long-term hormone therapy for perimenopausal and postmenopausal women (Review)**

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- 14 (epimestrol\* or estradiol\* or estriol\*).tw. (9476)
- 15 (oestradiol\* or Oestrone\*).tw. (1433)
- 16 ?estrogen-gest?gen.tw. (36)
- 17 (Cenestin\* or Estrace\* or Estratab\* or Ogen\*).tw. (94)
- 18 (estrofem\* or estrogel\*).tw. (31)
- 19 (Progynova\* or Ovestin\*).tw. (59)
- 20 (Estradot\* or Climara\*).tw. (26)
- 21 exp Estrogens/ad, ae, tu, th [Administration & Dosage, Adverse Effects, Therapeutic Use, Therapy] (87)
- 22 "estrogens, conjugated (usp)"/ or "estrogens, esterified (usp)"/ or estrone/ (1458)
- 23 (Premarin\* or Premique\* or prempo).tw. (235)
- 24 FemHRT.tw. (5)
- 25 (sequential adj1 (therap\* or regimen\*)).tw. (1301)
- 26 (combined adj1 (therap\* or regimen\*)).tw. (6394)
- 27 (continuous adj1 (therap\* or regimen\*)).tw. (662)
- 28 (sequential adj1 (therap\* or regimen\*)).tw. (1301)
- 29 (cyclical adj1 (therap\* or regimen\*)).tw. (42)
- 30 (ethinylestradiol or ethinyloestradiol).tw. (897)
- 31 (EstroGel or evamist or femtrace\* or Femring\*).tw. (15)
- 32 (Divigel\* or Estrasorb\*).tw. (16)
- 33 (Alora\* or Estraderm\* or Vivelle-Dot\* or Combipatch\* or Menostar\*).tw. (121)
- 34 (Cenestin\* or Enjuvia\* or Estratab\* or Menest\*).tw. (15)
- 35 Dienestrol/ (4)
- 36 Dienestrol\*.tw. (3)
- 37 estriol/ or estetrol/ (279)
- 38 or/9-37 (42982)
- 39 (tibolone or tibilone).tw. (547)
- 40 17 hydroxy.tw. (49)
- 41 17 alpha.tw. (250)
- 42 (boltin or livial).tw. (52)
- 43 (org od 14 or org od 4).tw. (29)
- 44 17 beta hydroxy\$.tw. (15)
- 45 or/39-44 (869)
- 46 exp selective estrogen receptor modulators/ or exp raloxifene/ or exp tamoxifen/ or exp toremifene/ (3841)
- 47 selective ?estrogen receptor modulator\*.tw. (489)
- 48 SERM\*.tw. (421)
- 49 bazedoxifene\*.tw. (191)
- 50 (Evista\* or Nolvadex\* or Fareston\*).tw. (159)
- 51 SR-16234\*.tw. (1)
- 52 (clomiphene\* or tamoxifen\* or toremifene\*).tw. (6861)
- 53 (femarelle\* or ormeloxifene\* or lasofoxifene\*).tw. (84)
- 54 (Raloxifene\* or Arzoxifene\*).tw. (981)
- 55 (Ospemifene\* or Levormeloxifene\*).tw. (98)
- 56 or/46-55 (8782)
- 57 38 or 45 or 56 (48839)
- 58 8 and 57 (15033)
- 59 exp BRCA2 Protein/ or exp Genes, BRCA1/ or exp Genes, BRCA2/ or exp BRCA1 Protein/ (353)
- 60 exp receptors, estrogen/ or exp receptors, progesterone/ (1596)
- 61 BRCA\*.ti. (807)
- 62 ?oestrogen\* receptor\*.ti. (124)
- 63 progesterone receptor\*.tw. (1354)
- 64 59 or 60 or 61 or 62 or 63 (3561)
- 65 58 not 64 (14081)
- 66 limit 65 to yr="2022 -Current" (1019)

### Appendix 3. MEDLINE search strategy

Ovid Platform

Searched from 1946 to september 2024

- 1 menopause/ or menopause, premature/ or perimenopause/ or postmenopause/ (59908)
- 2 (postmenopaus\* or menopaus\* or perimenopaus\*).tw. (108638)
- 3 exp Ovariectomy/ (27877)

4 Ovariectom\*.tw. (31584)  
 5 (oophorectom\* or hysterectom\*).tw. (49820)  
 6 (climacteri\* or vasomotor).tw. (19086)  
 7 (cognition or cognitive).tw. (552460)  
 8 or/1-7 (759657)  
 9 exp Hormone Replacement Therapy/ (26853)  
 10 (hormon\* adj1 (therap\* or substitut\* or replace\*)).tw. (48949)  
 11 replace\* therap\*.tw. (67886)  
 12 (Oestrogen\* or estrogen\*).tw. (181027)  
 13 (estrone\* or mestranol\* or quinestrol\*).tw. (10842)  
 14 (epimestrol\* or estradiol\* or estriol\*).tw. (92281)  
 15 (oestradiol\* or Oestrone\*).tw. (14566)  
 16 ?estrogen-gest?gen.tw. (173)  
 17 (Cenestin\* or Estrace\* or Estratab\* or Ogen\*).tw. (1681)  
 18 (estrofem\* or estrogel\*).tw. (23)  
 19 (Progynova\* or Ovestin\*).tw. (87)  
 20 (Estradot\* or Climara\*).tw. (28)  
 21 exp Estrogens/ad, ae, tu, th [Administration & Dosage, Adverse Effects, Therapeutic Use, Therapy] (36000)  
 22 "estrogens, conjugated (usp)"/ or "estrogens, esterified (usp)"/ or estrone/ (13350)  
 23 (Premarin\* or Premique\* or prempo).tw. (564)  
 24 FemHRT.tw. (5)  
 25 (sequential adj1 (therap\* or regimen\*)).tw. (2826)  
 26 (combined adj1 (therap\* or regimen\*)).tw. (24090)  
 27 (continuous adj1 (therap\* or regimen\*)).tw. (1937)  
 28 (sequential adj1 (therap\* or regimen\*)).tw. (2826)  
 29 (cyclical adj1 (therap\* or regimen\*)).tw. (82)  
 30 (ethinylestradiol or ethinyloestradiol).tw. (3772)  
 31 (EstroGel or evamist or femtrace\* or Femring\*).tw. (9)  
 32 (Divigel\* or Estrasorb\*).tw. (13)  
 33 (Alora\* or Estraderm\* or Vivelle-Dot\* or Combipatch\* or Menostar\*).tw. (143)  
 34 (Cenestin\* or Enjuvia\* or Estratab\* or Menest\*).tw. (32)  
 35 Dienestrol/ (188)  
 36 Dienestrol\*.tw. (234)  
 37 estriol/ or estetrol/ (6301)  
 38 or/9-37 (379865)  
 39 (tibolone or tibilone).tw. (1093)  
 40 17 hydroxy.tw. (685)  
 41 17 alpha.tw. (5954)  
 42 (boltin or livial).tw. (66)  
 43 (org od 14 or org od 4).tw. (44)  
 44 17 beta hydroxy\$.tw. (1752)  
 45 or/39-44 (9024)  
 46 exp selective estrogen receptor modulators/ or exp raloxifene/ or exp tamoxifen/ or exp toremifene/ (30145)  
 47 selective ?estrogen receptor modulator\*.tw. (3529)  
 48 SERM\*.tw. (2902)  
 49 bazedoxifene\*.tw. (495)  
 50 (Evista\* or Nolvadex\* or Fareston\*).tw. (223)  
 51 SR-16234\*.tw. (4)  
 52 (clomiphene\* or tamoxifen\* or toremifene\*).tw. (31643)  
 53 (femarelle\* or ormeloxifene\* or lasofoxifene\*).tw. (247)  
 54 (Raloxifene\* or Arzoxifene\*).tw. (3711)  
 55 (Ospemifene\* or Levormeloxifene\*).tw. (271)  
 56 or/46-55 (44470)  
 57 38 or 45 or 56 (408594)  
 58 8 and 57 (76153)  
 59 randomized controlled trial.pt. (622395)  
 60 controlled clinical trial.pt. (95610)  
 61 randomized.ab. (662224)  
 62 randomised.ab. (130002)  
 63 placebo.tw. (259026)  
 64 clinical trials as topic.sh. (203377)  
 65 randomly.ab. (442861)

66 trial.ti. (318868)  
67 (crossover or cross-over or cross over).tw. (103863)  
68 or/59-67 (1717098)  
69 exp animals/ not humans.sh. (5261313)  
70 68 not 69 (1583411)  
71 58 and 70 (12230)  
72 exp BRCA2 Protein/ or exp Genes, BRCA1/ or exp Genes, BRCA2/ or exp BRCA1 Protein/ (15996)  
73 exp receptors, estrogen/ or exp receptors, progesterone/ (61905)  
74 BRCA\*.ti. (12449)  
75 ?oestrogen\* receptor\*.ti. (1973)  
76 progesterone receptor\*.tw. (24939)  
77 or/72-76 (90596)  
78 71 not 77 (11139)

#### Appendix 4. Embase search strategy

Ovid Platform

Searched from 1980 to September 2024

1 exp climacterium/ (9945)  
2 exp "menopause and climacterium"/ or exp menopause/ or exp postmenopause/ or exp premenopause/ (144834)  
3 (postmenopaus\* or post-menopaus\* or post menopaus\*).tw. (106011)  
4 (perimenopaus\* or peri-menopaus\* or peri menopaus\*).tw. (9173)  
5 (climacter\* or menopaus\*).tw. (92342)  
6 exp ovariectomy/ (40185)  
7 Ovariectom\*.tw. (35222)  
8 oophorectom\*.tw. (17223)  
9 vasomotor.tw. (16550)  
10 (cognition or cognitive).tw. (742281)  
11 or/1-9 (262123)  
12 11 and 10 (5185)  
13 11 or 12 (262123)  
14 exp hormone substitution/ (60570)  
15 exp estrogen therapy/ (18812)  
16 (hormon\* adj1 (therap\* or substitut\* or replace\*)).tw. (73090)  
17 replace\* therap\*.tw. (106439)  
18 (Oestrogen\* or estrogen\*).tw. (216459)  
19 (estrone\* or mestranol\* or quinestrol\*).tw. (10147)  
20 (epimestrol\* or estradiol\* or estriol\*).tw. (105842)  
21 ?estrogen-gest?gen.tw. (144)  
22 (Cenestin\* or Estrace\* or Estratab\* or Ogen\*).tw. (3039)  
23 (estrofem\* or estrogel\*).tw. (486)  
24 (Progynova\* or Ovestin\*).tw. (1118)  
25 (Estradot\* or Climara\*).tw. (403)  
26 FemHRT.tw. (67)  
27 (sequential adj1 (therap\* or regimen\*)).tw. (4788)  
28 (combined adj1 (therap\* or regimen\*)).tw. (32971)  
29 (continuous adj1 (therap\* or regimen\*)).tw. (3030)  
30 (sequential adj1 (therap\* or regimen\*)).tw. (4788)  
31 (cyclical adj1 (therap\* or regimen\*)).tw. (107)  
32 (ethinylestradiol or ethinyloestradiol).tw. (4504)  
33 (EstroGel or evamist or femtrace\* or Femring\*).tw. (291)  
34 (Divigel\* or Estrasorb\*).tw. (160)  
35 (Alora\* or Estraderm\* or Vivelle-Dot\* or Combipatch\* or Menostar\*).tw. (1248)  
36 (Cenestin\* or Enjuvia\* or Estratab\* or Menest\*).tw. (185)  
37 exp dienestrol/ (566)  
38 Dienestrol\*.tw. (194)  
39 exp estriol/ or exp estradiol plus estriol/ or exp estradiol plus estriol plus norethisterone acetate/ (6798)  
40 or/14-39 (478530)  
41 (tibolone or tibilone).tw. (1497)  
42 exp tibolone/ (3186)  
43 17 hydroxy.tw. (661)

44 17 alpha.tw. (2144)  
45 (boltin or livial).tw. (450)  
46 (org od 14 or org od 4).tw. (105)  
47 17 beta hydroxy\$.tw. (519)  
48 or/41-47 (6503)  
49 exp selective estrogen receptor modulator/ (9100)  
50 selective ?estrogen receptor modulator\*.tw. (4871)  
51 (SERM or SERMS).tw. (3825)  
52 bazedoxifene\*.tw. (820)  
53 (Evista\* or Nolvadex\* or Fareston\*).tw. (2456)  
54 SR-16234\*.tw. (18)  
55 (clomiphene\* or tamoxifen\* or toremifene\*).tw. (46780)  
56 (femarelle\* or ormeloxifene\* or lasofoxifene\*).tw. (402)  
57 (Raloxifene\* or Arzoxifene\*).tw. (5266)  
58 (Ospemifene\* or Levormeloxifene\*).tw. (403)  
59 or/49-58 (59959)  
60 40 or 48 or 59 (514381)  
61 13 and 60 (96270)  
62 Clinical Trial/ (1080406)  
63 Randomized Controlled Trial/ (837904)  
64 controlled clinical trial/ (473876)  
65 multicenter study/ (403114)  
66 Phase 3 clinical trial/ (77995)  
67 Phase 4 clinical trial/ (7710)  
68 exp randomization/ (100147)  
69 Single Blind Procedure/ (56158)  
70 Double Blind Procedure/ (220409)  
71 Crossover Procedure/ (79401)  
72 Placebo/ (403569)  
73 Randomi?ed controlled trial\$.tw. (357788)  
74 Rct.tw. (59307)  
75 (random\$ adj2 allocat\$).tw. (58292)  
76 Single blind\$.tw. (33603)  
77 Double blind\$.tw. (248410)  
78 ((treble or triple) adj blind\$).tw. (2188)  
79 placebo\$.tw. (379121)  
80 prospective study/ (939509)  
81 or/62-80 (3086662)  
82 case study/ (102849)  
83 case report.tw. (573288)  
84 abstract report/ or letter/ (1297909)  
85 Editorial.pt. (811590)  
86 Letter.pt. (1309583)  
87 Note.pt. (1001691)  
88 or/82-87 (3850268)  
89 81 not 88 (2940599)  
90 61 and 89 (21221)  
91 exp gene mutation/ or exp BRCA1 protein/ or exp BRCA2 protein/ (979567)  
92 exp estrogen receptor/ (114927)  
93 exp progesterone receptor/ (49734)  
94 BRCA\*.ti. (18511)  
95 progesterone receptor\*.tw. (33290)  
96 ?estrogen\* receptor\*.tw. (81472)  
97 or/91-96 (1122503)  
98 90 not 97 (16660)

## Appendix 5. PsyclINFO search strategy

Ovid Platform

Searched from 1806 to September 2024

1 exp menopause/ (4333)

**Long-term hormone therapy for perimenopausal and postmenopausal women (Review)**

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2 (postmenopaus\$ or post-menopaus\$ or post menopaus\$).tw. (3411)  
3 (perimenopaus\$ or peri-menopaus\$ or peri menopaus\$).tw. (916)  
4 (climacter\$ or menopaus\$).tw. (5965)  
5 or/1-4 (7918)  
6 exp hormone therapy/ (2420)  
7 HT.tw. (13517)  
8 exp progestational hormones/ (2687)  
9 Progestin\$.tw. (716)  
10 progestogen.tw. (136)  
11 progesterone\$.tw. (4823)  
12 (medroxyprogesterone acetate or MPA).tw. (825)  
13 dydrogesterone.tw. (10)  
14 (norethisterone or norethindrone).tw. (53)  
15 HRT.tw. (719)  
16 (hormone adj1 therap\$).tw. (1501)  
17 (estrogen\$ or oestrogen\$).tw. (9515)  
18 estradiol\$.tw. (6861)  
19 CEE.tw. (254)  
20 conjugated equine estrogen\$.tw. (85)  
21 premarin.tw. (44)  
22 estriol.tw. (69)  
23 oestradiol.tw. (557)  
24 or/6-23 (31552)  
25 5 and 24 (2714)  
26 (mice or rats or murine).tw. (228676)  
27 25 not 26 (2395)

## Appendix 6. CINAHL search strategy

Searched from 1982 to 5 September 2016 (later CINAHL references are included in the CENTRAL output; 26 September 2024)

Ebsco platform

| #   | Query                                                                                                                                                                                                | Results   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| S38 | S25 AND S37                                                                                                                                                                                          | 3,344     |
| S37 | S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36                                                                                                                            | 1,071,060 |
| S36 | TX allocat* random*                                                                                                                                                                                  | 5,230     |
| S35 | (MH "Quantitative Studies")                                                                                                                                                                          | 14,815    |
| S34 | (MH "Placebos")                                                                                                                                                                                      | 9,793     |
| S33 | TX placebo*                                                                                                                                                                                          | 39,352    |
| S32 | TX random* allocat*                                                                                                                                                                                  | 5,230     |
| S31 | (MH "Random Assignment")                                                                                                                                                                             | 41,527    |
| S30 | TX randomi* control* trial*                                                                                                                                                                          | 109,343   |
| S29 | TX ( (singl* n1 blind*) or (singl* n1 mask*) ) or TX ( (doubl* n1 blind*) or (doubl* n1 mask*) ) or TX ( (tripl* n1 blind*) or (tripl* n1 mask*) ) or TX ( (trebl* n1 blind*) or (trebl* n1 mask*) ) | 848,664   |

(Continued)

|     |                                                                                                                     |         |
|-----|---------------------------------------------------------------------------------------------------------------------|---------|
| S28 | TX clinic* n1 trial*                                                                                                | 189,022 |
| S27 | PT Clinical trial                                                                                                   | 79,712  |
| S26 | (MH "Clinical Trials+")                                                                                             | 202,243 |
| S25 | S6 AND S24                                                                                                          | 9,967   |
| S24 | S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17<br>OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 | 58,243  |
| S23 | TX mestranol or estrone                                                                                             | 242     |
| S22 | TX oestriol                                                                                                         | 15      |
| S21 | TX estriol                                                                                                          | 326     |
| S20 | TX premarin                                                                                                         | 53      |
| S19 | TX conjugated equine estrogen*                                                                                      | 246     |
| S18 | TX CEE                                                                                                              | 812     |
| S17 | TX estradiol                                                                                                        | 4,345   |
| S16 | TX estrogen* or TX oestrogen*                                                                                       | 14,167  |
| S15 | TX hormone N1 therap*                                                                                               | 19,141  |
| S14 | TX norethisterone or TX norethindrone                                                                               | 182     |
| S13 | TX dydrogesterone                                                                                                   | 39      |
| S12 | TX medroxyprogesterone acetate or TX MPA                                                                            | 16,317  |
| S11 | TX progesterone                                                                                                     | 4,342   |
| S10 | TX progestogen                                                                                                      | 459     |
| S9  | (MM "Progestational Hormones+") OR (MM "Medroxyprogesterone Acetate")<br>OR (MM "Estrogens+")                       | 6,252   |
| S8  | TX HT or TX HRT                                                                                                     | 10,069  |
| S7  | (MM "Hormone Replacement Therapy+") OR (MM "Hormone Therapy+")                                                      | 5,753   |
| S6  | S1 OR S2 OR S3 OR S4 OR S5                                                                                          | 25,078  |
| S5  | TX hot flush* or TX hot flash*                                                                                      | 2,572   |
| S4  | TX climacter* or TX menopaus*                                                                                       | 14,243  |
| S3  | TX perimenopaus* or TX peri-menopaus* or TX peri menopaus*                                                          | 3,419   |
| S2  | TX postmenopaus* or TX post-menopaus* or TX post menopaus*                                                          | 13,673  |

(Continued)

|    |                                                                                                                           |        |
|----|---------------------------------------------------------------------------------------------------------------------------|--------|
| S1 | (MM "Climacteric+") OR (MM "Perimenopause") OR (MM "Perimenopausal Symptoms+") OR (MM "Menopause+") OR (MM "Hot Flashes") | 10,339 |
|----|---------------------------------------------------------------------------------------------------------------------------|--------|

## Appendix 7. Data extraction domains

We extracted data from the primary studies for the following study characteristics.

### Trial characteristics

1. Method of randomisation
2. Method of allocation concealment
3. Use of stratification
4. Adequacy of double-blinding (i.e. an explicit statement that therapies could not be distinguished by appearance or administration route, or both)
5. Number of participants screened for eligibility, randomised, analysed, excluded, lost to follow-up or dropped out (i.e. withdrew from the study but were followed up)
6. Level of adherence to therapy
7. Whether an intention-to-treat analysis was done
8. The use of a power calculation to estimate sample size
9. Duration, timing and location of the study
10. Study design (e.g. parallel or cross-over, single-centre or multi-centre)
11. Source of funding

### Characteristics of study participants

1. Inclusion and exclusion criteria
2. Age and any other recorded characteristics of women in the study
3. Menopausal status (i.e. peri- or postmenopausal and how status was defined, surgical or natural menopause) of the women in the study
4. Baseline equality of treatment groups
5. Means of recruitment

## Appendix 8. Trial registry search strategies

Web platforms

Searched 5 September 2022

ClinicalTrials.gov:

hormone and menopausal and random

The World Health Organisation International Trials Registry Platform search portal:

hormone + menopausal

## WHAT'S NEW

| Date             | Event                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 September 2026 | Amended                                                | Correction of data error affecting two analyses: 1.12: Death from breast cancer, oestrogen-only hormone therapy, and analysis 1.13 Death from breast cancer, combined continuous hormone therapy. Data reported was yearly mortality and not breast cancer mortality at the specific follow up time. These outcomes are not reported in summary of findings tables; analyses and main text have been edited. |
| 4 September 2026 | New citation required but conclusions have not changed | conclusions not changed with this amendment                                                                                                                                                                                                                                                                                                                                                                  |

## HISTORY

Protocol first published: Issue 2, 2003

Review first published: Issue 3, 2005

| Date             | Event                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 November 2025 | New citation required and conclusions have changed | We added two new studies that did not change the conclusions, but the certainty was revised and conclusions changed slightly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 27 November 2025 | New search has been performed                      | Added two new studies (Ran 2017; REPLENISH 2018) and 30 new publications of already included studies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6 January 2017   | New search has been performed                      | <p><b>New included studies:</b></p> <ul style="list-style-type: none"> <li>• <a href="#">KEEPS 2012</a></li> <li>• <a href="#">ELITE 2014</a></li> </ul> <p><b>New excluded studies:</b></p> <ul style="list-style-type: none"> <li>• <a href="#">AHT 2015</a>; <a href="#">Paoletti 2015</a>; <a href="#">Rasgon 2014</a> <a href="#">Schierbeck 2012</a> <a href="#">Mirkin 2016</a> (new exclusions)</li> <li>• <a href="#">Haines 2003</a>; <a href="#">Nielsen 2006</a> (previously included, as quality of life is no longer a review outcome)</li> <li>• <a href="#">Pefanco 2007</a> (previously included, now excluded, as specific (not global) measures of cognition are no longer a review outcome)</li> </ul> <p><b>WHI changes:</b></p> <ol style="list-style-type: none"> <li>1. Updated <a href="#">WHI 1998</a> to include data from follow-up publications</li> <li>2. Mean intervention period of combined <a href="#">WHI 1998</a> corrected from 5.2 years to 5.6 years (corresponding with the date when women were instructed to stop taking the medication) (see <a href="#">Chlebowski 2003</a>)</li> </ol> |
| 6 January 2017   | New citation required and conclusions have changed | The addition of two new studies and changes in review outcomes have led to changes in the conclusions of this review.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 31 October 2012  | Amended                                            | <p>Correction to 'Characteristics of included studies' table and description of studies, related to sample numbers for <a href="#">ESPRIT 2002</a> and <a href="#">HERS 1998</a></p> <p>No change to data or analyses</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3 February 2012  | New citation required and conclusions have changed | <p>Search updated to February 2012</p> <p>New studies: <a href="#">Greenspan 2005</a> <a href="#">Nielsen 2006</a> <a href="#">Pefanco 2007</a> <a href="#">Tierney 2009</a></p> <p>Updated <a href="#">WHI 1998</a> for the following outcomes: lung cancer (<a href="#">Chlebowski 2009</a>, <a href="#">Chlebowski 2010ca</a>), breast cancer (<a href="#">Chlebowski 2010c</a>), cognition in older women (<a href="#">Espeland 2010</a>; <a href="#">Resnick 2009</a>), colorectal cancer (<a href="#">Prentice 2009ba</a>; <a href="#">Ritenbaugh 2008</a>), clinical outcomes at 7.1 and 10.7 years in oestrogen-only HT arm (<a href="#">LaCroix 2011</a>) and at 7.9 years in combined arm (<a href="#">Heiss 2008</a>)</p> <p>Updated EPHT for quality of life (<a href="#">Veerus 2008</a>)</p>                                                                                                                                                                                                                                                                                                                           |

| Date             | Event                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                                        | Updated WISDOM for quality of life (Welton 2008)<br><br>Reclassified equivalent HT doses in line with Australian Menopause Society recommendations: amended table of comparisons for studies using oral oestradiol (n = 10)<br><br>Added summary of findings table and absolute risks, updated Discussion and Conclusions sections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 11 February 2009 | New citation required but conclusions have not changed | Review updated June 2008                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 31 May 2008      | New search has been performed                          | Added the following new studies:<br><br><a href="#">Barakat 2006</a> , <a href="#">EPHT 2006</a> , <a href="#">WISDOM 2007</a> , <a href="#">Yaffe 2006</a> .<br><br><a href="#">WHI 1998</a> oestrogen-only arm, data added:<br><br>Venous thromboembolism at 2 and 7.1 years' follow-up<br><br>Coronary heart disease, venous thromboembolism, stroke, breast cancer, fracture and quality of life at 7.1 years' follow-up<br><br><a href="#">WHI 1998</a> combined arm, data added:<br><br>Subgroup analysis of breast cancer risk by prior hormone summarised in the text (Discussion section)<br><br>Data on main outcomes after 3 years post intervention (Discussion section)<br><br>Results from <a href="#">WHI 1998</a> (WHISCA) on specific cognitive functions in older women added to the text<br><br>No substantial changes to overall findings of this review. Statistically significant risk of venous thromboembolism for WHI oestrogen-only arm now evident at follow-up "up to 2 years" |
| 18 May 2008      | Amended                                                | Converted to new review format                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 20 November 2003 | New citation required and conclusions have changed     | Substantive amendments made                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## CONTRIBUTIONS OF AUTHORS

For the 2023 update, five authors (AL, MB, CB, SM and CF) screened retrieved records for new studies. LNY extracted data and drafted the text. MB extracted and entered data and drafted the text. All authors commented on the final version.

For the 2017 update of the review, Jane Marjoribanks (JM) extracted and entered data and drafted the text, and Jasmine Lee (JL) checked study selection and data extraction. Helen Roberts (HR), Cindy Farquhar (CF) and Anne Lethaby (AL) commented on and contributed to the drafts.

For the 2008 and 2012 updates of the review, JM extracted and entered data and drafted the text, and CF checked study selection and data extraction. HR, CF and AL commented on and contributed to the drafts.

CF and AL developed the original protocol and circulated it to members of the Cochrane HT Study Group for comment. The following people contributed specifically to the protocol: Professor Shah Ebrahim, Dr Peter Tugwell, Teresa Moore and Maria Judd. For the original version of the review, JM and Jane Suckling searched for relevant studies and selected studies for inclusion, and JM extracted and entered data that were checked by Quirine Lamberts. JM drafted the review, circulated it to other members of the Cochrane HRT Study Group for comment and edited the draft.

The following individuals commented on the draft of the original review: Breast Cancer Group: Sue Carrick, Sue Lockwood (Editor); Dementia and Cognitive Improvement Group: Professor Leon Flicker (Editor), Professor Lon Schneider (Editor); Heart Group: Lee Hooper (Editor), Theresa Moore (Review Group Coordinator); Gynaecology and Fertility Group: Cindy Farquhar (Co-ordinating Editor), Anne Lethaby (Editor); Stroke Group: Professor Ale Agra (Editor), Steff Lewis (Statistical Editor).

## DECLARATIONS OF INTEREST

Cindy Farquhar is an editor of the Cochrane Gynaecology and Fertility Group and declares that she has taken no part in the editorial process for this update.

Sanja Mirkov has declared two fiduciary officer roles (unpaid voluntary work) with the Society of Hospital Pharmacists of Australia. She receives royalties for her contribution to a pharmacology textbook: *McKenna's Pharmacology for Nursing and Other Health Professionals* 3rd edition, 2021. She has received payments for editorial work and seminars.

MB, CB, LNY and AL have no interests to declare.

## SOURCES OF SUPPORT

### Internal sources

- None, Other

None

### External sources

- None, Other

None

## DIFFERENCES BETWEEN PROTOCOL AND REVIEW

For the 2017 update, we decided to omit quality of life as an outcome and to focus on adverse events only, to make the review more concise in terms of the long-term risks of hormone therapy. Quality of life is more likely to be affected during the period that hormone therapy is taken. This meant that the 2017 version excluded two previously included studies ([Haines 2003](#); [Nielsen 2006](#)).

We also decided in the 2017 update to limit the outcome of 'cognitive function' to studies using global measures of cognition, with the same motivation to keep the review as concise as possible. This did not change any of our data.

For the 2017 update, given that some studies have over 100 publications, we did not include further publications of included studies that did not report outcomes included in the review. Also, we did not include data from studies where there were no events in either group for a particular outcome in the meta-analysis for that outcome because they did not add useful data ([Higgins 2017](#)).